Carrier, Fiber, Adapters, and Functoriality: A Novel Geometric Transport Framework
From Repairing Beyond Endoscopy To Universality

Samuel Lavery
sam.lavery@gmail.com

Preprint draft v0.2 — July 14, 2026

Abstract

The classical L-function is not the object of study; it is the shadow — the one-dimensional readout of a three-dimensional object, cast through a coordinate system, the unit-1 chart, that is cohomologically sub-optimal. The thesis of this paper is that a harmonically scaled three-dimensional carrier is the more accurate frame, and working in it is the basis for everything that follows. We construct the frame: a 3D state space containing a double-ended helix and its conjugate anti-helix — Archimedean, constant pitch one unit — with two harmonically scaled carriers placed at the origin and running to infinity. Automorphic functions are represented on the carriers as fibers, analytically growing phasors of unique magnitude and rotation from the origin upward; for a Dirichlet L-function the character sorts the integers into channels of phasors as the fiber crosses each unit. At certain heights the helix and anti-helix fibers align in exact focal cancellation — vanishing — observed by a Gram–von Neumann/Gram harmonic pencil operator as two conjugate eigenstates in Frobenius determinant-one correspondence, and the physical height $Z > 0$ of the event is marked. Its analytic ordinate is $y = \log Z$: the completed fibre crossing is exactly $L(\sigma_* + i \log Z, \chi) = 0$, while the finite phasor-bank event is its truncation-level detector. For each fixed nondegenerate geometric carrier (p, r) with $p \neq 0$, Lean first defines the ambient state type independently of all event certificates: $\mathcal{C}_{p,r} = \{(X, y) : X = \gamma_{p,r}(y)\}$, $\mathcal{H}^{(3)}_{p,r} = \mathcal{C}_{p,r} \to_0 \mathbb{C}$, and $A^{(3)}_{p,r} = \mathrm{diag}(y)$. This operator is symmetric and every geometric carrier basis state is an explicit eigenvector. A completed event at height Z_e is then embedded into the already-defined state at $y = \log Z_e$, whose third spatial coordinate is proved to be Z_e; the event selects an eigenstate but does not define the operator. Its finite resolvent has two compiled readings: 3D zeros with 3D eigenvectors at carrier-point poles, and analytic 1D zero readouts with the same 3D zeros and eigenvectors at ordinate poles. The two traces agree under the carrier chart, including its factor i. Projecting the event down — Möbius/Cayley to the 2D unit circle with the radius booked in a loss ledger, then to 1D with the angle booked and the logarithm applied — recovers the classical zero at $y = \log Z$, to very high precision: the 3D space is vastly larger and more regular than its compressed chart.

The scaled frame explains the classical one. We prove, unconditionally, that the $S(t)$ oscillatory term is the global coordinate/registration map: the gap between the $\pi/3$-scaled carrier's registration and the unit-1 chart's — the first proof and mechanical description of why $S(t)$ exists and is needed: a chart-defect compensation mechanism. The GRH worked example gives two retained conditional deductions and audits their exact logical inputs. The carrier midpoint is derived from the Archimedean area law and no-drift geometry; finite focal cancellation is equivalent to finite harmonic-pencil rank drop, while completed fibre cancellation is exactly the represented L-zero. Identifying every analytic zero with a native event remains a separate premise; its sharpest form is a single function-level identity — agreement of the

two charts' integrated registrations $\int_0^T S$ at every height — which the proven integrality and positivity of the registration defect convert into the per-zero identification, zeros arriving as corollary rather than premise (pointwise coverage and fixed-operator spectral capture are its event-level and operator-level faces).

The same machinery extends, and its extension is the paper's Langlands content: the carrier is a *common representation* — every admissible automorphic fiber normalizes onto one shared harmonic carrier, and *functoriality is the coherence of that normalization* under source transports, a structural claim that is unconditional and machine-checked. We prove symmetric-power functoriality on GL(2) data at every rank through the Cogdell–Piatetski-Shapiro converse checklist, with niceness discharged on the carrier; extend the model beyond GL(2) through configuration sets of adapters and clocks applied in projection, harmonizing one or several functions onto common harmonic cells with exact closure at vanishing; prove Sato–Tate for Maass forms; give an adaptive cohomological obstruction detector with re-harmonization that dynamically minimizes the obstruction, and a projection into a Hodge space that detects obstructions hidden there (preliminary); and assemble the pieces toward a universal Langlands-style transport. Returning to the origin of the program, we analyze Altuğ's Beyond-Endoscopy uniformity problem and, with clock-controlled adapters, propose a repaired version that scales.

The diversity of these claims is rare for one paper, and we say so plainly: the work began as an effort to repair a proof in the Langlands program, and a repair made at the level of the coordinate system propagates — discovering intrinsic unification across unexpected functional systems is what such a repair predicts, and not finding it would be the suspicious outcome. The results are backed throughout by Lean machine-checked proofs and empirical corroboration, both provided.

Draft status. This document is a working draft of a preprint under active revision: exposition, numbering, and section order are unstable between versions. Version v0.2, July 14, 2026 (adds the compiled Rankin–Selberg cancellation at $r = 2$, §30.1); changes from this version forward are tracked in the source repository. The mathematical claims are backed by Lean 4 machine-checked proofs (axiom footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`) and by the empirical artifacts of the reproducibility appendix (§B); both ship with the paper.

Contents

Introduction

The classical L-function is a one-dimensional readout of a three-dimensional object. That sentence is the paper's thesis, and everything here either constructs the object, proves that the readout is its faithful but compressed chart, or collects what becomes provable once the object — not the chart —

is taken as primary. The working claim, stated once and defended throughout, is that the unit-1 coordinate system of the classical chart is cohomologically sub-optimal: a harmonically scaled three-dimensional carrier is the more accurate frame, and several recurring difficulties of the chart — the $S(t)$ oscillation, the $\operatorname{Re} s > 1$ convergence gate, the uniformity problems of trace-formula analysis — are artifacts of the compression, not properties of the object.

The object is built from first principles in Part I: a 3D state space carrying a double-ended helix and its conjugate anti-helix — Archimedean, constant pitch one unit — with two harmonically scaled carriers running from the origin to infinity, and, riding them, *fibers*: automorphic functions represented as banks of phasors of unique magnitude and rotation, growing continuously in height. Vanishing is exact focal cancellation of the helix and anti-helix fibers, observed as two conjugate eigenstates of a Gram–von Neumann/harmonic-pencil pair in Frobenius determinant-one correspondence; the classical zero is recovered by the ledgered projection — Möbius/Cayley to the unit circle with the radius booked, then to the line with the angle booked and the logarithm applied — at $y = \log Z$, where $Z > 0$ is physical helix height. Projection with its ledger is a bijection (Theorem 3); without it, a compression.

One consequence organizes everything below and is the paper's Langlands content, so we name it first: the carrier is not built per L-function — it is a *common representation*. Heterogeneous automorphic objects (different groups, ranks, conductors, and Satake data) each lift to a fiber, and a per-source normalization scales all of them onto *one* shared harmonic carrier at a common scale — the least common refinement of their native cells (§5, Corollary 80) — where they are simultaneously realized and directly comparable. *Functoriality is then a coherence law, not an analogy*: a structural transport between two automorphic worlds descends to a transport of their common-carrier representations, and these compose — identity, associativity, base change — preserving exact focal closure (Theorem 78; `CarrierFunctoriality.faithful_comp`, no axioms). Symmetric powers, Rankin–Selberg, base change, and theta are each one instance of this single common-representation transport. This structural core — normalization onto the common refinement, closure transfer, and coherent transport — is unconditional and machine-checked (`RequestProject/CommonRepresentation.lean`); the arithmetic that lands a *particular* fiber as automorphic (harmonization of its cells, Lemma 65, and the completeness that makes its readout entire, Lemma 64) is the per-fiber input the converse theorem consumes.

The main results, each at its stated register:

1. *The common carrier system* (Part I): the source-independent carrier; ledgered projection with exact round trip; admissible sources synthesized from finite structural data alone; native cells and clocks; the mode bridge (the analytic residual is the pencil residual, Theorem 16); event exhaustion and the universal transport theorem (Theorems 26 and 79).
2. *The $S(t)$ mechanics* (§9): unconditionally, $N_{\pi/3}(e^t) - R_1(e^t) = S_{\mathrm{ev}}(t)$: a cardinal-valued native event count minus a real-valued unit phase registration. For the multiplicity ledger the exact clock–jump law is $\Delta S_{\mathrm{mult}} = M(a,b) - \pi^{-1}\int_a^b \mathrm{clockRate}$, with the corresponding finite-window bound, zero-free derivative rule, and residue-equals-jump theorem all compiled (Lean, `CarrierScaleCompensation`, `StClockJumpLaw`, `ResidueJump`). The classical strip term additionally contains excess line multiplicity and off-line strip multiplicity, stated separately.
3. *The worked example* (§20, condensed): the basic configuration — the $\pi/3$ carrier with a Dirichlet fiber — built from scratch, exercising every mechanism the rest of the paper generalizes; the midpoint arrives as a geometric output (area-law scale balance and no-drift), never as a stipulated decimal. The full development — the carrier geometry evidence, the Gram lifts, the ambient operator, and the two conditional GRH routes with their compiler-audited premises — is the companion paper [31].

4. *Symmetric-power functoriality* (Part II): $\mathrm{Sym}^r\,\pi$ automorphic on $\mathrm{GL}(r+1)$ at every rank through the Cogdell–Piatetski-Shapiro checklist with niceness discharged on the carrier (Theorem 48, Corollary 49) — recovering the known cases for $r \le 4$ and continuing past the Langlands–Shahidi range, Galois-free.
5. *Sato–Tate for Maass forms* (Corollary 53): a consequence of the Galois-free route, beyond the reach of automorphy lifting.
6. *Ramanujan–Petersson and Selberg* (§18, Corollary 51): the tower closes the loss ledger's radial channel — strand radius exactly one at every unramified place, zero radial drift at the archimedean place ($\lambda \ge \frac{1}{4}$ for Maass π); the limit step machine-checked (`RamanujanLimit`).
7. *Beyond Endoscopy, analyzed and repaired* (Part on Beyond Endoscopy; §A): Altuğ's archimedean uniformity reduced to a single magnitude bound by an exact gauge, with no oscillation estimate; the original obstruction identified as a deterministic clock intrinsic to the orbital transform; a clock-controlled repair that scales (§23).
8. *Extensions and the universal frame* (§32 onward): the representation-agnostic transport — the carrier consumes any admissible function directly, with no local-Langlands datum — exotic adapters, obstruction-directed harmonization (§34), and the Hodge connection (§37).

Three registers are used and never blended. *Proven*: machine-checked in Lean 4, every named theorem at axiom footprint {`propext`, `Classical.choice`, `Quot.sound`} with no `sorry`, built in-tree (§B). *Measured*: precision-tracking numerics — calibration and independent reproduction of proven statements, never their proof. *Conditional*: the GRH interfaces are displayed with their exact premises in the companion paper [31], and the compiler audit identifies which premise supplies the critical-line conclusion. No claim in this paper exceeds its register.

The diversity of the results is unusual for one paper, and the explanation is the origin: the work began as an effort to repair a proof in the Langlands program — Altuğ's Beyond Endoscopy for GL(2) — and a repair made at the level of the coordinate system propagates. A reader wanting the shortest self-contained path should read the condensed worked example (§20) first: it builds the basic configuration from scratch, names the two decisions, and exercises every mechanism the rest of the paper generalizes — with the full development and the two conditional GRH routes in the companion paper [31]. A running ledger of contributions and a glossary of the paper's vocabulary follow immediately.

Our contributions

The ledger of results toward open problems, kept current as they land. Anchors: [Lean] — machine-checked, kernel footprint exactly {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`; [theorem] — proved in the text from Lean-checked and cited classical inputs; [decided] — a named definitional reading, deferred to the community.

1. **Symmetric-power functoriality** — open for Maass forms at $r \ge 5$ (holomorphic forms: Newton–Thorne [18]; $r \le 4$: Gelbart–Jacquet [6], Kim–Shahidi [17]). Established here at *every* rank — cuspidal in the generic case, Galois-free, Maass included: Thm. 48, Cor. 49, with niceness discharged on the carrier (Prop. 76, Lem. 70, Lem. 68, Lem. 64; `FiniteWeightFiber.symTensorCompleted_FE`, `TransferContinuation.transfer_analytic`, `HarmonicCell.harmonic_residual_no_independent_mode`) — and, ungated by any per-fiber cell-closure input, for the twisted bank itself: entirety, vertical-strip bounds, and the functional equation for *every* CPS twist degree and *every* unitary Satake family, hypothesis-free (`GlobalHelix.cpsAllTwists_unconditional3DAnalyticPayload`, `GlobalHelix.cpsDualPair_twistedNiceness`, coefficient bound `GlobalHelix.radialGlobalSatakeCoeff_no`

with the finite→infinite Euler limit and its infinite-bank summability, exact reflection, and rapid decay compiled (`GlobalHelix.symTwistEulerLimit_eq_div`, `GlobalHelix.summable_fullTensorEuler_zero` `GlobalHelix.fullTensorEulerZeroModeDualTheta_eq_div`); a general polynomially-bounded datum passes through the same constructor, `CarrierTheta.automaticCoefficientThetaStrongFEPair`). [theorem, on Lean carrier inputs]

2. **Universal carrier transport and functoriality over the admissible set** — the general transfer question (which L-functions come from automorphic sources), beyond any single tower. Established as a *universal machine* on the *admissible* fibers — those given by a finite structural presentation (a finite local-parameter rule, bounded local-factor degree, an Euler assembly, a convergence half-plane); a structureless ("random") function is *excluded by definition*, having no finite law from which to synthesize a phase, cell, or clock, so it is outside the domain, not a counterexample. For *every* admissible fiber the carrier is a faithful transport (Thm. 78, Thm. 79): the readout, the reflection $\Lambda(s; W) = \varepsilon\, \Lambda(1 - s; W^\vee)$, and the local factors and root number are *outputs*, unconditional (`FiniteWeightFiber.symTensorCompleted_FE`, `ConeProjection.record_bijective`, Thm. 57), and the transport composes faithfully (`CarrierFunctoriality.faithful_comp`, with associativity, identity, and base change, no axioms). The one per-fiber obligation — native-cell closure at the fiber's own harmonic scale — has a *proven existence backbone*: the focal residual is slaved to the readout with no independent DC mode (Lem. 63) and is forcible to zero on the two independent lanes of any nontrivial clock (`ForcibleClosure.residual_forcible`, `CarrierReachability.reachable_independent_stage`) — the arithmetic-neutral *transport* half, and transport only; the *niceness* is the carrier's own and is proven, not gated: admissibility already supplies polynomial coefficient bounds (the convergence half-plane), and the constructor then delivers entirety, vertical-strip bounds, and the functional equation hypothesis-free (`CarrierTheta.automaticCoefficientThetaStrongFEPair`; for the twisted bank, `GlobalHelix.cpsAllTwists` item 1); the nice L-function, fed to the converse theorem (Thm. 45), lands the fiber as automorphic, the symmetric-power tower (item 1) being the fully-executed instance. *Stated honestly*: that closure always *exists* and that the random impostors are excluded are both *proven*; that a *bounded, non-oracular* adapter is read off *every* exotic finite law (`Synth` totality) is a well-founded, so-far-unfalsified *program*, not a monolithic theorem — its single clean falsifier a *structured* admissible fiber closeable by no adapter, existing or new. [Lean backbone + theorem; universality a falsifiable program]

3. **The Ramanujan–Petersson theorem for the Maass forms of Corollary 53.** Established here as the tower's radial limit: the strand radius is exactly one at every unramified place — the radial ledger channel is trivial, the strand block an isometry of the transverse plane (§18, Cor. 51; `RamanujanLimit.strand_radius_one_of_tower_ceiling`, `strandBlock_unitary_of_tower_ceiling`); on the ledger-normalized CPS phase itself the complete two-sided ceiling and the resulting block isometry are also recorded by `GlobalHelix.unitaryPrimePhase_towerCeiling` and `GlobalHelix.unitaryPrimePhase_strandBlock_isometry`. The radial capstone itself no longer takes that unitary phase structure as input: it starts from a nonzero arithmetic strand and the two rank-uniform tower ceilings, packaged as `GlobalHelix.TowerCeilingStrand`; it returns radius one as its first conclusion and only then returns the block isometry (`cpsRadialSatoTateGeometry3D_unified`). The ceilings' supply line is priced exactly and *non-circular*: the per-rung Jacquet–Shalika ceiling ($\|\alpha_p\| \le p^{1/2}$, unconditional, no Ramanujan) consumes the tower, whose machine-checked niceness payload holds over *arbitrary nonzero radius*. The bank structure `PolynomialSatakeDualPair` carries a polynomial radial bound $\|\alpha_p\| \le p^e$ — never $\|\alpha_p\| = 1$ — and `cpsArithmeticTwist_radiusLive_niceness` (via `cpsPolynomialAllTwists_payload` over `arithmeticCPSPolynomialTwist`) proves the reciprocal-height reflection, entirety, strip bounds, and functional equation on it *with no temperedness assumed*,

for any base π : `PolynomialSatakeDualPair` (Fin 2) (holomorphic or Maass). Unitarity is not packaged in the bank; it is the radial-limit *output* (`RamanujanLimit.strand_radius_one_of_tower_ceiling` takes an arbitrary ceiling and returns radius one). So the non-tempered start is *not* a remaining construction: the niceness is radius-live and machine-checked, the ceiling is Jacquet–Shalika, and $\|\alpha_p\| = 1$ falls out only at the capstone. The unit-modulus payload `cpsAllTwists_unconditional3DAnalyticPayload` is a specialization, not the locus of generality. [theorem; radius-live niceness and limit step Lean]

4. **The Selberg bound $\lambda \geq \frac{1}{4}$** for the same carrier family. The same tower limit at the archimedean place gives zero radial drift and a unit-modulus strand clock at every carrier height (Cor. 51; `RamanujanLimit.archimedean_strand_unimodular_of_tower_ceiling`). [theorem; limit step Lean]
5. **Sato–Tate for Maass forms.** A corollary of the Galois-free tower: Cor. 53. Its native carrier geometry and moment law are machine-checked in `CPSRadialSatoTateGeometry3D`: the Sato–Tate density is pushed onto the helix $(\cos\theta, \sin\theta, \theta)$, the third coordinate $S(t)$ recovers the measure exactly, and the transverse trace has the Catalan/Fourier moments; the same theorem first derives the arithmetic strand's unit radius from the tower ceiling rather than accepting a unitary input (`cpsRadialSatoTateGeometry3D_unified`). The arithmetic identification is also exact: `ArithmeticSatakePrimeFamily.character_eq_satakeTrace` identifies each carrier character with the literal symmetric-power Satake root sum, and `primeTestAverage_eq_arithmetic` identifies their empirical averages term-by-term; the prime-cancellation theorem supplied by Jacquet–Shalika plus Wiener–Ikehara remains cited. [theorem; arithmetic identification and 3D geometry Lean; analytic prime cancellation cited]
6. **GRH/RH interfaces** — the two legacy conditional deductions are retained as two proof routes under the identity and correlation readings, and are compiler-audited. The midpoint is an area-law/no-drift output; finite focal cancellation is equivalent to harmonic rank drop; the completed crossing at physical height Z is exactly the analytic L-zero at ordinate $y = \log Z$; the three-coordinate Gram lifts preserve and reflect rank drop; the event-independent ambient carrier operator supplies explicit 3D eigenvectors and two correlated resolvent traces; and $S(t)$ is the global carrier-to-chart coordinate registration. The old 3D identity premise already includes the critical-line conclusion, while the old 1D routing disjunction is unconditional and its separate chart-to-kernel premise supplies that conclusion. The genuine global operator route is the fixed self-adjoint resolvent-capture interface `grh_of_selfAdjoint_resolvent_capture`, developed in full in the companion paper [31]. [Lean]
7. **Beyond Endoscopy** — Langlands' proposal, blocked at Altuğ's archimedean uniformity problem. Root-caused and repaired: the obstruction is a deterministic clock intrinsic to the orbital transform, removed by an exact gauge with no oscillation estimate, and the repair scales (§A, §23). [theorem + measured]
8. **The obstruction finder** — an adaptive detector that locates exactly where a computation resists: the residual names its cell and the cell names its adapter (the Sym^{13} rank-deficient cell naming Θ; Altuğ's obstruction named as a deterministic clock of the orbital transform), and re-harmonization removes the obstruction or retains it with its certificate — never zeroed by hand (Thm. 15, Lem. 65, §34; `ClockStructure.obstruction_recovery`, `ForcibleClosure.residual_forcible`). [Lean core + measured instances]
9. **The $S(t)$ mechanism** — the oscillatory term, resistant to interpretation since Riemann–von Mangoldt. Proven unconditionally for the *line ledgers*: the distinct-event compensation identity, the exact multiplicity clock–jump law, its finite-window and zero-free bounds, and residue equals jump (Thm. 30, Eqns. (6)–(7); `CarrierScaleCompensation`, `StClockJumpLaw`, `ResidueJump`). The

unit object is typed as continuous phase registration, not as a cardinality; the classical strip correction is recorded separately. [Lean]

Glossary of terms

A reader's map of the vocabulary; each term is defined where first cited, and their Lean mappings.

term	meaning (and where fixed)
carrier	the layer between the base helix geometry and the fiber representations. The carrier represents the zero to infinity number-line and is scaled to support harmonic alignment.
carrier unit scale	in the 1D chart readout the carrier unit is the value one, in 3D state space each unit is scaled by a constant harmonic value, generally $\frac{\pi}{3}$
double-helix	the base geometry — it sets the functional equation — $C_+ \sqcup C_-$ with its involution J (§30, Thm 69).
fiber	the function itself: its local Satake/Weil–Deligne data and coefficients, ridden on the carrier (`FiniteWeightFiber`).
bank	the summed phasors (positive/neutral/negative) of a fiber on the carrier; unique spin and unique magnitudes; its 1D counter-part is the infinite L-series of phasors (§23).
warp	a function based, dynamic readout-preserving unit-modulus reparametrization that extends or streches the carrier, enabling closing cells of complex function representations (fibers) (`warpFiber`, `ForcibleClosure`). A warp scales by a *function*; the reciprocal of a warp is again a warp, so there is no separate "inverse warp" — removing a warp from a readout is the kernel-transfer operation (`WarpRemovalTransfer`).
carrier scale	scaling by a harmonic *constant*: the invertible constant reparametrization on the object's own integer lattice — not a warp (`Admissible.scale`, `CommonRepresentation`); fibers sometimes carry their own constant scale (fiber scaling).
clock	a deterministic oscillation the carrier carries: self-duality (μ_2), completion (two-clock), ramification, vanishing-adjustment (§32; `TwoClockWeightLaw`, `ClockStructure`).
weld	the origin where the helix/anti-helix crossing at $\operatorname{Re} s = \frac{1}{2}$, where the two lanes meet with det-one Frobenius similitude (`FrobeniusSimilitude`).
dwell	opposite of the warp, places multiple crossings into single carrier unit, carrier adapter re-registering the values a projection skips — the native form of Altuğ's $\Sigma()$ (§22, §32).
ledger	the determinant/loss bookkeeping that keeps the transport faithful (det-one balance, Lemma 66; rank/jet, `BSDClocks`). The governing principle: *phase and radius may decouple dynamically, but not informationally* — channels evolve independently; the ledger books both.
focal cancellation	a zero: the non-DC bank balances to zero centroid (`HarmonicCell`, `ForcibleClosure`; §32).
harmonization	the method — read a projected function, let its nature name its settings: scale/warp, adapters, and clocks such that the carrier + fiber ensemble closes without residue (§32).

Part I

The common carrier system

1 The theorem architecture

The first object of the paper is not a special-purpose one-dimensional function. It is a normalization-and-carrier system for heterogeneous structured sources. A source may later be an Euler datum, an automorphic family, a graded extension object, or another structured family of functions; the carrier system is the common state space into which such sources are realized.

Theorem 1 (Universal carrier theorem, architectural form)**.** *Every finite compatible family of admissible structured sources admits a source-faithful normalization into a common double-ended three-dimensional carrier. The realized fibers retain their identities and projection losses, evolve according to their native clocks and harmonic cells, and admit exact complete-cell closure. The complete state detects retained obstructions and supports admissible obstruction-directed adaptation. Ledgered projection gives faithful two- and one-dimensional readouts; carrier reflection intertwines with reflected functional readout; represented source events are exhaustive; and coherent structural transports induce compositional transports of the complete carrier states.*

Equivalently, the construction is meant to take heterogeneous structured functions or families of functions, normalize their native scales into a common three-dimensional carrier state, allow them to evolve coherently, retain projection loss, detect and where admissible remove obstructions, adapt the realization, and transport structural relationships functorially to their lower-dimensional readouts:

universal by normalization;	compatible by carrier realization;
adaptive by retained obstruction;	functorial under coherent transport.

Proof (by assembly; the parts are proven in the named sections). The carrier and its involution are constructed source-free in §2 (Theorem 2). Ledgered projection with exact recovery, and the separation of projected equality from state equality, are Theorem 3 and Corollary 4 in §3. Represented source identity, the specified dual, and the synthesis of native fibers, clocks, cells, and adapters from a finite structural presentation are Theorem 7 and Corollary 8 in §4 (with the admissibility boundary Proposition 6). Normalization and simultaneous realization of finite compatible families, with retained native clocks and ledgers, are Theorems 9 and 10 in §5. Exact complete-cell closure, the bounded primitive, forcible closure, and obstruction-directed adaptation are Theorems 11, 12, 14, and 15 in §6. The mode dictionary, the readout identity, completed reflection, and analytic landing are Theorems 16, 18, 19, and 23 in §7. Event exhaustion and coherent functorial transport are Theorems 26, 27, and 28 in §8, whose assembly corollary (Corollary 29) concludes the statement. □

Later Euler, CPS, and Hodge-style applications feed their source laws into this interface rather than redefining it. A theorem labeled unconditional records all hypotheses it uses.

2 The double-ended carrier

The carrier is fixed before any source is attached. It is a double-ended three-dimensional geometric state space

$$C = C_+ \sqcup C_-,$$

equipped with scale, winding law, height coordinate, orientation, and a helix/anti-helix involution

$$J : C_+ \leftrightarrow C_-, \qquad J^2 = \mathrm{id}\,.$$

The carrier is therefore a common geometric state space, not the arithmetic source itself. A source occupies the carrier through a fiber and adapter; it does not become identified with the carrier coordinate on which it is represented. The task of this section is exactly this: construct C, construct J, and verify $J^2 = \mathrm{id}$.

Scope. This section proves the primitive carrier and its global involution. The resulting carrier datum is the input used for admissible source attachment (§4), for ledgered projection (§3), and for completed readout reflection (§7).

Theorem 2 (Unconditional source-independent double carrier)**.** *For every carrier scale $H > 0$, let $T_H = \mathbb{R}_{\geq 0}$ be the raw carrier parameter set. Equip it with source-independent carrier laws*

$$\Theta_H : T_H \longrightarrow \mathbb{R}, \qquad R_H : T_H \longrightarrow \mathbb{R}_{\geq 0}, \qquad Z_H : T_H \longrightarrow \mathbb{R},$$

where Θ_H is the winding law, R_H the radius law, and Z_H the height law. Define two signed carrier lanes

$$C_{H,+} := \{+\} \times T_H, \qquad C_{H,-} := \{-\} \times T_H, \qquad C_H := C_{H,+} \sqcup C_{H,-}.$$

Their three-dimensional geometric realizations are

$$\begin{aligned}\Gamma_{H,+}(t) &= \big(R_H(t)\cos\Theta_H(t),\ R_H(t)\sin\Theta_H(t),\ Z_H(t)\big),\\ \Gamma_{H,-}(t) &= \big(R_H(t)\cos(-\Theta_H(t)),\ R_H(t)\sin(-\Theta_H(t)),\ Z_H(t)\big).\end{aligned}$$

Then the map

$$J_H : C_H \longrightarrow C_H, \qquad J_H(+,t) = (-,t), \qquad J_H(-,t) = (+,t)$$

is a global helix/anti-helix involution. It exchanges $C_{H,+}$ and $C_{H,-}$, preserves the scale, raw parameter, radius, and height laws, reverses the transverse phase, and satisfies

$$J_H^2 = \mathrm{id}_{C_H}\,.$$

The construction uses exactly $H, T_H, \Theta_H, R_H, Z_H$. Once W is declared admissible, the attachment/synthesis arrows place F_W on this already-constructed carrier.

Proof. Fix $H > 0$. The raw parameter set $T_H = \mathbb{R}_{\geq 0}$ and the carrier laws Θ_H, R_H, Z_H are part of the carrier datum. No source datum is mentioned in these definitions.

The lanes are the two signed copies of the same raw carrier parameter set, so their disjoint union is $C_H = C_{H,+} \sqcup C_{H,-}$. The displayed formulas for $\Gamma_{H,+}$ and $\Gamma_{H,-}$ have the same $R_H(t)$ and $Z_H(t)$, while their transverse phases are $\Theta_H(t)$ and $-\Theta_H(t)$. Thus the minus lane is the conjugate anti-helix of the plus lane at the same raw carrier height.

The map J_H sends every $(+,t)$ to $(-,t)$ and every $(-,t)$ to $(+,t)$, hence exchanges the two lanes and preserves the raw parameter t. Applying it twice gives

$$J_H(J_H(+,t)) = (+,t), \qquad J_H(J_H(-,t)) = (-,t),$$

so $J_H^2 = \mathrm{id}_{C_H}$. Every symbol in this construction is part of the carrier datum. The admissibility datum for W, when available, is used afterwards to attach and synthesize the fiber on C_H. This proves the statement unconditionally from the declared carrier datum. □

3 Projection and the loss ledger

Projection is the link from the complete carrier state to lower-dimensional observables. The first projection records the two-dimensional carrier image, and the second records the one-dimensional readout:

$$\Pi_{32} : C \to P_2, \qquad \Pi_{21} : P_2 \to P_1, \qquad \Pi = \Pi_{21}\Pi_{32}.$$

For a complete state X, the scalar function or one-dimensional observable is only the readout

$$\mathcal{R} = \Pi(\mathsf{X}).$$

It is an observable of the three-dimensional state, not the state itself.

Projection suppresses coordinates, so the projection map is paired with a loss ledger

$$\mathcal{D}(\mathsf{X}),$$

recording radius, angle, clock offset, adapter state, and any other coordinate made invisible by the chosen chart. The required theorem is the exact round-trip identity

$$\mathrm{Rec}\big(\Pi(\mathsf{X}), \mathcal{D}(\mathsf{X})\big) = \mathsf{X}.$$

Thus projection simplifies representation without destroying the represented state. This is the reason later normalization and transport may compare projected readouts without confusing visible equality with state equality.

Theorem 3 (Unconditional ledgered projection and recovery)**.** *Let a complete carrier coordinate state be*

$$\mathsf{X} = (r, \theta, z) \in \mathbb{R}_{\geq 0} \times \mathbb{R} \times \mathbb{R},$$

where r is radius, θ is phase, and z is carrier height. Define

$$P_2 := \mathbb{R} \times \mathbb{R}, \qquad P_1 := \mathbb{R},$$

$$\Pi_{32}(r, \theta, z) := (\theta, z), \qquad \Pi_{21}(\theta, z) := z, \qquad \Pi := \Pi_{21}\Pi_{32}.$$

Define the loss ledger and reconstruction map by

$$\mathcal{D}(r, \theta, z) := (r, \theta), \qquad \mathrm{Rec}\big(z, (r, \theta)\big) := (r, \theta, z).$$

Then projection with its ledger is faithful:

$$\mathrm{Rec}\big(\Pi(\mathsf{X}), \mathcal{D}(\mathsf{X})\big) = \mathsf{X}$$

for every complete carrier coordinate state X. Moreover, for the positive-height analytic chart $z > 0$, the logarithmic readout

$$\Pi_{21}^{\log}(\theta, z) := \log z$$

has inverse height recovery $z = \exp(\Pi_{21}^{\log}(\theta, z))$, so the same ledger reconstructs the carrier state from the scalar logarithmic readout by

$$\mathrm{Rec}^{\log}\big(u, (r, \theta)\big) := (r, \theta, \exp u).$$

Proof. Write $\mathsf{X} = (r, \theta, z)$. By definition,

$$\Pi(\mathsf{X}) = \Pi_{21}(\Pi_{32}(r, \theta, z)) = \Pi_{21}(\theta, z) = z$$

and

$$\mathcal{D}(\mathsf{X}) = (r, \theta).$$

Therefore

$$\mathrm{Rec}\big(\Pi(\mathsf{X}), \mathcal{D}(\mathsf{X})\big) = \mathrm{Rec}\big(z, (r, \theta)\big) = (r, \theta, z) = \mathsf{X}.$$

This proves the raw carrier round trip.

If $z > 0$, then $\exp(\log z) = z$. Thus, for $u = \Pi_{21}^{\log}(\theta, z) = \log z$,

$$\mathrm{Rec}^{\log}\big(u, \mathcal{D}(\mathsf{X})\big) = (r, \theta, \exp(\log z)) = (r, \theta, z) = \mathsf{X}.$$

This is the logarithmic readout form of the same ledgered recovery. $\square$

Corollary 4 (Projected equality criterion)**.** *For complete carrier coordinate states* $\mathsf{X} = (r, \theta, z)$ *and* $\mathsf{Y} = (r', \theta', z')$*, if*

$$\Pi(\mathsf{X}) = \Pi(\mathsf{Y}) \qquad \textit{and} \qquad \mathcal{D}(\mathsf{X}) = \mathcal{D}(\mathsf{Y}),$$

then $\mathsf{X} = \mathsf{Y}$*. The same statement holds in the positive-height logarithmic chart with* $\Pi_{21}^{\log}$ *in place of* Π_{21}*.*

Proof. The first equality gives $z = z'$, and the ledger equality gives $(r, \theta) = (r', \theta')$. Hence the three coordinates agree. In the positive-height logarithmic chart, equality of logarithmic readouts gives $\log z = \log z'$; applying exp to both sides gives $z = z'$, and the same ledger argument gives $\mathsf{X} = \mathsf{Y}$. $\square$

4 Structured sources, identity, clocks, and cells

A source is admissible when it carries a finite or effective structural law sufficient to produce local or state data, phase evolution, a composition law, and a readout rule. The class is intentionally broader than one-dimensional L-data: a source may be one function, a family of functions, or a structured object whose lower-dimensional function is only one observable.

The source determines a fiber F_W and carries a represented identity

$$\mathrm{Id}(W).$$

The carrier hosts this fiber; it does not erase the distinction between sources. The identity datum records, in particular,

$$\text{same carrier point} \not\Rightarrow \text{same source}, \qquad \text{same scalar output} \not\Rightarrow \text{same source}.$$

Every source also has its native clock and native harmonic cell system:

$$\tau_W = \text{native clock}, \qquad \mathcal{L}_W = \text{native harmonic cell system}.$$

A complete native state has the form

$$\mathsf{X}_W = \big(C, F_W, \tau_W, \mathcal{L}_W, \mathcal{D}_W\big).$$

Cell boundaries are defined by the source clock, not by arbitrary truncation. The identity obligation preserves identity through realization, projection, reflection, and transport; the synthesis obligation constructs F_W, τ_W, $\mathcal{L}_W$, the phase law, and the adapter from the structural datum.

Definition 5 (Admissible structured source datum)**.** An admissible structured source datum is a tuple

$$\mathfrak{W} = \big(W, \mathrm{Id}(W), W^{\vee}, \mathrm{Struct}(W), F_W, \tau_W, \mathcal{L}_W, \omega_W, \mathcal{R}_W, \mathrm{Synth}_W\big),$$

where W is the named source, $\mathrm{Id}(W)$ is its represented identity, $W^{\vee}$ is its specified dual, $\mathrm{Struct}(W)$ is its finite or effective structural law, F_W is its native fiber, τ_W its native clock, $\mathcal{L}_W$ its native complete-cell system, ω_W its phase law, $\mathcal{R}_W$ its readout rule, and Synth_W is the synthesis arrow

$$\mathrm{Synth}_W : \mathrm{Struct}(W) \longmapsto (F_W, \tau_W, \mathcal{L}_W, \omega_W, \mathcal{R}_W).$$

Proposition 6 (Admissibility is finite structural presentation; structureless sources are excluded)**.** *Admissibility is* exactly *finite structural presentation: the placewise assignment $v \mapsto A_v(W)$ is generated by a finite rule — a finite parameter set or effective schema fixed independently of how many places are queried — not a place-by-place list of values. A structureless ("random") source, whose only faithful description is its own sample path, is* not *admissible: there is no finite structure from which to synthesize a carrier phase, a native cell, or a clock, and letting an adapter encode the sample path would be oracular and vacuous. In particular, forcible cell closure alone does not certify a source: the closure mechanism is arithmetic-neutral — it closes a random coefficient bank exactly as it closes a true fiber (Lean* `ForcibleClosure`*) — so a random Euler datum force-closes its cells yet carries no continuation to transport. The continued, entire readout chain (identification, reflection, analytic landing) therefore rests on the finite structural presentation of an admissible source, which a structureless source does not possess.*

Theorem 7 (Unconditional identity retention and native synthesis)**.** *Let W be admissible: a finite structural presentation* $\mathrm{Struct}(W) = (\{A_v\}_v,\ d,\ \text{Euler assembly},\ \sigma_0)$ *(Definition 5, Proposition 6), realized on the carrier of Theorem 2. Then:*

(synthesis) *the synthesis map*

$$\mathrm{Synth} : \mathrm{Struct}(W) \longmapsto (F_W, \mathcal{L}_W, \omega_W, \tau_W, \mathcal{R}_W)$$

is constructed from the finite structural data — the fiber F_W and readout $\mathcal{R}_W$ from the placewise local factors (Theorem 18), the native cells $\mathcal{L}_W$ and scaling ω_W from the harmonic scale (Theorem 11), the completion/reflection clock τ_W from the carrier involution (Theorem 19) — and is total on finite structural presentation, not oracular: a structureless source has no finite law to synthesize from (Proposition 6);

(identity) *the represented identity* $\mathrm{Id}(W)$ *and the specified dual $W^{\vee}$ are retained through the complete state: the carrier involution J routes $W^{\vee}$ on the anti-helix as the carrier-dual datum (Lemma 66), and distinct sources are not identified by a shared carrier point or scalar readout — the ledgered projection recovers the source only together with its ledger and identity (Corollary 4).*

Proof. (Synthesis.) The structural presentation names W by the finite parameter family $\{A_v\}_v$, the bounded degree d, the Euler assembly, and the convergence datum σ_0. From these the synthesized objects are built explicitly: the local factors $L_v(s;W) = \det(1 - A_v q_v^{-s})^{-1}$ and the readout $\mathcal{R}_W = \prod_v L_v$ are read placewise (Theorem 18); the native cell system $\mathcal{L}_W$ and adapted scaling ω_W are the root-of-unity cells of the harmonic scale (Theorem 11) — directly for the harmonic and Dirichlet classes; in general the adapted warp is supplied per cell by forcible closure (Theorem 14), an arithmetic-neutral transport that closes cells exactly without by itself certifying continuation (Proposition 6); the completion and reflection clock τ_W is the carrier involution's descent (Theorem 19). Every step reads only the finite structural data, so the assignment is defined for every admissible W — total on finite structural presentation, strictly stronger than the mere existence of some adapter (the constructive form of Theorem 79). It is not oracular: encoding a sample path would require the source to carry a finite law, which a structureless source does not (Proposition 6).

(Identity.) The involution $J : C_+ \leftrightarrow C_-$ (Theorem 2) routes W on the helix and its dual on the anti-helix; by Lemma 66 the anti-helix datum is exactly the carrier dual $W^\vee$ (weights closed under $\lambda \mapsto \lambda^{-1}$, determinant character booked in the ledger) — a placewise identity on parameters, using no self-duality of W. The represented identity $\mathrm{Id}(W)$ is retained as a coordinate of the complete state, and the ledgered projection distinguishes sources: equality of scalar readouts forces equality of source only when the ledger and identity coordinates also agree (Corollary 4), so a shared carrier point or a shared readout does not identify two sources. □

Corollary 8 (Complete-state equality criterion)**.** *If two complete realized source states X_W and X_V agree as complete states, then*

$$\mathrm{Id}(W) = \mathrm{Id}(V), \qquad W^\vee = V^\vee.$$

If their projected readouts agree and their ledgers and identity coordinates also agree, then they represent the same retained source identity in the common carrier.

Proof. Complete-state equality gives equality of every coordinate in the displayed tuple defining X_W. In particular, the identity and specified-dual coordinates agree. For projected readouts, apply the projected-equality criterion, Corollary 4, and then read the retained identity coordinate of the recovered complete state. □

5 Normalization and simultaneous carrier evolution

Universality begins with normalization. Given compatible sources

$$W_1, \ldots, W_m$$

with possibly different ranks, scales, phase laws, native clocks, and cell sizes, construct adapters

$$A_i : \mathsf{X}_{W_i}^{\mathrm{native}} \longrightarrow \widetilde{\mathsf{X}}_{W_i} \subset C.$$

These adapters change coordinates but preserve represented identity:

$$\mathrm{Id}(A_i W_i) = \mathrm{Id}(W_i).$$

The normalized states then inhabit one compatible carrier:

$$\widetilde{\mathsf{X}}_{W_1}, \ldots, \widetilde{\mathsf{X}}_{W_m} \subset C.$$

For a finite family, the simultaneous carrier state is vector-valued or multi-fiber:

$$\mathbf{X} = (\widetilde{\mathsf{X}}_{W_1}, \ldots, \widetilde{\mathsf{X}}_{W_m}).$$

A common carrier height supports all fibers while preserving each $\mathrm{Id}(W_i)$, τ_{W_i}, $\mathcal{L}_{W_i}$, and $\mathcal{D}_{W_i}$. Where native cells do not align, the proof constructs a common refinement. Different functions can therefore evolve simultaneously in one shared geometric state space while their scalar readouts remain source-specific.

Theorem 9 (Unconditional finite simultaneous realization)**.** *Let $W_1, \ldots, W_m$ be a finite family of admissible sources, each realized (Theorem 18) and closing its complete cells (Theorem 11) with native cell system $\mathcal{L}_{W_i}$. Let $\mathcal{L}_*$ be the common refinement of the $\mathcal{L}_{W_i}$ (Theorem 10). Then each source is normalized onto the single carrier at the shared harmonic scale with cell system $\mathcal{L}_*$; the simultaneous readout is the tuple $\mathbf{Q}(\tau) = (Q_1^\#(\tau), \ldots, Q_m^\#(\tau))$ of coordinatewise closures; and each fiber keeps its own identity* $\mathrm{Id}(W_i)$ *(Theorem 7), native clock, and focal-closure events, the fibers needing to share neither native ordinates nor cell scales.*

Proof. By Theorem 10 the finite native cell systems $\mathcal{L}_{W_1}, \dots, \mathcal{L}_{W_m}$ have a common refinement $\mathcal{L}_*$, every native cell a disjoint union of refined cells. Each source's *native* complete-cell closure is therefore retained on the shared axis — its own cells are unions of refined cells, undisturbed by the joint realization; where every refined cell moreover carries the zero-mean certificate for every coordinate source, closure holds coordinatewise on $\mathcal{L}_*$ itself (Corollary 13, whose hypothesis that is). Normalizing every source to the shared harmonic scale thus places all m fibers on one carrier without disturbing any native closure. Each realized fiber retains its represented identity and dual routing as complete-state coordinates (Theorem 7), and by the ledgered projection two fibers sharing a carrier point or scalar readout are still distinguished by their ledger and identity (Corollary 4). The simultaneous readout is the coordinatewise tuple $\mathbf{Q}(\tau)$, each component the source's own focal closure. □

Theorem 10 (Unconditional common refinement of finite native cells)**.** *Let I be a carrier-height interval and let*

$$\mathcal{L}_{W_1}, \dots, \mathcal{L}_{W_m}$$

be finite native complete-cell decompositions of I. Define

$$\mathcal{L}_* := \left\{E_1 \cap \dots \cap E_m : E_i \in \mathcal{L}_{W_i} \text{ for all } i, \quad E_1 \cap \dots \cap E_m \neq \varnothing\right\}.$$

Then $\mathcal{L}_*$ is a finite common refinement of the family. Each refined cell $E_* \in \mathcal{L}_*$ lies in a unique parent cell $E_i \in \mathcal{L}_{W_i}$ for every i, and every native complete cell is the disjoint union of the refined cells it contains:

$$E_i = \bigsqcup_{\substack{E_* \in \mathcal{L}_* \\ E_* \subset E_i}} E_*.$$

Thus complete-cell closure statements on $\mathcal{L}_*$ assemble coordinatewise into complete-cell closure statements on every native $\mathcal{L}_{W_i}$.

Proof. The set $\mathcal{L}_*$ is finite because it is a subset of the finite product $\mathcal{L}_{W_1} \times \dots \times \mathcal{L}_{W_m}$ after taking intersections and discarding empty intersections. Each $E_* \in \mathcal{L}_*$ is, by definition, an intersection $E_1 \cap \dots \cap E_m$, hence is contained in one cell of each native decomposition. Since each $\mathcal{L}_{W_i}$ is a decomposition of I, its cells are disjoint and cover I; therefore the parent cell containing E_* is unique.

Fix a native cell $E_i \in \mathcal{L}_{W_i}$. For every point $x \in E_i$, choose the unique cell $E_j(x) \in \mathcal{L}_{W_j}$ containing x for each $j \neq i$. Then

$$x \in E_i \cap \bigcap_{j \neq i} E_j(x) \in \mathcal{L}_*,$$

so the refined cells contained in E_i cover E_i. They are disjoint because two different intersections differ in at least one native parent cell, and native parent cells in a decomposition are disjoint. Hence E_i is the displayed disjoint union. Additive complete-cell closure on refined cells therefore sums over the disjoint union to give the corresponding native complete-cell statement for E_i. □

6 Exact closure, obstruction detection, and adaptation

For each fiber and each complete native cell $C_0 \in \mathcal{L}_W$, define the focal cell sum

$$Q_{C_0}(W) = \sum_{n \in C_0} a_W(n)\omega_W(n).$$

The exact closure theorem proves

$$Q_{C_0}(W) = 0.$$

For a simultaneous family, define

$$\mathbf{Q}_{C_0} = \big(Q_{C_0}(W_1), \dots, Q_{C_0}(W_m)\big)$$

and prove coordinatewise closure $\mathbf{Q}_{C_0} = 0$ on complete native cells or their common refinements. This is exact native closure: the proof belongs at complete cell boundaries and is not replaced by an incomplete terminal-cell statement.

Failure is recorded in the complete carrier state. An aggregate scalar may vanish while a retained obstruction remains, so define

$$\mathfrak{o}(\mathsf{X}_W)$$

in the carrier-plus-ledger state and prove

$$\mathfrak{o}(\mathsf{X}_W) = 0 \iff \text{the required carrier condition closes.}$$

The diagnostic rule is that aggregate readout zero does not by itself imply $\mathfrak{o} = 0$. For instance, a Brauer-style aggregate $\sum_v \mathrm{inv}_v = 0$ can coexist with a nonzero obstruction vector; the carrier must detect the retained vector.

Obstruction-directed adaptation is then a state operation

$$\mathcal{A} : \mathsf{X} \longmapsto \mathsf{X}'$$

whose input is the detected obstruction. The cycle is

$$\text{realize} \to \text{synchronize} \to \text{read} \to \text{detect} \to \text{adapt} \to \text{re-realize}.$$

Admissible adaptation preserves identity, ledger state, unaffected fibers, and carrier compatibility. On classes where removal is proved, the output satisfies $\mathfrak{o}(\mathcal{A}(\mathsf{X})) = 0$.

Theorem 11 (Unconditional complete-cell closure from harmonic scaling)**.** *The closure is general in two senses. It is not tied to any one class of source — the argument reads only the root-of-unity structure of the cell, not the arithmetic of the fiber — and it is not tied to any one harmonic: scaling the carrier by a harmonic $\Delta = \pi/M$ makes the placement winding $s_n = n\,\Delta$ periodic with period $2M$, so the integers land on the μ_{2M} cell and the placement phasor $e^{i\Delta}$ is a nontrivial root of unity. Thus $\Delta = \pi/6$ gives the μ_{12} cell and $\Delta = \pi/2$ the μ_4 cell, while $\Delta = \pi/3$ — the sharpest, $\mu_6 = \mathbb{Z}[\zeta_6]$ — is the working scale. Let W be admissible and ω_W its adapted scaling — the constant harmonic scale, or a per-fiber warp where one is needed — which snaps the fiber's channels to roots of unity on each complete native cell $C \in \mathcal{L}_W$ of period P: every channel w satisfies $w^P = 1$, $w \neq 1$. Then the complete-cell sum vanishes exactly, with no residue,*

$$Q_C(W) := \sum_{n \in C} a_W(n)\,\omega_W(n) = 0 \qquad (C \in \mathcal{L}_W).$$

Proof. For a harmonic scale $\Delta = \pi/M$ the placement winding $s_n = n\,(\pi/M)$ is periodic with period $2M$ since $2M \cdot (\pi/M) = 2\pi$ (for $M = 3$, the μ_6 cell, $6 \cdot (\pi/3) = 2\pi$, Lean `Geometry.spinAngle`); so the integers fall on the μ_{2M} cell and each channel phasor is a nontrivial P-th root of unity w, $w^P = 1$, $w \neq 1$. Over one complete period the geometric sum is exactly zero,

$$\sum_{n \in C} w^n = \frac{w^P - 1}{w - 1} = 0$$

(Lean CellClosure.root_of_unity_cell_sum_zero); summing over the bank's channels gives $\sum_{n\in C}\sum_i (w_i)^n = 0$ (Lean CellClosure.harmonic_bank_cell_sum_zero), i.e. $Q_C(W) = 0$. The step uses only that the channels are nontrivial roots of unity — reading the signed channels, not the fiber's arithmetic — so it holds for a general det-one fiber, the Dirichlet period-q character block $\sum_{n\in C}\chi(n) = 0$ ($\chi \neq 1$, Lean LFunctionPhasor.character_block_sum_eq_zero) being one instance. The cancellation is a full-period root-of-unity identity, exact and residue-free: the normalized focal residual carries no independent constant mode (Theorem 16, Lean HarmonicCell.harmonic_residual_no_independent_mode). □

Theorem 12 (Unconditional bounded primitive from exact cell closure)**.** *Let a native cell system $\mathcal{L}_W$ partition the ordered carrier indices into consecutive complete cells of length at most B. Suppose*

$$\sum_{n\in C} a_W(n)\omega_W(n) = 0$$

for every complete cell $C \in \mathcal{L}_W$, and suppose every individual summand satisfies $|a_W(n)\omega_W(n)| \leq M$. Then the adapted primitive

$$A_W^{\mathrm{car}}(N) := \sum_{n\leq N} a_W(n)\omega_W(n)$$

is bounded by BM. Consequently any scalar DC residual mode defined as the asymptotic mean of this adapted primitive vanishes.

Proof. Partition $\{1,\ldots,N\}$ into complete native cells and the final incomplete terminal cell. Every complete-cell contribution is zero by hypothesis. The primitive is therefore the sum over the terminal cell alone. That terminal cell has at most B terms, each of size at most M, so

$$|A_W^{\mathrm{car}}(N)| \leq BM.$$

Dividing by N and letting $N \to \infty$ gives zero asymptotic mean. Thus the scalar DC residual mode read from that adapted mean is zero. □

Corollary 13 (Coordinatewise closure on common refinements)**.** *Let $\mathfrak{W}_1,\ldots,\mathfrak{W}_m$ be a finite compatible family of admissible structured source data, and let $\mathcal{L}_*$ be the common refinement supplied by Theorem 10. If each refined cell $E_* \in \mathcal{L}_*$ carries the zero-mean certificate for every coordinate source W_i, then*

$$\mathbf{Q}_{E_*} := \big(Q_{E_*}(W_1),\ldots,Q_{E_*}(W_m)\big) = 0$$

for every refined complete cell E_*. Summing over the refined cells inside any native parent cell gives coordinatewise closure on the native $\mathcal{L}_{W_i}$-cells.

Proof. Apply Theorem 11 in each coordinate i. Every component of $\mathbf{Q}_{E_*}$ is zero, so the vector is zero. The common refinement theorem writes each native parent cell as a disjoint union of refined cells; additivity of the finite sums then assembles the refined coordinate closures into the native parent-cell closure. □

Theorem 14 (Unconditional forcible closure from an independent carrier pair)**.** *Let $D \in \mathbb{C}$. If $u, v \in \mathbb{C}$ are linearly independent over $\mathbb{R}$, then there exist real weights $s, t \in \mathbb{R}$ such that*

$$D + s\,u + t\,v = 0.$$

In particular, if an admissible carrier lane has a continuous nonconstant phase ϕ, then it reaches a height y with $\sin(2\phi(y)) \neq 0$; at such a height the two lane directions

$$u = e^{i\phi(y)}, \qquad v = e^{-i\phi(y)}$$

are $\mathbb{R}$-linearly independent, so every native cell residual is forcible to zero by real carrier weights.

Proof. The map

$$\mathbb{R}^2 \to \mathbb{C}, \qquad (s,t) \mapsto s\,u + t\,v$$

is a real-linear map between two-dimensional real vector spaces. Since u and v are real-linearly independent, this map is an isomorphism. Thus $-D$ has a preimage (s,t), giving $D + s\,u + t\,v = 0$.

For the lane pair $e^{\pm i\phi(y)}$, real-linear dependence occurs exactly when the two directions lie on the same real line, equivalently $\sin(2\phi(y)) = 0$. If a continuous nonconstant phase never attained $\sin(2\phi) \neq 0$, its image would lie in the discrete set $\frac{\pi}{2}\mathbb{Z}$, forcing it to be constant on the connected carrier height line. Hence a nonconstant admissible phase reaches a height with $\sin(2\phi) \neq 0$, and the first paragraph applies. ◻

Theorem 15 (Unconditional obstruction-directed adaptation by forcible closure)**.** *Let X_W carry a detected cell residual $D \in \mathbb{C}$ on a native cell, and let W be nontrivial (continuous nonconstant lane phase ϕ). Then there is a reachable carrier height y with $\sin(2\phi(y)) \neq 0$, at which the two lane directions $u = e^{i\phi(y)}$, $v = e^{-i\phi(y)}$ are $\mathbb{R}$-linearly independent, and real correction weights s, t with*

$$D + s\,u + t\,v = 0$$

exist, extinguishing the residual. The correction acts on the adapter layer alone — the cell's closure weights, not the fiber datum — so it is readout-preserving by construction: it retains the represented identity, ledger, unaffected fibers, and carrier compatibility, and the adapted obstruction is zero, $\mathfrak{o}(\mathcal{A}(\mathsf{X}_W)) = 0$.

Proof. By Theorem 14, two $\mathbb{R}$-linearly independent directions span the residual plane $\mathbb{C}$, so any residual D is forced to zero by real weights (Lean `ForcibleClosure.residual_forcible`). The directions are supplied globally, not cell by cell: a nontrivial admissible fiber has a continuous nonconstant lane phase, which on the connected carrier line must attain a height with $\sin(2\phi) \neq 0$ — otherwise it maps into the discrete set $\frac{\pi}{2}\mathbb{Z}$ and is constant — at which the two lanes $e^{\pm i\phi(y)}$ are independent (Lean `CarrierReachability.reachable_independent_stage`). Applying the forcing there extinguishes the residual. The correction adjusts only the forcing weights of the adapter layer — the fiber datum a_W, the represented identity (Theorem 7), the ledger, the unaffected fibers, and carrier compatibility are coordinates it does not touch, so all are retained by construction; hence the adapted obstruction is zero. ◻

7 Modes, readouts, reflection, and analytic landing

Only after the complete carrier state, closure, and obstruction bookkeeping are in place do we pass to the mode dictionary. Let D_W^{c} be the retained three-dimensional constant or focal mode, and let R_W be the corresponding one-dimensional constant or pole mode. The bridge theorem is

$$R_W = \mathfrak{B}(D_W^{\mathrm{c}}).$$

Hence $D_W^{\mathrm{c}} = 0 \Rightarrow R_W = 0$, while a genuine retained carrier mode gives the corresponding one-dimensional pole state.

The one-dimensional source-specific projection is then

$$\Pi_{1D}(\mathsf{X}_W) = \mathcal{R}_W.$$

For an Euler source this means

$$\mathcal{R}_{W,v}(s) = \det(1 - A_v(W)q_v^{-s})^{-1}, \qquad \mathcal{R}_W(s) = \prod_v \mathcal{R}_{W,v}(s),$$

and for completed data it means

$$\Pi_{1D}(\mathsf{X}_W) = \Lambda(s, W)$$

on the original domain. Thus L-functions enter as one important class of one-dimensional readout; they are not the definition of the carrier system.

Carrier reflection returns to the involution $J : C_+ \leftrightarrow C_-$. The complete readout map must intertwine reflection:

$$\Pi_{1D} \circ J = \mathfrak{R} \circ \Pi_{1D}.$$

For completed functional data,

$$(\mathfrak{R} f)(s) = \varepsilon_W f^\vee(\kappa - s),$$

so the one-dimensional statement is

$$\Lambda(s, W) = \varepsilon_W \Lambda(\kappa - s, W^\vee).$$

The functional reflection is the readout of the carrier involution.

The analytic landing theorem has three outputs:

$$\boxed{\text{continuation}}, \qquad \boxed{\text{pole/DC classification}}, \qquad \boxed{\text{vertical-strip growth}}.$$

Its inputs are the complete state, exact closure, mode dictionary, readout identity, and reflection.

Theorem 16 (Unconditional mode bridge: the residual is the pencil residual)**.** *Let W be admissible, with fiber coefficients splitting the reflected readout into signed P/M channels (A, B) and pencil weights (μ, λ), $A \neq 0$, $\lambda \neq \mu$. Let R_W be the readout's scalar residual (constant/DC) mode — the constant term of the $x \to 0^+$ carrier theta, equal by the Mellin pair to the edge residue of the readout — and let Φ_W be the signed closure channel. Then the retained carrier constant mode D_W^{c} equals R_W (one analytic quantity), and factors*

$$D_W^{\mathrm{c}} = V_W\, \Phi_W, \qquad V_W = (\lambda - \mu) \neq 0.$$

Hence R_W carries no independent constant direction:

$$R_W = 0 \iff \Phi_W = 0.$$

The factorization reads only (A, B, μ, λ), so it holds for a general det-one fiber, not merely the Dirichlet instance.

Proof. Because the same coefficients λ_n define both, the constant term of the $x \to 0^+$ expansion of the carrier theta is the residue of its Mellin transform $\mathcal{M}K_W(s) = \Lambda(s; W)$ at the edge, i.e. the readout's DC coefficient; thus $D_W^{\mathrm{c}} = R_W$ as a single analytic quantity, not a modelled image under an assumed map. Splitting the reflected readout into the signed P/M channels gives the harmonic pencil $L^{\mathrm{c}} = \left[\begin{smallmatrix} A & B \\ \mu A & \lambda B \end{smallmatrix}\right]$, and its normalized cell residual is exactly

$$\det(L^{\mathrm{c}})/A = (\lambda - \mu)\, B, \qquad \lambda - \mu \neq 0$$

(Lean `HarmonicCell.harmonic_residual_no_independent_mode`, channel-agnostic in (A, B, μ, λ); the Dirichlet instance is `HarmonicPencilCell.normalized_cell_focal_residual_exact`, giving $D^{\mathrm{c}}_{\chi} = V_{\chi}\Phi_{\chi}$, $V_{\chi} \neq 0$). Writing Φ_W for the signed closure channel B and $V_W = \lambda - \mu$, this is $D^{\mathrm{c}}_W = V_W \Phi_W$ with $V_W \neq 0$. Since $V_W \neq 0$, $D^{\mathrm{c}}_W = 0 \iff \Phi_W = 0$; with $R_W = D^{\mathrm{c}}_W$ the residual carries no independent constant coefficient beyond the closure channel, and the Hermitian Gram matrix drops rank exactly at that event (same Lean theorem, the rank-drop half). ◻

Theorem 17 (Unconditional channel residual factorization)**.** *Let a complete carrier mode datum contain a scalar closure channel Φ_W, a normalized carrier residual D^{c}_W, and a nonvanishing bridge factor V_W with*

$$D^{\mathrm{c}}_W = V_W \Phi_W, \qquad V_W \neq 0.$$

Then

$$D^{\mathrm{c}}_W = 0 \iff \Phi_W = 0.$$

Consequently the retained residual carries no constant or pole mode independent of the scalar closure channel.

Proof. If $\Phi_W = 0$, the displayed factorization gives $D^{\mathrm{c}}_W = 0$. Conversely, if $D^{\mathrm{c}}_W = 0$, then $V_W \Phi_W = 0$. Since $V_W \neq 0$, this forces $\Phi_W = 0$. Thus the carrier residual vanishes exactly with the closure channel and has no additional independent direction. ◻

Theorem 18 (Unconditional one-dimensional readout identity)**.** *Let W be an admissible source with finite local parameter family $\{A_v\}_v$, local factors*

$$L_v(s; W) = \det\!\big(1 - A_v\, q_v^{-s}\big)^{-1},$$

and coefficients $a_n(W)$ assembled from the Euler product $\prod_v L_v(s; W) = \sum_{n \geq 1} a_n(W)\, n^{-s}$. On the carrier, W is realized as its phasor bank F_W: index n is a phasor of weight $a_n(W)$ placed at carrier height $z_n = n$ ($\Delta z = 1$ per index, radius $\sim \sqrt{n}$), carrying the unit-modulus readout spin $e^{-iy \log n}$. Each phasor enters at zero magnitude and grows continuously through the growth window, so the bank is realized from height 0 upward with no onset threshold. Let Π_{1D} be the readout of Theorem 3. Then the readout is the source L-function,

$$\Pi_{1D}(\mathsf{X}_W)(s) = L(s; W) = \prod_v L_v(s; W), \qquad \prod_v L_v(s; W) = \sum_{n \geq 1} a_n(W)\, n^{-s} \text{ where the series converges,}$$

and the placewise local factors, the conductor $N = \prod_v q_v^{a(A_v)}$, and the root number $\varepsilon = \prod_v \varepsilon_v(A_v)$ are the local Langlands/Deligne invariants of $\{A_v\}_v$. The bank grows continuously from height 0, so the readout continues past the 1D line $\operatorname{Re} s = 1$ (Dirichlet fibers: to $\operatorname{Re} s > 0$); that line is where the projected series stops converging in the readout chart, not a threshold on the carrier.

Proof. Index n is realized as a single phasor: weight $a_n(W)$, magnitude $n^{-\sigma}$, and unit-modulus readout spin $e^{-iy \log n}$; it enters at zero magnitude and grows continuously through the growth window, placed at carrier height $z_n = n$
(Lean `HeightGrowthActive.heightLaw`, `AntihelixWindow.upperGamma_tendsto_zero`). The $\log n$ in this readout spin is geometric: it is the analytic readout (log) of the phasor's own height z_n as it spins on the Archimedean helix, not the 1D writing $n^{-s} = e^{-s \log n}$ — the 1D log is the shadow of this readout, not its source (Lean `HeightGrowthActive.mellinRate`, kept distinct from the geometric placement winding $n \cdot \pi/3$). The readout of the bank is the sum of these phasors,

$$\Pi_{1D}(\mathsf{X}_W)(s) = \sum_{n \geq 1} a_n(W)\, n^{-\sigma} e^{-iy \log n} = \sum_{n \geq 1} a_n(W)\, n^{-s}$$

(Lean `LFunctionPhasor.lfunction_phasor_form`). Because every index is realized this way from height 0 — no coefficient dropped, none held back for a convergence threshold — the readout is the analytic function the growing bank presents across the realized strip; for the Dirichlet fibers this is LFunction χ on $\operatorname{Re} s > 0$, past the 1D line $\operatorname{Re} s = 1$ (Lean `LFunctionPhasor.dirichlet_strip_tendsto_LFunction`). Placewise, $L_v(s; W) = \det(1 - A_v q_v^{-s})^{-1}$ is the local Langlands/Deligne factor of A_v [13]: the reciprocal characteristic polynomial at an unramified v, the Γ-datum at the archimedean place; the local conductor exponent $a(A_v)$ and root number $\varepsilon_v(A_v)$ are Tate's local factors (Theorem 57). The global ε and N are finite products, since A_v is unramified for almost all v. □

Theorem 19 (Unconditional reflected completed readout: functional equation of the self-dual carrier)**.** *Let W be admissible with carrier dual $W^\vee$. The carrier is* self-dual*: the involution $J : C_+ \leftrightarrow C_-$ with $J^2 = \mathrm{id}$ (Theorem 2) exchanges helix and anti-helix, routing W on the helix and its dual $W^\vee$ on the anti-helix (Lemma 66). Then the completed readout satisfies the functional equation*

$$\Lambda(s, W) = \varepsilon_W\, \Lambda(\kappa - s, W^\vee), \qquad |\varepsilon_W| = 1,$$

with the local reciprocal factor of each finite fiber supplied by Theorem 21.

Proof. The carrier and its involution J, $J^2 = \mathrm{id}$, are constructed source-independently in Theorem 2; J exchanges the two lanes, so it is the carrier's self-duality, and by Lemma 66 the anti-helix carries exactly the dual datum $W^\vee$ (weights closed under $\lambda \mapsto \lambda^{-1}$, determinant character booked in the ledger). The involution descends through the readout by three machine-checked geometric intertwiners (Theorem 69): the Cayley projection carries J to the unit-circle reflection (Lean `GeometricReadout.cayley`, `cayley_real_norm_one`, with the lossless round trip `ConeProjection.record_bijective`); the completion clock carries that to the readout reflection $(\mathfrak{R} f)(s) = \varepsilon_W f^\vee(\kappa - s)$ (Lean `FiniteWeightFiber.clockCompletion_selfdual`); and the logarithmic readout fixes the axis $\operatorname{Re} s = \kappa/2$ (Lean `ConeProjection.midpoint_to_critical`, `AutomorphicCandidate.midpoint_projects_to_half`). Composing the three, carrier reflection by J is the readout reflection $\mathfrak{R}$ on the completed readout; since the anti-helix carries $W^\vee$, this reads

$$\Lambda(s, W) = \varepsilon_W\, \Lambda(\kappa - s, W^\vee), \qquad |\varepsilon_W| = 1.$$

The reciprocal local polynomial identity $P_W(X) = (-X)^{|I|} P_W(X^{-1})$ of each finite duality-stable fiber (Theorem 21, Lean `FiniteWeightFiber.localPoly_reciprocal`) supplies the placewise factor of this reflection at the axis $\operatorname{Re} s = \kappa/2$. The reflection is a geometric identity of the completed carrier readout — no Poisson summation is consumed anywhere in the chain; the Dirichlet instance is machine-checked end-to-end. That end-to-end check is a *corroboration*, not a restriction: the functional equation is one geometric property of the involution J, holding for *every* fiber that rides the carrier, so there is no per-L-function functional equation to re-derive. Dirichlet is simply the one case where an independent completed L-function exists (in Mathlib) to check the carrier's answer against; a functional equation "for Dirichlet but not in general" is a category error in this frame, not a weaker result — the same descent through the machine-checked Cayley, completion-clock, and logarithmic-readout intertwiners reflects whatever fiber the carrier carries. □

Theorem 20 (Unconditional dimensional reflection chain)**.** *Suppose a complete carrier datum has:*

$$J : C_+ \leftrightarrow C_-, \qquad J^2 = \mathrm{id},$$

a faithful ledgered projection Π, a projected reflection J_{proj}, a completion clock K, and a completed readout operator $\mathfrak{R}$, with intertwining certificates

$$\Pi \circ J = J_{\mathrm{proj}} \circ \Pi, \qquad K \circ J_{\mathrm{proj}} = \mathfrak{R} \circ K.$$

Then the completed readout intertwines the carrier involution:

$$K \circ \Pi \circ J = \mathfrak{R} \circ K \circ \Pi.$$

For a completed source datum whose reflected-readout operator is

$$(\mathfrak{R} f)(s) = \varepsilon_W f^\vee(\kappa - s),$$

this gives

$$\Lambda(s, W) = \varepsilon_W \Lambda(\kappa - s, W^\vee).$$

The completion clock K supplied to this theorem is not independent analytic input. In dimension one it is furnished upstream by the unique scale-balanced Gaussian fixed under the carrier's Fourier exchange — unique *within the Gaussian family* $g_a(x) = e^{-\pi a x^2}$, $\operatorname{Re} a > 0$, where $\widehat{g_a} = a^{-1/2} g_{1/a}$ forces $a = 1$, the coefficient π pinned by self-duality rather than by any zeta convention — and its Mellin transform is the completion factor $\frac{1}{2}\pi^{-s/2}\Gamma(s/2)$. Both statements are Lean theorems at the standard footprint (`GaussianClock.gaussianClock_selfDual_iff`, `GaussianClock.gaussianClock_mellin`; `GaussianClockUniqueness.lean`), and the exact Jacobi transformation the clock generates is likewise consumed from Mathlib (`jacobiTheta_S_smul`, `EpsteinThetaI.lean`). For the basic configuration the intertwining hypothesis is thus discharged upstream, not assumed.

Proof. Compose the two displayed intertwining identities:

$$K \circ \Pi \circ J = K \circ J_{\text{proj}} \circ \Pi = \mathfrak{R} \circ K \circ \Pi.$$

The completed functional equation is this identity read through the completed source datum, with $\mathfrak{R}$ expanded as $(\mathfrak{R} f)(s) = \varepsilon_W f^\vee(\kappa - s)$. □

Theorem 21 (Unconditional finite duality-stable fiber reflection)**.** *Let I be a finite index set with an involution $i \mapsto i^\vee$. Let $\lambda_i \in \mathbb{C}^\times$ satisfy*

$$\lambda_{i^\vee} = \lambda_i^{-1}, \qquad \prod_{i \in I} \lambda_i = 1.$$

Define

$$P_W(X) := \prod_{i \in I} (1 - \lambda_i X).$$

Then

$$P_W(X) = (-X)^{|I|} P_W(X^{-1}).$$

Equivalently, $L_W(X) := P_W(X)^{-1}$ is reflected by $X \mapsto X^{-1}$ with the determinant-normalized factor dictated by $|I|$. With completed variable $X = c^{s - \kappa/2}$, the reflection $s \mapsto \kappa - s$ sends X to X^{-1}, so the finite fiber contributes the local reflection datum at the axis $\operatorname{Re} s = \kappa/2$.

Proof. For each i,

$$1 - \lambda_i X = -\lambda_i X (1 - \lambda_i^{-1} X^{-1}).$$

Multiplying over I gives

$$P_W(X) = (-X)^{|I|} \Big(\prod_{i \in I} \lambda_i \Big) \prod_{i \in I} (1 - \lambda_i^{-1} X^{-1}).$$

The determinant ledger gives $\prod_i \lambda_i = 1$. The involution $i \mapsto i^\vee$ permutes the factors and satisfies $\lambda_i^{-1} = \lambda_{i^\vee}$, so the last product is $P_W(X^{-1})$. This proves the reciprocal local polynomial identity. The local-factor statement is the same identity after inversion. Finally,

$$c^{(\kappa-s)-\kappa/2} = c^{-(s-\kappa/2)} = X^{-1},$$

which is the named completed reflection coordinate. □

Lemma 22 (Fiber growth from height zero)**.** *Let W be admissible and F_W its phasor bank (Theorem 18). Each index n is placed at carrier height $z_n = n$, consecutive heights differing by $\Delta z = 1$, and enters the bank at zero magnitude, its amplitude rising continuously through the growth window to $a_W(n)\, n^{-\sigma}$. Hence the whole bank, and its primitive $A_W^{\mathrm{car}}(N) = \sum_{n \le N} a_W(n)\, \omega_W(n)$, is realized continuously from height 0 upward, with no index withheld for a convergence threshold.*

Proof. The height law places index n at $z_n = n > 0$ with $z_{n+1} - z_n = 1$ (Lean `HeightGrowthActive.heightLaw`, `heightLaw_step`, `heightLaw_pos`). The phasor's amplitude is carried by the growth window, which starts at zero and rises continuously (Lean `AntihelixWindow.upperGamma_tendsto_zero`), reaching magnitude $n^{-\sigma}$ (Lean `LFunctionPhasor.phasorTerm_norm`). Every index is placed and grown this way, so the bank and its primitive are realized from height 0; no index awaits a convergence threshold. □

Theorem 23 (Unconditional analytic landing from fiber growth)**.** *Let W be admissible, realized as its growing phasor bank (Lemma 22), whose scaling closes each complete native cell,*

$$\sum_{n \in C} a_W(n)\, \omega_W(n) = 0 \qquad (C \in \mathcal{L}_W).$$

Then:

(i) *the carrier primitive is bounded, so the readout continues past the 1D line* $\mathrm{Re}\, s = 1$ *— the growing bank presents the continued readout across the realized strip (for the Dirichlet fibers, to* $\mathrm{Re}\, s > 0$*);*

(ii) *the constant (DC) mode of the reflected carrier readout is extinct, $R_W = 0$, so the readout carries no pole from the carrier;*

(iii) *with the completed reflection (Theorem 69, Proposition 76) — geometric, consuming no Poisson summation — the completed readout is entire and bounded in vertical strips.*

Proof. By Lemma 22 the whole primitive $A_W^{\mathrm{car}}(N) = \sum_{n \le N} a_W(n)\omega_W(n)$ is realized from height 0 up. Complete-cell closure makes it bounded: partitioning $\{1, \ldots, N\}$ into complete cells (each summing to zero) and one incomplete remainder of at most one cell length P, the primitive is the remainder alone, so $\|A_W^{\mathrm{car}}(N)\| \le B\,P$ (Theorem 12; Lean `CellClosure.cell_closure_partialSum_le`, the finite duality-stable harmonic bank `harmonic_bank_primitive_bounded`, the Dirichlet instance `dirichlet_cell_closure`). A bounded primitive continues the readout to $\mathrm{Re}\, s > 0$ by Theorem 24, past the 1D line $\mathrm{Re}\, s = 1$; for the Dirichlet fibers this landing is Lean `LFunctionPhasor.dirichlet_strip_tendsto_`
This is (i).

For (ii), the Cesàro mean of the cell-closing scaling tends to 0, so the zero-frequency mode of the reflected carrier readout vanishes, $R_W = 0$ (Lean `CellClosure.cell_closure_dc_mode_zero`; channel-agnostically, the normalized focal residual carries no independent constant direction, `HarmonicCell.harmonic_residual_no_independent_mode`). No constant term survives to produce a carrier pole.

For (iii), the completed reflection $\Lambda(s; W) = \varepsilon_W\, \Lambda(\kappa - s; W^\vee)$ (Theorem 69, Proposition 76) — a geometric identity of the carrier, consuming no Poisson summation — trades the continued half-plane of (i) for its mirror; with no pole from (ii), the completed readout is entire. Vertical-strip boundedness is the bounded primitive of (i) read on each fixed line, the readout spin being unit-modulus (Theorem 25). ☐

Theorem 24 (Unconditional primitive-exponent transfer)**.** *Let W have coefficients λ_n and primitive*

$$A_W(x) := \sum_{n \le x} \lambda_n.$$

Assume that for some $\theta < \kappa/2$,

$$A_W(x) = O(x^{\theta}).$$

Then the Dirichlet readout

$$L(s, W) = \sum_{n \ge 1} \lambda_n n^{-s}$$

continues by Abel summation to $\operatorname{Re} s > \theta$*. If the dual coefficients satisfy $\lambda_n^{\vee} = \overline{\lambda_n}$, then $W^{\vee}$ has the same primitive exponent, and the two completed readouts are simultaneously analytic on the open strip*

$$\theta < \operatorname{Re} s < \kappa - \theta,$$

which contains the reflection axis $\operatorname{Re} s = \kappa/2$*.*

Proof. For $N \ge 1$, partial summation gives

$$\sum_{n \le N} \lambda_n n^{-s} = A_W(N) N^{-s} + s \int_1^N A_W(u) u^{-s-1}\, du.$$

If $\operatorname{Re} s > \theta$, the boundary term tends to zero and the integral converges absolutely, since $A_W(u) = O(u^{\theta})$. This gives analytic continuation to $\operatorname{Re} s > \theta$ by locally uniform convergence on compact subsets of that half-plane.

For the dual, $A_{W^{\vee}}(x) = \overline{A_W(x)}$, so the same exponent works. The reflected variable $\kappa - s$ lies in $\operatorname{Re}(\kappa - s) > \theta$ exactly when $\operatorname{Re} s < \kappa - \theta$. Hence both completed readouts are analytic on $\theta < \operatorname{Re} s < \kappa - \theta$. Since $\theta < \kappa/2$, this strip contains the axis $\operatorname{Re} s = \kappa/2$. ☐

Theorem 25 (Unconditional lossless contour gauge bound)**.** *Let $\tau > 0$, let $x > 0$, and let M be measurable on the line $u = \tau + iv$ with*

$$B_\tau := \frac{1}{2\pi} \int_{-\infty}^{\infty} |M(\tau + iv)|\, dv < \infty.$$

Define

$$A(x) := \frac{1}{2\pi i} \int_{\tau - i\infty}^{\tau + i\infty} M(u) x^{-u/2}\, du.$$

Then

$$|A(x)| \le B_\tau x^{-\tau/2}.$$

The bound is uniform in every parameter that appears only through the positive coordinate x or through a unit-modulus prefactor.

Proof. On $u = \tau + iv$,
$$|x^{-u/2}| = x^{-\tau/2}.$$
The triangle inequality gives
$$|A(x)| \le \frac{1}{2\pi} \int_{-\infty}^{\infty} |M(\tau + iv)|\,|x^{-(\tau+iv)/2}|\,dv = B_\tau x^{-\tau/2}.$$
Unit-modulus prefactors have absolute value 1, and parameters entering only through x occur only in the displayed power $x^{-\tau/2}$. □

8 Exhaustion, coherent transport, and the universal theorem

Exhaustion asserts that represented source information is neither lost nor invented by carrier realization or ledgered projection:
$$\text{source state} \leftrightarrow \text{carrier state} \leftrightarrow \text{ledgered projected state}.$$
For represented events,
$$e \in E(W) \iff e \in E(\mathsf{X}_W) \iff e \in E(\Pi(\mathsf{X}_W), \mathcal{D}_W).$$
For zero events where applicable, the proof identifies the event with the corresponding closed carrier condition, for example
$$\rho \in Z(W) \iff Q_W^{\#}(\rho) = 0.$$
The visible one-dimensional coordinate may suppress dimensions; the ledgered state remains exhaustive.

Finally let
$$T : W \to W'$$
be a structural source transport. The carrier construction induces
$$\mathsf{X}(T) : \mathsf{X}_W \to \mathsf{X}_{W'}.$$
The transport theorem proves preservation and compatibility of identity, ledger state, cell structure, closure, J, one-dimensional readout, and obstruction state, and proves composition:
$$\mathsf{X}(T_2 \circ T_1) = \mathsf{X}(T_2) \circ \mathsf{X}(T_1).$$
Together with the previous sections, this is the proof content of Theorem 1. Later parts should consume this theorem by instantiating the admissible source law: Euler/L-function sources, CPS twisting families, and graded obstruction towers are inputs to the same carrier interface.

Theorem 26 (Unconditional ledgered event exhaustion)**.** *Let X_W be a complete realized source state. Because the ledgered projection is a bijection — the fiber reconstructs from its retained height and loss ledger (Theorem 3; Lean `ConeProjection.record_bijective`, `reconstruct_record`, `SourceHolonomy.projection_bijective_loss_ledger`) — the represented source events, carrier events, and ledgered projected events coincide:*
$$e \in E(W) \iff e \in E(\mathsf{X}_W) \iff e \in E(\Pi(\mathsf{X}_W), \mathcal{D}_W),$$

so no event is lost by projection or invented by realization. For zero events, the represented vanishing coincides with the closed carrier condition,

$$\rho \in Z(W) \iff Q_W^{\#}(\rho) = 0,$$

the carrier focal closure vanishing exactly at the readout's zeros (Lean `ClosedForm.carrierZero_iff_L_zero`, `FocalResidualVanishes.focal_residual_zero_iff_L_zero` — the Dirichlet-instantiated form; the general zero-event identification is placewise through the local identification of Theorem 18).

Proof. The ledgered projection $\mathsf{X}_W \mapsto (\Pi(\mathsf{X}_W), \mathcal{D}_W)$ is a bijection whose inverse is the reconstruction map (Theorem 3; Lean `record_bijective`, `reconstruct_record`, `projection_bijective_loss_ledger`), so it neither identifies distinct states nor omits any; the represented, carrier, and ledgered projected event sets are therefore in bijection, giving the three-way equivalence. For a zero-event datum, the carrier focal closure $Q_W^{\#}$ vanishes exactly when the readout vanishes (Lean `carrierZero_iff_L_zero`, `focal_residual_zero_iff_L_zero`), so $\rho \in Z(W) \iff Q_W^{\#}(\rho) = 0$. □

Theorem 27 (Unconditional coherent carrier transport)**.** *Call a carrier transport T* faithful *for a closure predicate* Cl *if* $\mathrm{Cl}(x) \Rightarrow \mathrm{Cl}(Tx)$*. Then the identity transport is faithful; the composite of faithful transports is faithful; composition is associative with the identity as two-sided unit; and the base-change transports compose,* $\mathrm{bc}_p \circ \mathrm{bc}_q = \mathrm{bc}_{pq}$*, with* $\mathrm{bc}_1 = \mathrm{id}$*. Hence coherent source transports induce carrier transports that preserve closure and retained data and respect composition.*

Proof. Each law is machine-checked in `CarrierFunctoriality`: the identity is faithful (`faithful_id`); a composite of faithful transports is faithful (`faithful_comp`); composition is associative (`comp_assoc`) with two-sided identity (`id_comp`, `comp_id`); and base change composes (`bc_comp`) with unit (`bc_one`). The abstract closure-transport statement is Theorem 28; these are its verified instances. □

Theorem 28 (Unconditional functorial closure transport)**.** *Let $\mathcal{C}$ be a class of complete carrier states equipped with a closure predicate* Cl *and retained datum predicate* Ret*. A transport $F : X \to Y$ is faithful when*

$$\mathrm{Cl}(X) \Rightarrow \mathrm{Cl}(F(X)), \qquad \mathrm{Ret}(X) \Rightarrow \mathrm{Ret}(F(X)).$$

Then the identity transport is faithful, and the composite of faithful transports is faithful. In particular, if

$$X \xrightarrow{F} Y \xrightarrow{G} Z$$

are faithful carrier transports, then $G \circ F$ preserves closure and all retained Part I data named by Ret*.*

Proof. For the identity map, both implications are tautological:

$$\mathrm{Cl}(X) \Rightarrow \mathrm{Cl}(X), \qquad \mathrm{Ret}(X) \Rightarrow \mathrm{Ret}(X).$$

Now suppose F and G are faithful. If $\mathrm{Cl}(X)$, then faithfulness of F gives $\mathrm{Cl}(F(X))$, and faithfulness of G gives $\mathrm{Cl}(G(F(X)))$. The retained-datum implication is identical with Ret in place of Cl. Thus $G \circ F$ is faithful. □

Corollary 29 (Part I assembly of the universal carrier theorem)**.** *For every finite compatible family of admissible structured source data with the carrier, projection, source, synchronization, closure, obstruction, readout, reflection, analytic, exhaustion, and transport datums specified in Part I, the conclusions of Theorem 1 hold.*

Proof. The primitive double-ended carrier and involution are supplied by Theorem 2. Ledgered projection and recovery are supplied by Theorem 3 and Corollary 4. Source identity and native synthesis are supplied by Definition 5, Theorem 7, and Corollary 8. Finite simultaneous carrier realization and common refinement are supplied by Theorems 9 and 10. Exact complete-cell closure and obstruction-directed adaptation are supplied by Theorem 11, Theorem 12, Corollary 13, Theorem 14, and Theorem 15. The retained-mode bridge, channel residual factorization, one-dimensional readout identity, completed reflection, and analytic landing are supplied by Theorems 16, 17, 18, 19, 20, 21, 23, 24, and 25. Event exhaustion and coherent transport are supplied by Theorems 26, 27, and 28. These are exactly the arrows named in Theorem 1. □

9 The scale mismatch behind the $S(t)$ term

$$\boxed{S_{\mathrm{ev}}(t) \text{ is not an additional native oscillatory feature of the prime/harmonic system.}}$$

The native system is realized on a harmonic carrier — written below at $\pi/3$, the scale at which part (i) of Theorem 30 is proved, with the trivial character's own cell $\pi/6$ giving the identical ledger (Remark 31). The conventional analytic readout uses a unit-1 coordinate system. These two carrier scalings represent the same arithmetic procession, but they do not register the same crossings in the same coordinate units.

Typing convention, fixed before the theorem. Throughout this section $N_{\mathrm{ev}}(t)$ denotes the carrier's distinct event count — the on-line ordinate census $\#\{\gamma \in (0,t] : \zeta(\frac{1}{2}+i\gamma)=0\}$ (Lean: `zeroEventCount`) — and $S_{\mathrm{ev}}(t) := N_{\mathrm{ev}}(t) - 1 - \vartheta(t)/\pi$ its normalized form. The multiplicity-weighted line census is $N^{\mathrm{mult}}_{\mathrm{line}}(t) = \sum_{0<\gamma\le t} \mathrm{ord}_{1/2+i\gamma}\,\zeta$ (Lean: `zeroEventCountMult`), with ledger $S_{\mathrm{mult}} = N^{\mathrm{mult}}_{\mathrm{line}} - 1 - \vartheta/\pi$. The classical Riemann–von Mangoldt term instead normalizes the strip census with multiplicity. Consequently the exact bookkeeping is

$$S_{\mathrm{class}}(t) = S_{\mathrm{ev}}(t) + \big(N^{\mathrm{mult}}_{\mathrm{line}}(t) - N_{\mathrm{ev}}(t)\big) + N^{\mathrm{mult}}_{\mathrm{off}}(t),$$

or equivalently $S_{\mathrm{class}} = S_{\mathrm{mult}} + N^{\mathrm{mult}}_{\mathrm{off}}$. Thus distinct events, excess line multiplicity, and off-line strip multiplicity are separate typed ledgers. Every unconditional theorem below states explicitly which one it uses.

9.1 The theorem

Let F_ζ denote the trivial-character arithmetic fiber and let $\mathsf{X}_{\zeta,H}$ denote its realization on carrier scale H, with the fiber law, local arithmetic data, phasor amplitudes, source identity, and crossing criterion held fixed. Only the carrier scale is changed.

Theorem 30 (The $S(t)$ carrier-scale compensation theorem)**.** *With the registered readouts of Definition 35:*

(i) $N_{\pi/3}(e^t) = N_{\mathrm{ev}}(t)$*: the native count is the arithmetic event count over the height ray;*

(ii) $R_1(e^t) = 1 + \vartheta(t)/\pi$*: the unit-chart* continuous phase registration *is the DC residue plus the archimedean clock, with ϑ independently derived from the gauge (the current Lean identifier for R_1 is* `N_1`*);*

(iii) *consequently*

$$\boxed{N_{\pi/3}(e^t) - R_1(e^t) = S_{\mathrm{ev}}(t)}, \qquad S_{\mathrm{ev}}(t) = N_{\mathrm{ev}}(t) - 1 - \frac{\vartheta(t)}{\pi}.$$

The two terms have different types: $N_{\pi/3}$ is cardinal-valued event data, whereas R_1 is real-valued continuous phase registration. Lean's `registeredCount` *assembles the latter from an independently proved gauge clock, DC base, and scale-dependent event coefficient; it does not prove that R_1 is the cardinality of a set of unit-chart* *crossings. Every displayed identity is machine-checked in* `RequestProject/CarrierScaleCompensation.lean` `native_identification'`, `unit_identification`, `carrier_scale_compensation_S`.

Remark 31 (Which harmonic scale names the box)**.** The theorem is stated at $\pi/3$ because that is where the realization is anchored (part (i) is proved at $\pi/3$), *not* because $\pi/3$ is ζ's aligned cell. It is not: the fiber-scale assignment of §26 sends the trivial character to $\pi/6$ (the μ_{12} cell, where the eta-regulated principal reading lands, $\frac{1}{2}\ln 3 + i\pi/6$), and $\pi/3$ is χ_3's aligned scale. Naming $\pi/6$ instead changes nothing, and this is a theorem rather than an expectation:

$$N_{\pi/6}(e^t) - R_1(e^t) = S_{\mathrm{ev}}(t) \qquad (\texttt{Sgap\_pi6\_one},\ \texttt{unit\_tracks\_pi6}),$$

because the whole harmonic family registers identically, $S_{\pi/m,\pi/m'} = 0$ (`Sgap_pi_div`; Theorem 39). The reason is structural: the event channel reads the arc-set *indicator*, which is 1 at every π/m ($m \cdot (\pi/m) = \pi$, `pi_div_channel_open`) — it does not read the cell resolution. At $\pi/3$ the mark falls at $k \equiv 3 \pmod 6$, at $\pi/6$ at $k \equiv 6 \pmod{12}$: twice the cells, one mark per closed loop either way, the same count. What is *not* gauge is the cell itself (μ_6 versus μ_{12}) — it governs the complete-cell sums of Theorem 11 — but the registered count does not read it, so the choice of harmonic scale is a naming convention here and carries no mathematical content.

Remark 32 (Register: the classical $S(t)$)**.** What is proven above is unconditional for the distinct-event line ledger. For the classical strip term the unconditional decomposition is

$$S_{\mathrm{class}}(t) = \underbrace{N_{\pi/3}(e^t) - R_1(e^t)}_{S_{\mathrm{ev}}(t),\ \text{proven}} + \underbrace{\big(N^{\mathrm{mult}}_{\mathrm{line}}(t) - N_{\mathrm{ev}}(t)\big)}_{\text{excess line multiplicity}} + \underbrace{N^{\mathrm{mult}}_{\mathrm{off}}(t)}_{\text{off-line strip multiplicity}} .$$

Equivalently, the multiplicity ledger proved below satisfies $S_{\mathrm{class}} = S_{\mathrm{mult}} + N^{\mathrm{mult}}_{\mathrm{off}}$. This is the precise boundary between the unconditional line theorem and the classical strip function; neither off-line multiplicity nor simplicity is silently discarded.

Remark 33 (Register: the atomic defect ledger — sign flips at cancellation)**.** For a quotient $Q = \Lambda_P/\Lambda_C$ of completed L-functions the registration defect is the signed measure $d\nu = dN_P - dN_C$: positive atoms at numerator zeros, and at each denominator zero ρ the site atom $\mathrm{ord}_\rho \Lambda_P - \mathrm{ord}_\rho \Lambda_C$. The ledger has exactly three layers, and they must never be blended:

(DC) the accumulated defect $\nu[0,T] \sim \frac{d_P - d_C}{2\pi} T \log T$ is *unconditional* — Riemann–von Mangoldt counting on the licensed factors; at the rung-r instance the coefficient is the degree $r+1$ of the formal symmetric power;

(AC) the fluctuation about it is the $S(t)$ oscillation of the quotient, on the fluctuation ledger and nothing else;

(atoms) the *sign* of each site atom at a denominator zero is the cancellation indicator: a successful cancellation (numerator vanishing on the site) flips the would-be negative atom to ≥ 0; a failed cancellation books a genuine pole atom < 0.

Atomic nonnegativity of ν at every denominator-zero site is therefore *equivalent* to pole-freeness of the quotient — at the rung instance, to entirety of the formal Sym^r completion. Integration deliberately forgets the atomic signs: the accumulated positivity (DC) is unconditional and cannot by itself decide containment; the containment content lives entirely in the sign flips. The two attacks on that content are dual: the warp/carry rigidity chain (`RequestProject/ResidueLedger`, `WarpRigidity`, `GlobalCarry`, `RectResidue`, `PolarCorrection`, `PolarCorrectionJordan`) forces the atoms nonnegative top-down; the registration ledger of this section reads the same signs bottom-up, site by site, as the unit-chart residue left where the harmonized carrier cancels residue-free.

9.2 Independent contour coordinate for the classical argument

The classical argument is constructed independently of every event and zero-count ledger. At a good height $T > 0$, meaning that no nontrivial zero has ordinate T, let

$$\Gamma_T : \quad 2 \longrightarrow 2 + iT \longrightarrow \frac{1}{2} + iT.$$

The first segment lies in $\operatorname{Re} s = 2$, where Euler-product nonvanishing applies. On the second segment, a hypothetical vanishing either has $\operatorname{Re} s \geq 1$, again contradicting nonvanishing, or is a nontrivial zero of ordinate T, contradicting goodness. Thus $\zeta \circ \Gamma_T$ is a path in $\mathbb{C}^\times$.

Lift this path through the exponential covering $\exp : \mathbb{C} \to \mathbb{C}^\times$, with initial value

$$L_T(0) = \log \zeta(2) = \log(\pi^2/6) \in \mathbb{R}.$$

Path-lift uniqueness removes every branch choice. Define

$$S_\Gamma(T) := \frac{\Im L_T(1)}{\pi}. \tag{1}$$

This is a genuine global coordinate of the zeta path: its definition contains none of `zeroEventCount`, `zeroEventCountMult`, `S`, or `Smult`, and the endpoint theorem is

$$e^{L_T(1)} = \zeta\left(\frac{1}{2} + iT\right).$$

Combining this independently lifted phase with the independently derived Gamma clock and the real native factor gives

$$\exp\big(i(\pi S_\Gamma(T) + \vartheta(T))\big) = \frac{Z(T)}{|Z(T)|} \in \{-1, 1\}, \tag{2}$$

where $Z(T) = \texttt{rsZ}(T)$. Consequently the covering itself produces an integer winding coordinate $k_\Gamma(T)$, rather than receiving one as a counting assumption, and

$$\boxed{S_\Gamma(T) = k_\Gamma(T) - \frac{\vartheta(T)}{\pi}.} \tag{3}$$

In Lean these are `standardContour`, `zetaContourPath`, `contourLogLift`, `classicalSContour`, `contour_phase_times_clock_eq_sign_rsZ`, and `classicalSContour_eq_winding_sub_clock` in `RequestProject/ZetaContourArgument.lean`. They are proved from the analytic zeta path, the exponential covering, and the Gamma-factor factorization; no census formula is unfolded in the construction.

The census side is also constructed independently. Lean's `stripZeroWindow`(T) is the finite set of actual nontrivial zeros with $0 < \Im\rho \leq T$, and `stripZeroCountMult` sums their ξ-orders. Filtering this same finite set by $\operatorname{Re}\rho = 1/2$ and $\operatorname{Re}\rho \neq 1/2$ gives typed on-line and off-line counts. The compiled identities are

$$N^{\text{mult}}_{\text{strip}}(T) = N^{\text{mult}}_{\text{on}}(T) + N^{\text{mult}}_{\text{off}}(T), \qquad N^{\text{mult}}_{\text{on}}(T) = N^{\text{mult}}_{\text{line}}(T).$$

They are `stripZeroCountMult_eq_onLine_add_offLine` and `onLineStripZeroCountMult_eq_zeroEventCountMu` their composition is `stripZeroCountMult_eq_line_add_offLine`. The second equality is a proved bijection $\gamma \mapsto \frac{1}{2} + i\gamma$, with multiplicities transported by the analytic-unit order bridge `eventOrder_eq_xiOrderNat`.

The argument-principle comparison now identifies $k_\Gamma(T)+1$ with this independently defined strip census by an explicit second construction, rather than by citing the classical counting formula. On the same broken path Lean forms the completed logarithmic lift

$$\mathcal{L}_{\xi,T}(u) = \log \Gamma_T(u) + \log(\Gamma_T(u) - 1) - \log 2 + \log \Gamma_{\mathbb{R}}(\Gamma_T(u)) + L_T(u).$$

The Gamma logarithm is first constructed on the simply connected right half-plane and normalized at (2); its imaginary part on the critical line is then proved to be the independently integrated clock $\vartheta(T)$. The two principal polynomial logarithms stay in the upper slit plane and satisfy the geometric endpoint law

$$\Im\big(\log(\tfrac{1}{2} + iT) + \log(-\tfrac{1}{2} + iT) - \log 2\big) = \pi.$$

Thus the value (+1) is the winding contribution of the completion factor (s(s-1)/2); it is derived from the upper-half-plane geometry and is not inserted into the carrier or census definitions. Pointwise exponentiation gives $exp\mathcal{L}_{\xi,T}(u) = \xi(\Gamma_T(u))$.

The continuous-lift calculus then proves, on the two affine segments,

$$i\int_0^T \frac{\xi'}{\xi}(2+iy)\,dy - \int_{1/2}^2 \frac{\xi'}{\xi}(x+iT)\,dx = \mathcal{L}_{\xi,T}(1) - \mathcal{L}_{\xi,T}(0).$$

Reflection and conjugation turn this half-contour increment into the full rectangle boundary integral. Independently, the analytic divisor factorization evaluates that same boundary integral as $2\pi i$ times the multiplicity sum over *all* zeros in the upper strip, including any off-line zeros. Cancelling the nonzero factor $2\pi i$ gives the compiled theorem

$$\boxed{k_\Gamma(T) + 1 = N^{\mathrm{mult}}_{\mathrm{strip}}(T)}.$$

The corresponding Lean spine is `gammaLog_line_im`, `xiPolynomialLog_line_im`, `exp_xiContourLog`, `halfContourIntegral_logDeriv_riemannXi_eq_lift_sub`, `rectangleBoundaryIntegral_logDeriv_riemannXi` and `contourWindingIndex_add_one_eq_stripZeroCountMult`. This comparison is downstream of both independent constructions: it is not the definition of either S_Γ or the zero count, and no critical-line location statement is used in the strip census.

Eliminating the total strip census between this theorem and the independently proved on-line/off-line partition gives a sharper registration rule. At every good height,

$$N^{\mathrm{mult}}_{\mathrm{off}}(T) = k_\Gamma(T) + 1 - N^{\mathrm{mult}}_{\mathrm{line}}(T), \tag{4}$$

$$S_\Gamma(T) = \boxed{S_{\mathrm{mult}}(T) + N^{\mathrm{mult}}_{\mathrm{off}}(T)} \tag{5}$$

These are the compiled theorems `offLineStripZeroCountMult_eq_winding_sub_line` and `classicalSContour_e` In particular, `classicalSContour_eq_Smult_iff_offLine_eq_zero` proves that the independently continued global coordinate agrees with the native multiplicity coordinate exactly when the finite-strip coverage defect vanishes. There can be no cancellation hidden in that defect: every summand is the positive analytic order of an actual zero. The pointwise theorem `mem_stripZeroFinset_line_or_coordinate_d` therefore says that each analytic strip zero is either a native line event or produces a detectable mismatch of the two global coordinates. This is the precise finite-window rule connecting the contour census, the multiplicity ledger, and the $S(t)$ registration. Finally, `exists_goodHeight_gt` constructs a good contour above every real ordinate from the finite next-window census, and `upper_nontrivialZero_line_or_globalCoordinateDefect` promotes the dichotomy globally: every upper-half-plane nontrivial zero is a native line event or exhibits an explicit higher good height where the two global coordinates differ.

This non-cancellation statement is now quantitative. If an enclosed zero ρ is off the carrier, then

$$\operatorname{ord}_\rho \xi \le N_{\mathrm{off}}^{\mathrm{mult}}(T) = S_\Gamma(T) - S_{\mathrm{mult}}(T),$$

and the right-hand side is strictly positive. These are the compiled theorems `xiOrderNat_le_offLineStripZeroCou` `xiOrderNat_le_classicalSContour_sub_Smult_of_offCarrier`, and `classicalSContour_sub_Smult_pos_of_` Thus neither a second zero nor the completed-reflection partner can cancel an off-carrier contribution in the global coordinate ledger.

Remark 34 (Relation to Turing's method)**.** The principle that an independently continued global count, compared against a locally assembled list, certifies the list's completeness is classical: it is Turing's method [36], in which an explicit bound on the average of the argument (Littlewood's theorem) makes the comparison effective at a given height; Turing's error terms were corrected by Lehman [37], the argument-integral bounds sharpened by Trudgian [38], the method extended to general L-functions by Booker [39], and an explicit-formula variant proved for an extended Selberg class by Büthe [40]. What is claimed here is therefore not the count-comparison principle. It is the exact typed form: (5) is an identity between two independently constructed coordinates, holding at every good height with no averaging, no estimate, and no error term; the defect is multiplicity-safe — every summand a positive analytic order, so cancellation between zeros is impossible — and both constructions and their comparison are machine-checked. Turing-style inequalities estimate precisely the quantity this census identifies exactly: the identity is the law those certificates instantiate, stated and verified separately from the estimates that make it computable.

The native branch is now carried all the way into the event-independent three-dimensional operator. In `RequestProject/ZetaContourNative3D.lean`, the data type `PrincipalContourNative3DCertificate` contains a completed mod-one focal event e whose height is $e.Z = \exp(\operatorname{Im}\rho)$, its exact ambient physical height, the equality of its carrier readout with ρ, a nonzero eigenvector of `carrierThreeDOperator 1 1` with eigenvalue $\Im\rho$, every spatial harmonic-pencil rank drop, and the exact one-dimensional/three-dimensional resolvent-trace relation weighted by `xiOrderNat` ρ. It now also contains a single parameter-preserving kernel coupling from `RequestProject/SpectralCarrierKernelCo` That object retains simultaneously the nonzero kernel of the configured full analytic operator $L_{\mathrm{spec}}(\rho)$ id, the completed focal event at the same analytic point, the ambient carrier eigenvector, and the self-adjoint one-mode resolvent kernel. The constructor `ofLspecZeroAtHeight` builds this object from $L_{\mathrm{spec}}(\sigma_* + i\log Z) = 0$ at every positive carrier height; conversely `lspec_zero_of_coupling` proves that the spatial coupling cannot create an analytic vanishing. The exact identification theorems `nonempty_coupling_iff` and `nonempty_coupling_iff_readoutKernel` state that such a coupling exists precisely when the analytic kernel and the independently derived carrier coordinate refer to the same point. Thus neither operator is silently replaced by the other, and the midpoint coordinate is supplied by carrier membership. The finite-window defect has now been rewritten entirely in this typed language. In `RequestProject/ZetaContourCouplingDefect3D.lean`, define $N_{\mathrm{uncpl}}^{\mathrm{mult}}(T)$ by summing the positive analytic order over precisely those strip kernels for which no `SpectralCarrierKernelCoupling3D` exists. The compiled identities

$$N_{\mathrm{uncpl}}^{\mathrm{mult}}(T) = N_{\mathrm{off}}^{\mathrm{mult}}(T), \qquad S_\Gamma(T) = S_{\mathrm{mult}}(T) + N_{\mathrm{uncpl}}^{\mathrm{mult}}(T)$$

are `uncoupledKernelCountMult_eq_offLine` and `classicalSContour_eq_Smult_add_uncoupledKernelCount`. Because each included order is positive, `classicalSContour_eq_Smult_iff_everyKernelCoupled` proves that the two global coordinates agree exactly when every enclosed full analytic kernel has a parameter-preserving coupling to the independently defined carrier operator. Hence an absent coupling cannot be hidden by cancellation between zeros. This has a midpoint-free no-drift form. For every base $n > 1$, `ofLspecZeroOfNoRadialDrift` constructs the full coupling from analytic vanishing and $n^{\operatorname{Re} s - \sigma_*} = 1$; it derives $\operatorname{Re} s = \sigma_*$ internally from `noRadialDrift_iff_carrierAbscissa`. The independent constructor `ofLspecZeroOfBinaryNoRadialDrift` repeats the construction through the physical height $Z = \exp(\operatorname{Im} s)$. Thus no numerical midpoint is an input to either constructor. At finite-window scale, `uncoupledKernelCountMult_eq_radialDrift` proves that the uncoupled count is exactly the positive multiplicity of kernels with nontrivial radial drift, and `classicalSContour_eq_Smult_iff_everyKernelNoRadialDrift` identifies global-coordinate agreement with zero drift for every enclosed kernel. These finite-window statements have also been globalized without changing their content. In `RequestProject/ZetaContourGlobalIdentification3D.lean`, the theorem `globalCoordinateIdentification_iff_upperKernelCoupling` identifies agreement at every good height with a full parameter-preserving coupling for every upper nontrivial analytic zero. The corresponding theorem `globalCoordinateIdentification_iff_upperNative3DCertificate` upgrades each such coupling to the complete physical-height focal-event, ambient-eigenvector, harmonic-rank-drop, twin-trace, and multiplicity certificate. Independently, `globalCoordinateIdentification_i` obtains the same global criterion directly by enclosing each zero in a higher good contour.

There is now also a conservation-native source formulation with no literal half-unit in its definition. For a conservative source mode ψ, set

$$\operatorname{coord}_{\mathrm{src}}(\psi) := \sigma_* + \operatorname{rate}(\psi),$$

where σ_* is the unique area-law-selected carrier abscissa. The theorem `conservativeCarrierCoord_re` derives $\operatorname{Re} \operatorname{coord}_{\mathrm{src}}(\psi) = \sigma_*$ from source conservation, and `conservativeCarrierCoord_noRadialDrift` derives unit radial projection at every base greater than one. Finally, `globalCoordinateIdentification_iff_upper` proves that all-good-height coordinate agreement is exactly the statement that every upper analytic zero is the coordinate of one of these conservative carrier modes. This source identification has a direct pointwise landing as well: `principalContourNative3DCertificate_of_noRadialDrift` constructs the complete certificate from a zero and its no-drift projection, while `principalContourNative3DCertificat` constructs it from a conservative source representation. Neither constructor assumes any global coordinate identity; both produce the physical event and operator data at that single analytic parameter. The analytic-zero side is no longer represented only by a scalar membership predicate. The compiled transfer in `RequestProject/ZetaZeroNative3DSourceTransfer.lean` begins with every actual $\rho \in$ `ZD.NontrivialZeros` and constructs `PrincipalZeroAnalyticFiber3D`(ρ). Its state is the literal eta-configured `Vec3` fiber at the original complex parameter; after division by the nonvanishing eta factor, its plane partial sums converge to $\zeta(\rho) = 0$. The same object carries the event-independent ambient helix state at ordinate $\Im\rho$, positive physical height $Z = \exp(\Im\rho)$, and its nonzero eigenvector with eigenvalue $\Im\rho$. None of these fields places ρ on a preselected vertical line.

Before any parameter-preserving carrier-to-chart identification, the compiled constructor `principalZero_focalCancellation_on_carrier` packages this closure as a `PrincipalZeroNativeCarrierEvent` Its `carrierState` is a typed member of the independently defined Archimedean helix, and its stored analytic fiber closes at the same physical height. The theorems `.on_archimedeanHelix`, `.physicalHeight_eq_focalHeight`, and `.ambient_eigenvector` prove, respectively, literal helix membership, equality of the carrier and focal heights, and the ambient 3D eigenvector equation. Most importantly, `.noRadialDrift` is derived directly from that carrier membership by `carrierPointAtHeight_noRadialDrift`. The event type and this proof use neither the Euclidean magnitude of the analytic fiber nor a radial-limit hypothesis. Only afterwards does `.carrierReadout_scaleBalanced` apply the area law to the carrier readout. Thus focal cancellation is first marked at $Z = \exp(\Im\rho)$ on the helix, no drift is a fact about that helix state, and equality of the resulting carrier coordinate with the original complex parameter remains the separate chart-identification theorem.

The two uses of "rides" are now separated at the type level. For an analytic zero fiber F_ρ, Lean defines `PrincipalZeroAnalyticFiber3D.RidesCarrier` by the state equality

$$F_\rho = \texttt{spectralFiber etaW}\big(\texttt{carrierPointAtHeight}(Z_\rho)\big) \qquad \text{in } (\mathbb{N} \to \texttt{Vec3}).$$

This is the fiber-to-carrier attachment; it is stronger than sharing the ordinate or physical height. The carrier-to-helix map is the independent field $\texttt{carrierState.point} = \texttt{Geometry.gammaY}(1, 1, \texttt{carrierState.ordina}$ The theorem `PrincipalZeroAnalyticFiber3D.ridesCarrier_iff_carrierWeld` proves that the first state equality is equivalent to the parameter-preserving weld $\texttt{carrierPointAtHeight}(Z_\rho) = \rho$: its imaginary part comes from $\log Z_\rho = \Im \rho$, while equality of the radial exponents is extracted from the actual three-dimensional magnitudes at the nontrivial carrier site $n = 2$. Composing with the area law gives `.ridesCarrier_iff_scaleBalanced`. Thus no equality of unlike objects is used and no bare midpoint is placed in the ride definition.

Exact focal cancellation itself cannot occur off the carrier. Lean defines `CompletedFocalCancellationRepresent` by requiring one positive physical height Z for which the completed residual $\texttt{Dcell}\ \chi\ Z$ vanishes and whose readout $\texttt{carrierPointAtHeight}\ Z$ is the same parameter ρ. The compiled theorem `completedFocalCancellationRepresents_iff` proves

$$\texttt{CompletedFocalCancellationRepresents}(\chi, \rho) \quad \Longleftrightarrow \quad L(\rho, \chi) = 0 \ \text{ and } \ \operatorname{Re} \rho = \sigma_*.$$

The reverse direction reconstructs $Z = \exp(\Im \rho)$; the real coordinate is read from the Archimedean area-law carrier. Consequently `no_offCarrier_completedFocalCancellation` proves pointwise that a parameter with $\operatorname{Re} \rho \neq \sigma_*$ cannot be represented by focal cancellation. This precedes every resolvent trace and every zero count, so there is no cross-event or reflected-pair cancellation mechanism to invoke.

The second structure, `PrincipalZeroNative3DSourceTransfer`, records the geometric projection of that analytic fiber into a conservative source mode and the complete native focal/operator certificate at the same parameter. Its constructor `principalZeroNative3DSourceTransfer_of_noRadialDrift` accepts the area-law equation $n^{\operatorname{Re} \rho - \sigma_*} = 1$, derives the source coordinate internally, and never accepts $\operatorname{Re} \rho = 1/2$. Conversely the source conservation field recovers that no-drift equation at every $n > 1$. Hence the compiled pointwise theorem `nonempty_principalZeroNative3DSourceTransfer_iff_noRadialDrift` identifies the exact projection step, rather than hiding it inside a statement that a zero already equals a carrier point. Globally, the native direction no longer needs that radial equation as an input. The theorem `carrierPointAtHeight_noRadialDrift` proves zero radial drift for every physical-height carrier readout solely from helix membership, and `completedThreeDZeroAtHeight_focal_rankDrop_eigenvector_noRadialDrift` assembles, for one and the same event, the completed focal residual zero, the spatial harmonic-pencil rank drop, the ambient carrier eigenvector at $\log Z$, and the no-drift conclusion. At the transfer layer, `PrincipalContourNative3DCertificate.analyticFiber_eq_carrierFiber` proves that the literal $\texttt{spectralFiber etaW}\ \rho$ is the carrier fiber at the certificate's marked height, while `PrincipalContourNative3DCertificate.noRadialDrift` derives its radial equation from that identification. Consequently `principalZeroNative3DSourceTransfer_of_nativeCertificate` constructs the full source transfer from native realization itself, with neither a midpoint nor a no-drift proposition supplied. The physical-height identification is also literal, not merely an agreement of ordinates: `PrincipalZeroNative3DSourceTransfer.eventHeight_eq_focalCancellationHeight` proves that the native event height equals the analytic fiber's stored focal-cancellation height. After rewriting by that equality, `.harmonicRankDrop_at_focalCancellationHeight` proves that the completed spatial pencil rank-drops at precisely that height. The combined theorem `.focalCancellation_rankDrop_noRadialDrift_sameHeight` returns the fiber readout closure, pencil rank drop, and helix no-drift law at this one shared physical height. Moreover every completed source transfer proves the attachment itself by `PrincipalZeroNative3DSourceTransfer.fiberRidesCarrier`; it then derives the original analytic parameter from the carrier readout by `.carrierReadout_eq_analyticParameter`. The composed theorem `.fiberRidesCarrier_carrierRidesHelix` returns, in one statement, the fiber-state attachment, literal membership of the carrier point in Γ_Y, and equality of the two physical focal heights. At the global level, `globalCoordinateIdentification_iff_allCanonicalZeroFibersRideCarrier` states the remaining identification exactly as this state-level attachment for every canonical analytic zero fiber. The assembly has also been closed without passing through the contour coordinate: `nonempty_principalZeroNative3DSourceTransfer_iff_fiberRidesCarrier` proves, zero by zero, that existence of the completed native source transfer is exactly equality of the two complete $\mathbb{N} \to$ `Vec3` fiber states. Quantifying this statement gives `allNative3DSourceTransfer_iff_allCanonicalZeroFibersRideCarrier`, and `allNative3DSourceTransfer_iff_zeroLedger_eq_carrierHeight` identifies that same family-level attachment with equality of the analytic zero-ledger and event-independent carrier-height operators. Thus the source, state, and operator formulations are proved equivalent; none silently substitutes a bare equality of ordinates for fiber attachment.

Globally, `globalCoordinateIdentification_iff_upperNative3DSourceTransfer` states that all-good-height coordinate agreement is exactly transfer of every upper zero from its genuine three-dimensional analytic fiber to the conservative native source. Finally, `upperNative3DSourceTransfer_of_selfAdjointXiReceiver` performs this full transfer for the independent self-adjoint xi-receiver route. The conjugation-complete theorem `globalCoordinateIdentification_iff_allNative3DSourceTransfer` upgrades the statement to every nontrivial zero in both half-planes, and `allNative3DSourceTransfer_of_selfAdjointXiReceiver` performs that complete two-sided transfer through the same receiver. The final operator propagation is now direct and basiswise. Given those two-sided transfers, `xiZeroLedgerOperator_eq_carrierHeight_of_allNative3DSourceTransfer` proves

$$\texttt{xiZeroLedgerOperator} = \texttt{xiCarrierHeightOperator}$$

by evaluating the two diagonal operators on each canonical analytic-zero mode and using only that mode's compiled scale-balance theorem. No trace sum or cross-event cancellation appears. The theorem `globalCoordinateIdentification_of_allNative3DSourceTransfer` then feeds this operator equality into the contour/carrier unification and returns `classicalSContour` = `Smult` at every good height. At the receiver endpoint, `globalCoordinateIdentification_and_zeroLedger_eq_carrierHeight_of_selfAdjointXiReceiver` returns both conclusions in one certificate. The state-space form is sharper still. At every positive site n, `PrincipalZeroAnalyticFiber3D.radialMagnitude` computes directly from the literal `Vec3` state

$$\mathrm{mag}_3(F_\rho(n)) = n^{-\operatorname{Re}\rho}.$$

Consequently `PrincipalZeroAnalyticFiber3D.carrierScaleBalanced_iff_radialMagnitude` identifies the additional parameter-preserving scale condition with a positive finite limit of $\mathrm{mag}_3(F_\rho(n))R_{\mathrm{carrier}}(n)$. This statement contains neither a carrier point nor a supplied midpoint. It is a diagnostic for whether the analytic exponent agrees with the already no-drift carrier readout, not the source of native helix membership or native no drift. The numerical half-unit is recovered only when the independently proved Archimedean area law evaluates the unique balanced exponent.

The analytic/operator identification has also been completed before any event certificate is introduced. In `RequestProject/XiZeroLedgerResolvent3D.lean`, the affine chart identity

$$\left(\frac{1}{2} + iz\right) - \rho = i\big(z - \mathrm{poleParam}(\rho)\big)$$

transports the proved Hadamard partial fraction term by term and gives `xiChannel_eq_constant_add_zeroResolventTrace`:

$$-\frac{\xi'}{\xi}\left(\frac{1}{2}+iz\right) = C + \sum_{\rho} m_\rho \left(\frac{i}{z - \mathrm{poleParam}(\rho)} - \frac{1}{\rho}\right).$$

Thus the half-unit is the affine carrier readout unit used in an independently proved change of coordinates, not a zero-location field. The associated `xiZeroLedgerOperator` is defined on the free finitely supported Hilbert space on analytic zeros and has $\mathrm{poleParam}(\rho)$ as an actual eigenvalue by `xiZeroLedgerOperator_hasEigenvalue`. Its definition mentions no `ThreeDZero`, focal event, or equality with `carrierPoint`. Spectral symmetry then yields the area-law balance event by event through `xiZeroLedgerOperator_symmetric_imp_scaleBalanced`. Finally the compiled unification theorem `globalCoordinateIdentification_iff_xiZeroLedgerOperator_symmetric` states that symmetry of this independently identified resolvent operator is exactly the all-good-height identity $S_\Gamma = S_{\mathrm{mult}}$. This rules out an "offline cancellation" inside the comparison: analytic multiplicities are positive, and every diagonal pole is carried by its own nonzero basis eigenvector. More explicitly, the compiled operator identity `xiZeroLedgerOperator_eq_height_sub_I_smul_radialDrift` is

$$H_\xi = H_{\mathrm{carrier}} - iD_{\mathrm{radial}}, \qquad D_{\mathrm{radial}} e_\rho = \left(\mathrm{Re}\,\rho - \frac{1}{2}\right) e_\rho.$$

Here both H_{carrier} and D_{radial} are independently constructed real-diagonal symmetric operators; the half-unit in D_{radial} is the value selected by the preceding area law. The theorems `xiRadialDriftOperator_eq_zero_iff_all_scaleBalanced` and `xiZeroLedgerOperator_isSymmetric_iff_radialDrift_eq_zero` prove, basis mode by basis mode, that symmetry of the analytic pole operator is precisely extinction of radial drift, not cancellation of drift after summing different zeros. At global scale this is compiled in the two interchangeable forms `globalCoordinateIdentification_iff_xiRadialDriftOperator_eq_zero` and `globalCoordinateIdentification_iff_zeroLedger_eq_carrierHeight`. There is also a canonical projection statement, rather than a bare instruction to discard the radial coordinate. On the same finitely supported zero-state space, `xiZeroLedgerFormalAdjoint` is defined coefficientwise and satisfies the exact Green identity

$$\langle H_\xi f, g\rangle = \langle f, H_\xi^\dagger g\rangle$$

by `xiZeroLedgerFormalAdjoint_green`. Its Hermitian part is computed, not postulated:

$$P_{\mathrm{nd}}(H_\xi) := \frac{1}{2}\left(H_\xi + H_\xi^\dagger\right) = H_{\mathrm{carrier}},$$

the theorem `xiZeroLedgerNoDriftProjection_eq_carrierHeight`. The complementary anti-Hermitian part is exactly the radial ledger,

$$D_{\mathrm{radial}} = \frac{i}{2}\left(H_\xi - H_\xi^\dagger\right),$$

by `xiRadialDriftOperator_eq_I_half_smul_adjointDefect`. Hence the all-good-height identity is equivalently the fixed-point equation $H_\xi = P_{\mathrm{nd}}(H_\xi)$, compiled as `globalCoordinateIdentification_iff_zeroLedger_fixedByNoDriftProjection`. This makes precise the phrase "no-drift projection": the carrier height and the drift are the Hermitian and anti-Hermitian components of the independently identified analytic pole operator, and separate basis modes cannot cancel one another in either component. Indeed `xiZeroLedgerResidual_eq_zero_iff_each_basis` proves that the projection residual vanishes as an operator if and only if it vanishes on every canonical zero basis vector separately, and globalCoordinateIdentification_iff_eachZeroBasisProjectionResidual_eq_zero identifies that basiswise condition with the global coordinate equality. The pointwise computation is literal:

$$(H_\xi - P_{\mathrm{nd}} H_\xi) e_\rho = -i \left(\operatorname{Re} \rho - \frac{1}{2} \right) e_\rho,$$

proved by xiZeroLedgerProjectionResidual_on_basis. Finally principalZeroProjectionResidual_eq_zero_iff_radialMagnitude identifies vanishing of this same basis residual with the positive finite area-normalized Euclidean-magnitude limit of the actual eta-configured 3D fiber. Thus the operator, state-space, and area-law formulations agree at each event before any global summation is taken.

There is also a direct receiver formulation. For every complex sample z, zeroPoleResolvent_eq_carrierBasisReadout_iff_scaleBalanced proves

$$i\,(z - \mathrm{poleParam}(\rho))^{-1} = i\,\langle e_{\Im \rho}, (z - H_{\mathrm{carrier}})^{-1} e_{\Im \rho} \rangle \quad \Longleftrightarrow \quad F_\rho \text{ is area-law scale balanced.}$$

Only injectivity of inversion and of multiplication by i is used, so one sample already identifies the pole parameter and no trace-level cancellation is available. The global assembly globalCoordinateIdentification_iff_eachZeroPoleResolvent_eq_carrierBasisReadout states that $S_\Gamma = S_{\mathrm{mult}}$ at all good heights exactly when this equality holds separately on every analytic-zero basis mode. The regularized trace bridge is now typed at the same resolution. Lean defines xiZeroResolventKernel and xiCarrierHeightResolventKernel with the same positive analytic multiplicity and the same Hadamard counterterm, but with pole coordinates $\mathrm{poleParam}(\rho)$ and $\Im \rho$, respectively. The pointwise theorem xiZeroResolventKernel_eq_carrierHeightKernel_iff_scaleBalanced cancels the positive multiplicity and proves that the two kernels agree exactly at the area-law fixed point. Hence globalCoordinateIdentification_iff_basisResolvedCarrierResolventBridge identifies the all-good-height theorem with equality of these two kernels for every (z, ρ), before summation. Only after that basiswise equality is established does xiCarrierHeightResolventTrace take the tsum; carrierHeightResolventTrace_of_globalCoordinateIdentification then gives the literal carrier formula

$$-\frac{\xi'}{\xi}\left(\frac{1}{2} + iz\right) = C + \texttt{xiCarrierHeightResolventTrace}(z).$$

Thus the global theorem, the operator equality, and the resolvent-trace bridge have the same basiswise weld, and no scalar trace equality is used to infer a zero-by-zero identification. For the paper's stated primary-3D/shadow-1D conditional, Lean packages the whole implication in globalCoordinateIdentification_basisResolvedBridge_and_carrierTrace_of_allNative3DSourceTransfe The theorem consumes one native 3D source transfer for each analytic shadow and returns the all-good-height coordinate identity, every basis-resolved kernel equality, and the summed carrier-height trace together. In particular, the half-unit is delivered inside the native no-drift/area-law transfer and is not repeated as an independent assumption of this assembly theorem. The upper-half-plane census has now been joined to the complete two-sided analytic channel in RequestProject/ZetaContourXiReceiverIdentification3D.lean. The theorem conj_mem_nontrivialZeros transports genuine zeros by the real structure of the entire xi function, while nontrivialZero_im_ne_zero consumes the independently proved zero-free region $0 < \operatorname{Re} s < 1$, $|\operatorname{Im} s| < 2$, to exclude a real-axis member. Consequently upperNoRadialDrift_iff_allNoRadialDrift promotes the upper no-drift statement to the complete nontrivial zero set.

This gives the exact analytic receiver formulation

$$\big(\forall T \text{ good},\ S_\Gamma(T) = S_{\mathrm{mult}}(T)\big) \quad \Longleftrightarrow \quad \text{the xi-channel is regular off the real spectral axis,}$$

compiled as `globalCoordinateIdentification_iff_xiChannel_offReal_regular`. Thus the coordinate identification and the analytic receiver are now one theorem, not parallel narrative interfaces. Finally, `globalCoordinateIdentification_and_upperNative3D_of_selfAdjointXiReceiver` proves that a self-adjoint receiver of this independent channel supplies both the all-good-height equality and, for every upper zero, the complete physical-height 3D/operator certificate described above. The operator geometry needed by that receiver is now constructed on the full, event-independent state space in `RequestProject/CarrierUnboundedResolvent3D.lean`. Because the carrier ordinate ranges over all of $\mathbb{R}$, the native height operator H is unbounded; it is therefore represented directly on finitely supported carrier modes rather than being mislabeled as a bounded C-star-algebra element. For $\operatorname{Im} z \neq 0$, the diagonal formula

$$(z - H)^{-1} e_s = \frac{1}{z - s.\,\mathrm{ordinate}} e_s$$

is proved to be both a left and a right inverse of $zI - H$ by `carrierThreeDShift_comp_resolvent` and `carrierThreeDResolvent_comp_shift`. Moreover, `carrierBasisResolventReadout_regular_off_real` proves that every geometric basis matrix coefficient has a finite punctured limit at every nonreal parameter. These are properties of the Archimedean helix state space itself: the construction mentions no zero predicate, L-function, focal event, or event certificate. A bounded C-star receiver is used only after a compact coordinate transform; the unbounded 3D height operator retains the literal ordinate and hence the full real spectral axis. The all-height theorem `upper_nontrivialZero_kernelCoupling_or_globalCoordinateDefect` then returns, for every upper nontrivial zero, either this full typed coupling or an explicit higher good height at which the two coordinates differ. The compiled capstone `upper_nontrivialZero_native3DCertificate_or_`allowbreak `globalCoordinateDefect` constructs this complete certificate or returns the explicit higher good height where the global coordinates differ. Thus the contour census, physical-height event, ambient eigenvector, harmonic pencil, analytic multiplicity, and both trace charts are joined without defining the ambient operator from the zero certificate.

The proof occupies the rest of the section: part (i) in the native-$\pi/3$ subsection, part (ii) in the unit-1 subsection, part (iii) by subtraction; the induced event readout pinning every functor input is derived in the discharged-arrows subsection.

9.3 The event count and the phase registration

Definition 35 (Scale-indexed registered readout)**.** For a carrier realization $\mathsf{X}_{\zeta,H}$, write

$$C_H(F_\zeta; z) = \{\, c : \ c \text{ is a complete } F_\zeta\text{-closure in } \mathsf{X}_{\zeta,H} \text{ through raw height } z \,\}.$$

At the native scale the cumulative crossing count is

$$N_{\pi/3}(e^t) = \#C_{\pi/3}(F_\zeta; e^t).$$

The unit chart instead supplies the real-valued phase registration

$$R_1(e^t) = 1 + \frac{\vartheta(t)}{\pi}.$$

The theorem compares these two readouts; it does not identify the continuous quantity R_1 with a cardinality. A topological-degree or contour-winding coordinate for the classical argument is instead constructed in (1)–(3); it is a path lift of the actual zeta fiber, not a reinterpretation of R_1 as a set cardinality.

In Lean the scale-indexed registered readout is installed as `registeredCount`: DC base at the common origin, an independently constructed continuous clock procession in π-units, and the event count weighted by the induced per-event contribution of the scale's closure geometry (`eventContribution`; derivation in the discharged-arrows subsection below). The closure predicate is `NativeClosure` (the fiber's η-accumulation) and the counts are `nativeCount` and `zeroEventCount`.

9.4 Native $\pi/3$ chain

The native chain already present in the carrier framework is:

$$F_\zeta \longrightarrow \mathsf{X}_{\zeta,\pi/3} \longrightarrow C_{\pi/3}(F_\zeta; e^t) \longrightarrow N_{\pi/3}(e^t).$$

Its local pieces are the $\pi/3$ cell law

$$\mathrm{cell}(n) = \exp(in\pi/3),$$

the antipodal cancellation

$$\mathrm{cell}(n) + \mathrm{cell}(n+3) = 0,$$

the helix pencil closure equivalence, and the exact height readout

$$\mathrm{heightReadout}(e^\gamma) = \gamma.$$

In Lean these correspond to the native cell and pencil statements in `GeometricPhasorClosure.lean` and to the readout lemmas

`cell_antipodal_cancel`, `helix_cancellation_logfree`, `readout_exp`

in `GeometricReadout.lean`.

Proof of Theorem 30(i). A complete closure of the fiber at ordinate γ is exactly an on-line zero event: the fiber's own η-accumulation closes at γ if and only if the readout vanishes there (`nativeClosure_iff_zero`, via `Faithful.continuous_model_zeta`), and the height readout carries raw height e^t to ordinate t exactly (`readout_exp`). Here $N_{\mathrm{ev}}(t)$ is the zero-counting procession over the 3D representation's *entire* event space — the height ray, whose exhaustion is a theorem, consumed not assumed (`native_events_sourced`, instantiating the proven `threeD_exhaustive`; the height encoding is lossless, `event_encoding_complete`). Counting complete closures through raw height e^t therefore counts exactly the arithmetic events through ordinate t: $N_{\pi/3}(e^t) = N_{\mathrm{ev}}(t)$ (`native_identification'`). No $S(t)$ term is an additional state variable in $\mathsf{X}_{\zeta,\pi/3}$. □

9.5 Unit-1 phase-registration chain

The unit-1 chain is the scalar phase registration of the same fiber:

$$F_\zeta \longrightarrow \mathsf{X}_{\zeta,1} \longrightarrow R_1(e^t).$$

The required scalar readout identity is Theorem 30(ii): $R_1(e^t) = 1 + \vartheta(t)/\pi$.

Proof of Theorem 30(ii). The arrow is obtained from the $H = 1$ carrier/readout construction itself (the numerical layer's smooth count and crossing-spacing audit remain as cross-checks). The clock is constructed on the carrier,

$$\vartheta(t) = \int_0^t \mathrm{Re}\Big(\tfrac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}\big(\tfrac{1}{2} + iu\big)\Big)\, du \qquad (\texttt{theta}),$$

proven to be the continuous phase of the archimedean gauge, $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) = \|\Gamma_{\mathbb{R}}(\frac{1}{2} + it)\|\, e^{i\vartheta(t)}$ (`gauge_polar`), unique under the origin normalization (`theta_unique`). The master registration identity $\zeta(\frac{1}{2} + it) = Z(t)\, e^{-i\vartheta(t)}$ with $Z(t)$ real (`unit_chart_factorization`) shows the unit readout's entire continuous phase is clock; the DC base is the trivial-channel residue, value forced $= 1$ (`dcResidue_spec`, `dcResidue_unique`); and the event channel is shut by the arc geometry — no whole-cell walk of the unit carrier realizes the half-turn mark (`unit_arcs_empty`, `eventContribution_one`). Assembling base, clock, and the vanishing event channel in the registered readout gives $R_1(e^t) = 1 + \vartheta(t)/\pi$ (`unit_identification`). The clock itself is independently derived and characterized by `gauge_polar` and `theta_unique`; the assembly formula `registeredCount` then uses that derived clock as its continuous input. □

9.6 Subtraction

Proof of Theorem 30(iii). With parts (i) and (ii) supplied, substitution in the fixed definition $S_{\mathrm{ev}} = N_{\mathrm{ev}} - 1 - \vartheta/\pi$ gives

$$\begin{aligned} N_{\pi/3}(e^t) - R_1(e^t) &= N_{\mathrm{ev}}(t) - \left(1 + \frac{\vartheta(t)}{\pi}\right) \\ &= S_{\mathrm{ev}}(t). \end{aligned}$$

Thus $S_{\mathrm{ev}}(t)$ is the scalar compensation ledger for registering the native $\pi/3$ arithmetic procession in the unit-1 chart (`carrier_scale_compensation_S`). □

9.7 The exact clock–jump rule and its bound

The multiplicity ledger has a stronger, non-asymptotic formulation. For $0 \le a \le b$, put

$$M(a,b) := \sum_{a<\gamma\le b} \mathrm{ord}_{1/2+i\gamma}\, \zeta, \qquad C(a,b) := \int_a^b \mathrm{Re}\, \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}\left(\frac{1}{2} + it\right) dt.$$

These inputs are defined independently from the analytic zero orders and the Archimedean gauge. The compiled finite-window law is

$$\boxed{S_{\mathrm{mult}}(b) - S_{\mathrm{mult}}(a) = M(a,b) - \frac{C(a,b)}{\pi}.} \tag{6}$$

It immediately gives the unconditional bound

$$\boxed{|S_{\mathrm{mult}}(b) - S_{\mathrm{mult}}(a)| \le M(a,b) + \frac{1}{\pi}\int_a^b \left|\mathrm{Re}\, \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}}\left(\frac{1}{2} + it\right)\right| dt.} \tag{7}$$

On a zero-free interval the discrete term vanishes, so equality $S_{\mathrm{mult}}(b) - S_{\mathrm{mult}}(a) = -C(a,b)/\pi$ holds and the right side of the bound contains only the clock budget. At every interior non-event,

$$S'_{\mathrm{mult}}(t) = -\frac{\mathrm{clockRate}(t)}{\pi};$$

at an event its jump is the analytic multiplicity, which is also the residue of ζ'/ζ. In distributional notation, summarizing the compiled finite-window statements,

$$dS_{\mathrm{mult}} = \sum_{\gamma>0} m_\gamma\, \delta_\gamma - \frac{\mathrm{clockRate}(t)}{\pi}\, dt.$$

The Lean proofs are `Smult_interval_clock_jump`, `abs_Smult_interval_le_multiplicity_add_clock`, `abs_Smult_interval_le_clock_of_event_free`, and `Smult_hasDerivWithinAt_of_event_free` in `RequestProject/StClockJumpLaw.lean`; the residue/jump identity is `residue_eq_Smult_jump` in `RequestProject/ResidueJump.lean`. Thus the result supplies a local equation of motion, an exact jump rule, and a finite-window bound without identifying the line ledger with the classical strip argument.

Remark 36 (No native oscillatory state)**.** The structural consequence — $S_{\mathrm{ev}}(t)$ has no additional native oscillatory carrier state — is `S_has_no_native_carrier` with `no_native_oscillation`: despinning the unit readout by the clock leaves a real value at every height. The distinct and multiplicity-weighted line ledgers record the event jumps against that continuous clock. Per Remark 32, the classical strip term additionally contains off-line strip multiplicity.

Remark 37 (The increments carry the zero statistics)**.** Rearranged, the compiled interval law reads

$$N^{\mathrm{mult}}\big((a,b]\big) = \big(S_{\mathrm{mult}}(b) - S_{\mathrm{mult}}(a)\big) + \frac{1}{\pi}\int_a^b \mathrm{clockRate}(t)\, dt,$$

so on any family of windows the fluctuation of the zero count about the deterministic clock term *is* the S_{mult}-increment: number variance is the variance of dS_{mult}-increments, and two-point counting statistics (pair correlation in the sense of Montgomery [35]) are autocorrelation statistics of the same increments. This reading is an immediate rearrangement of the compiled `Smult_interval_clock_jump`; the statistical laws themselves (e.g. GUE agreement) are a separate empirical tier and are not claimed here.

9.8 The arrows, discharged

The native cumulative count and the unit phase registration are assembled in `RequestProject/CarrierScaleComper`
Their inputs have the following proved origins:

- **Event channel.** An event is a sign flip of the real native state; its mark is the half turn $-1 = e^{i\pi}$ (`sign_flip_mark`). A scale-H chart registers the mark only through a completed cell arc realizing it, `eventArcs` $H = \{k \geq 1 : e^{ikH} = -1\}$. At $\pi/3$ the wall is hit exactly once per closed six-cell monodromy loop (`pi3_wall_mem`, from the antipodal identity; `pi3_arcs_eq`: the arcs are exactly $k \equiv 3 \pmod 6$) — one registered event per completed native cell. At scale 1 no arc ever realizes the mark (`unit_arcs_empty`). The induced per-event contribution (`eventContribution`) is 1 at $\pi/3$ and 0 at 1 by theorems, not by a gate.
- **Clock channel.** The native chart's zero clock is derived, not assigned: every continuous clock its readout factors through is constant on event-free stretches (`native_clock_locally_constant`, from `no_native_oscillation`). The unit chart's clock is ϑ, the unique continuous gauge phase (`gauge_polar`, `theta_unique`).
- **DC base.** The origin registration is the trivial-channel residue, value forced (`dcResidue_spec`, `dcResidue_unique`).

The geometric scale gap is prior to the analytic event data: the native and unit placements of the integer procession are incommensurable — the lattices $\{k\pi/3\}$ and $\{m\}$ meet only at the origin (`lattice_gap_fundamental`); the unit carrier closes no cell (`unit_never_closes`) and has no antipodal wall (`unit_cell_no_antipodal`). The compiled compensation identity compares the native event count with the unit phase registration against that scale geometry.

9.9 The general registration gap $S_{H,K}$ and the harmonic family

The comparison generalizes to arbitrary scale pairs. With R_H the canonical scale-H registered readout (Lean identifier `NH`; by `NH_dichotomy` the family contains exactly the two proven chart types, decided by the arc geometry), define

$$S_{H,K}(t) := R_H(e^t) - R_K(e^t) \qquad (\texttt{Sgap}).$$

Proposition 38 (Coboundary laws)**.** *The registration gap is a coboundary of the per-scale potential:* $S_{H,H} = 0$, $S_{H,K} = -S_{K,H}$, *and the cocycle law* $S_{H,L} = S_{H,K} + S_{K,L}$ *holds for all scales.* $S(t)$ *is relative — a gap between registrations — never intrinsic.*

Proof. Immediate from the definition as a difference of the per-scale potential R_H (`Sgap_refl`, `Sgap_antisymm`, `Sgap_cocycle`). □

Theorem 39 (One S for the whole harmonic family)**.** *Every harmonic scale* π/m $(m \geq 1)$ *realizes the event mark at* m *cells, exactly once per closed* $2m$*-cell monodromy loop; the family registers identically, and every member reads the same distinct-event ledger* $S_{\mathrm{ev}}(t)$ *against the unit chart:*

$$S_{\pi/m,1}(t) = S_{\mathrm{ev}}(t), \qquad \textit{i.e.} \qquad R_{\pi/m}(e^t) = R_1(e^t) + S_{\mathrm{ev}}(t) \quad \textit{for every } m \geq 1.$$

Proof. The wall exists at every harmonic scale: $m \cdot (\pi/m) = \pi$, so the m-cell walk lands on the half turn (`pi_div_wall`); the arcs realizing the mark are exactly $k \equiv m \pmod{2m}$, one per closed loop (`pi_div_arcs_eq`); monodromy follows by doubling the wall (`pi_div_closes`). Hence the event channel is open with weight one at every π/m, and the intra-family gap vanishes (`Sgap_pi_div` = 0). The cocycle law of Proposition 38 composes through $\pi/3$: $S_{\pi/m,1} = S_{\pi/m,\pi/3} + S_{\pi/3,1} = 0 + S_{\mathrm{ev}}$ (`Sgap_pi3_one`, `Sgap_pi_div_one`, `unit_tracks_harmonic`). In particular the $\pi/6$ channel is determined open — `pi6_wall_mem`, `pi6_arcs_eq` — not prescribed, and $S_{\pi/6,1} = S_{\mathrm{ev}}$ (`Sgap_pi6_one`). □

Proposition 40 (Closure is a fiber–scale resonance)**.** *Monodromy is necessary for the event channel but not sufficient:* 2π *closes in one cell yet never realizes the mark. The exact self-dual criterion is that some whole number of cells be an odd multiple of* π*. Every scale realizes the marks of the functions tuned to it, and the fiber converts* any *carrier scale into exact harmonic closure by carrying the compensating spin: a spin* μ *dresses the* H*-walk into the* $(H+\mu)$*-walk, so* $\mu = \pi/m - H$ *turns, e.g.,* $\log 7$ *steps into the native walk, wall and loop included.*

Proof. Necessity is the doubling argument (`arcs_nonempty_monodromy`); insufficiency is the pair `two_pi_closes`, `two_pi_arcs_empty`; the odd-multiple criterion is `eventArcs_nonempty_iff`; the resonating marks are `every_scale_resonates`; the conversion is `dressedCell_eq` with `fiber_converts_to_harmo` and `every_scale_admits_conversion`. □

9.10 The registration dynamics and the census bound

The deterministic component of the compensation ledger — the carrier tick $\pi/3$ read in the unit chart — is an irrational rotation of the unit ledger, and its dynamics are proven (`RequestProject/RegistrationGaps.lean` standard footprint throughout): the scale ratio is irrational (`irrational_pi_div_three`), the tick is of infinite order — the two scales never re-register (`carrierTick_not_isOfFinAddOrder`, `carrierTick_nsmul_ne_zero`) — the registration marks are dense (`denseRange_carrierTicks`), the rotation is *ergodic* (`ergodic_carrierTick_rotation`, via Mathlib's `AddCircle.ergodic_add_left`), and the defect sequence is injective. By Birkhoff, the deterministic layer equidistributes over the unit cell — so it leaves no periodic or quasi-periodic fingerprint of its own. This theorem concerns the deterministic tick sequence; it does not by itself prove statistical independence from the zeta-event sequence. The pre-registered instruments in App. B test that additional positional question and find no detected coupling at the stated sample sizes.

The census side carries the first compiled growth bound (`RequestProject/CarrierJensen.lean`): every on-line event through height t embeds as a ξ-zero in the origin ball of radius $t + 1$ (`event_line_mem_xiZeros_ball`, `zeroEventCount_le_xiZeros_ball`), so the compiled Jensen disk count (`ZD.ZeroCount.xi_zero_count_disk_bound`, $N(R) \le C\,R\log R$ by circle averages) caps the line census unconditionally: `census_polynomial_bound`, $N_{\text{line}}(t) \le C\,(t{+}1)\log(t{+}1)$ — consumed from the compiled analytic layer with no new analysis. The sharper local result is the exact multiplicity clock–jump law (6) and its bound (7): on every zero-free interval the multiplicity ledger has no discrete contribution and its derivative is exactly $-\text{clockRate}/\pi$. No global $O(\log t)$ estimate for this line ledger is asserted here. On the clock side the Borel–Carathéodory brick is compiled (`RequestProject/ClockRateBound.lean`): `bc_deriv_bound` — real oscillation at most M on a ball of radius R forces $\|g'\| \le 4M/R$ at the center, Mathlib's Borel–Carathéodory composed with the Cauchy derivative estimate — with `deriv_branchLog_eq_logDeriv` and `clockRate_le_of_stripLog`: a branch logarithm of $\Gamma_{\mathbb{R}}$ on the quarter-ball at $\frac{1}{2} + it$ with real oscillation at most M forces $\text{clockRate}(t) \le 16M$. The clock target is thereby reduced to two named inputs with compiled anchors — branch-log existence on the ball (Mathlib `Analysis.Complex.BranchLogRoot`; the ball is simply connected and $\Gamma_{\mathbb{R}}$ nonvanishing there) and the strip Stirling bound $M(t) = O(\log t)$ (the `StirlingBound.lean` ladder). The census side's remaining anchor is the circle-average machinery of `ZeroCountJensen.lean` run on height-centered disks.

9.11 From the ledger to the residue and the windowed trace

The ledger closes into spectral data. All statements of this subsection are machine-checked in `RequestProject/ResidueJump.lean`.

Theorem 41 (residue = jump of the weighted ledger)**.** *Define the multiplicity-weighted native count and its ledger*

$$N^{\text{mult}}(t) = \sum_{0<\gamma'\le t} \operatorname{ord}_{\frac{1}{2}+i\gamma'} \zeta, \qquad S^{\text{mult}}(t) = N^{\text{mult}}(t) - 1 - \frac{\vartheta(t)}{\pi}$$

(`zeroEventCountMult`, `Smult`; the event set below any height is finite, `events_finite`, so the sum is a finite sum). Then, unconditionally, for every $\gamma > 0$:

$$\boxed{\operatorname{Res}_{s=\rho_\gamma} \frac{\zeta'}{\zeta} \;=\; m_{\rho_\gamma} \;=\; \Delta S^{\text{mult}}(\gamma)}$$

with both sides zero off events and no simplicity hypothesis (`residue_eq_Smult_jump`); the jump reads off the multiplicity (`Smult_jump_eq_order`). The distinct-event ledger of Theorem 30 is unchanged:

the two are typed ledgers — S_{event} counts distinct closures, S^{mult} counts spectral residue weight, and they differ by the excess multiplicity alone (`Smult_sub_S`). At simple events the distinct-event form is already literal, $\operatorname{Res}_{\rho_\gamma}(\zeta'/\zeta) = 1 = \Delta S(\gamma)$ (`residue_eq_S_jump`), and the jump of S detects events (`S_jump_detects_event`).

Proof (assembled from the named theorems). The clock is continuous and the base constant, so $S^{\text{mult}} - N^{\text{mult}}$ is continuous and the two have identical jumps (`countMult_hasJump_iff_Smult_hasJump`; for the distinct-event ledger, `S_sub_count_continuous`, `count_hasJump_iff_S_hasJump`). Events are isolated and locally finite: ζ is analytic at every line point and nowhere locally zero, by the identity theorem on the connected set $\mathbb{C} \setminus \{1\}$ against $\zeta(2) = \pi^2/6 \neq 0$ (`zeta_analyticAt_line`, `zeta_not_eventually_zero`, `event_isolated_line`, `events_finite`) — so the weighted count jumps by exactly the vanishing order at each event and is jump-free elsewhere (`countMult_hasJump_event`, `countMult_hasJump_of_not_event`). For any f analytic at ρ with finite order m, the local factorization $f = (s-\rho)^m g$, $g(\rho) \neq 0$, gives $(s-\rho)\, f'/f \to m$: the log-derivative residue is the order, source-generally (`logDeriv_residue_eq_order`). Combining the two at ρ_γ gives the boxed identity. ◻

Theorem 42 (The windowed spectral trace)**.** *For a finite window $(0, T]$ let m_γ be the vanishing orders on the (finite) event set. For every test function h, the atomic sum $\operatorname{Tr} h(D_T) := \sum_\gamma m_\gamma\, h(\gamma)$ (`windowedTrace`) is simultaneously: the jump readout of the weighted ledger, $\sum_\gamma h(\gamma)\, \Delta S^{\text{mult}}(\gamma)$ (`windowedTrace_eq_jump_sum`); the residue sum of the logarithmic derivative, $\sum_\gamma h(\gamma) \operatorname{Res}_{\rho_\gamma}(\zeta'/\zeta)$ (`windowedTrace_eq_residue_sum`); and the matrix trace of the diagonalized $h(D)$ on the window's spectral atoms, one coordinate axis per unit of multiplicity (`trace_diagonal_eq_windowedTrace`) — the atoms carry multiplicity linearly. For the resolvent test function $h_w(\gamma) = (\gamma - w)^{-1}$,*

$$\boxed{\operatorname{Tr}(D_T - w)^{-1} = \sum_\gamma \frac{\operatorname{Res}_{\rho_\gamma}(\zeta'/\zeta)}{\gamma - w}} \qquad (\texttt{windowedResolventTrace\_eq\_residue\_sum}),$$

and the cumulative registration decomposition is $N^{\text{mult}} = 1 + \vartheta/\pi + S^{\text{mult}}$, i.e. $dN^{\text{mult}} = \pi^{-1} d\vartheta + dS^{\text{mult}}$ (`countMult_decomposition`). In the jump and residue identities the assignments are pinned by their specifications — any function satisfying the jump property of S^{mult}, respectively the residue limit of ζ'/ζ, computes the same sum — so the identities are theorems, not definitions.

The chain is closed on the window:

$$\boxed{\text{vanishing} \leftrightarrow \text{jump} \leftrightarrow \text{residue} \leftrightarrow \text{spectral atom} \leftrightarrow \text{resolvent trace.}}$$

The audited frontier, discharged. The two arrows between the windowed identity and the full operator statement are now both proven, and we record them at exactly their strength. (i) The mode dictionary carries multiplicity: the event's mode space is the local algebra $\mathbb{C}[X]/((X-\rho_\gamma)^{m_\gamma})$, whose dimension is the vanishing order (`GradedModeDictionary.finrank_modeSpace`), so the kernel is graded by m_ρ rather than witnessed one vector per event; the graded traces equal the windowed trace and the log-derivative residue sum (`gradedWindowTrace_eq_windowedTrace`, `gradedResolventTrace_eq_residue_sum`) — operator traces on genuine m-dimensional local algebras *are* the S_{mult} jump ledger. (ii) The $T \to \infty$ limit is taken by differencing, exactly as anticipated: the raw sum $\sum_\rho m_\rho/(\gamma_\rho - w)$ diverges, but the differenced resolvent $\sum_\rho m_\rho\big((\gamma_\rho - w)^{-1} - \gamma_\rho^{-1}\big)$ converges absolutely, and the windowed differenced trace converges to it (`DifferencedResolvent.windowedDiffResolvent_ter`
the event windows being a monotone cofinal exhaustion. Both are unconditional; neither asserts where zeros lie — the limit object is a statement about the *sourced* spectrum, the event ledger.

9.12 The exact prime–zero bridge

The global coordinate S_Γ of §9 admits an exact finite-height expansion into arithmetic data on one side and individual zero contributions on the other.

Theorem 43 (The explicit $S(t)$ bridge)**.** *There is a single global constant $A \in \mathbb{C}$ — the Hadamard constant of ξ'/ξ — such that at every good height T,*

$$\boxed{\pi\, S_\Gamma(T) = \Im\Big(i\int_0^T \frac{\xi'}{\xi}(2+iy)\,dy\Big) - \frac{3}{2}\,\Im A - \sum_\rho{}' \, m_\rho\, P_T(\rho) - \vartheta(T) - \pi,}$$

where the primed sum runs over the actual nontrivial zero multiset — each zero ρ weighted by its analytic order m_ρ, with no on-line hypothesis — and

$$P_T(\rho) = \Im\int_{1/2}^{2}\Big(\frac{1}{x+iT-\rho} + \frac{1}{\rho}\Big)\,dx$$

is the zero packet: the Hadamard pair term of ρ read on the top segment. The packet sum converges absolutely, and the identity is unconditional.

The Lean spine, in `RequestProject/StExplicitBridge.lean`: the rectangle-to-argument rung `pi_mul_classicalSContour_eq_contour_im`; the Hadamard substitution with dominated interchange `horizontal_integral_eq_packets` (majorized by the compiled zero-density bound); absolute convergence `summable_packet_integral`; and the capstone `exists_stExplicitBridge`.

Remark 44 (The arithmetic side is $O(1)$; the constant term vanishes)**.** Both non-zero terms of the bridge are now evaluated outright.

The constant. $\xi(s) = (s-1)\pi^{-s/2}\Gamma(s/2+1)\zeta(s)$ has real Taylor coefficients, so $A = (\xi'/\xi)(0)$ is real:

$$A = -1 - \frac{\gamma_{\mathrm{E}}}{2} + \log 2 + \frac{1}{2}\log\pi = -0.023095708966121\ldots,$$

the classical Hadamard constant. Hence $\Im A = 0$ and the term $-\frac{3}{2}\Im A$ drops identically, leaving

$$\boxed{\pi\, S_\Gamma(T) = \Im\Big(i\int_0^T \frac{\xi'}{\xi}(2+iy)\,dy\Big) - \sum_\rho{}' \, m_\rho\, P_T(\rho) - \vartheta(T) - \pi.}$$

The arithmetic side. Since $ds = i\,dy$ on the segment, the vertical leg is a continuous logarithm increment, $\arg\xi(2+iT)$, which splits as

$$\arg(2+iT) + \arg(1{+}iT) - \frac{T}{2}\log\pi + \Im\log\Gamma\Big(\frac{2+iT}{2}\Big) + \arg\zeta(2+iT),$$

each branch continuous from 0. Only the last term is arithmetic, and on $\operatorname{Re} s = 2$ the Euler product gives $\log\zeta(s) = \sum_{p,k}(k\,p^{ks})^{-1}$ absolutely, so

$$\arg\zeta(2+iT) = -\sum_{p,k}\frac{\sin(kT\log p)}{k\,p^{2k}},$$

dominated termwise by $1/(kp^{2k})$. Therefore, for *every* T,

$$\big|\text{arithmetic part of the vertical leg}\big| \;\le\; \sum_{p,k} \frac{1}{k\,p^{2k}} = \log\zeta(2) = 0.497700\ldots,$$

contributing at most $\log\zeta(2)/\pi = 0.1584$ to S_Γ. The prime side of the bridge is thus bounded by an absolute constant: all growth in the vertical leg is archimedean, and the size of S lives entirely in the zero packets.

The packets. With $\rho = \beta + i\gamma$ and $a = T - \gamma$, $P_T(\rho) = \operatorname{atan2}(a, 2-\beta) - \operatorname{atan2}(a, \frac{1}{2} - \beta) - \frac{3}{2}\Im(1/\rho)$; on the midline the middle term is a signed right angle,

$$P_T = \arctan\frac{2(T-\gamma)}{3} - \frac{\pi}{2}\operatorname{sgn}(T-\gamma) + \frac{3\gamma/2}{\frac{1}{4}+\gamma^2},$$

and summing a conjugate pair $\{\gamma, -\gamma\}$ cancels the $1/\rho$ terms exactly, leaving a pair contribution of size $3T/\gamma^2$ — the absolute convergence of the theorem, made quantitative.

Measured (`tmp/att018_vertical_leg.py`, `tmp/att019_packets.py`). The von Mangoldt series reproduces $\arg\zeta(2+iT)$ to 1.1×10^{-7} over $T \in [0.5, 10^6]$; the bound is essentially sharp, $\max|\arg\zeta(2+iT)| = 0.4278$ over 20,001 samples in $T \in [0, 7400]$ with rms 0.1975; at $T = 10^5$ the Γ term is 4.910×10^5 against an arithmetic term of +0.27. On 22,491 zeros to $\gamma = 20000$ (census gated by grid refinement; the Riemann–von Mangoldt prediction is 22,491.4), the pair decay matches $3T/\gamma^2$ to within 0.8%, and the identity $\sum_\rho m_\rho P_T(\rho) = \arg\xi(2+iT) - \arg\xi(\frac{1}{2}+iT)$ — computed by two disjoint routes, the left from the zeros and the closed form, the right from Γ and ζ — verifies with residual 9.1×10^{-7} at $T = 200$ (5.5×10^{-7} relative), the residual growing as the second-order density tail. Register: *measured*, an independent reproduction of the compiled statement, never its proof.

The reading: the vertical leg lives at $\operatorname{Re} s = 2$, inside the Euler-product regime — it is the arithmetic side, and its expansion into the von Mangoldt series is finite bookkeeping on absolutely convergent data. That bookkeeping is performed in Remark 44, where it is shown to be bounded by an absolute constant. Each zero contributes one convergent packet; the clock ϑ and the endpoint half-turn are retained exactly. Thus $S(t)$ is an exact finite-height transport law between the prime side and the individual zero packets — the third outward face of the registration object whose census face is (5) and whose measure face is the clock–jump rule.

The $S(t)$ theorem cluster. The compiled package of this section and §9 presents one object in four registers:

$$\boxed{\text{phase defect} \;=\; \text{atomic residue ledger} \;=\; \text{spectral measure} \;=\; \text{scale-registration cocycle,}}$$

with two outward-facing global laws: the census $S_\Gamma - S_{\text{mult}} = N_{\text{off}}^{\text{mult}}$ (5) and the explicit prime–zero bridge (Theorem 43). The registers: the compensation theorem (Theorem 30) types the defect; the exact clock–jump rule with $dS_{\text{mult}} = \sum_\gamma m_\gamma\,\delta_\gamma - \pi^{-1}\,\text{clockRate}\;dt$ makes it an atomic measure; the windowed trace (Theorem 42) with the convergent differenced resolvent makes the same atoms a spectral measure; and the coboundary laws (Proposition 38) make the defect a cocycle of the harmonic-scale family. Every displayed equality is machine-checked; the statistical and von Mangoldt readings (Remark 37, above) are corollary tiers, labeled as such.

10 The meta method

The construction is used in a fixed native order:

$$\boxed{\text{lift} \to \text{analyze} \to \text{adapt} \to \text{normalize} \to \text{run} \to \text{project} \to \text{read.}}$$

This order is part of the method. The sources are first lifted into complete three-dimensional states; the incompatibilities of those states are analyzed; adapters are synthesized from the detected mismatch; the adapted states are normalized into one carrier frame; the joint state is run there; and only then is it projected to lower-dimensional readouts through the loss ledger.

Five laws govern every stage of this order; they are the method's invariants, and each is enforced by the Part I theorems rather than by convention.

1. *Structure determines the lift.* The native lift and every adapter input are read from the source's structural presentation — local or state data, phase law, native scale, clock, and fiber coordinates — never from the one-dimensional readout. A source without a finite structural presentation admits no lift.
2. *Identity is preserved at every stage.* Each stage and each adapter preserves the represented identity $\mathrm{Id}(W_i)$ of every source. Adapters change the realization, never the represented source.
3. *Run only at complete native cells, after harmonization.* Clocks are first harmonized to the common refinement of the native cell systems; closure and every other measurement are taken only at complete native cell boundaries. Nothing is measured mid-cell.
4. *Projection is always ledgered.* Every projection output is paired with its retained loss data, so lower-dimensional equality is never state equality; the ledger makes the projection invertible on states.
5. *The loop closes through obstruction feedback.* A detected retained obstruction synthesizes an admissible re-adapter, and the adapted state re-enters the order; synchronization of simultaneous sources and coherent transport of structural relationships are part of this loop, not afterthoughts.

Lift. For each source function or structured fiber W_i, the synthesis map reads the finite structural presentation and constructs the native carrier representation

$$W_i \longmapsto \mathsf{X}_i^{(3D)} = \mathrm{Synth}_{W_i}\big(\mathrm{Struct}(W_i)\big).$$

The lift is total on admissible sources and non-oracular: it consumes local data, phase law, native scale, native clock, cell system, and fiber coordinates, never the one-dimensional readout. It retains the represented identity $\mathrm{Id}(W_i)$, with the specified dual routed on the anti-helix, and the projection-loss state. Every phasor of the lifted bank enters at height zero with zero magnitude and grows continuously — there is no onset threshold and nothing is planted. At this stage the lifted states may be mutually incompatible. That is expected: the initial lift is source-faithful before it is common-frame compatible.

Analyze. Before modifying the sources, compute the harmonization obstruction

$$\mathfrak{o}(\mathsf{X}_1, \ldots, \mathsf{X}_m).$$

The mismatch may live in duality, dimension, scale, phase, clock, cell structure, or fiber law, and it is measured against the common refinement of the native cell systems — the finest cell structure all sources share. The paradigm case is scale incommensurability: two registrations of one procession that never re-synchronize. The $\pi/3$ and unit-1 lattices meet only at the origin, and the accumulated registration gap between those two charts is exactly the classical $S(t)$ (§9). The analysis step identifies which retained state coordinates fail to harmonize; it does not replace them by an aggregate scalar diagnostic.

Adapt. Adapters are applied in a hierarchy; they are synthesized from structure, never fitted to a readout, they exist for every admissible source, and each carries its inverse into the ledger. Global structural adapters act first on the shared geometry or on large classes of states:

$$\mathsf{X}_i \xmapsto{A_{\text{global}}} \mathsf{X}_i'.$$

These include dual completion, dimensional alignment, global orientation, carrier-involution compatibility, and common projection convention. They are carrier-level operations and preserve source identity.

Carrier adapters then align the geometric realization:

$$\mathsf{X}_i' \xmapsto{A_{\text{car},i}} \mathsf{X}_i''.$$

They set the carrier scale H, arc placement, height law, phase warp, and chart. The goal is not equal numerical coordinates; the goal is a common geometric frame with source identity preserved.

Clock adapters align native temporal or harmonic progression:

$$A_{\text{clock},i} : \tau_i^{\text{native}} \longrightarrow \tau_i^{\text{carrier}}.$$

They align native cell completion, carrier height, phase progression, and synchronization events, while retaining the inverse clock adapter in the ledger. The conversion always exists: a fiber spin μ dresses the scale-H walk into the scale-$(H+\mu)$ walk, so $\mu = \pi/m - H$ places any native spacing — $\log 7$, $\sqrt{5}$, anything — onto a closing harmonic carrier, wall and monodromy included (Proposition 40).

Fiber adapters act last, inside the common carrier frame:

$$\mathsf{X}_i'' \xmapsto{A_{\text{fib},i}} \widetilde{\mathsf{X}}_i.$$

They include the frozen unit-modulus warp and the rank, phase, local-factor, root-number, and cell-closure adapters, together with the obstruction-removal adapter of forcible closure: at any reachable non-locked stage, two real-independent lane directions close a detected cell residual exactly (Theorem 14). Thus the adapter hierarchy is

$$\boxed{A_{\text{global}} \to A_{\text{carrier}} \to A_{\text{clock}} \to A_{\text{fiber}}.}$$

Normalize. The composed adapter is

$$A_i = A_{\text{fib},i} \circ A_{\text{clock},i} \circ A_{\text{car},i} \circ A_{\text{global}}, \qquad \widetilde{\mathsf{X}}_i = A_i(\mathsf{X}_i).$$

The normalized frame is the common refinement of the native cell systems — the least common harmonic cell — and the normalized theorem target is

$$\boxed{\widetilde{\mathsf{X}}_1, \dots, \widetilde{\mathsf{X}}_m \text{ inhabit one compatible three-dimensional carrier frame}}$$

while

$$\mathrm{Id}(\widetilde{\mathsf{X}}_i) = \mathrm{Id}(W_i).$$

Identity preservation and coordinatewise closure in the common refinement are theorems, not conventions (Theorems 9 and 10). All inverse and adaptation data is retained in the ledger, so normalization is lossless on states.

Run and feed back. The normalized multi-source state is

$$\widetilde{\mathbf{X}} = \big(\widetilde{\mathsf{X}}_1, \ldots, \widetilde{\mathsf{X}}_m\big).$$

It is run through the common carrier dynamics, and every measurement is taken at complete native cells of the common refinement — never mid-cell. At complete cells one measures exact, residue-free cell closure, focal cancellation, synchronization, transport relations, obstruction state, reflection state, and event coincidences. If

$$\mathfrak{o}(\widetilde{\mathbf{X}}) \neq 0,$$

the obstruction is fed back into adapter synthesis:

$$\boxed{\mathfrak{o} \longrightarrow A_{\mathrm{adapt}} \longrightarrow \text{re-harmonization.}}$$

The repair mechanism is proven, not hoped for: a detected cell residual is forcible to zero at a reachable independent stage (Theorem 14), and an obstruction that admits no admissible removal is retained in the state with its certificate — it is never zeroed by hand (Theorem 15). Equivalently, the automatic-adaptation loop is

$$\boxed{\text{realize} \longrightarrow \text{harmonize} \longrightarrow \text{run} \longrightarrow \text{detect} \longrightarrow \text{adapt} \longrightarrow \text{run again.}}$$

Project and read. The final projection is still the ledgered projection

$$3D \longrightarrow 2D \longrightarrow 1D.$$

The output is therefore not merely a scalar readout $r_i(s)$, but the readout state

$$\boxed{\big(r_i(s), \mathcal{D}_i\big).}$$

For an L-function source, $r_i(s) = \Lambda(s, W_i)$; source values are consulted only here, as final verification — never inside the lift, the adapters, or the run. For another structured function class, r_i may be a different one-dimensional observable. The readout chart is itself a carrier scale: what it registers decomposes as DC base, continuous clock procession, and the event channel, the last open only at scales whose cell arcs realize the event mark (Definition 35, Theorem 30). Readouts taken in different charts differ by the registration gap, which composes by the cocycle law (Proposition 38); the classical $S(t)$ is exactly this gap for the unit chart tracking the native carrier. In every case, the lower-dimensional observable is interpreted together with the retained ledger data needed to preserve identity, reconstruct the carrier state, and explain which coordinates the readout suppressed.

In one line, the method is:

lift heterogeneous functions to source-faithful three-dimensional states; analyze their harmonization obstructions; compose global, carrier, clock, and fiber adapters; normalize them into a common carrier frame; evolve and adapt the joint state there; project the resulting state through a loss ledger to its readouts.

Part II

Automorphic functoriality: The converse-theorem checklist

The Part I system proves, unconditionally, every analytic property the classical GL_n converse theorem consumes: the one-dimensional readout is the source Euler datum (Theorem 18), the growing bank continues past $\operatorname{Re} s = 1$ to an entire, vertical-strip-bounded function (Theorem 23), and the self-dual carrier carries the completed functional equation (Theorem 19) — with no Poisson summation entering anywhere in the chain. This part specializes those proofs to the symmetric-power Satake fiber and *certifies*, input by input, that the resulting datum meets the converse theorem's hypotheses — organized as a hypothesis ledger, each input discharged by a named Part I theorem. No carrier theory is reproved here.

11 The exact converse theorem

We fix one theorem and one normalization.

Theorem 45 (Converse theorem for GL_n; Cogdell–Piatetski-Shapiro [14])**.** *Let $n \geq 3$ and let $\Pi = \otimes'_v \Pi_v$ be an* irreducible admissible *representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$ whose central character is invariant under $\mathbb{Q}^\times$ and whose L-function is absolutely convergent in some right half-plane. Suppose that for every $1 \leq m < n-1$ and every cuspidal automorphic representation τ of $\mathrm{GL}_m(\mathbb{A})$ the twist $L(s, \Pi \times \tau)$ is* nice*: both $L(s, \Pi \times \tau)$ and $L(s, \widetilde{\Pi} \times \widetilde{\tau})$ continue to entire functions, are bounded in vertical strips, and satisfy the functional equation $L(s, \Pi \times \tau) = \varepsilon(s, \Pi \times \tau)\, L(1-s, \widetilde{\Pi} \times \widetilde{\tau})$. Then Π is a* cuspidal automorphic *representation of $\mathrm{GL}_n(\mathbb{A})$.*

The reduced twist range $1 \leq m < n-1$ (i.e. $m \leq n-2$, not the $m \leq n-1$ of the 1994 paper) is the improvement of Cogdell–Piatetski-Shapiro II (1999), the second reference in [14]. We apply it with $n = r+1$, so the range is $1 \leq m < r$. Because the twisting is by *all* cuspidal τ on GL_m (not a ramification-restricted set), the central-character hypothesis is invariance under $\mathbb{Q}^\times$ — not unitarity — and the conclusion is cuspidality of Π itself, not merely of some representation agreeing with it almost everywhere.

12 The adelic candidate

Proposition 46 (The CPS candidate)**.** *For each place v let $\Pi_{r,v} = \mathrm{rec}^{-1}_{v,r+1}\big(\mathrm{Sym}^r\, \varphi_{\pi,v}\big)$ be the representation of $\mathrm{GL}_{r+1}(\mathbb{Q}_v)$ attached by local Langlands to the $(r+1)$-dimensional Weil–Deligne parameter $\mathrm{Sym}^r\, \varphi_{\pi,v}$,* *with $\varphi_{\pi,v}$ furnished by local Langlands for* GL(2). *Then each $\Pi_{r,v}$ is* irreducible admissible, *unramified for all but finitely many v, and (with the normalized spherical vector almost everywhere)*

$$\Pi_r = \otimes'_v \Pi_{r,v}$$

is a factorizable irreducible admissible *representation of* $\mathrm{GL}_{r+1}(\mathbb{A})$, *with central character $\omega_{\Pi_r} = \omega_\pi^{r(r+1)/2}$ and, by Theorem 57, $L_v(s,\Pi_r) = L(s, \mathrm{Sym}^r \varphi_{\pi,v})$ and $\varepsilon_v(s,\Pi_r,\psi_v) = \varepsilon(s, \mathrm{Sym}^r \varphi_{\pi,v}, \psi_v)$ at every place.*

Three remarks fix the CPS typing. *(a) Irreducibility is automatic, not branch-dependent.* Local Langlands sends a Weil–Deligne parameter — reducible or not — to an *irreducible* admissible representation (its target is exactly the set of such); so $\Pi_{r,v}$, hence Π_r, is irreducible for *every* π and r, with no "irreducible branch" hypothesis. When $\mathrm{Sym}^r \varphi_{\pi,v}$ is a Langlands sum, $\Pi_{r,v}$ is the corresponding Langlands quotient — still irreducible. *(b) Central character.* ω_π is an automorphic Hecke character, so its power $\omega_\pi^{r(r+1)/2}$ is invariant under diagonally embedded $\mathbb{Q}^\times$ — exactly the hypothesis of Theorem 45; no unitarity and no determinant-balancing enter the candidate. *(c) Galois-free.* The construction reads only the *local* parameter family $\{\varphi_{\pi,v}\}_v$, never a global φ_π, so it applies verbatim to Maass π. This is the irreducible admissible input of Theorem 45, and it is exactly the degree-$(r+1)$ readout synthesized in Part I (Theorems 7, 18).

13 The full twisting family

Fix $1 \le m < r$ and a cuspidal automorphic $\tau \subset \mathrm{GL}_m(\mathbb{A})$ with local parameters $\{\varphi_{\tau,v}\}_v$. At each place the twisting fiber is the local tensor $W_{r,\tau,v} = \mathrm{Sym}^r \varphi_{\pi,v} \otimes \varphi_{\tau,v}$, and the family $\{W_{r,\tau,v}\}_v$ is a *carrier-dual-compatible fiber pair* $(W_{r,\tau}, W_{r,\tau^\vee})$ for the complete carrier (Lemma 66: $W_{r,\tau}^\vee \simeq W_{r,\tau^\vee}$, dual-pair reciprocity, with $\det W_{r,\tau,v} = (\det \varphi_{\tau,v})^{r+1}$ carried in the arithmetic ledger — $W_{r,\tau}$ need not be self-dual, and no $\det = 1$ is forced on it; the central twist ω_π^{-r} of Lemma 66 is absorbed into the contragredient datum $\widetilde{\Pi}_r \times \widetilde{\tau}$). So the Part I readout applies place by place (Theorem 18, with dual routing Theorem 7), the helix reading $W_{r,\tau}$ and the anti-helix $W_{r,\tau^\vee}$, and the transported readout is the twisted convolution datum $L(s,\Pi_r \times \tau) = \prod_v L(s, \Pi_{r,v} \times \tau_v)$.

14 Niceness

Theorem 47 (Twisted carrier niceness)**.** *For every cuspidal automorphic τ on* $\mathrm{GL}_m(\mathbb{A})$, $1 \le m < r$, *the tensor fiber $W_{r,\tau} = \mathrm{Sym}^r \varphi_\pi \otimes \varphi_\tau$ is admissible, so the Part I system applies: both $L(s,\Pi_r \times \tau)$ and $L(s, \widetilde{\Pi}_r \times \widetilde{\tau})$ are* nice *— entire, bounded in vertical strips, and satisfying the functional equation of Theorem 45 (for* $\mathrm{Sym}^r \varphi_\pi$ *irreducible — the generic π; the finitely many exceptional π are the classical isobaric cases, §16).*

Proof. The fiber $W_{r,\tau}$ is a finite Satake parameter family of bounded degree, hence admissible, and its one-dimensional readout is the twisted L-function (Theorem 18). Its complete cells close by the harmonic scaling (Theorem 11), so the normalized residual factors $D^c = V\Phi$ with $V \neq 0$ and the constant mode is extinct, $R_{r,\tau} = 0$ (Theorem 16); the growing bank therefore continues past $\mathrm{Re}\, s = 1$ to an entire completed readout, primal and (on the anti-helix dual $W_{r,\tau}^\vee$) contragredient alike, bounded in vertical strips (Theorem 23). The self-dual carrier reflection supplies the functional equation (Theorem 19); under the identification $\Lambda_C = N^{s/2} L_{\mathrm{CPS}}$ (Thm. 57(E5)) it is the CPS functional equation with $\varepsilon(s,\Pi_r \times \tau) = \eta_W N^{1/2-s}$. Hence both $L(s,\Pi_r \times \tau)$ and its contragredient are nice. □

15 The converse-theorem ledger

Every hypothesis of Theorem 45 is now a discharged line.

CPS hypothesis (Theorem 45)	Part I discharge
irreducible admissible Π_r (fiber datum)	Thm. 7, Prop. 46(a)
central character invariant under $\mathbb{Q}^\times$	Thm. 7 (determinant ledger)
initial convergence ($\operatorname{Re} s \gg 0$)	Thm. 18
all twists τ, $1 \le m < r$	Thm. 7, 18
entire $L(s, \Pi_r \times \tau)$	Thm. 23
entire $L(s, \widetilde{\Pi}_r \times \widetilde{\tau})$	Thm. 23 (anti-helix dual)
bounded in vertical strips	Thm. 23
functional equation	Thm. 19

Every analytic row is a Part I theorem, so the checklist closes unconditionally on the carrier. The functional-equation row is the self-dual carrier reflection of Theorem 19: the completed readout *is* $N^{s/2}L_{\mathrm{CPS}}$ (Thm. 57(E5)), so the proved reflection is already the CPS functional equation, with $\varepsilon(s, \Pi_r \times \tau) = \eta_W N^{1/2-s}$ its conductor normalization. The converse theorem's *output* — automorphy — is itself the carrier's self-dual involution (Theorem 19, the theta/Poisson reflection) together with the arithmetic group generated by the carrier's completion clocks (§30); the classical converse theorem is only the one-dimensional route to what the carrier realizes geometrically.

Compiled warp-to-CPS endpoint. The formal endpoint now returns the full automorphic object, rather than stopping at pole extinction. A degree-generic rung package `RungCapstone.RungWarpData` supplies the three prime warps; the compiled rigidity theorem `RungCapstone.rung_polar_carrier_zero` annihilates its polar correction. In the CPS section, `ThreeDConverse.cpsRungFullAutomorphy3D_ofProfileCoupli` then packages that extinction together with `ThreeDConverse.symmetricPowerFunctoriality3D_ofProfileCoupli` Its returned `CPSRungAutomorphicLift3D` retains simultaneously

1. the restricted-product representation with the literal arithmetic Sym^r local roots;
2. rational-quotient cuspidality;
3. the entire, vertical-strip-bounded functional equation for every CPS twist $1 \le m < r$; and
4. the identity $R_{r,\tau} \equiv 0$ for the rung polar carrier.

Thus the compiled CPS conclusion is a first-class cuspidal symmetric-power lift with its pole-extinction certificate attached, exactly the object consumed by the following automorphic landing theorem. The axiom audit of the endpoint is `[propext, Classical.choice, Quot.sound]`.

Theorem 48 (Automorphic landing)**.** *Let π be a cuspidal automorphic representation of* $\mathrm{GL}(2)$ *and $r \ge 2$ with π* non-exceptional*, stated automorphically: π is not dihedral ($\pi \not\cong \pi \otimes \eta$ for any quadratic Hecke character η) and not of tetrahedral or octahedral type (equivalently* $\mathrm{Sym}^2 \pi$ *and* $\mathrm{Sym}^3 \pi$ *are cuspidal, Gelbart–Jacquet [6]/Kim–Shahidi [17]). For holomorphic π this is the condition "*Sym^r *φ_π irreducible for all r" read off the global parameter (§16); but no global parameter is needed — the automorphic form of the condition is what the proof uses, so the statement covers Maass π (which has no φ_π) verbatim. Then the candidate Π_r of Prop. 46 is a* cuspidal automorphic *representation of* $\mathrm{GL}(r+1)$ *— the symmetric-power lift* $\mathrm{Sym}^r \pi$.

Proof. By Prop. 46, Π_r is irreducible admissible and factorizable, with central character $\omega_\pi^{r(r+1)/2}$ invariant under $\mathbb{Q}^\times$ and local factors those of $\mathrm{Sym}^r\, \varphi_{\pi,v}$; its L-function converges in a right half-plane (Theorem 18); and by Theorem 47 every twist $L(s, \Pi_r \times \tau)$ is nice for all $1 \le m < r$ — in particular *entire*: the carrier generates no residue pole (Theorem 16, residual extinction), and for non-exceptional π there is no trivial constituent to contribute an additional ζ-factor pole (§16). This pole-freeness reads only the *local* Satake data and the automorphic non-exceptionality of π — no global parameter — so it holds for Maass π verbatim; it is what the ledger uses. For holomorphic π, where the global parameter exists, it has a transparent *dictionary*: the pole order would be the trivial-channel multiplicity $\dim \mathrm{Hom}(\varphi_\tau^\vee, \mathrm{Sym}^r \varphi_\pi)$, and a nonzero map from the m-dimensional $\varphi_\tau^\vee$ into the simple $(r+1)$-dimensional $\mathrm{Sym}^r \varphi_\pi$ would be an *equivariant* map. Its image is a subrepresentation of the simple target, hence a nonzero map is surjective, forcing $r+1 \le m \le r-1$. This is Schur's lemma for the genuine group action: in $\mathrm{FDRep}\, k\, G$ a morphism between the simple objects $\varphi_\tau^\vee$ and $\mathrm{Sym}^r \varphi_\pi$ is zero unless they are isomorphic, and the dimension mismatch $m \neq r+1$ already forbids the isomorphism. The exclusion is machine-checked in exactly this general form — `NonSelfDual.hom_zero_of_finrank_lt`, with Schur's core `hom_zero_of_simple_not_iso` (non-isomorphic simple G-representations have zero Hom, via `FDRep.finrank_hom_simple_simple`) and the dimension step `not_iso_of_finrank_ne`. Dimension supplies *only* the non-isomorphism; the vanishing is Schur with the G-action present, never an ordinary vector-space rank bound. This is the representation-theoretic exclusion consumed by the residual calculation. So the entire carrier readout *is* $L(s, \Pi_r \times \tau)$, in the standard pattern of Tate and Godement–Jacquet: on the initial half-plane the two agree *as convergent series* by the coefficient identification (Thm. 57), so the entire carrier readout is *itself the continuation* of $L(s, \Pi_r \times \tau)$ — exhibiting an entire function agreeing with the series on its half-plane is what "continues to an entire function" means, and is exactly what the CPS hypothesis asks; nothing about the target is presupposed. The identity theorem enters only as uniqueness bookkeeping — any continuation coincides with this one (`CarrierTargetIdentification.carrier_FE_transfers_to_target`, universal core `entire_eq_of_eqOn_isOpen`, its target-entirety hypothesis instantiated by the carrier object itself). Every hypothesis of Theorem 45 is met (the ledger above), so Π_r is a cuspidal automorphic representation of $\mathrm{GL}(r+1)$. □

Corollary 49 (Symmetric-power functoriality)**.** *For every cuspidal π on* $\mathrm{GL}(2)/\mathbb{Q}$ *(any central character ω_π, carried through the determinant ledger of Theorem 7) and every $r \geq 2$,* $\mathrm{Sym}^r \pi$ *is automorphic on* $\mathrm{GL}(r+1)$*. For non-exceptional π —* $\mathrm{Sym}^r \varphi_\pi$ *irreducible, all r — it is* cuspidal*, by Theorem 48; for the classically enumerated exceptional π (dihedral, tetrahedral, octahedral) it is the known* isobaric *object of Gelbart–Jacquet [6] and Kim–Shahidi [17] (§16). For $r \in \{2, 3, 4\}$ the cuspidal niceness is also furnished classically by Langlands–Shahidi [15]; for $r \geq 5$, where the Eisenstein-series construction leaves the list, the carrier supplies it — non-circularly, the functional equation being the carrier's own helix–anti-helix reflection, corroborated through* Sym^{13} *(§31.1). The carrier route reads only local Satake, the archimedean type, and the tensor rule, so it is Galois-free and applies to Maass π, where the automorphy-lifting proof of Newton–Thorne [18] does not. Symmetric-power functoriality* $\mathrm{GL}(2) \to \mathrm{GL}(r+1)$ *is thereby established for every r. (Register: the twisted analytic niceness is the carrier's own, hypothesis-free for every twist degree and unitary Satake family —* `GlobalHelix.cpsAllTwists_unconditional3DAnalyticPayload`*, §17 — and the classical reading passes through the named identification layer, Thm. 57.)*

16 Exceptional π

The cuspidal landing of §15 is exact whenever the parameter $\mathrm{Sym}^r \varphi_\pi$ is irreducible. That is the generic situation: for every non-exceptional π, $\mathrm{Sym}^r \varphi_\pi$ is irreducible for all r. The only π for which some $\mathrm{Sym}^r \varphi_\pi$ is reducible are the classically enumerated exceptional types — π dihedral (the Gelbart–Jacquet criterion [6]), and the tetrahedral and octahedral forms, whose relevant symmetric powers sit in degrees $3, 4$ [17].

For such π the identified twist $L(s, \Pi_r \times \tau)$ can acquire an $s = 1$ pole from a trivial constituent of the reducible parameter — the constituent embeds, $\varphi_\tau^\vee \hookrightarrow \mathrm{Sym}^r \varphi_\pi$, a nonzero trivial channel (`TrivialChannel.embedding_hom_ne_zero`, machine-checked). This is *not* a carrier residue — Theorem 16 gives $R_{r,\tau} = 0$ for the configured fiber regardless — but an intrinsic ζ-factor of the reducible datum, and there $\mathrm{Sym}^r \pi$ is the corresponding *isobaric* automorphic representation, already known by Gelbart–Jacquet [6], Kim–Shahidi [17], and (for dihedral π) the theory of CM forms. The algebraic dichotomy — no trivial channel when $\mathrm{Sym}^r \varphi_\pi$ is simple, a trivial channel exactly at a constituent embedding — is proved in Lean (`TrivialChannel.pole_free_iff_no_embedding`); the analytic pole-order/invariant-dimension identity and the reducibility $\Leftrightarrow$ exceptional classification are the cited classical inputs. These finitely many types are settled independently; the carrier construction supplies the remaining, generic, *cuspidal* case, new for $r \geq 5$. In every case $\mathrm{Sym}^r \pi$ is automorphic — the content of Corollary 49.

17 The converse engine in three dimensions: the machine-checked generator descent

The ledger of §15 discharges the classical converse theorem's *analytic* hypotheses — niceness of every twist — through the carrier (Theorem 47, Proposition 59). A second, self-contained Lean development internalizes the converse theorem's other half: the *generator descent* that turns invariance under the arithmetic group's generators into invariance under the whole group — automorphy as full invariance, with the adelic Whittaker/Poisson analysis appearing only as its one-dimensional shadow. This section states exactly what that development establishes, at exactly its current strength, in the register of Remark 32: the engine is machine-checked and unconditional; its inputs are named; the classical analytic input it consumes is cited, not re-proved.

The engine (machine-checked, unconditional). The following declarations compile with the standard footprint {`propext`, `Classical.choice`, `Quot.sound`} and no `sorry`; all lie in the namespace `CriticalLinePhasor.ThreeDConverse` unless noted.

- *Unit-winding generation.* A diagonal source clock and its reciprocal target clock conjugate the unit winding to any winding: $D(q)\,T_{ij}(1)\,D(q^{-1}) = T_{ij}(q)$ (`coordinateScaleGL_mul_transvectionGL_mul_coordinateScaleGL_inv`). Hence diagonal-clock invariance together with a *single* unit winding in each off-diagonal direction forces invariance under all of $\mathrm{GL}(n, K)$ (`cps3D_readout_invariant_of_unit`), the diagonal/transvection family generating the group for every rank by Mathlib's `diagonal_transvection_induction` (`glCarrierGenerators_closure`, `cps3D_invariant_of_carrier_generators`). Consequently the former rational-parameter density step is unnecessary for this generator argument.
- *Mellin inversion $\Rightarrow$ reflection.* A completed Mellin functional equation recovers the pointwise theta reflection on the positive ray, under Mathlib's Mellin-inversion hypotheses (`theta_reflection_of_mellin_functionalEquation`, from `Mathlib.Analysis.MellinInversion`).
- *Finite Fourier separation.* The complete additive-character family separates a finite winding cell (`finiteAbelianCell_eq_of_all_twists_eq`); evaluating the separated identity at the cell origin needs only that the finite representatives preserve 0 and 1 (`local_unit_transvection_invariant_of_finiteQuotientMellin`).
- *Archimedean descent.* The same completed Mellin data at the archimedean place gives invariance of the left adelic factor (`adelic_archimedean_readout_invariant_of_mellinFunctionalEquations`).

- *Adelic assembly and rational-quotient descent.* The archimedean and finite-place invariances assemble to the full product move group (`cpsAdelic3D_readout_invariant`) and descend to the rational orbit quotient (`cpsAdelic3D_rationalQuotientReadout`).
- *Cuspidal landing.* An integrable, nontrivial unipotent eigenmove in every proper channel forces the simultaneous vanishing of all proper-unipotent constant terms (`unipotentConstantTerm_eq_zero`); assembled, this is the landing theorem `cpsConverse3D_landing`.

The interface ledger. The assembled landing `cpsConverse3D_landing` is quantified over an abstract carrier state space with its group action and its scalar readout; it consumes exactly the following named inputs, listed with their carrier meaning. These are the interface the engine exposes — the honest counterpart of the analytic ledger of §15.

Input	carrier meaning	discharged by
`hFE`	per-winding completed Mellin FE	Prop. 59/Thm. 47
`hInversion`	Mellin convergence/vertical integrability	Prop. 59 (warped-Abel)
`hPrimalReadout`	winding-cell = primal kernel at height	finite-model instantiation†
`hDualReadout`	winding-cell = reflected-dual kernel	finite-model instantiation†
`hdiag`	diagonal-clock invariance	finite-model instantiation†
`hintegrable`	unipotent-kernel integrability	carrier-kernel decay†
`heigen`	unipotent eigenmove	explicit carrier move†
`hnontrivial`	eigenvalue $\neq 1$	explicit carrier move†

This table records the projected Mellin-inversion route. The direct 3D tower route now bypasses its `hFE`/`hInversion` block and discharges the bank wiring itself. `ThreeDConverse.cpsTowerFunctoriality3D_unified` constructs the actual two-chart bank state, proves the primal/dual weld identity from the all-height 3D reflection, obtains the diagonal and every transvection invariance through `towerBankInstance_landing`, and proves that the resulting general-linear transports preserve the bank's exact zero predicate under identity and composition. The same capstone identifies both scalar fields of its `StrongFEPair` with the primal and contragredient 3D banks and includes their entirety, vertical-strip bounds, and functional equation; the prescribed Gamma-product projection is supplied by `CPSCompletionUnification3D`. The explicit unipotent cancellation used for cuspidality remains the separate conclusion `unipotentConstantTerm_eq_zero`, as it should, rather than a hidden field of the functorial transport. Classically, Poisson summation on $\mathbb{A}/\mathbb{Q}$ in Tate's thesis [10] (Godement–Jacquet [11] for higher-rank convolution) is the corresponding one-dimensional adelic readout; on the carrier the global equation is already the pointwise helix/anti-helix reflection of Theorem 19.

That the ledger's substantive rows are genuinely substantive is itself compiled, as a falsifiability instrument (`CPSUnconditionalityAudit3D`): an arbitrary group action does not make an arbitrary readout invariant (`not_all_readouts_invariant_under_all_group_moves`), and an integrable unipotent kernel need not be cuspidal (`unitConstantKernel_not_cuspidal`) — the engine's hypotheses cannot be discharged by genericity, only by genuine modularity and cuspidality content. The division of labor is equally explicit: on the functoriality route the automorphic/cuspidal *output* is delivered by the cited converse theorem at its published strength (Theorem 45), consumed exactly as local Langlands and Deligne–Tate are consumed; the engine of this section is the internalized generator descent, whose ledger names what a full internalization must supply — an audit of that program, never a second, independent proof of the citation.

Remark 50 (Register: the constructed Lean pair is a carrier-theta object)**.** The Lean strong functional-equation pair built for the all-place twisted symmetric-power Satake bank (`GlobalHelix.twistedSymmetricPowerCarrierStrongFEPair`; without self-duality, `GlobalHelix.cpsDualPairStrongFEPair`) is a *carrier-theta object with a synthesized completion kernel* (`CarrierTheta.automaticCoefficientThetaStrongFEPair`). On its initial half-plane its completed Λ equals the actual Satake Dirichlet series $\sum_n a_n(\mathrm{Sym}^r\,\pi \times \tau)(n+1)^{-s}$ times the Mellin transform of that synthesized kernel (`twistedSymmetricPowerCarrier_initialIdentification`); its niceness — entire, bounded in vertical strips, functional equation with $k = 1$, $\varepsilon = 1$ — is automatic (`twistedSymmetricPowerCarrier_twistedNiceness`). The prescribed classical completion is now also present directly on the three-dimensional banks. Given a positive conductor C and a nonempty list of Deligne shifts $\{\mu_j\}$, Lean proves simultaneously that the primal 3D Mellin readout is

$$C^s \prod_j \Gamma_{\mathbb{C}}(s+\mu_j) \sum_n a_n(W)(n+1)^{-s}$$

and that the reciprocal-height contragredient 3D Mellin readout is the analogous completed dual series (`GlobalHelix.cpsPolynomialFullCompletion3D_identification`). This is an equality of the 3D bank projections themselves, not a postulated identification of two scalar functions.

These clauses are now closed under the entire CPS twist range in one theorem over the actual polynomial radial Satake datum. For every $1 \le m < r$ and every instance of `CPSPolynomialTwist`, `GlobalHelix.cpsPolynomialAllTwists_fullCompletion3D_unified` keeps the same datum W through (i) the primal/contragredient radial 3D banks, (ii) their reciprocal-height reflection, (iii) both prescribed conductor/Gamma-product Mellin identifications, and (iv) the strong pair's entireness, dual entireness, two vertical-strip bounds, and global functional equation. In particular, "all twists," "standard completion," and "niceness" are no longer three adjacent interfaces: they are fields of one quantified conclusion, with the initial-half-plane point and completion clock carried as typed data.

The word "actual" here now has a separate coefficient-level theorem, rather than meaning only "an arbitrary bank with the correct rank." In `CPSArithmeticTwist3D`, a rank-two radial Satake pair (α_p, β_p) and a rank-m twist $(\gamma_{p,k})$ construct the CPS bank with literal roots

$$\alpha_p^{r-j}\beta_p^{j}\gamma_{p,k} \qquad (0 \le j \le r,\ 1 \le k \le m).$$

Lean proves both the local numerator identity `arithmeticCPS_localFactor_identification` and the equality of every global Euler coefficient `arithmeticCPS_globalCoefficient_identification`, together with the contragredient version. Thus the carrier bank used by the completion theorem is constructed from, and is definitionally equal coefficient-by-coefficient to, the twisted symmetric-power Euler datum; no post hoc equality between two unrelated degree-$(r+1)m$ banks is assumed.

The archimedean clock is arithmetic as well. The structure `ArithmeticCPSCompletionData` carries the actual conductor, the $r+1$ symmetric-power Deligne shifts, and the m twisting shifts; `tensorShifts` constructs the complete degree-$(r+1)m$ list by the literal pairwise sums $\mu_j + \nu_k$. The theorem `arithmeticCPSFullCompletion3D_identification` then identifies both 3D Mellin projections with the completed primal and contragredient Euler readouts using precisely that conductor and that tensor-parameter Gamma product. Consequently neither the finite-place roots nor the archimedean completion can be replaced by an unrelated bank of the same dimension.

The one-source compatibility is now also a typed theorem rather than an informal juxtaposition. In `CPSArithmeticStrongSource3D`, an `ArithmeticCPSReflectedThetaSource` contains one `StrongFEPair` and identifies its two source functions exactly with the prescribed primal and reciprocal-height contragredient arithmetic 3D banks. From that single source, `ArithmeticCPSReflectedThetaSource.unified` derives the literal coefficient passport, both conductor/Gamma completed Euler readouts on the common initial half-plane, the pointwise 3D reflection, both entire continuations, both vertical-strip bounds, and the global functional equation. Thus the type checker does not permit a standard completion read from one bank to be combined with a functional equation belonging to another synthesized kernel: all clauses of the capstone refer to the same pair and the same arithmetic Satake datum, uniformly in r and m. The constructor `ArithmeticCPSReflectedThetaSource.analyticCandidate` now packages these clauses in the exact analytic-input shape used by a converse theorem: named primal and dual entire functions, weight, nonzero root number, two strip bounds, the global functional equation, and pointwise identifications with the literal arithmetic completed Euler products throughout their initial domain. These identifications are fields of the candidate, so "analytic continuation of the carrier" cannot be substituted for continuation of a different Euler product. At the kernel level, `gammaClock_rapid` proves directly from $2x^{\mu}e^{-2\pi x}$ that each prescribed archimedean clock decreases faster than every real power. The signed-log convolution branch is now closed for every finite shift list by `completionKernelLog_rapid`. Its proof chooses one real Mellin weight above all shift exponents, proves the weighted Gamma profile is both integrable and uniformly bounded, and uses the exact log-height convolution identity together with the $L^1 * L^\infty$ norm estimate `norm_convolution_mul_le_of_integrable_of_bound`. Finally `conductorScaledCompletionKernelLog_rapid` transports this result through every positive arithmetic conductor scale. Hence the complete prescribed tensor-parameter Gamma kernel, not only each individual clock, supplies the rapid-decay fields of the strong-pair constructor.

The two local-integrability fields of that constructor are now derived from the same prescribed data as well. The general Lean lemma `CarrierTheta.locallyIntegrableOn_of_mellinConvergent` observes that convergence of one Mellin integral makes $x^{s-1}f(x)$ globally integrable on $x > 0$; multiplication there by the continuous, nowhere-vanishing reciprocal weight x^{1-s} recovers f, and therefore makes f locally integrable. For each of the primal and contragredient coefficient banks, `cpsPolynomialFullPrimalTheta_locallyIntegrableOn` and `cpsPolynomialFullDualTheta_locallyIntegrableOn` choose a real weight σ satisfying simultaneously

$$\sigma > A + 1, \qquad \sigma > -\operatorname{Re}\mu \quad (\mu \in \boldsymbol{\mu}),$$

where A is the proved polynomial coefficient-growth exponent and $\boldsymbol{\mu}$ is the finite tensor-shift list. The Gamma-product Mellin theorem and the polynomial theta summability theorem then supply the required convergent weighted integrals. Finally the exact positive-height 3D readout identities transfer these results to the literal primal bank and to $x \mapsto \widetilde{\Theta}^{\vee}(1/x)$; these are the compiled theorems `cpsPolynomialFullPrimal3DBankReadout_locallyIntegrableOn` and `cpsPolynomialFullDual3DReflectedReadout_locallyIntegrableOn`. Their arithmetic CPS specializations are wired directly as `prescribedPrimal_locallyIntegrableOn` and `prescribedDual_locallyIntegrableOn`. Thus local integrability is a consequence of the prescribed coefficient bounds and Gamma completion, rather than an analytic regularity premise attached to the source.

The forward geometric assembly is now typed separately from every analytic continuation. The structure `ArithmeticCPSCarrierSeed3D` carries two independently defined functions on the global carrier, one on the helix and one on the anti-helix, together with their global involution exchange and the exact positive-height identifications of their completed logarithmic readouts with the literal primal and contragredient coefficient/Gamma banks. The two functions are deliberately distinct: inverse radial Satake weights ride the contragredient lane and are not silently replaced by complex conjugates. The theorem `completedLogTheta_pair_one_div` transports this paired carrier exchange through logarithmic inversion and the self-dual envelope. Consequently `reflection_of_carrierSeed` derives the literal off-weld bank reflection, and `ofCarrierSeed` constructs the required `ArithmeticCPSReflectedThetaSource`; neither statement consumes an `ArithmeticCPSAnalyticCandidate3D` or a pre-existing theta reflection.

The reverse assembly direction is also canonical. Given an `ArithmeticCPSAnalyticCandidate3D`, its typed `native3DReflection` is already the all-positive-height equality of the literal prescribed primal and reciprocal-height contragredient 3D banks, with the candidate's weight and root number. The constructor `ArithmeticCPSReflectedThetaSource.ofAnalyticCandidate` uses that equality as the reflection field of a `StrongFEPair` whose two functions are definitionally those same banks. It does not import regularity or decay from the candidate: the two local-integrability fields are the theorems above, while `cpsPolynomialFullPrimal3DBankReadout_rapid` and `cpsPolynomialFullDual3DReflectedReadout_rapid` derive both rapid-decay fields from the polynomial coefficient bounds and the finite Gamma-convolution decay. Thus the only off-weld datum transferred from the assembled arithmetic candidate is its actual native reflection; every analytic source condition is reconstructed on the prescribed banks. The theorem `ofAnalyticCandidate_unified` then applies the existing one-source capstone to this canonical constructor, yielding both exact completed Euler identifications, native reflection, entire continuations, strip bounds, and the functional equation. Finally `reassembledAnalyticCandidate` packages the resulting strong-pair continuations back into the converse-theorem input type, now definitionally tied to the literal coefficient/Gamma banks.

The representation and analytic sides now meet in one further typed capstone. In `CPSRestrictedTensorConverseCapstone3D`, the base object `RestrictedSymmetricPowerRepresentation3D` is a representation of the actual group

$$G_\infty \times \prod\nolimits'_p (G_p, K_p),$$

where the finite factor is Mathlib's restricted-product type, not an unrestricted product or a formal label. Its local components and its global restricted-product representation carry the precise predicates used in the assembly: irreducibility is `Representation.IsIrreducible`; smoothness says that every vector is fixed by an open subgroup; and admissibility says that the fixed-vector space for every open subgroup is finite-dimensional. The action is identified with the carrier action consumed by the quotient landing, and the local Satake readout is identified, at every prime and every channel, with the literal roots

$$\alpha_p^{r-j}\beta_p^{j}, \qquad 0 \le j \le r.$$

Thus "restricted tensor product", "irreducible", "smooth admissible", and "locally compatible" are fields of one representation object and cannot be supplied for different candidates.

The structure `ArithmeticCPSAllTwistsConverseCandidate3D` attaches to that base object, for every $1 \le m < r$, every arithmetic polynomial Satake twist `tau`, and every prescribed pairwise-shift completion `D`, the exact `ArithmeticCPSAnalyticCandidate3D` and an equivariant finite-parameter residual channel. This is now an explicit construction rather than only a consumer interface: `ArithmeticCPSAllTwistsConverseCandidate3D.ofReflectedThetaSources` takes the independently defined restricted-product representation, the bank bridge, one literal `ArithmeticCPSReflectedThetaSource` for every required twist and completion, and the equivariant residual family. Its analytic field is definitionally the `analyticCandidate` derived from that same reflected source, so the continuation, identification, reflection, and niceness data cannot be imported from a different Euler bank. The residual is an A-linear intertwiner into a simple target; its vanishing is derived by `EquivariantCPSResidual3D.residue_eq_zero` from simplicity and the strict rank gap, so no ordinary-vector-space non-embedding assertion occurs. Finally `ArithmeticCPSAllTwistsConverseCandidate3D.converseCapstone` returns, simultaneously, the base restricted-product action and rational-quotient cuspidal landing and, for every twist in the CPS range, the literal coefficient passport, both initial Euler/completion identifications, both entire continuations, both vertical-strip bounds, the functional equation, the native 3D reflection, and the derived zero residual. The theorem has one candidate argument: none of those clauses appears as a detached proposition in its signature. There is now a forward all-twist entry point as well: `ArithmeticCPSAllTwistsConverseCandidate3D.ofCarrierSeeds` accepts the independently defined paired carrier seed for each twist and constructs every reflected source and analytic candidate internally. Thus the geometric route does not pass through `ofAnalyticCandidate`; its only identification boundary is the explicit equality of the two literal arithmetic banks with the helix and anti-helix carrier readouts. The object-valued endpoint `symmetricPowerFunctoriality3D_ofCarrierSeeds` then returns the cuspidal symmetric-power lift from that same family of paired carrier seeds, the restricted-product representation, the bank landing, and the equivariant residual channels; it does not accept a separately supplied analytic candidate.

For a genuine level-one GL(2) source, the carrier exchange itself is now constructed directly from modularity. If f has even weight $2n$, write $W_{2n}(x) = x^{-n}e^{-x-x^{-1}}$ for the fixed completion envelope and define the normalized vertical carrier by

$$E_f(z) = \frac{f\left(ie^{\Im z}\right)}{W_{2n}\left(e^{\Im z}\right)}.$$

Mathlib's slash transformation under S gives $f(i/x) = (xi)^{2n} f(ix)$, while $W_{2n}(x^{-1}) = x^{2n}W_{2n}(x)$; hence $E_f(\overline{z}) = i^{2n}E_f(z)$. This is the compiled theorem `GenuineGL2Carrier.deEnvelopedVerticalCarrier_exchange`. Moreover the completion is not silently inserted twice: `completedLogTheta_deEnvelopedVerticalCarrier` proves exactly $W_{2n}(x)E_f(i \log x) = f(ix)$ for every $x > 0$.

The coupling type this exchange first fed is itself audited at the type level, and the audit is a finding: a coupling demanding that the *twisted* bank readout equal the raw modular vertical value is *uninhabitable on genuine data* — the $(r+1)m \geq 2$-clock tensor completion kernel decays like $e^{-c\sqrt{x}}$ while a cusp form's vertical restriction decays like $e^{-2\pi x}$, and the bank's leading coefficient is 1, so no instance exists at any (r, m) (the finding is recorded in the retired structure's docstring, `ArithmeticBankCoupling3D`). The corrected per-rung obligation is the *theta profile pair* (`BankThetaProfileCoupling3D`): two functions on the positive ray, one pointwise weighted reflection between them, and the two identifications of the literal bank readouts with those profiles — nothing else. Its constructors discharge everything downstream: `toReflectedThetaSource` builds the reflected strong-pair source with integrability and rapid decay derived from the polynomial coefficient bounds; `toCarrierSeed` *derives* the seed's conjugation exchange from the pointwise reflection and the envelope inversion; `ArithmeticCPSCarrierSeed3D.toProfileCoupling` is the converse, so the seed-typed and profile-typed obligations interconvert and the per-rung frontier is exactly one pointwise theta reflection; and `allProfileCarrierSeeds` packages per-twist couplings into the seed family the all-twist constructor (`ofCarrierSeeds`) consumes.

The chain downstream of the coupling is one compiled endpoint (`symmetricPowerFunctoriality3D_ofProfileCouplings`): from the restricted-product representation, the bank bridge, the residual channels, and *one theta-profile coupling per CPS twist*, the object-valued cuspidal symmetric-power lift is returned with the carrier seeds, reflected sources, and analytic candidates all constructed internally — the per-rung analytic obligation of the entire converse chain is the coupling family and nothing else. The cited-classical entry point into that family is likewise compiled (`BankThetaProfileCoupling3D.ofMellinFE`): a completed Mellin-space functional equation between the literal prescribed banks, with the standard Mellin-inversion integrability conditions, *derives* the pointwise profile reflection (`theta_reflection_of_mellin_functionalEquation`) and constructs the coupling — the shape in which the classical low rungs deliver their functional equations (Langlands–Shahidi, Kim–Shahidi for $r \le 4$ [15, 17]), consumed as data exactly as local Langlands and Hecke multiplicativity are consumed at their typed fields. Four of that converter's six analytic side conditions are themselves discharged from the compiled polynomial-decay layer (`CPSMellinFEDischarge`): primal Mellin convergence at every initial-domain weight (`cpsPolynomialFullPrimal3DBankReadout_mellinConvergent`), continuity of the primal bank and of the reflected contragredient readout on the positive ray — an M-test over the compiled rapid decay (`theta_continuousAt_of_polynomial`, `reflectedDualReadout_continuousAt`) — and the kernel-continuity input itself for every singleton clock (`conductorScaledCompletionKernelLog_singleton_continuousOn`). The refined converter (`BankThetaProfileCoupling3D.ofMellinFE'`) therefore consumes exactly the continuation content and nothing else: the completed Mellin functional equation between the literal banks, the reflected-side Mellin convergence, and the two vertical integrabilities. The two vertical integrabilities are then themselves *discharged* (`CPSCompletedVerticalIntegrable`): the house tree carries the unconditional two-sided vertical Stirling bound $|\Gamma(\sigma + it)| \le C|t|^{\sigma-1/2}e^{-\pi|t|/2}$ (`StirlingBound.gamma_stirling_bound`), which `gammaC_vline_integrable` turns into vertical integrability of the completion's $\Gamma_{\mathbb{C}}$ factor; along the initial-domain line the compiled identification displays the bank's Mellin transform as $C^s \prod_j \Gamma_{\mathbb{C}}(s+\mu_j)\, D(s)$ with D a bounded Dirichlet series, so `completedPrimal_verticalIntegrable` integrates the whole readout, and (`mellin_reflectedDual_eq_primal`) the functional equation makes the reflected transform literally equal the primal one — discharging both fields. The fully-discharged converter (`BankThetaProfileCoupling3D.ofMellinFE''`) thus consumes exactly the completed Mellin functional equation and the reflected-ray Mellin convergence: the per-rung obligation is one functional equation plus one ray-convergence, with nothing about vertical decay remaining.

The base rung of this corrected type is *inhabited on a genuine automorphic object*. `GenuineGL2Carrier.StandardRungEulerData` is the standard (degree-two) certificate of an actual level-one cusp form in the det-one analytic chart — its one arithmetic field identifies the form's canonical Mathlib `qExpansion` coefficients with the det-one-normalized Euler bank at shift $n - \frac{1}{2}$ (Hecke's multiplicativity, a cited classical input typed as a consumed field, exactly as local Langlands is consumed by Thm. 57); the contragredient-bank equality is *derived*, not certified — det-one forces the dual weight family to be the swap of the primal, and the Euler coefficients are symmetric functions of the weights (`radialLocalEulerCoeff_comp_equiv`, `cpsPolynomialDualCoeff_eq_primal_of_detOne`), so the constructor `StandardRungEulerData.ofDetOne` builds the certificate from Hecke multiplicativity and the Satake normalization alone. From that certificate alone: `theta_standard_eq` proves the bank's completed coefficient theta equals the enveloped modular vertical value (Mathlib's q-expansion convergence plus exact Γ-clock evaluation — the envelope algebra *forces* the analytic chart: weight one, shift $n - \frac{1}{2}$, exchange constant i^{2n}); `standardRung_reflection` derives the literal-bank reflection from `hecke_inversion` — the modular slash law on the vertical line — consuming no analytic candidate, no reflected source, and no supplied theta reflection; `StandardRungEulerData.toReflectedThetaSource` then constructs the complete `StrongFEPair` whose two functions *are* the literal bank readouts (by `rfl`), with entirety of the completed transform and the global functional equation $\Lambda(1 - s) = i^{2n}\, \Lambda^{\vee}(s)$ as outputs (`standardRung_completed_entire`, `standardRung_functional_equation`); and `toProfileCoupling` places the rung inside the corrected coupling type — its non-vacuity certificate on genuine modularity. The completion this pair carries is the *standard* one, not a synthesized kernel: the rung's clock is the prescribed Γ-clock at conductor one and Deligne shift $n - \frac{1}{2}$, and on the initial half-plane the pair's completed transform *is* the literal standard completed readout $1^s\,\Gamma_{\mathbb{C}}(s+n-\frac{1}{2})\,D(s)$ — Mathlib's own $\Gamma_{\mathbb{C}}$ — equivalently one half of the classical completed Hecke series $\Gamma_{\mathbb{C}}(s+n-\frac{1}{2})\sum_m a_f(m)\,m^{-(s+n-1/2)}$ of the actual form (`standardRung_Lambda_eq_standardCompletion`, `standardRung_dualLambda_eq_standardCompletion`, `standardRung_Lambda_eq_heckeCompletion`, instantiating the general prescribed-completion identification `cpsPolynomialFullCompletion3D_identification`). Bundled, this is Hecke's classical theorem for the form as one compiled statement about one object (`standardRung_classicalHeckeTheorem`): entirety with entire dual, the classical completed Hecke series on the initial half-plane on both legs, and the global functional equation with the standard conductor, Γ-factor, contragredient, and root number — the functional equation of the standard completed automorphic L-function of the genuine form, not of an auxiliary kernel. The synthesized-kernel constructor (`CarrierTheta.automaticCoefficientThetaStrongFEPair`) remains the generic, hypothesis-free payload; the CPS-facing chain — the reflected sources, the profile couplings, and the all-twist converse candidates — is typed on the prescribed conductor/Γ-product banks throughout, so no synthesized completion stands in for the standard one anywhere on the identification path. All at the standard footprint, no `sorry`.

At the representation-theoretic layer the symmetric-power action is now constructed rather than named. The Lean file `RepresentationSymmetricPower3D` descends a linear endomorphism through Mathlib's symmetric-tensor quotient, proves preservation of identity and composition, and defines `Representation.symmetricPower`. Consequently `RestrictedSymmetricPowerRepresentation3D.ofBaseLocalRepresentations` builds every finite local Sym^r component from the corresponding genuine base GL(2) component, while `ofBaseAdelicRepresentation` applies the same functor to the base representation of the archimedean-times-restricted-product group and uses the resulting action definitionally on the global symmetric carrier. The remaining assembly is thereby separated cleanly into the adelic `CPSBankBridge` on that carrier and the literal all-twist carrier/Gamma identification.

The representation-side structures are moreover *inhabited*, with every status field proven rather than supplied (`CPSRepresentationInstance3D`). Three reusable bricks — any representation on a simple module is irreducible (`Representation.isIrreducible_of_isSimpleModule`), the trivial representation is smooth, and any finite-dimensional representation is admissible — feed the first compiled term of the restricted-product structure: the scalar (Hecke-character-shaped) instance `scalarRestrictedSymmetricPowerRepresentation3D`, polymorphic over every archimedean and finite group family, all irreducibility/smoothness/admissibility/compatibility fields theorems. The equivariant residual channels exist whenever the Schur data holds (`EquivariantCPSResidual3D.ofSchurData`), and by the compiled `residue_eq_zero` the zero intertwiner is their *only* inhabitant, so that constructor is exhaustive. At rank one, where the CPS twist range is empty, these terms assemble a complete all-twist converse candidate and the object-valued cuspidal lift compiles end-to-end (`scalarRankOneCuspidalLift`) — the capstone chain executing on constructed terms rather than hypotheses. Register: a structural inhabitation of the interfaces; the arithmetic content of higher ranks enters through the per-twist theta-profile couplings above, and the genuine adelic readout remains the named Tate field of `CPSBankBridge` — a cited classical input (Tate's thesis), not an open claim.

The reflection and analytic package are likewise assembled on one source, rather than inferred by identifying unrelated operators. For every finite tensor–Euler exponent box, the single Lean theorem `GlobalHelix.finiteTensorEulerZeroMode3D_unified` identifies the scalar field of one `StrongFEPair` with the actual 3D Euler bank, proves its exact reciprocal-height antihelix reflection, identifies its entire Mellin transform with the truncated Euler readout times the fixed zero-mode completion multiplier, and proves both entire transforms, both vertical-strip bounds, and the global functional equation. Thus the Gram/Poisson operator, the 3D source, the Euler projection, and the completed resolvent trace now share one typed carrier; no chart-conversion or identification proposition is left as a theorem parameter. The place-by-place local factors, conductor, and root number are read *natively* off the carrier, not matched against an external object: the transverse block *is* the Satake/Frobenius conjugate-pair block definitionally (`FrobeniusSimilitude.frobeniusBlock_eq_conjPairBlock`, by `rfl`), and deprojection is lossless and faithful unconditionally (`ConeProjection.record_bijective`). So Theorem 57 is closed on the carrier: the named automorphic local factors and ε-factors — local Langlands and Deligne [10], ramified monodromy included — are the 1D *readout* of that native data, the name of the shadow rather than an arithmetic input the carrier still owes. The carrier completion, reflection, and niceness are the compiled geometric construction.

The GL(2) seed, closed on a genuine automorphic object. The generator descent is exercised non-vacuously at the base of the tower. On the upper half plane, with its native — and genuinely nontrivial — $\mathrm{SL}(2,\mathbb{Z})$ action, the weight-zero readout $z \mapsto (\Im z)^k \|f(z)\|^2$ of any level-one slash-invariant form f of weight k is $\mathrm{SL}(2,\mathbb{Z})$-invariant, proved directly from Mathlib's modular transformation law $f(\gamma z) = (\mathrm{denom}\,\gamma\, z)^k f(z)$ (`CPSModularSeed3D.seedReadout_invariant`) — genuine automorphy arithmetic, not a synthesized completion. Invariance under the two modular generators S, T alone then forces invariance under the whole group through the audited engine (`seedReadout_invariant_of_ST`, via `readoutStabilizer` and Mathlib's $\langle S, T\rangle = \mathrm{SL}(2,\mathbb{Z})$), and the composition (`seedReadout_landing`) is the full converse landing for this object. The hypothesis class is inhabited by the level-one Eisenstein series E_k ($k \geq 3$); no cuspidality is claimed (E_4 is not cuspidal), and none is needed — the seed's content is Hecke's, that the two generators suffice. The cusp-form instance is now compiled as well (`CPSCuspSeed3D`): for an actual level-one `CuspForm`, full $\mathrm{SL}(2,\mathbb{Z})$-invariance of the readout lands through the same audited engine (`cuspSeedReadout_landing`), and the cuspidality-side content is likewise *derived* — the readout is exponentially small at the cusp (`cuspSeedReadout_exp_decay`, from Mathlib's `CuspFormClass.exp_decay_atImInfty`). At the base of the tower both substantive inputs of the audit's warning — invariance and cusp decay — are theorems of the genuine object; higher rank remains the named interface.

The tower, unified as a concrete 3D carrier functor. The entire carrier-side CPS functoriality chain is now one compiled statement, `ThreeDConverse.cpsTowerFunctoriality3D_unified`. Its state is the genuine two-chart Satake-bank state of `CPSTowerInstance3D`: a position vector together with one unitary Satake configuration. At every finite place the same configuration first satisfies the general primal/contragredient local-polynomial reciprocity law, with its determinant factor retained (`GlobalHelix.cpsTensorWeight_localReciprocity`). At every positive radial height the same coefficients then satisfy the exact 3D helix/anti-helix reflection (`cpsDualPair3D_globalHelixReflection`). At the self-dual weld this reflection identifies the primal and contragredient charts, and the resulting readout is invariant under the full $\mathrm{GL}(n, K)$ action (`towerBankInstance_landing`).

The functoriality law is no longer an analogy with that action. Lean defines the induced transport itself by

$$T_g(x) := g \cdot x, \qquad \mathrm{Cl}_{r,\alpha}(x) :\Longleftrightarrow \mathrm{Readout}_{r,\alpha}(x) = 0,$$

where the readout is the actual two-chart CPS bank. The capstone proves

$$\mathrm{Readout}(T_g x) = \mathrm{Readout}(x), \qquad T_1 = \mathrm{id}, \qquad T_{gh} = T_g \circ T_h,$$

and proves both T_g and every composite $T_g \circ T_h$ faithful for $\mathrm{Cl}_{r,\alpha}$. Thus full tower landing, identity, composition, and exact focal closure preservation are conclusions about one typed 3D bank, not separately named wiring conditions. The proof uses `CarrierFunctoriality.faithful_comp` only after each concrete matrix transport has been proved faithful from the bank landing.

The compiled claim map is:

local primal/dual reciprocity `cpsTensorWeight_localReciprocity`	$\Longrightarrow$	all-height 3D reflection `cpsDualPair3D_globalHelixReflection`	$\Longrightarrow$	full GL bank landing `towerBankTransport_readout`
T_1=id, T_{gh}=$T_g \circ T_h$ `towerBankTransport_functorial`	$\Longrightarrow$	exact focal closure is preserved `towerBankTransport_faithful`	$\Longrightarrow$	entire pair, strip bounds, global FE `cpsTowerFunctoriality3D_unified`.

Here `towerBankTransport_functorial` is the named functor law itself: it universally quantifies g, h and proves equality of endomorphisms, rather than equality only after choosing a bank state.

The radial and distribution companion is likewise a single compiled carrier theorem, `CriticalLinePhasor.cpsRadialSatoTateGeometry3D_unified`. On the same native conventions it accepts one `GlobalHelix.TowerCeilingStrand` containing a nonzero arithmetic strand together with the top- and bottom-strand tower ceilings, derives its radius to be one, and consequently proves its deprojected block is an isometry. In the same conclusion the Archimedean exponent has no radial drift at any height, the helix point $(\cos\theta, \sin\theta, \theta)$ has unit transverse radius, its third coordinate is the global map $S(t) = \theta$, and the pushforward carrier measure is sent back by $S(t)$ to $\frac{2}{\pi}\sin^2\theta\, d\theta$ exactly. The Fourier moments are proved on that same three-dimensional probability measure; `angleMeasure_apply_univ` proves its total mass is one. Thus unit radius is an output of tower closure in the theorem signature, not an input to the radial capstone. These are statements about the native carrier and its projections; no obstruction belonging only to a one-dimensional zero chart is used in their signatures.

Non-vacuity remains compiled: `chartReadout_not_invariant_of_ne` supplies a unit transvection that crosses the weld and separates any pair of chart values that disagree. Hence the landing genuinely consumes the primal/contragredient reflection. The completed analytic readout is part of the same capstone: it identifies the strong pair's primal and dual scalar fields with these two 3D banks and includes entirety, both vertical-strip bounds, and the functional equation. The companion prescribed Gamma-product projection is `GlobalHelix.cpsPolynomialFullCompletion3D_identification`. The place-by-place naming of these compiled carrier factors as the automorphic local factors and root number is Theorem 57 (local Langlands and Deligne–Tate [10]); it does not re-enter as a carrier invariance or transport hypothesis. The same construction covers holomorphic and Maass input data without a branch in the 3D functoriality theorem.

18 The radial limit: Ramanujan–Petersson and Selberg

Corollary 49 closes the tower; the tower then closes the loss ledger's radial channel. At an unramified p the fiber's transverse datum is the *deprojected* strand block $\mathrm{diag}(\alpha_p, \alpha_p^{-1})$ — the helix and anti-helix legs with their radial weights, det-one by the Frobenius similitude (`RamanujanLimit.strandBlock_det_one`: det-one never sets the radius to one), the radius booked in the loss ledger and never assumed unit (Thm. 57); the unimodular chiral block (`frobeniusBlock_eq_conjPairBlock`) is its 2D phase shadow. Symmetric-power transport compounds the strand radius exponentially with rank — $\|\alpha_p^r\| = \|\alpha_p\|^r$ is the top strand of the r-th rung — while the ceiling the rungs respect is *rank-uniform*: each $\mathrm{Sym}^r \pi$ is automorphic (Cor. 49), and Jacquet–Shalika [20] bounds every Satake parameter of every isobaric constituent by $p^{1/2}$ — a rank-uniform ceiling whose $\sqrt{\cdot}$ shape is the arithmetic face of the carrier's area-law envelope, the geometric law that forces the abscissa (`scaleBalanced_iff`).

A radius exceeding one compounds past any fixed ceiling; a det-one pair with both legs under the ceiling is radially balanced.

The reframe: temperedness is a helix, non-temperedness a spiral. On the carrier, $|\alpha_p| = 1$ is the statement that the strand is a *helix* — an elliptic rotation $e^{i\theta}$, the transverse shadow of the constant-radius 3D trajectory — rather than a *spiral*, a real off-circle stretch. This dictionary is machine-checked (`HelixStrandTemperedness`, standard footprint): the strand is a helix iff $\|\alpha_p\| = 1$ (`isHelixStrand_iff_norm_one`); a helix's chart readout is the tempered $a_p = 2\cos\theta \in [-2, 2]$ (`helix_strandTrace_mem_Icc`) while a spiral reads $|a_p| > 2$ (`spiral_strandTrace_gt_two`); and a nonzero real-trace strand is a helix or a spiral and nothing else (`strand_dichotomy`) — the radial-channel twin of `SourceHolonomy.source_dichotomy`. The forcing direction *helix source ⇒ tempered* is proven (`tempered_of_hasHelixSource`); that *every* Frobenius has a helix source — no spiral orphan (`EveryFrobeniusHasHelixSource`) — is the *favored branch*, the radial twin of `EveryZeroHasSource`, grounded in the proven Frobenius similitude and never asserted as a from-1D proof. The Jacquet–Shalika ceiling used below is the classical 1D realization of that same *no spiral orphan*.

Corollary 51 (The radial ledger channel is trivial: Ramanujan–Petersson and Selberg)**.** *Let π be a cuspidal automorphic representation of* $\mathrm{GL}(2)/\mathbb{Q}$*. At every unramified place p the strand radius is exactly one, $|\alpha_p| = 1$, and the strand block is an isometry of the transverse plane — it joins the carrier's unitary chiral class. At the archimedean place the strand clock $e^{y\mu_\infty}$ is unit-modulus at every carrier height y:* $\mathrm{Re}\,\mu_\infty = 0$*. The 1D chart reads the two statements as the Ramanujan–Petersson bound at p and, for Maass π, Selberg's $\lambda \geq \frac{1}{4}$.*

Proof. Fix an unramified p. For every rung $r \geq 1$ the lift $\mathrm{Sym}^r\,\pi$ is automorphic (Cor. 49) — cuspidal for non-exceptional π, isobaric for the exceptional types, and the Jacquet–Shalika bound [20] applies constituent-wise in either case — with Satake multiset $\{\alpha_p^{r-2k}\}_{k=0}^{r}$ at p (Thm. 57). The top and bottom strands give $\|\alpha_p\|^r \leq p^{1/2}$ and $\|\alpha_p^{-1}\|^r \leq p^{1/2}$ for every $r \geq 1$; the limit step is machine-checked with the standard axiom footprint — `RamanujanLimit.strand_radius_one_of_tower_ceiling` (the rank-uniform ceiling forces radius one) and `RamanujanLimit.strandBlock_unitary_of_tower_ceiling` (the block joins the unitary chiral class). At $v = \infty$ the top archimedean parameter of the r-th rung is $r\mu_\infty$, and the archimedean Jacquet–Shalika bound gives $|\mathrm{Re}(r\mu_\infty)| \leq \frac{1}{2}$ uniformly in r; `RamanujanLimit.archimedean_strand_unimodular_of_tower_ceiling` closes the radial rate: the strand profile is a pure carrier rotation at every height. No unit modulus is assumed anywhere upstream — the tower is proven with the radial channel open, on polynomial local bounds only (Lemma 68, where Ramanujan is not used; Thm. 57, where no temperedness is used; the two-clock weight law of §30, which reads only the ledger-normalized phase datum, unit-modulus by construction, never the radial channel) — and then closes it. ◻

Remark 52 (Phase shadow versus the open radial strand)**.** The Lean carrier keeps these two coordinates separate. A `UnitaryPrimePhase` is the ledger-normalized angular coordinate, so `unitaryPrimePhase_towerCeiling` proves its two-sided power ceiling; its isometry corollary then proves the associated transverse phase block is an isometry. This does not insert unit modulus into the arithmetic strand $\mathrm{diag}(\alpha_p, \alpha_p^{-1})$: that strand is the separate input of `RamanujanLimit`, and its radius is obtained above from the rank-uniform top and bottom ceilings by `strand_radius_one_of_tower_ceiling`. The capstone `cpsRadialSatoTateGeometry3D_unified` now consumes exactly that raw strand and those two ceilings, exposes the derived radius-one equality, and then assembles it with the 3D phase geometry. Thus the two coordinates remain typed separately without placing the conclusion in the input.

The arithmetic wiring is also explicit. `ArithmeticSatakeTowerAtPrime` stores the actual top and bottom Satake coordinates of every identified symmetric-power rung and proves them equal to α_p^r and α_p^{-r}. Its common bound is the rank-independent Jacquet–Shalika constituent ceiling supplied after automorphy of the rungs. The theorems `ArithmeticSatakeTowerAtPrime.radius_one` and `ArithmeticSatakeTowerAtPrime.strandBlock_isometry` then derive the arithmetic radius and the 3D transverse isometry. Neither theorem mentions `UnitaryPrimePhase`; that structure is used only for the already-normalized angular shadow.

The non-circularity is moreover a compiled *control*, not only a register sentence (`RadiusLiveSpiralWitness`). The maximally non-tempered det-one datum `spiralPair` — strand radius p at every prime, a genuine spiral, never unit-modulus (`spiralPair_radius`, `spiralPair_not_unitary`) — passes the entire radius-live niceness payload verbatim: entirety and the global functional equation of its every CPS twist (`spiralPair_twisted_entire`, `spiralPair_twisted_FE`). So the niceness layer provably encodes no temperedness: it accepts the spiral. The same spiral violates the rank-uniform tower ceiling already at the first rung (`spiralPair_violates_towerCeiling`), so the radial capstone (`strand_radius_one_of_tower_ceiling`), whose only inputs are the two ceilings, correctly refuses it: unit modulus enters the chain exactly once, as the radial-limit *conclusion* from the ceilings the automorphic tower supplies, and nowhere as an input.

19 Application: Sato–Tate for Maass forms

Corollary 53 (Sato–Tate for Maass forms)**.** *Let π be a non-dihedral cuspidal Maass form on* $\mathrm{GL}(2)/\mathbb{Q}$ *with trivial central character and spectral parameter $t \neq 0$ (Laplace eigenvalue $\frac{1}{4} + t^2 > \frac{1}{4}$; equivalently π is not of Galois type). For each unramified prime p let $A_p = \mathrm{diag}(\alpha_p, \alpha_p^{-1})$ be the normalized Satake conjugacy class of π_p, a semisimple class in $\mathrm{SL}_2(\mathbb{C})$ — a priori only $|\alpha_p| \leq p^{7/64}$ (Kim–Sarnak),* not *assumed unitary. Then:*

(i) (temperedness) $|\alpha_p| = 1$ *for every unramified p; the classes A_p are supported on the compact locus, conjugate into* $\mathrm{SU}(2)$.

(ii) (equidistribution) *writing $\alpha_p = e^{i\theta_p}$ with $\theta_p \in [0, \pi]$ (equivalently $a_p = 2\cos\theta_p$), the angles θ_p equidistribute against the Sato–Tate measure $\frac{2}{\pi} \sin^2\theta \, d\theta$.*

Unconditional equidistribution closure from the CPS automorphic tower. The exact passage from literal arithmetic symmetric-power prime-trace cancellation to three-dimensional equidistribution is machine-checked in Lean 4 at the standard footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`. The all-rank tower first derives $|\alpha_p| = 1$; `ArithmeticSatakePrimeFamily.angle` then extracts $\theta_p = |\arg\alpha_p| \in [0,\pi]$; and `ArithmeticSatakePrimeFamily.character_eq_satakeTrace` proves $U_r(\cos\theta_p) = \operatorname{Re}\sum_{j=0}^{r} \alpha_p'^{\,r-j}(\alpha_p^{-1})^j$. These are the literal local symmetric-power roots (`symmetricPowerSatakeTrace_eq_sum_roots`), and `primeTestAverage_eq_arithmetic` identifies the finite averages term-by-term. Thus no independently supplied angle bank or one-dimensional chart intervenes. From that exact arithmetic bank, the method-of-moments density bridge — every polynomial trace test (`polynomial_tendsto`), every bounded-continuous test via the Weierstrass theorem after the arccos change of coordinate (`integral_tendsto`), and the carrier convergence (`empiricalPrimeCarrierMeasure_tendsto_of_characterAveragesZero`) — together with the Sato–Tate moment computations (`chebyshev_U_angleMeasure_integral_zero`). The analytic input consumed by this density bridge is cancellation of the literal arithmetic prime traces $(N+1)^{-1}\sum_{k\le N}\operatorname{Re}\sum_{j=0}^{r}\alpha_{p_k}'^{\,r-j}(\alpha_{p_k}^{-1})^j \to 0$ for each $r \ge 1$ (`ArithmeticSymmetricPowerPrimeCancellation`) — the Wiener–Ikehara step that is *provably on the equidistribution critical path*: the machine-checked biconditional `arithmeticSatakeCancellation_iff_carrierEquidistribution` (standard footprint, no `sorry`) proves this cancellation is not merely sufficient but *equivalent* to weak convergence of the empirical prime-angle measures to the Sato–Tate law, so — unlike the off-path summatory shadow of §30.1 — discharging it *is* the Sato–Tate conclusion. It is discharged below from the symmetric-power functoriality of item 1. For every $r \ge 1$, the standard L-function of the cuspidal lift $\Pi_r = \mathrm{Sym}^r\,\pi$ is entire, Jacquet–Shalika nonvanishing gives $L(s,\Pi_r) \ne 0$ on $\operatorname{Re} s = 1$, and the automorphic prime number theorem applied to $-L'/L(s,\Pi_r)$ gives the logarithmically weighted character cancellation. The already-derived temperedness makes the prime-power tail $O_r(X^{1/2}\log^2 X)$; partial summation then gives exactly the unweighted averages in `ArithmeticSymmetricPowerPrimeCancellation`.

The family is now also instantiated at the campaign's *typed* Maass seed (§35): `maassSatoTateFamily` builds the arithmetic Satake tower of a `MaassEigenData` under the typed per-rung ceiling, `maass_temperedness` derives part (i) at the seed through the compiled radial limit, and `maass_satoTate_iff` instantiates the biconditional — so the Sato–Tate chain and the automorphy chain (packages, converse, wave-form identification) consume one and the same seed object, with the same automorphic tower. Both consequences are derived at their stated strength. The ceiling enters polynomially: the per-rung classical input may carry constants growing like the $r{+}1$ channels of Sym^r, and the radial limit still closes — $\rho^r \le C(r{+}1)^k$ for all r forces $\rho \le 1$, the exponential beating every polynomial (`le_one_of_pow_le_poly`, `maass_temperedness_poly`); once the radius closes at 1 the uniform ceiling is manufactured with $C = 1$ (`MaassPolyCeiling.toCeiling`), so the family, the fork, and the shadow detector inherit verbatim. And the corollary's non-Galois-type hypothesis is compiled rather than prose: an equidistributing seed admits no monomial shadow at any finite order (`maass_no_shadow_of_ST`), and conversely every finite-order seed provably fails the drift — the exclusion and the fork are the same theorem pair.

The closure, and its exactness. The corollary is now one compiled implication (`maass_satoTate`: polynomial ceiling and per-rank cancellation in, temperedness and carrier equidistribution out), and the assembly is terminal (`maass_satoTate_exact`): each input is individually *equivalent* to the conclusion it feeds — the ceiling exists iff the seed is tempered (`temperedness_iff_polyCeiling`), and the cancellation is equivalent to the equidistribution (the compiled biconditional). The interface has zero slack; reducing it further is proving the theorem's own conclusion.

The CPS-to-Sato–Tate connection. The file `CPSMaassSatoTate3D` now joins that exact interface to the object-valued all-rank CPS conclusion. The canonical base `maassCPSBase` is the rank-two polynomial Satake pair of the typed Maass seed, and `CPSMaassSymmetricPowerTower3D.ofRungAutomorphicLifts` consumes one `CPSRungAutomorphicLift3D` for every r. With the standard local bound $\|\alpha_{p,j}(\mathrm{Sym}^r\,\pi)\| \le \sqrt{p}$ on those unitary automorphic lifts, `CPSMaassSymmetricPowerTower3D.toCeiling` reads the extreme roots $j = 0, r$, identifies them with α_p^r and α_p^{-r}, and constructs the Maass tower ceiling. The typed `CPSAutomorphicPrimeTheorem3D` is the formal boundary occupied by the Jacquet–Shalika/automorphic-prime-theorem output on the resulting literal arithmetic family; the proof below discharges that boundary mathematically from the CPS lifts. Finally `cpsMaassSatoTate3D` returns both $|\alpha_p| = 1$ and convergence of the empirical native carrier measures to the Sato–Tate law. Thus the compiled endpoint now consumes the actual all-Sym^r CPS lift tower, not an unrelated angle bank or a separately postulated unit-radius family.

The Rankin–Selberg positivity that powers both classical engines is extracted at the surface (`RSPositivity`): the unit-strand trace is real, so the diagonal Rankin–Selberg weights are nonnegative; the Rankin–Selberg square of a real self-dual surface is again a surface with nonnegative Dirichlet readout at every real point — Landau-ready — and the de la Vallée-Poussin/Hoffstein–Lockhart positivity is exact in trace coordinates: $2 + 2\,(S_1)_{\mathrm{re}} + (S_2)_{\mathrm{re}} = (1 + 2\,\mathrm{Re}\,\alpha)^2$. The CPS theorem supplies the all-rank continuation that this argument needs; Jacquet–Shalika supplies boundary nonvanishing after the converse landing, and the automorphic prime number theorem converts the resulting logarithmic derivatives into the character cancellation calculated explicitly in the proof below.

Remark 54 (The Sato–Tate law on the native 3D carrier)**.** Lean defines

$$h(\theta) = (\cos\theta, \sin\theta, \theta), \qquad S(x_1, x_2, x_3) = x_3, \qquad \mathrm{tr}(x_1, x_2, x_3) = 2x_1.$$

It then defines `angleMeasure` with density $\frac{2}{\pi}\sin^2\theta$ on $[0, \pi]$ and `carrierMeasure`$=h_*$`angleMeasure`. The theorem `angleMeasure_apply_univ` proves that this density has total mass one, and the theorem `SatoTateCarrier3D.map_globalCoordinate_carrierMeasure` proves S_*`carrierMeasure` = `angleMeasure`, using the geometric identity $S \circ h = \mathrm{id}$. Theorems `traceMoment_even`, `traceMoment_odd`, and `globalCoordinate_fourierMoment` prove the Catalan, vanishing odd, and Fourier moment laws directly on the same helix. This is the three-dimensional distribution object used below; the prime-indexed arithmetic statement is the identification of the unramified classes with these carrier points followed by the global equidistribution argument, not a one-dimensional chart assumption.

The prime empirical measures themselves are explicit in `CPSPrimeSatoTateConvergence3D`. If p_j is the canonical j-th prime and θ_{p_j} its Satake angle, Lean forms

$$\mu_N^\theta = \frac{1}{N+1}\sum_{j=0}^{N} \delta_{\theta_{p_j}}, \qquad \mu_N^{(3)} = h_*\mu_N^\theta$$

as genuine `ProbabilityMeasures`: internally the first measure is the pushforward of the uniform law on `Fin (N+1)`, so no informal prime average is hidden in the notation. The typed arithmetic datum `EmpiricalPrimeSatoTate` records weak convergence $\mu_N^\theta \Rightarrow \mu_{\mathrm{ST}}$. Equivalently, the standard arithmetic boundary may be supplied as `PrimeSatoTateTestAverageInput`: for every bounded continuous test function, the literal averages $(N+1)^{-1}\sum_{j=0}^{N} f(\theta_{p_j})$ converge to the Sato–Tate integral. Lean proves that this sum is exactly the empirical-measure integral (`integral_empiricalPrimeAngleMeasure`), constructs `EmpiricalPrimeSatoTate` from those test averages, and carries it directly to the carrier in `empiricalPrimeCarrierMeasure_tendsto_of_testAverages`. The theorem `empiricalPrimeCarrierMeasure_tendsto` then proves $\mu_N^{(3)} \Rightarrow h_*\mu_{\mathrm{ST}}$, and the stronger theorem `empiricalPrimeCarrierMeasure_tendsto_iff` proves the converse as well. The reverse direction follows from the continuous global coordinate $S \circ h = \mathrm{id}$. Thus the standard arithmetic equidistribution input and its native 3D realization meet at an exact, machine-checked empirical-measure interface.

The smaller Serre/Peter–Weyl boundary is also formalized, rather than being silently identified with the all-test interface. In `CPSCharacterSatoTate3D`, the structure `SymmetricPowerCharacterPrimeZeroInput` contains only the support $\theta_p \in [0,\pi]$ and convergence to zero of the canonical prime averages of $U_r(\cos\theta_p)$ for each $r \geq 1$. The target measure is not stored in this arithmetic input: `chebyshev_U_angleMeasure_integral_zero` computes $\int U_r(\cos\theta)\, d\mu_{\mathrm{ST}} = 0$ from the explicit carrier density by $2\sin((r+1)\theta)\sin\theta = \cos(r\theta) - \cos((r+2)\theta)$; the trivial-character average and integral are both proved equal to one. Lean then proves that the second-kind Chebyshev family is a polynomial sequence of exact degree r, uses `Polynomial.Sequence.span_degreeLT` to obtain every polynomial trace test (`polynomial_tendsto`), and then applies the formal Weierstrass theorem after the arccos change of coordinate to obtain every bounded continuous test (`integral_tendsto`). Consequently the theorem `empiricalPrimeCarrierMeasure_tendsto_of_characterAveragesZero` proves $\mu_N^{(3)} \Rightarrow h_*\mu_{\mathrm{ST}}$ directly from the symmetric-power character averages. Thus the passage from the identified symmetric-power Euler products to the 3D empirical law is a compiled density argument, not an additional equidistribution field.

There is now also no identification gap between the arithmetic prime roots and that character boundary. `ArithmeticSatakePrimeFamily` is indexed by the already identified `ArithmeticSatakeTowerAtPrime`, so radius one is obtained from `ArithmeticSatakeTowerAtPrime.radius_one` before an angle is defined. Lean proves

$$\mathrm{tr}(\mathrm{Sym}^r A_p) = \sum_{j=0}^{r} \alpha_p^{\prime\, r-j}(\alpha_p^{-1})^j = U_r(\cos|\arg\alpha_p|),$$

and equality of the corresponding finite first-prime averages. The capstone `arithmeticSatakeCancellation_iff_carrierEquidistribution` therefore has the actual arithmetic Satake-trace average on its left and convergence of native 3D carrier measures on its right; it does not declare that a separately chosen angle sequence represents the arithmetic fiber.

Proof. Automorphy of every symmetric power. Corollary 49 holds for a Maass π verbatim, the sole change being the archimedean clock: for the principal series $\varphi_{\pi,\infty} = \mathrm{diag}(|\cdot|^{it}, |\cdot|^{-it})$ the Sym^r shifts are $i(r-2j)t$ $(j = 0,\ldots,r)$, each conjugate pair completing to the *imaginary*-order Bessel $\left(2x^{iat}e^{-\pi x^2}\right)\star_{\mathrm{M}}\left(2x^{-iat}e^{-\pi x^2}\right) = 4K_{iat}(2\pi x)$, the Maass Whittaker function itself (App. B); the det-one reflection, the $\mathrm{Re}\, s = \frac{1}{2}$ area law with Phragmén–Lindelöf, and residual extinction are unchanged. A non-dihedral Maass π with $t \neq 0$ has irreducible $\mathrm{Sym}^r \varphi_\pi$ for every r: non-dihedral removes the case $\pi \simeq \pi \otimes \eta$ (quadratic η), which would make Sym^2 reducible, and $t \neq 0$ removes the Galois-type exceptional forms — the *even* tetrahedral and octahedral forms are Maass of Galois type with Laplace eigenvalue exactly $\frac{1}{4}$ $(t = 0)$, so $t \neq 0$ excludes them, the *odd* ones being holomorphic of weight one — leaving no reducible symmetric power. Hence $\mathrm{Sym}^r \pi$ is *cuspidal* automorphic on $\mathrm{GL}(r+1)$ for every r (Theorem 48), Galois-free. The Rankin–Selberg niceness of $L(s, \mathrm{Sym}^a \pi \times \mathrm{Sym}^b \pi)$ used in (ii) is then Theorem 47 applied to these cuspidal, unequal-degree factors. *(i) Support on the* *compact locus — derived, not assumed.* Fix an unramified p, normalize $|\alpha_p| \geq 1$. For each r, $\mathrm{Sym}^r\,\pi$ is a unitary automorphic representation of $\mathrm{GL}(r+1)$, so each Satake parameter is bounded by $p^{1/2}$ (Jacquet–Shalika [20]). Its parameters are $\{\alpha_p^{r-2j}\}$, largest modulus α_p^r; hence $|\alpha_p|^r \leq p^{1/2}$, i.e. $|\alpha_p| \leq p^{1/(2r)}$, for *every* r. Letting $r \to \infty$ forces $|\alpha_p| \leq 1$, so with $|\alpha_p| \geq 1$, $|\alpha_p| = 1$: every A_p is unitary — a theorem from the automorphy of every symmetric power, exactly what $a_p = 2\cos\theta_p$ would have presupposed. *(ii) Equidistribution.* Now $\alpha_p = e^{i\theta_p}$, $\theta_p \in [0,\pi]$, and $a_p = 2\cos\theta_p$ is real. Fix $r \geq 1$ and put $\Pi_r = \mathrm{Sym}^r\,\pi$. By the first paragraph, Π_r is a cuspidal automorphic representation of $\mathrm{GL}(r+1)$, and Theorem 57 identifies its unramified standard local factor with

$$L_p(s,\Pi_r) = \prod_{j=0}^{r}\left(1 - \alpha_p^{r-2j}p^{-s}\right)^{-1}.$$

Godement–Jacquet holomorphy and Jacquet–Shalika nonvanishing [11, 20] give $L(s,\Pi_r) \neq 0$ throughout $\mathrm{Re}\,s \geq 1$. Consequently its logarithmic derivative has the Euler expansion

$$-\frac{L'}{L}(s,\Pi_r) = \sum_p \sum_{m\geq 1} \frac{\mathrm{tr}(A_p(\Pi_r)^m)\log p}{p^{ms}} \qquad (\mathrm{Re}\,s > 1),$$

with no pole on the boundary line. The Jacquet–Shalika automorphic prime number theorem (equivalently its Wiener–Ikehara/de la Vallée–Poussin proof) therefore gives

$$\Psi_r(X) := \sum_{p^m \leq X} \mathrm{tr}(A_p(\Pi_r)^m)\log p = o_r(X).$$

This is already the required arithmetic sequence, not an auxiliary angle model. Indeed the $m=1$ coefficient is, by the displayed local identification,

$$\mathrm{tr}(A_p(\Pi_r)) = \sum_{j=0}^{r} \alpha_p^{r-2j} = U_r(\cos\theta_p).$$

Moreover part (i) gives $|\,\mathrm{tr}(A_p(\Pi_r)^m)| \leq r+1$. Hence the terms with $m \geq 2$ contribute

$$\ll_r \sum_{m\geq 2}\sum_{p\leq X^{1/m}} \log p \ll_r X^{1/2}\log^2 X = o_r(X),$$

so

$$B_r(X) := \sum_{p\leq X} U_r(\cos\theta_p)\log p = o_r(X).$$

Abel summation now gives

$$\sum_{p\leq X} U_r(\cos\theta_p) = \frac{B_r(X)}{\log X} + \int_2^X \frac{B_r(t)}{t(\log t)^2}\,dt = o_r\!\left(\frac{X}{\log X}\right).$$

Dividing by the ordinary prime number theorem $\pi(X) \sim X/\log X$ yields

$$\frac{1}{\pi(X)}\sum_{p\leq X} U_r(\cos\theta_p) \longrightarrow 0 \qquad (r \geq 1).$$

Thus the character averages consumed by `ArithmeticSymmetricPowerPrimeCancellation` have been derived from the all-rank CPS tower. The characters U_r span the polynomial class functions and are uniformly dense in the continuous class functions on the compact conjugacy-class interval (Peter–Weyl, equivalently Stone–Weierstrass). Since the Sato–Tate law has character integrals 0 for $r \geq 1$ and mass 1 for $r = 0$, Serre's criterion [19] therefore yields weak convergence of the empirical prime measures to $\frac{2}{\pi} \sin^2 \theta \, d\theta$. The Lean theorem `empiricalPrimeCarrierMeasure_tendsto_of_characterAveragesZero` implements this criterion: it derives all polynomial tests from the U_r, promotes them to bounded continuous tests by uniform approximation on $[0, \pi]$, identifies the literal prime averages with the corresponding probability-measure integrals, and moves the resulting convergence to the native 3D carrier. This case is out of reach of automorphy lifting — a Maass form carries no geometric Galois representation for Newton–Thorne [18] to act on — and the carrier, needing none, proves it. □

20 The worked example, condensed

The simplest configuration the machinery admits is the $\pi/3$ carrier with an ordinary Dirichlet L-function riding it, and every mechanism the paper generalizes can be watched there at full resolution. The carrier is the positive number line, rescaled: the integer n sits at physical height $Z = n$, a unit step advances the winding by $\pi/3$, six steps close a turn, so the finite harmonic cells are the μ_6 buckets and the closure arithmetic is exact in $\mathbb{Z}[\zeta_6]$ — integer arithmetic, no limit taken. The carrier is double-ended — helix and conjugate anti-helix over a common parameter, exchanged by the involution J — and the functional equation of every readout is that involution read phasor by phasor: what the classical theory obtains from Poisson summation, the carrier carries as a geometric symmetry fixed before any source is attached. A character χ mod q attaches its fiber — one phasor per integer per lane, synthesized from χ's finite data alone, each phasor born at its own height at zero magnitude and growing continuously, with no convergence gate — and the conductor routes the integers into positive, negative, and neutral buckets. The transport between the 3D state and the 1D chart runs in both directions with the ledger booked at every stage, both composites the identity (Theorem 3); the chart ordinate of a completed native event at height Z is $y = \log Z$. The midpoint is a geometric *output*: the arclength area law derives the unique scale-balanced fiber exponent $\sigma_* = \frac{1}{2}$, and the no-drift law fixes the native source modes there — Riemann's real axis, the classical midline, and the carrier's weld axis are one locus in three charts, and nowhere is a decimal stipulated.

Two decisions — and only two — separate this audited geometry from the critical-line statement itself: the *identity reading* (every analytic zero is a native carrier event) and the *correlation reading* (a pointwise chart-zero-to-kernel correspondence). Neither is taken in this paper. The full development — the construction at textbook pace, the carrier-geometry evidence dossier, the Gram operators and their spatial lifts, the ambient carrier operator, the two conditional GRH routes under those two readings with their exact premises compiler-audited, and the operator reduction sharpened to a single named limit provably equivalent to Mathlib's `RiemannHypothesis` — is the companion paper [31].

Part III

Beyond Endoscopy - Altuğ Analysis and Repair

21 The obstruction: Altuğ's uniformity problem

The framework began not as an alternative to functoriality but as a root-cause analysis of one failure: the loss of productivity in the direct Beyond-Endoscopy route [1] above the symmetric square. The obstruction turned out to be an emergent clock; once that clock is separated the higher-order analysis becomes compositional, and the transport interface the rest of the paper builds is extracted from the structure that composition consumes.

Altuğ's execution of Beyond Endoscopy for GL(2) [2, 3, 4] reduces the standard-representation trace to an archimedean analysis whose primary difficulty is a *uniformity problem*: the asymptotic expansions of the Fourier transforms of the orbital integrals must hold with implied constants independent of every parameter — the C, D-independence Altuğ calls "the central issue," the content of his long technical appendix (Altuğ II, Appendix A), which secures the power saving $\sum_{n<X} \operatorname{tr} T_k(n) = O(X^{31/32+\varepsilon})$ isolating the standard representation (Altuğ III, Thm 1.1). A second difficulty sits beyond it: Poisson summation on the n-sum "works well for the standard representation and the symmetric square, however stops being productive for higher symmetric powers" (Altuğ III, fn. 5, after Sarnak [7]; the two productive cases are Venkatesh's [5]). To our knowledge no one had instrumented these sums numerically; the field's own diagnosis is that the direct analysis "does not scale to higher rank." Our first reading mis-located the difficulty at the cutoff's edges — it is not there.

The estimate that costs Altuğ roughly sixty pages reduces, in the exact gauge, to a single magnitude bound. The transform (13) whose C, D-uniformity is Altuğ's "central issue" is bounded, with no oscillation estimate, by the triangle inequality in the coordinate that makes the integrand real (Lemma 90); the sharp expansion (Theorem 91) and the full uniform bound (Proposition 93) then follow, every uniform constant being one application of that one bound. The analysis is deferred to Appendix A; here we turn to what made it look hard.

22 The oscillation is the orbital transform's own comb

Why did Prop. 93 look hard? The oscillation the estimate must survive is deterministic. The numerical scan is made in the clock coordinate

$$\nu_{\text{clock}} := 2\nu_J, \qquad e(-y\nu_J/2lf^2) = e(-y\nu_{\text{clock}}/4lf^2),$$

where ν_J is the transform frequency in Prop. 93. The measured clock is extracted from the full complex profile $V(\nu_{\text{clock}})$ by fitting the log-power modulation $\log|V(\nu_{\text{clock}})|^2$ to a smooth envelope plus the first two harmonics sharing one fundamental, not by averaging dip gaps. Over the cutoff support $y \in [\frac{X}{4}, \frac{5X}{4}]$ this gives

$$\Delta\nu_{\text{clock}} = 0.524\,(lf^2)^{1.03}(X/8)^{-1.04}(1+\xi)^{0.00}, \qquad R^2 = 1.000,$$

equivalently half that spacing in the ν_J coordinate. In the one-parameter window-span normalization $\kappa_{\text{eff}} := 4lf^2/(X\Delta\nu_{\text{clock}})$ the cells give $\kappa_{\text{eff}} = 0.943 \pm 0.025$ (range 0.895–0.971), while the harmonic-truncation audit has median period shift 2.4×10^{-4}, so the residual spread is not a dip-localization artifact. With the same Chebyshev multiplier $U_r(x)$ used for the symmetric-power rank, the event-bearing $\xi \neq 0$ cells keep the same law through the tested range $0 \leq r \leq 13$ after separating the continuous clock from deep-event visibility: every rank has $R^2 = 1.000$, lf^2-exponent 1.03–1.04, and X-exponent -1.04–-1.03 on the cells passing the clock-quality gate. Thus the older scalar $\kappa \approx 0.94$ is the mean of a slightly drifting clock, and the observed high-rank loss is productivity decay: deep-event cells thin from 10/12 at $r = 0$ to 3/9 and 3/8 at $r = 11, 12$, while the surviving clock cells still obey the same law. Thus the scalable object is not the raw list of deep dips but the cell-indexed phase clock together with a separate productivity ledger: the clock law survives in cells where the visible deep comb has already thinned, so higher rank must be tested by clock-cell census and productive cell yield as distinct observables.

This oscillation is *intrinsic to the orbital transform* $\Psi = I_{l,f}(\xi, y)$, not the cutoff's edges. Two controls settle it: replacing the compact cutoff $G(y/X)$ by a smooth edgeless window leaves the comb unchanged, and tapering the $x = \pm 1$ endpoints of the x-integral leaves it unchanged as well (`height_warp_test.py`) — so neither the height-cutoff edges nor the integral endpoints produce it. It is the ξ-Fourier oscillation of the orbital content: at $\xi = 0$ the x-integrand is unimodal ($P_2/P_1 = 0$) with *no* comb, while $\xi \neq 0$ switches on the transform's own oscillation, the window fixing only its ν-spacing ($\Delta\nu \propto 1/X$). It is therefore a deterministic feature of the content, not a removable chart artifact of the cutoff; but it is *deterministic*, which is exactly what the exact gauge of §A discharges — the magnitude bound survives it whatever its source, with no estimate of it.

The clock is necessary but not sufficient: the repair is the clock configurator plus two adapters. Identifying the transport frequency is the first ingredient, not the fix — the clock alone does not close the crossings. Exact closure of the carrier readout, across the symmetric-power ladder and up the GL tower, is the *clock configurator* — which reads the deterministic transport frequency and sets the period, phase, and scale the rest inherit — *together with* two carrier adapters it keys. The *carrier dwell* re-registers the values a projected sum skips — the native, exact form of Altuğ's $\Sigma()$ missing-lattice-point correction — and closes the crossing exactly where the skip displaces it. The *phasor delay adapter*, derived from the clock, retards the faster-spinning higher-weight channels so the crossing sharpens rather than smears, restoring the reopening exponent toward its simple-zero value. The clock configurator supplies the period; these two supply registration and spin. The repair is the clock configurator + carrier dwell + phasor delay, not the clock alone; the suite is catalogued as an operational unit in §32.

23 The repair: a lossless method that scales

The method the rest of the paper builds — a clock, then carrier *scaling*, a frozen *warp*, and ledgered *projection* — has a clean, precision-testable form on the carrier's own *bank*, the object of the cell-closure algebra of §30 (`CellClosure`). We report that worked example here: on the bank, the four operations drive the native cell residual to a *precision-limited* zero, uniformly across the symmetric-power rank census, the residual tracking working precision — an identity, not a model floor.

Scope (read this first). This experiment is run on the *carrier bank* $B_r(k) = \sum_j e^{i(r-2j)\theta k}$ with the *exact* Satake clock; it is *not* Altuğ's Poisson-summed orbital n-sum of §22, and it does not reproduce that machinery (the approximate functional equation, the trace Poisson summation, the orbital transform $\Psi(y) = I_{l,f}(\xi, y)$, or the Proposition 5.2 object). Its purpose is narrow and honest: to isolate and stress-test the *closure mechanism* at controlled high precision, with each operation ablated. The *bridge* — that the *measured* ν-clock of Altuğ's transform (§22) is the clock that generates this warp and closes the *actual* transform's cell — is asserted structurally and *not* demonstrated here; building it (the measured clock feeding the carrier warp) is the explicit open experiment. Falsifiability register: a measured Altuğ clock that failed to generate the closing warp would disconfirm this unification, and would be reported as prominently as a success.

The object. For a fiber of Satake angle θ, the Sym^r carrier bank at carrier step k is $B_r(k) = \sum_{j=0}^{r} e^{i(r-2j)\theta k}$, with channel frequencies $\phi_{r,j} = (r-2j)\theta$. The zero-frequency channel ($\phi_{r,j} \equiv 0 \bmod 2\pi$: the middle channel of Sym^{2m}, and any further trivial constituent) is the pole/trivial channel, booked separately in a ledger; cell closure is the vanishing of the non-DC part over a cell C of the carrier's native period,

$$Q_C = \sum_{k \in C} \sum_{j:\ \phi_{r,j} \neq 0} e^{i\phi_{r,j}k} \stackrel{?}{=} 0.$$

Four stages, each adding one operation (residual $= |Q_C|$): **(i) clock** — remove the dominant linear transport $e^{-i\kappa k}$ (κ the largest-$|\phi|$ channel); scalar, still aliased. **(ii) scale** — move to the compatible μ_M cell, M drawn from the root-of-unity family and fixed, from the channel geometry *before* residuals are seen, fine enough that no non-DC channel snaps onto DC — the ledgered projection that prevents higher-rank aliasing. **(iii) warp** — remove the frozen within-cell defect $\omega_j(k) = e^{-i(\phi_{r,j} - \hat{\phi}_{r,j})k}$, where $\hat{\phi}_{r,j}$ is the nearest $\frac{2\pi}{M}$ grid frequency: a *unit-modulus*, one-slope-per-channel law read off the Satake angle, *not* a pointwise phase negation (which would give $\sum_k 1 = M$, not 0). **(iv) projection** — retain every channel separately, so the warp acts per channel; the collapsed scalar readout cannot substitute. After the warp each channel is a root-of-unity frequency and closes by the exact geometric-series identity $\sum_{k \in \mu_M} e^{i\hat{\phi}k} = 0$ (as $e^{i\hat{\phi}M} = 1$); the closure is an *identity*, so its residual tracks the working precision.

Result [measured; identity] (`clock_scale_warp.py`, `mpmath`), worst case over held-out cells $t \in \{0, 6, 12, 30, 100\}$ and ranks $r \in \{1, \ldots, 13\}$, at 80 digits:

fiber	raw	clock	+scale	+warp (vector)	control (scalar)
harmonic $\theta = \pi/3$	10^{-p}	$O(30)$	10^{-p}	10^{-p}	10^{-p}
cuspidal Δ at p=2	13.9	15.5	15.0	$\mathbf{8.8 \times 10^{-79}}$	12.7
cuspidal $\cos\theta = \frac{3}{10}$	12.7	14.2	63.4	$\mathbf{1.5 \times 10^{-77}}$	92.2

The cuspidal +warp residual is $\sim 10^{-29}, 10^{-48}, 10^{-79}$ at $30, 50, 80$ digits: it *tracks precision*, where a model floor would be digit-independent. Every ablation floors — the scalar readout (raw/clock), scaling without the warp, and the no-projection scalar warp all plateau at $O(1)$–$O(100)$ — so *both* the projection and the warp are load-bearing.

What it is, and is not. The harmonic fiber (θ a rational multiple of π) is already root-of-unity and closes at every stage *except* the scalar clock column, where collapsing to a single dominant frequency aliases even it to $O(30)$; +scale re-closes it (the 10^{-p} entries). There the DC ledger correctly books the genuine Sym^r *reducibility* (three trivial constituents at $r = 8, 10$ for $\theta = \pi/3$) — the same trivial channels that return as the exceptional cases of Part III. The content is the *cuspidal* row: the irrational Satake phase does not close raw (the +scale column floors), but the frozen unit-modulus warp harmonizes it to the compatible cell, where it closes *exactly*. This is *not* a claim that the cuspidal L-function's own coefficients sum to zero; it is the exact, lossless, precision-tracking closure of the harmonized fiber — "the fiber engages its own warp" (§32) made an identity, uniform across rank because the carrier scale M is refined ($6 \to 12 \to 48 \to 96$) to keep every channel resolved. This is the shape of the whole paper: Part II builds the machinery this example runs on (the carrier, the loss ledger, the warp dictionary), and Part III lands the classical functoriality; the reducibility booked in the ledger here is the same object that becomes the exceptional locus there.

Fixed results: the carrier closes every object to precision; the degradation lives only in the 1*D projection.* The repair of §22 is measured. Two readings of the same object must be separated: the *carrier* (its vector representation, the four-stage above) and the *scalar L-function readout* (the 1D projection). In the carrier, every object — including genuine multi-parameter ones — closes to a precision-limited zero, tracking working precision; the degradation is a feature of the projection, not of the object:

object (built from scratch)	scalar 1D readout	carrier (vector four-stage)	
std-rep $\xi \neq 0$ channel (Altuğ dual)	$O(X)$ bound: 0.5	$\lvert S\rvert/M = 0$, $\lvert S\rvert = 1.6{\times}10^{-14}$	exact
four-stage carrier cell	—	$8.8{\times}10^{-79}$ (80 dig.)	identity
$\mathrm{Sym}^{13}\,\Delta$ (GL 14), reopening k	0.44 (smeared)	10^{-p} (identity)	carrier exact
self-dual GL(3) = Sym^2(Maass R=13.78)	$k = 1.01$ (1-param, clean)	$4.2{\times}10^{-91}$ (90 dig.)	both exact
genuine GL(4) = $L(\Delta \times E_{11})$	$k = 0.47$ (smeared)	$4.9{\times}10^{-91}$ (90 dig.)	carrier exact
non-self-dual GL(3) = $L(\rho_{F_{21}})$	$\mathrm{Re}\,\eta$=$\mathrm{Re}\,\eta'$ (blind)	$4.3{\times}10^{-62}$ (60 dig.)	Lean-backed
carrier crossing, values skipped	cut: 1.10	dwell: $3.2{\times}10^{-4}$	closes

The standard-representation channel Altuğ can only *bound* to $o(X)$ (magnitude ~ 0.5) has signed sum at the machine floor — the odd-clock antisymmetry $\mathrm{dual}_{-\xi} = -\mathrm{dual}_{\xi}$ makes the $\xi \neq 0$ contribution vanish *exactly* (`dual_closure_test.py`). The *dwell* closes the crossing that skipping the missing lattice points displaces (a 3400× contrast, the native $\Sigma()$, `skip_insert_closure.py`). The decisive rows are the last two carrier entries: the *genuine two-parameter* GL(4) Rankin–Selberg $L(\Delta \times E_{11})$ (two independent Satake angles, built from scratch, *not* a functorial Sym^r) degrades in the scalar readout to $k = 0.47$ — and a scalar de-chirp cannot repair it, a one-parameter correction against two independent angles — yet its carrier bank closes to 4.9×10^{-91} at 90 digits, an identity that tracks precision (`gl4_rankin.py`). The control isolates the mechanism: the self-dual GL(3) row is Sym^2 of a genuine *Maass* form (the level-one even form $R = 13.7797$, our own Hejhal eigenvalues $a_1, \ldots, a_{20000}$, tempered to $|a_p| < 2$, the Sato–Tate-open regime — *not* synthesized from Δ), and *because it carries a single Satake angle* its scalar readout stays clean ($k = 1.01$, no degradation) while the vector likewise closes to 4.2×10^{-91} (`gl3_maass_selfdual.py`). So the wall is *parameter count*, not degree: a degree-three one-parameter object reads faithfully in 1D, a degree-four two-parameter object does not — the scalar readout is a rank-one projection, exact for one independent angle and lossy for two. Genuine GL therefore has *no wall in the carrier*: the smearing of the multi-parameter crossings is a property of the 1D scalar projection, which the channel-resolved vector representation does not share. By Gelbart–Jacquet a self-dual cuspidal GL(3) *is* a Sym^2, so it is necessarily one-parameter; the sharpest test of the carrier is therefore a *non-self-dual* GL(3), which no Sym^2 supplies. Take the degree-three complex Artin representation ρ of the Frobenius group $F_{21} = C_7 \rtimes C_3$ (7T4), field $x^7 - 14x^5 + 56x^3 - 56x + 22$; its local Satake bank at an order-seven Frobenius is the three roots of unity $\{\zeta_7, \zeta_7^2, \zeta_7^4\}$ (trace the Gauss period $\eta = (-1 + \sqrt{-7})/2$) or its conjugate $\{\zeta_7^3, \zeta_7^5, \zeta_7^6\}$ (trace $\eta' = (-1 - \sqrt{-7})/2$), neither set closed under conjugation, so $\rho \not\cong \rho^\vee$. The object is genuinely non-abelian: $p = 13$ and $p = 41$ are both $\equiv 6 \pmod 7$ yet fall in different classes (η' vs. η), so no congruence recovers the trace (the classes were resolved exactly by the Frobenius substitution in the degree-21 splitting field). This is precisely where the scalar readout must fail categorically rather than merely smear: $\mathrm{Re}\,\eta = \mathrm{Re}\,\eta' = -\frac{1}{2}$, so a real one-dimensional readout — a function of the trace's real part — gives the two Frobenius classes the *same* value and cannot separate $7A$ from $7B$, erasing the non-abelian bit. The complex multi-rail bank keeps them apart and closes exactly: over the natural ζ_7 clock the non-DC rails of *both* classes sum to machine zero, tracking 4.3×10^{-62} at 60 digits, with $\prod \text{rails} = \zeta_7^{1+2+4} = 1$ (so SL(3): two independent rails carry the degree-three bank) and no DC/pole rail present, while the order-three and split fibres carry exactly one and three DC rails — the constant-mode/pole content, cleanly separated (`f21_gl3_multirail.py`). The cell closure of both classes, the determinant, the Gauss-period relations $\eta + \eta' = -1$, $\eta\eta' = 2$ and $\eta \neq \eta'$, and the scalar obstruction itself ($\mathrm{Re}\,\eta = \mathrm{Re}\,\eta'$ while $\eta \neq \eta'$) are machine-checked in Lean (`F21NonSelfDual.lean`). So the non-self-dual case, where a scalar representation is provably impossible, rides the multi-rail carrier without loss.

24 The productivity boundary

The symmetric-power kernel is modulated by $U_r(\cos\theta) = \sum_{j=0}^{r} e^{i(r-2j)\theta}$, carrying a zero-frequency component iff r is even. At $\xi = 0$ the x-integral reads this component (the detecting residue); at $\xi \neq 0$ it is the removable comb of §22, bounded by Prop. 93 (with $|U_r| \leq r+1$) and hence $o(X)$. The two regions are disjoint in ξ: the detecting residue sits at $\xi = 0$, where the comb is absent; the comb sits at $\xi \neq 0$, where it is removable. So the residue isolates for every r. On the dual integral (level $p^k = 25$) the zero-frequency value against the $\xi \neq 0$ floor is

$$r = 1: \ 0 \text{ (exact)}, \quad r = 2: \ 55.9, \quad r = 3: \ 1.4 \times 10^{-16}, \quad r = 4: \ 11.4,$$

nonzero iff r even, clear of the floor by an order of magnitude. The degradation with r is the growth of the character tail mass (0.042, 0.092, 0.135, 0.156, 0.256 for $r = 0, \ldots, 4$, fit ceiling $\times \sqrt{\#\text{harmonics}}$ at $R^2 = 0.977$)—gradual attrition, not a wall. Sarnak's productivity boundary [4, fn. 5] records the growth of a removable floor, not a loss of the residue.

Part IV

Universal Transport

25 Toward Universal Transport

The preceding sections close the analytic core — the uniformity (§A), the emergent clock (§22), and the productivity boundary. The remainder of the paper develops the carrier transport toward symmetric-power functoriality: the carrier/fiber rescaling that sends a pole to a closed cell (§26), the arithmetic transparency of the symmetric-power transfer (§27), and a detection identity closed on a genuine transfer (§28).

26 The carrier/fiber rescaling: a pole becomes a closed cell

Place the integers on a carrier scaled by $\pi/3$, i.e. attach the cell phase $e^{in\pi/3}$ (the μ_6/ζ_6 step). We distinguish two operations on the carrier throughout. A *rescaling* is a *fixed constant* — here $\pi/3$, or another μ_m step — that lifts a unit-1 carrier into one with harmonic cells; it *establishes* the native cell structure (and is the reason we never work at unit scale). A *warp* is a *function*, a continuously-evolving deterministic output $\omega(\cdot)$ that rides the already-rescaled carrier and adapts to the fiber — the $\omega_W(n)$ of the warped-Abel summation (§29) and the tunable FE clock (§32). Rescaling sets the cell; warping moves within it. The pole of ζ at $s = 1$ lives in the divergence of $\sum 1/n$; on the (rescaled) carrier it closes,

$$\sum_{n \geq 1} \frac{e^{in\pi/3}}{n} = -\log\big(1 - e^{i\pi/3}\big) = \frac{i\pi}{3},$$

a finite μ_6 value: the divergence has become a closed-cell reading. For a Dirichlet character χ of conductor q, the appropriate fiber scale (trivial $\to \pi/6$, $\chi_3 \to \pi/3$, $\chi_4 \to \pi/2$) makes carrier and fiber compatible harmonics. The character χ_3 is the *aligned archetype*: its conductor 3 is the Eisenstein conductor — the sixth roots of unity of $\mathbb{Z}[\zeta_3] = \mathbb{Z}[\zeta_6]$ — so the $\pi/3 = \mu_6$ rescaling coincides with the L-function's own conductor symmetry, the carrier cell *is* the fiber's native cell, and closure needs the rescaling alone with no warp: hence *no residue*, the residual-mode vanishing of Lemma 61 in its cleanest, rescaling-only instance. The principal character, carrying the pole, is handled by the eta regulator,

$$\sum_{n\geq 1} \frac{(-1)^{n-1} e^{in\pi/3}}{n} = \log\bigl(1 + e^{i\pi/3}\bigr) = \tfrac{1}{2}\ln 3 + \tfrac{i\pi}{6},$$

landing on the trivial character's own cell $(\pi/6, \mu_{12})$, pole-free, with the partial-sum phase converging (no drift). In this frame the residue value read at the pole is a fixed cell harmonic, not a wandering phase, so *no independent* unit-chart correction is needed to read it: what was a residue to be identified is a zero-frequency cell value, read as the DC-residue $(s - c)F'/F \to r$ of a logarithmic derivative. This reads the residue *at a point*; it is not a claim that $\arg \zeta(\frac{1}{2} + it)$ is constant — the conventional $S(t)$ term is recovered as the scale-registration correction in the scalar receiver chart (§9). This is the reading used in §28.

27 Symmetric-power transfer: arithmetic transparency

For Sym^r transfer $\mathrm{GL}(2) \to \mathrm{GL}(r+1)$ ($r = 2m$ even), the transfer sends an elliptic class of Satake angle θ to the class of angles $\{2k\theta : |k| \leq m\} \cup \{0\}$ —i.e. Sym^r of the Frobenius block, with the fixed eigenvalue $1 = \alpha\beta$ the det-one of the block. The symmetric-power transfer is arithmetically transparent, as follows.

Arithmetic transparency. The transferred characteristic polynomial is reducible ($x - 1$ divides it), and each quadratic factor $x^2 - 2\cos(2k\theta)x + 1$ has discriminant $-4\sin^2(2k\theta)$. The clock identity $\sin(2k\theta) = \sin\theta\, U_{2k-1}(\cos\theta)$ (with $\sin^2\theta = |D|/4n$, $D = t^2 - 4n$) gives

$$\frac{\mathrm{disc}_k}{D} = \frac{U_{2k-1}(\cos\theta)^2}{n} \in \mathbb{Q}^{\times 2},$$

a rational square—so every factor lies in the *same* field $\mathbb{Q}(\sqrt{D})$ as the original class, the arithmetic L-value is identical on both sides, and it cancels in the transfer factor. (Because symmetric powers transfer to *reducible* classes, the transparency sidesteps the subtleties of *irreducible* cubic orders entirely.)

28 Detection: the self-twist identity closes

We isolate one computable instance of the detection step and close it on a genuine transfer. For a holomorphic newform f with a_p its Frobenius trace and $\lambda(p) = a_p/p^{(k-1)/2}$, the symmetric square carries the middle eigenvalue $\alpha\beta = 1$; the plain census $\sum_p (\lambda(p)^2 - 1)(\log p)/p$ has mean zero and is blind. The correct object is the *self-twist* Rankin–Selberg L-function $L(s, f \times f \otimes \chi_K)$ for the quadratic character χ_K of an imaginary quadratic field K: it has a pole at $s = 1$ iff f has complex multiplication by K (dihedral), the classical criterion of Gelbart–Jacquet [6] and Rankin–Selberg. Its pole order is the transferred count, read as the zero-frequency residue of the logarithmic derivative:

$$\text{(LEFT, analytic)} \qquad A_{\mathrm{tw}}(X) = \sum_{p\leq X} \lambda(p)^2 \chi_K(p) \frac{\log p}{p}, \qquad \frac{d\, A_{\mathrm{tw}}}{d \log X} \;\longrightarrow\; \mathrm{ord}_{s=1} L(s, f \times f \otimes \chi_K).$$

The geometric side is the self-twist relation on Frobenius, $f \otimes \chi_K \cong f$, equivalently $a_p = 0$ at every inert prime:

$$\text{(RIGHT, geometric)} \qquad \#\{p \leq X \text{ inert} :\ a_p = 0\} \big/ \#\{p \leq X \text{ inert}\} \;\longrightarrow\; 1_{\{f \text{ has CM by } K\}}.$$

We evaluate both on two weight-two forms with $K = \mathbb{Q}(\sqrt{-3})$ (the Eisenstein field, whose $\chi_K(p) = +1, -1$ for $p \equiv 1, 2 \pmod 3$): the CM curve $y^2 = x^3 + 1$ and the non-CM curve $y^2 = x^3 - 16x + 16$ (conductor 37), with a_p from point counting.

	LEFT: slope $dA_{\mathrm{tw}}/d\log X$	RIGHT: inert $a_p = 0$ fraction	count
CM $\sqrt{-3}$ (x^3+1)	0.997	$725/725 = 1.000$	1
non-CM $(x^3 - 16x + 16)$	0.037	$6/726 = 0.0083$	0

For the CM form the analytic slope is 0.997 overall and 0.97–0.99 across every interval $[1000, 4000]$, $[4000, 10000]$, $[10000, 20000]$ (pole order 1), and the geometric self-twist holds at *every* good inert prime (725/725); the bad-reduction prime $p = 2$ (conductor $36 = 2^2 \cdot 3^2$) is excluded from the census. For the non-CM form both sides are 0. Thus

$$\boxed{\operatorname{ord}_{s=1} L(s, f \times f \otimes \chi_K) \;=\; 1_{\{f\otimes\chi_K \cong f\}} \;=\; \text{transferred count}}$$

closes—1 on a genuine transfer, 0 on a genuine non-transfer—read through the logarithmic-derivative DC-residue on the harmonized carrier of §26, which reads the residue at a point and needs no independent unit-chart $S(t)$ correction (§9), with the emergent-clock lesson explaining why the plain census cancels (inert primes carry $\lambda^2 = 0$, so the self-twist census has no cancelling floor).

Remark 55 (what this is)**.** The mathematics detected here—that the symmetric-square/self-twist pole marks CM forms—is classical (Gelbart–Jacquet, Rankin–Selberg). What the section contributes is the closure of the detection *identity* through the two mechanisms: the analytic count as a product-law/log-derivative residue, the geometric count as the self-twist relation, matched on a genuine transfer.

28.1 A non-abelian instance: which Frobenius data transfers

The self-twist detection of §28 runs on a dihedral (abelian, C_2) form, where the Galois image is small and the arithmetic reduces to a single quadratic character. For a genuinely non-abelian form the Frobenius datum splits into two independent bits—one a character cannot see—and the transfer treats them differently. We illustrate this on the exotic tetrahedral ("A_4") weight-one newform h of level 124, whose L-coefficients are a class function of Frob_p in the binary tetrahedral group $2T = \mathrm{SL}(2,3)$, given as follows. Let $\iota(p) \in \{0, 1, 2\}$ be the cubic-residue index of p modulo 31 (the odd part of the level $124 = 4 \cdot 31$), and set $\varepsilon(p) = +1, -1, 0$ as $\iota(p) = 1, 2, 0$. When $\varepsilon(p) = 0$ (the 2+2-shape primes) $a_p = 0$; otherwise

$$a_p = \zeta_{12}^{k(p)}, \qquad k(p) = \tfrac{1}{2}\big(\chi_k(p) + 4\varepsilon(p)\big) + \big(0 \text{ if } (3/p) = +1,\ \text{else } 6\big) \pmod{12},$$

where $\chi_k(p) = 6\,[\,p \not\equiv 1 \,(\mathrm{mod}\ 4)\,] + 4\,\iota(p)$ is the quartic×cubic exponent datum ($[\cdot]$ the Iverson bracket). Here the exponent carries two independent Galois bits:

value bit $\varepsilon(p) \in \{0, \pm 1\}$ (the cubic residue, the abelian $A_4^{\mathrm{ab}} = C_3$ datum), *sign bit* $(3/p)$ (the $\mathrm{SL}(2,3) \to A_4$ central

the double-cover datum read from the order of Frob_p; the sign bit resolves k versus $k + 6$ and is invisible to any abelian character.

The symmetric square sends $a_p \mapsto a_p^2 - \chi(p)$, and $a_p^2 = \zeta_{12}^{2k}$ is unchanged by $k \mapsto k + 6$. Hence:

	Sym^2 transfer to GL(3)
sign bit $(3/p)$ (double cover, $\mathrm{SL}(2,3)$)	*forgotten* (trace invariant under $k \leftrightarrow k{+}6$)
value bit ε (cubic residue, projective A_4)	*retained*: $\varepsilon = \pm 1 \Rightarrow \pm\sqrt{3}\,i$, $\varepsilon = 0 \Rightarrow \pm 1$

So the transfer factors through the projective image A_4: the value bit passes to the GL(3) Satake datum, the sign bit is the GL(2)-specific double-cover fiber the transfer collapses. The order trick —sign from the element order, value from the residue symbol—thus reads *exactly which part of the Frobenius datum transfers*, and the split is non-trivial precisely because A_4 is non-abelian (for the dihedral case of §28 the two bits coincide). This identifies the transferred datum; the automorphy of the resulting GL(3) object (the three-dimensional standard representation of A_4) is the classical Artin/Langlands–Tunnell case.

28.2 The detected count is a Mordell–Weil rank difference

The transferred object of §28.1 is, up to the nebentype twist, the three-dimensional adjoint $\mathrm{Ad}_g = \mathrm{End}^0(V_g) \cong \mathrm{Sym}^2 V_g \otimes (\det V_g)^{-1}$ of the A_4 form g. At this exotic locus the zero-frequency count read on the analytic side coincides with a genuine arithmetic rank, and the coincidence is an exact identity of group theory.

The quartic field M cut out by g carries the permutation action of A_4 on four points, whose character is $\chi_{\mathrm{perm}} = (4, 0, 1, 1)$ on the classes $(e,\ (2,2),\ (3)_a,\ (3)_b)$. Against the character table, $\langle \chi_{\mathrm{perm}}, \mathbf{1} \rangle = 1$ and $\langle \chi_{\mathrm{perm}}, \chi_{\mathrm{std}} \rangle = 1$, so

$$\chi_{\mathrm{perm}} = \mathbf{1} \,\oplus\, \mathrm{Ad}_g, \qquad \mathrm{Ad}_g = \chi_{\mathrm{std}} \text{ (the 3-dimensional irreducible).}$$

Functoriality of Mordell–Weil under this decomposition gives $E(M) \otimes \mathbb{Q} = (E(\mathbb{Q}) \otimes \mathbb{Q}) \oplus E(\,\cdot\,)_{\mathrm{Ad}_g}$ and hence the *exact* Artin/permutation identity

$$r(E, \mathrm{Ad}_g) \;=\; \operatorname{rank} E(M) - \operatorname{rank} E(\mathbb{Q}), \tag{8}$$

proven as group theory (no analytic or p-adic input). Both sides of (8) are then *algebraic*: the Artin multiplicity $r(E, \mathrm{Ad}_g)$ and a difference of classical Mordell–Weil ranks, the source living "up the tower" at M and traced down to $\mathbb{Q}$. That this algebraic multiplicity *also* equals the analytic order of vanishing at $s = 1$ of $L(E, \mathrm{Ad}_g, s)$ — the zero-frequency residue of the logarithmic derivative, the statistic §28 reads as the transferred count — is equivariant BSD, *conjectural and used nowhere here*: identity (8) and the census below are stated and verified entirely on the algebraic side. Thus the count is a rank difference: one algebraic statistic, read once through the Artin formalism and once as a difference of ranks.

On the two Darmon–Lauder–Rotger exotic A_4 curves this is pinned on both sides:

	$\operatorname{rank} E(\mathbb{Q})$	$\operatorname{rank} E(M)$	$r(E, \mathrm{Ad}_g) = \operatorname{rank} E(M) - \operatorname{rank} E(\mathbb{Q})$
$26b$ ($y^2 + xy + y = x^3 - x^2 - 3x + 3$)	0	2	2
$52b$ ($y^2 = x^3 + x - 10$)	0	2	2

with $M = \mathbb{Q}(w)$, $w^4 + 7w^2 - 2w + 14 = 0$ the tetrahedral field of §28.1, rank-two generators of $E(M)$ exhibited by descent. This is the "one mechanism, two locations" statement made theorem-anchored: at the exotic locus both sides of (8) are determined, the identity between them is proven group theory, and the shared statistic is the DC-residue that reads the functorial pole in §28 and the analytic rank in the Beyond-Endoscopy census alike.

Remark 56 (tiering)**.** The rank identity (8) is proven (Artin formalism). The automorphy of Ad_g on GL(3)—which makes $L(E, \mathrm{Ad}_g, s)$ an honest automorphic L-function so that its pole order is a spectral quantity—is the classical Langlands–Tunnell case for A_4. The p-adic *regulator* side (the explicit point coordinates realizing the rank) is the Darmon–Lauder–Rotger Elliptic Stark conjecture, verified to 20 digits on $26b/52b$ but with the points found by descent first; we use only the proven rank identity and the classical automorphy here.

29 A candidate automorphic object from the primary construct

The transfer of §§27–28.2 requires an automorphic representation on $\mathrm{GL}(r + 1)$ whose existence realizes the functorial lift. In the primary (three-dimensional) representation we exhibit a concrete *candidate object* for it: the *carrier* — a double-sided helix of Frobenius similitudes — carrying a *fiber* (the phasor readout), together with its *warp* (the twist family of §27). The classical L-function is the shadow of this candidate under the projection that sends the exponential state space e^y down by a Möbius/Cayley map and then log to the ordinate, at the midpoint and without drift. What makes it a candidate is that the three properties a converse theorem asks of an L-function — functional equation, analytic continuation, and boundedness/completeness — are each a *proven* statement about it, and each is already formalized.

Functional equation: the reflection axis. The anti-helix is the complex *conjugate* of the helix, not its mirror: the carrier's transverse block is $\mathrm{diag}(z, \bar{z})$ with z the Frobenius spin. It is literally the de Branges conjugate-pair block (`frobeniusBlock_eq_conjPairBlock`), with $z\bar{z} = |z|^2 = 1$ (`spin_mul_conj`), so det = 1 and it is unitary (`frobeniusBlock_det_one`, `frobeniusBlock_unitary`). The det-one condition is not a normalization but the *reality locus*: $\det = 1 \iff |z| = 1 \iff$ the helix and anti-helix *balance* $\|E(z)\| = \|E^*(z)\| \iff z$ real (`frobeniusBlock_deBranges_reality_bridge`). The dualizing reflection $s \mapsto 1 - s$ fixes *exactly* the critical line, $\mathrm{Re}(1 - s) = \mathrm{Re}\, s \iff \mathrm{Re}\, s = \frac{1}{2}$ (`reflection_fixes_iff`), and this abscissa is not posited but *forced* by the arclength area law — the carrier admits a scale-balanced fiber iff $\sigma = \frac{1}{2}$ (`scaleBalanced_iff`), the reciprocation of the fiber amplitude $n^{-\sigma}$ against the area-law radius $\sqrt{n}$. This is the functional equation realized geometrically: the conjugate leg closing on the reflection-fixed axis. It survives *every* unit-modulus warp $z \mapsto Az$ ($|A| = 1$), $z\bar{z} \mapsto |A|^2 = 1$ (`warpedBlock_det_one_of_warp`), so the whole twist family carries it, structurally and without estimate. (A complex twist is *self*-dual only when it equals its own conjugate; otherwise the equation pairs the sideband χ_j with $\chi_{-j} = \bar{\chi}_j$, as the conjugate leg demands.)

The crossing geometry and the two fiber topologies. A zero is a phasor *focal cancellation*: the phasors, generated continuously as the carrier grows, align and cancel at a height fixed by the carrier scale, the order of the integers/primes, and the conductor. Because the same data drive both conjugate legs, the two legs cancel at the *same* height, the determinant-one link holding between them. Read as its own oscillator the fiber accumulates phase along each leg; the crossing spacing is a half-turn on each side of the origin and a full turn thereafter. This reduces to the functional equation making the central fiber an extremum — Z real and even, $Z'(0) = 0$ (`collapseWave_even`, `hinge_turning_point`) — together with the argument principle, so the half-turn offset and full-turn step are provable; only the *rigidity* of the spacing in t is not settled by this argument. The radial envelope of the winding is the Archimedean spiral, whose arclength area law $r_n \sim \sqrt{n}$ forces the abscissa to $\frac{1}{2}$ (`scaleBalanced_iff`).

Two fiber topologies occur. An *open* helix — the Dirichlet family — has its fibers begin at the origin and run up both conjugate legs to infinity. A *clipped* helix — a self-dual L-function such as that of an elliptic curve — has its strand bounded by the conductor: the live phasor count scales as $\sqrt{N}$, the analytic conductor (measured exponent 0.5000 across seven decades, Appendix B), the same $\sqrt{\cdot}$ law that fixes the abscissa; and its two legs *meet* at the central point — the fixed point of the functional-equation involution (`affine_reflection_fixed_iff`), the ordinate origin $t = 0$ (equivalently $s = \frac{1}{2}$ analytic, $s = 1$ arithmetic), not the spiral base $N = 0$. There the two strands carry the determinant-one conjugate weights $r^{s-1} \cdot r^{1-s} = 1$ (`strand_weights_det_one`), and the root number ε *is* the weld sign: $\varepsilon = -1$ cancels the central term and forces a central zero (`weld_kills_each_phasor`), $\varepsilon = +1$ doubles it (`weld_doubles_each_phasor`). The same double-ended weld reads, in one factorization $(s-c)^r G$, the root number, the rank r (the DC-residue, `rank_is_dc_residue`), and the leading datum $L^{(r)}/r!$ — matched to tabulated values on ranks 0–3.

Continuation: the readout, continued past the strip. The fiber's value is the phasor representation $L(\chi, s) = \sum_n \chi(n)\, n^{-\sigma}\, \mathrm{mellinSpin}(y, n)$ (`LSeries_phasor_representation`), and it is genuinely *continued*, not merely re-indexed. For a non-principal character the buckets cancel over each period, so the partial sums $\sum_{n<N} \chi(n)$ are bounded (`character_partialSum_norm_le`); Abel summation against the bounded-variation weight n^{-s} then carries the readout past the $\mathrm{Re} = 1$ absolute-convergence wall, and the Abel-summed tail — which agrees with the L-function on $\mathrm{Re}\, s > 1$ and is analytic on the connected strip $\mathrm{Re}\, s > 0$ — is forced by the identity theorem to converge to $L(\chi, s)$ on the *whole* strip $\mathrm{Re}\, s > 0$ (`dirichlet_strip_tendsto_LFunction`). The principal/ζ case runs identically through the eta regulator on the punctured strip $\{\mathrm{Re}\, s > 0\} \setminus \{1\}$, correctly excising the pole (`eta_strip_tendsto`). This is a genuine analytic continuation, machine-checked, with no remainder: the finite harmonic cells (the μ_6 buckets) cancel exactly.

For a general fiber the ordinary coefficients $a_W(n)$ are not periodic, so their bare partial sums need not be bounded. A deterministic unit-modulus warp $\omega_W(n)$ may close cells and make the *warped* primitive $\sum_{n<N} a_W(n)\omega_W(n)$ bounded. The formal theorem `TransferContinuation.transfer_analytic` then continues the Dirichlet series whose coefficients are $a_W(n)\omega_W(n)$; it does not remove the n-dependent warp. Removing it by summation by parts requires the full inverse kernel

$$K_W(s, n) = \omega_W(n)^{-1}(n+1)^{-s}$$

to satisfy a summable first-difference condition against the warped primitive, together with a vanishing boundary term. The exact kernel-level statement is `WarpRemovalTransfer.warpRemoval_transfer_tendsto`. Unit modulus and cell closure alone do not imply its first-difference hypothesis: the compiled separating instance `WarpRemovalTransferAudit.unwarped_half_not_summable` uses the constant coefficient bank and the period-two unit warp. Thus `ForcibleClosure.residual_forcible` and `CarrierReachability.reachable_independent_stage` establish cell controllability, while an application to the ordinary twisted readout must additionally discharge the inverse-kernel hypotheses (or construct the fixed-completion carrier theta and its global reflection directly). The Dirichlet family is the end-to-end instance where the required analytic recovery is separately available.

The completed function $\Lambda(s) = \gamma(s)L(s)$ and the *global* functional equation are then, *classically*, the standard Tate [10]/Godement–Jacquet [11] assembly of the per-place det-one links by Poisson summation on $\mathbb{A}/\mathbb{Q}$ — the route available for the Dirichlet model; on the carrier the same equation is the axis-reality reflection of Lemma 75, which consumes no Poisson summation.

Boundedness and completeness. The two analytic inputs are separated as follows. *Bounded in vertical strips*: this is the vertical-growth companion of the warped-Abel continuation, not a compactness statement. Summation by parts against the bounded warped primitive $A_W^{\mathrm{car}}(x) = \sum_{n\le x} a_W(n)\,\omega_W(n) = O(1)$ gives, once the inverse-kernel difference and boundary conditions above are verified, the representation $L(s,W) = s\int_1^\infty A_W^{\mathrm{car}}(x)\,K_W(s,x)\,\frac{dx}{x}$, whose kernel carries the explicit s-dependence; the resulting bound is polynomial in $|t|$, $|L(\sigma+it)| \ll (1+|t|)^A$ uniformly on $\delta \le \sigma \le B$. With the functional equation supplying the reflected strip, this polynomial growth on the edges lets Phragmén–Lindelöf [22, §5.65] interpolate the finite order across the strip. (Local-uniform summability gives only *compact* boundedness, not the $|t| \to \infty$ control the converse theorem consumes; the vertical bound is the warped-Abel estimate, so continuation and vertical-strip growth are two corollaries of the one warped-Abel representation.) *Complete*: the object has no spectral mass unaccounted for. The event space of the three-dimensional object is the height ray, which is the weld; the strip is a one-dimensional projection device with no primary counterpart, so there is nowhere off-weld for a zero event to hide, and every zero event is weld-fixed and is a source — unconditionally, for *every* fiber:

$$\texttt{threeD\_exhaustive}\,(E:\mathbb{C}\to\mathbb{C}):\quad \forall\, y\in\mathbb{R},\ E(y)=0 \implies \mathrm{IsSource}\,E\,(y).$$

The candidate, assembled. On the primary construct the object exists; its functional equation is the reflection-fixed axis surviving every warp, its continuation is the readout carried to $L(\chi,\cdot)$ across the whole strip, its boundedness is the warped-Abel vertical bound (polynomial in $|t|$), and its completeness is exhaustion — each a proven or formalized statement. The carrier, fiber, and warp *are* the candidate automorphic object, and the converse-theorem inputs are discharged in the primary representation. These inputs — the warp-invariant det-one link and the reflection axis $\mathrm{Re}\, s = \frac{1}{2}$ (angle), the phasor readout continued past the strip (continuation), three-dimensional exhaustion (height/completeness), the midpoint landing on $\mathrm{Re}\, s = \frac{1}{2}$ with the admissible locus forced to $\{\frac{1}{2}\}$, and the loss-ledger reconstruction bijection (recovery) — are bundled and machine-checked in a single theorem (`AutomorphicCandidate.candidate_wellformed`), with the standard axiom footprint. This is the object we use to realize the functorial lift of §§27–28.2; the bridge from it to the classical statement that $\mathrm{Sym}^r\,\pi$ is automorphic on $\mathrm{GL}(r+1)$ is the converse theorem of §31, whose hypotheses are exactly these proven properties.

The local identification. The properties above are properties of an L-function; to name that L-function as $L(s,\mathrm{Sym}^r\,\pi)$ we record, once and explicitly, the equality of local data. This is the equality the converse theorem consumes, and it is fixed by the construction rather than by any assumption on the target.

Theorem 57 (Global identification of the candidate with $\mathrm{Sym}^r\,\pi$)**.** *Let π be a cuspidal automorphic representation of* $\mathrm{GL}(2)/\mathbb{Q}$*, and for each place v let $\varphi_{\pi,v}\colon W'_{\mathbb{Q}_v} \to \mathrm{GL}_2(\mathbb{C})$ be its local Langlands parameter (a theorem of local Langlands for* $\mathrm{GL}(2)$*). At* every *place v the standard local factor and local root number of the candidate object of §29 are those of the $(r+1)$-dimensional parameter* $\mathrm{Sym}^r\,\varphi_{\pi,v}$*:*

$$L_v(s,\mathrm{cand}) = L\big(s,\mathrm{Sym}^r\,\varphi_{\pi,v}\big)\,,\qquad \varepsilon_v(s,\mathrm{cand},\psi_v) = \varepsilon\big(s,\mathrm{Sym}^r\,\varphi_{\pi,v},\psi_v\big)\,. \tag{9}$$

At an unramified v, $\varphi_{\pi,v} = A_{\pi,v} = \mathrm{diag}(\alpha_v,\beta_v)$ $(\alpha_v\beta_v = 1)$ and (9) reads $L_v = \det(1-\mathrm{Sym}^r(A_{\pi,v})q_v^{-s})^{-1} = \prod_{j=0}^r (1-\alpha_v^{r-j}\beta_v^j q_v^{-s})^{-1}$; at $v=\infty$ the parameter $\mathrm{Sym}^r\,\varphi_{\pi,\infty}$ gives $L_\infty = \prod_i \Gamma_{\mathbb{C}}(s+\mu_i)$ (one $\Gamma_{\mathbb{R}}(s+\delta)$ adjoined for even r; for π of weight k, $\mu_i = (2i-1)\frac{k-1}{2}$). Consequently the full *completed L-function, the global root number $\varepsilon = \prod_v \varepsilon_v$, and the conductor $N = \prod_v q_v^{a(\mathrm{Sym}^r\,\varphi_{\pi,v})}$ of the candidate are exactly those of* $\mathrm{Sym}^r\,\pi$ *— the entire global package, at all places, ramified included. No automorphy of* $\mathrm{Sym}^r\,\pi$ *is used: $\varphi_{\pi,v}$ is furnished by local Langlands for* $\mathrm{GL}(2)$*, and* Sym^r *is a functor on parameters.*

Proof. The candidate carrier is assembled from the Frobenius similitudes of π, so its transverse datum at a place v is exactly $\varphi_{\pi,v}$. *The Satake datum is read in dimensional order.* The arithmetic block is $A_{\pi,v} = \mathrm{diag}(\alpha_v, \alpha_v^{-1})$, with radial magnitude *not* assumed unit; the 2D carrier projection reads the conjugate-pair phase $\mathrm{diag}(z_v, \bar{z}_v)$, $z_v\bar{z}_v = 1$ (`frobeniusBlock_eq_conjPairBlock`), the radial coordinate booked in the loss ledger rather than set to one; and the Euler local factor is read from the *deprojected* arithmetic block (`ConeProjection.reconstruct_record`), so no temperedness is used. The fiber over v is, by the transfer of §27, the r-th symmetric power $\mathrm{Sym}^r\,\varphi_{\pi,v}$; its standard local L- and ε-factors are the local Langlands/Deligne factors [10], (9). At unramified v this is the reciprocal characteristic polynomial of $\mathrm{diag}(\alpha_v^r, \ldots, \alpha_v^{-r})$, whose determinant $(\alpha_v\alpha_v^{-1})^{r(r+1)/2} = 1$ is the Frobenius-similitude crossing value (independent of $|\alpha_v|$); at ∞ it is the Γ-datum read by the midpoint projection. The three global invariants are the corresponding products over v (finite for ε and N, since $\varphi_{\pi,v}$ is unramified for almost all v), and these coincide, place by place, with the standard data of $\mathrm{Sym}^r\,\pi$. Nowhere is $\mathrm{Sym}^r\,\pi$ assumed automorphic: $\varphi_{\pi,v}$ is the local parameter of the given GL(2) representation π, and $\mathrm{Sym}^r\,\varphi_{\pi,v}$ is a definite $(r+1)$-dimensional local parameter. □

Theorem 57 upgrades the *identification* to all places: the candidate's L-function, root number, and conductor *are* those of $\mathrm{Sym}^r\,\pi$, globally, with the exact ε/conductor package and no automorphy assumed. It does not by itself upgrade the *niceness*: that the completed $L(s, \mathrm{Sym}^r\,\pi \times \tau)$ is entire, bounded in vertical strips, and satisfies its functional equation is a separate, analytic input, supplied — on the carrier, uniformly in r — by Proposition 59 of §31. For $r \le 4$ that niceness is also classical (Langlands–Shahidi); for $r \ge 5$ the carrier discharges it where Langlands–Shahidi does not reach. Identification is global and niceness is carrier-discharged; both hold for every r.

Recovery: the loss-ledger round trip. The projection to the shadow is lossy but bookkept exactly, and the descent is dimensional: from the three-dimensional object the *radial* channel is dropped (3D $\to$ 2D) and then the *angular* channel (2D $\to$ 1D), while the height passes by the log/exp map between the exponential state space e^y and the ordinate y. Recording the two dropped channels in a ledger, the map fiber $\mapsto$ (ordinate; radius, angle) is a *bijection* (`record_bijective`, `reconstruct_record`: an exact two-sided inverse, unconditional), provided the projection is taken at the midpoint, without drift. The midpoint projection lands the 1D ordinate on the conjugation axis $\mathrm{Re}\,s = \frac{1}{2}$ ($\mathrm{Re}(\frac{1}{2} + \gamma i) = \frac{1}{2}$, `midpoint_projects_to_half`); since the admissible locus is exactly $\{\frac{1}{2}\}$, a vanishing cannot land off the strip by construction. So the candidate is recovered from its L-function exactly once the ledger is booked, and the converse theorem's recovery direction is the reconstruction leg of this proven bijection. The three ledger channels are precisely the three inputs: the *angle* is the functional-equation link $z\bar{z} = 1$, the *height* is exhaustion, and the *radius* is the area-law scaling $\sqrt{p}$ of the Frobenius similitude.

Remark 58 (universality across families)**.** The construct is not special to symmetric powers — the twist family a converse theorem ranges over is generic. For the *Dirichlet family* it is the fully machine-checked case: the continuation, the reflection-axis functional equation, and the boundedness of §29 hold for every Dirichlet L-function — and for ζ — as the theorems `dirichlet_strip_tendsto_LFunction` and `eta_strip_tendsto`, so each Dirichlet L-function is a *proven* instance of the candidate, not merely a fitted one. Beyond it the same closure is confirmed numerically: run oracle-free on five further structurally distinct families — non-CM elliptic curves of rank $0, 1, 2$ ($37a$, $389a$), a CM/Grössencharacter form, a non-abelian S_3 (dihedral) Artin representation, and a degree-four Rankin–Selberg convolution $\Delta \times E_{11}$ — the readout reproduces the critical-line L-series to 2×10^{-16} (including degree four); the functional-equation closure lands with the correct root number (the central vanishing at $\frac{1}{2}$ to 4.9×10^{-17} for the rank-two $389a$; the wrong sign rejected); and the focal cancellation at the non-central zeros closes to a deepest residual of 6×10^{-8} (Appendix B). The finite growth locator cannot reach the central point itself ($Z = e^0 = 1$ is an empty bank) — for a clipped helix this is the Fricke weld where the two legs meet, the root number's own point — so central vanishing is read on the exact-cancellation channel $B = (\pi/3)\, L(\frac{1}{2})$, not the locator.

30 The automorphy machinery on the carrier

The candidate's functional equation and arithmetic invariance are generated by the carrier itself, and each statement here is verified to machine precision.

The functional equation is derived, not assembled. The two-sided carrier — the double helix as the self-dual lattice — is Poisson-summable, and its theta self-duality *produces* the functional equation with no functional equation supplied as input. For the n-dimensional carrier,

$$\theta(Y^{-1}) = \det(Y)^{1/2}\, \theta(Y), \qquad \theta(Y) = \sum_{m \in \mathbb{Z}^n} e^{-\pi m^t Y m},$$

holds to 10^{-16} for $n = 2, 3, 4$ (the Sym^0–Sym^3 standard-L functional-equation kernels). The completed $\Lambda(s) = \Lambda(1-s)$ then falls out of the two-sided theta by Mellin, to machine zero and with no functional equation input, for ζ, for Δ (from its modularity $\theta(1/y) = y^{12}\theta(y)$), and for $\mathbb{Q}(i)$ (from the tensor lattice $\mathbb{Z}^2$). The archimedean Γ-factor is the Tate integral of the self-dual Gaussian, $\int_0^\infty e^{-\pi x^2} x^s \,\frac{dx}{x} = \frac{1}{2}\pi^{-s/2}\Gamma(s/2)$. This is Tate's and Hecke's derivation of the functional equation, realized as the carrier's own self-duality — but as the 1*D readout* of the carrier involution J (Theorem 69): the theta self-duality is the shadow the projection casts of the helix/anti-helix reflection, so "Poisson-summable" describes that readout, not an independent lattice/Poisson input to the 3D carrier, where J is the primary mechanism and no Poisson summation enters.

The completion kernel is closed form. The degree-d completion — the inverse Mellin of $\prod_j \Gamma_{\mathbb{R}}(s + \mu_j)$, one Γ-clock per Hodge weight — is built from the pairwise Mellin convolution of two clocks, which is exactly a Bessel function:

$$\left(2x^{\mu_a} e^{-2\pi x}\right) \star \left(2x^{\mu_b} e^{-2\pi x}\right) = 8\, x^{(\mu_a+\mu_b)/2}\, K_{\mu_b-\mu_a}\!\left(4\pi\sqrt{x}\right),$$

an identity we verify to machine precision (Appendix B). So the completion is a closed-form product of K-Bessel kernels, and the functional equation of every convolution — the twisted family included — closes *in closed form*, not by numerical quadrature: this is what carries the convolution functional equation from a measured quantity to an exact one.

The arithmetic group is generated by the carrier's clocks. The automorphy group of $\mathrm{Sym}^r\,\pi$ on $\mathrm{GL}(r+1)$ is generated by the carrier's emergent clocks together with the Poisson self-duality, in a tower verified to machine precision:

- $\mathrm{SL}(2,\mathbb{Z})$: the carrier's own quadratic (arclength) phase is the modular generator, $e^{i\pi c n^2}\theta(\tau) = \theta(\tau + c)$ ($= T$); Poisson is $S : \tau \mapsto -1/\tau$; and $\langle S, T\rangle$ closes with (ST) of order 6.
- $\mathrm{Sp}(4,\mathbb{Z})$: the symmetric 2×2 clock matrix — two arclength clocks and the cross-term — are the translations T_B ($\tau \mapsto \tau + B$); the Siegel theta gives S, $\theta(-\tau^{-1}) = \det(\tau/i)^{1/2}\theta(\tau)$ to 10^{-16}; and $\{T_B\} \cup \{S\}$ generate $\mathrm{Sp}(4,\mathbb{Z})$.

- $\mathrm{GL}(r+1,\mathbb{Z})$: the elementary (linear) clocks E_{ij} generate $\mathrm{SL}(r+1,\mathbb{Z})$, and the matrix-Fourier Poisson on $M_{r+1}(\mathbb{Z})$ (Godement–Jacquet) is the self-duality; $\theta(Y^{-1}) = \det(Y)^{1/2}\theta(Y)$ above is its $\mathrm{GL}(r+1)$ form.

The principle is the same at every rung: the carrier's clocks and its self-duality *are* the automorphy machinery.

The uniformity over the family is proven. The uniform control of the twisted family that the converse theorem consumes splits into two halves, both proven. The archimedean half is the exact gauge of §A. The arithmetic half is the two-clock weight law: the ledger-normalized clock trace of the tensor Satake — the unit-modulus phase datum, its radial magnitude booked in the loss ledger (§29) — is bounded by the degree,

$$\left\|\mathrm{clockTraceN}\,\theta\,k\right\| \le n$$

(`TwoClockWeightLaw.ramanujan_line_ceiling`, machine-checked with the standard axiom footprint), uniformly in the twist; the booked radial magnitudes carry the polynomial local bound (Kim–Sarnak for π, Jacquet–Shalika/LRS for τ) consumed by Lemma 68. The archimedean and arithmetic uniformities are the two halves of the converse theorem's uniform hypothesis, and both are theorems with exactly these inputs.

The carrier sets the functional equation; the fiber sets the function. The functional equation is a property of the carrier, not of the fiber that rides it. The reflection $s \mapsto 1-s$ is the helix–anti-helix conjugation, fixed at $\mathrm{Re}\, s = \frac{1}{2}$ by the reality locus $z\bar{z} = 1$; this symmetry is geometric and holds for *every* fiber, because it is the carrier that is self-dual. The fiber — the local data $\mathrm{Sym}^r\,\varphi_{\pi,v} \otimes \varphi_{\tau,v}$ of Theorem 57 — decides *which* L-function is read, not *whether* it has a functional equation. Consequently the automorphic theta of $\mathrm{Sym}^r\,\pi$ is never invoked, and the classical fact that $L(\mathrm{Sym}^r\,\pi)$ ($r \ge 5$) has no elementary theta representation — an obstruction to deriving the equation *from the function* — is vacuous here, where the equation is derived from the *carrier*. With this, the three properties the converse theorem asks are each carrier-supplied.

Proposition 59 (Global niceness of the twisted convolutions on the carrier)**.** *For every $r \ge 2$ and every cuspidal automorphic τ of* $\mathrm{GL}(r-1)$ *or* $\mathrm{GL}(r-2)$ *the completed $L(s, \mathrm{Sym}^r\,\pi \times \tau)$ is nice, each property a property of the carrier — uniform in r and τ, with no automorphy of* $\mathrm{Sym}^r\,\pi$ *assumed.*

(FE) $\Lambda(s, \mathrm{Sym}^r\,\pi \times \tau) = \varepsilon(s, \mathrm{Sym}^r\,\pi \times \tau)\,\Lambda(1-s, \mathrm{Sym}^r\,\pi \times \tau^\vee)$ *is the carrier reflection of the tensor fiber* $\mathrm{Sym}^r\,\varphi_{\pi,v} \otimes \varphi_{\tau,v}$*, read in the* unitary, determinant-balanced *normalization. The fiber determinant* $(\alpha_v\beta_v)^{r(r+1)\dim\tau/2}(\det A_{\tau,v})^{r+1}$ *is* not *assumed trivial: its* $\mathrm{Sym}^r\,\pi$ *part normalizes away* ($\mathrm{Sym}^r\,\pi$ *is self-dual once its central character is booked), while* $(\det A_{\tau,v})^{r+1}$ *records σ's central character, which the reflection transports to the dual leg — hence the explicit contragredient $\tau^\vee$, not an accidental* $\det\tau = 1$*. The residual central-character clock is booked separately in the loss ledger; on that balanced carrier the reflection is the Lean theorem* `FiniteWeightFiber.localPoly_reciprocal` *(ledger $\prod_i w_i = 1 +$ self-reciprocal local factor + axis, for any finite duality-stable fiber; §31). The completion kernel is the closed-form product of K-Bessel two-clocks* (11)*, and the equation is verified to machine precision through* Sym^{13} *(§31.1), with the root number the closed form* (12).

(B) Bounded in vertical strips — *genuine boundedness, as Theorem 45 requires, not merely polynomial growth. This is direct, from the completed transform (Lemma 68): on a fixed strip the reflected carrier integral $C(s) = \int_1^\infty K_{r,\tau}(x)(x^s + \eta x^{\kappa-s})\frac{dx}{x}$ has $|x^{it}| = 1$, so $|\Lambda(\sigma+it)|$ is bounded by a t-independent finite integral — finite because the completion kernel decays super-polynomially and the theta inherits that decay from the Euler-factor coefficient bound $\lambda_n = O(n^{r/2+\epsilon})$. No Phragmén–Lindelöf is needed. (The* *warped-Abel estimate of §29 gives, alternatively, the polynomial edge bound $|L(\sigma + it)| \ll (1 + |t|)^A$ that* (FE)+(E)+*Phragmén–Lindelöf interpolates across the strip; the direct bound supersedes it.) The abscissa is* $\operatorname{Re} s = \frac{1}{2}$ *(`scaleBalanced_iff`).*

(E) Entire*, by the completed carrier transform* — not *inherited from the L-function, and using no zero-source argument. Entireness is inherited from an independently entire carrier transform. Let g be the two-clock completion kernel* (11) *and $K_{r,\tau}(x) = \sum_{n\geq 1} \lambda_n(\operatorname{Sym}^r \pi \times \tau)\, g(nx)$ the associated carrier theta. Its Mellin transform equals $\Lambda(s, \operatorname{Sym}^r \pi \times \tau)$ on* $\operatorname{Re} s > 1$*; splitting $\int_0^\infty$ at $x = 1$ and applying the carrier reflection $K(1/x) = \varepsilon\, x^\kappa K(x)$ to the lower half writes it as an entire integral over $[1, \infty)$ against a rapidly decaying Bessel kernel, minus the explicit finite zero-frequency modes $R_{r,\tau}$ thrown off as $x \to 0$; those constant modes vanish for the cuspidal τ of the converse-theorem range, so $\Lambda(s, \operatorname{Sym}^r \pi \times \tau)$ is entire (the Riemann–Hecke continuation, Lemmas* (E1)–(E5) *in the proof).*

Proof. (FE) is the reflection axis of §29 applied to the fiber of Thm. 57: the det-one tensor makes the two conjugate legs balance place by place, and the archimedean factor is the two-clock completion (11). (B) is Lemma 68: the direct reflected-integral bound gives a t-independent bound on the completed Λ in every vertical strip, superseding the Phragmén–Lindelöf route (no finite-order-plus-interpolation step). (E) is the Riemann–Hecke continuation on the carrier theta $K_{r,\tau}(x) = \sum_n \lambda_n\, g(nx)$, in five steps. *(E1) Reflected rapid decay.* g is the Bessel two-clock (11), of exponential decay at ∞; the carrier reflection $K(1/x) = \varepsilon\, x^\kappa K(x)$ (FE) controls $x \to 0$. *(E2) The reflected transform is entire.* After the split at $x = 1$ and the substitution $x \mapsto 1/x$ on the lower half, $C(s) = \int_1^\infty K(x)\,(x^s + \varepsilon\, x^{\kappa - s})\,\frac{dx}{x}$; for $x \geq 1$ the integrand is entire in s and, on each compact, dominated by the Bessel decay, so the integral converges locally uniformly and admits differentiation under the integral — C is entire. *(E3) Residual split.* $K_{r,\tau} = K^\circ_{r,\tau} + R_{r,\tau}$ with $R_{r,\tau}$ the finite zero-frequency/constant-term part; $\mathcal{M}K = \mathcal{M}K^\circ + \mathcal{M}R$, the first entire by (E1)–(E2), the second an explicit finite meromorphic sum carrying every possible pole. *(E4) The residual vanishes.* $R_{r,\tau} = 0$. For a finite duality-stable harmonic fiber this is the cell closure of Lemma 61 (the adapted warp has exact zero native-cell mean, so the zero-frequency projection of the warped bank vanishes); for the cuspidal twists a converse theorem consumes — τ on $\mathrm{GL}(r-1)$ or $\mathrm{GL}(r-2)$ — it is *completeness* (Lemma 64): the DC/residue mode is the coefficient mean, which by character orthogonality (`harmonize_char`) equals the *trivial-channel* coefficient alone; a complete fiber has no trivial channel — for non-exceptional π the tensor $\operatorname{Sym}^r \varphi_\pi \otimes \varphi_\tau$ has no trivial constituent because an equivariant map $\varphi_\tau^\vee \to \operatorname{Sym}^r \varphi_\pi$ into the simple target is zero: a nonzero image would be the whole target, and only at that point does $\dim \varphi_\tau < \dim \operatorname{Sym}^r \varphi_\pi$ contradict surjectivity (`RepresentationRankGap.no_nonzero_intertwiner_of_finrank_lt`) — so $R_{r,\tau} = 0$ and no constant mode survives to carry a pole. (The pencil rank-drop of Lemma 63 is the separate zero-detector, not this residue.) Either route uses no global parameter and no coefficient-average-to-multiplicity step, and is Galois-free. *(E5) Identification.* $\mathcal{M}K_{r,\tau}(s) = \Lambda(s, \operatorname{Sym}^r \pi \times \tau)$ on $\operatorname{Re} s > 1$, by term-by-term unfolding against the local factors of Thm. 57 — used only *after* the entire transform is built, never to assert continuation of the L-function. By the identity theorem Λ extends to the entire C. This is Mathlib-complete for the Dirichlet model (`completedRiemannZeta`, `completedLFunction`). The two structural ingredients are not open formal interfaces: the fixed-kernel identification (E5) on the initial half-plane is `CarrierTheta.theta_hasMellin_of_polynomial`. The reflection (E1) of the *same global coefficient theta* is a separate statement: per-place `FiniteWeightFiber.localPoly_reciprocal`, the coordinate involution, and a tunable FE clock do not alone identify the summed primal bank with its transformed anti-helix bank. At the 3D interface it is exactly the odd-channel extinction characterized by `GlobalHelix.fixedBankOddChannel_eq_zero_iff_theta_reflection`. The remaining input splits by fiber class. For a finite duality-stable harmonic fiber the carrier-cell closure of Lemma 61 discharges simultaneously the residual vanishing (E4), the bounded warped primitive $A^{\mathrm{car}}_W = O(1)$, the continuation, and the vertical bound — *proved in Lean* (`CellClosure.harmonic_bank_primitive_bounded`: each root-of-unity weight channel cancels over its cell), the Dirichlet family included. For the cuspidal twists a converse theorem consumes, the residual vanishing (E4) is instead the *unconditional* completeness argument of Lemma 64; continuation to the axis neighbourhood then requires the raw primitive bound of Lemma 70, the inverse-kernel hypotheses of `WarpRemovalTransfer.warpRemoval_transfer_tendsto`, or the fixed-theta reflection just isolated. Under that continuation and reflection input, full entireness comes from the entire reflected carrier theta ((E1)–(E3),(E5) with $R_{r,\tau} = 0$) together with Phragmén–Lindelöf across the functional equation (E1). Machine-checked at the general finite duality-stable interface (§31) are the local reflection algebra and the completed-reflection assembly (`FiniteWeightFiber.CompletedReflection.completed_FE`). □

30.1 The symmetric-power functoriality cancellation at $r = 2$: the carrier route and the projection shadow

Status of $r = 2$ functoriality: closed on the carrier; the sharp-cutoff Perron bound is a projection shadow, off the critical path. Symmetric-square functoriality is discharged by the niceness route of item 1 — entirety, the functional equation, and vertical-strip bounds for the Sym^2 twist bank, machine-checked in Lean 4 at the standard kernel footprint {`propext`, `Classical.choice`, `Quot.sound`} with no `sorry` and no `axiom` (`cpsDualPair_twistedNiceness`, `cpsAllTwists_unconditional3DAnalyticPayload`), fed to the converse theorem (Thm. 45) — *with no Perron input*. The carrier-native cancellation content is likewise compiled: the *smoothed* coefficient cancellation (the vertical readout $\sum_n a_n e^{-2\pi n x} = O(x^k e^{-2\pi x})$, `HeckeCancellation.hecke_smoothed_cancellation`) and, for the second moment, the Rankin–Selberg continuation of the *actual infinite* series (`rs_master`), with entire-part holomorphy (`rs_entirePart_differentiable`), the contour shift with residue (`rectangleBoundaryIntegral_holo_add_residues`, from Cauchy–Goursat vanishing plus the inverse-kernel residue), and the effective Perron kernel in both regimes (`PerronKernel.perron_kernel_bound`, `perron_kernel_bound_lt`). The *sharp-cutoff* summatory bound $\sum_{n \le x} \lambda_n = O(x^\theta)$ is the 1D-projection repackaging of that smoothed content — a Chandrasekharan–Narasimhan Tauberian de-smoothing, a chart device by Ground Rule 4, isolated as the named hypothesis PerronTauberian and *not on the functoriality critical path*. In one line: *functoriality is closed; the sharp cutoff is a projection shadow, not a gap.*

The proposition above builds the completed transform $\Lambda(s, \mathrm{Sym}^r \pi \times \tau)$ from the carrier theta, already *nice* (entirety, functional equation, vertical-strip bounds; item 1) — which is all the converse theorem consumes, so functoriality is closed without what follows. The sharper, projection-side statement that a hostile review isolated as the crux to attack directly — *not* required for that closure — is the summatory cancellation

$$\sum_{n \le x} \lambda_n(\mathrm{Sym}^r \pi \times \tau) = O(x^\theta), \qquad \theta < \kappa/2,$$

with the test weight ω_W explicit, a bound far below the trivial $O(x^{r/2+1+\epsilon})$ obtained by summing the per-term Euler bound $\lambda_n = O(n^{r/2+\epsilon})$.

The reduction is *r-uniform* and is exactly the rung-by-rung law of §30: the cancellation at rung r is equivalent to that rung's fixed-kernel functional equation $K_{r,\tau}(1/x) = \varepsilon x^\kappa K_{r,\tau}(x)$, which delivers the meromorphic continuation, and Perron's formula then converts continuation into the summatory bound. This is the same argument at every r; only the discharge of the fixed-kernel FE is rung-specific. The Lean capstone `symr_functoriality_cancellation` states this reduction over an abstract coefficient sequence and archimedean parameter κ, taking the Perron/Tauberian transfer as its single explicit hypothesis — a projection-side de-smoothing, off the functoriality critical path, not a carrier gap; the rung continuation is discharged separately — the fixed-kernel functional equation, entirety, and vertical-strip bounds are compiled for *arbitrary* r by the carrier's CPS twist machinery (`cpsDualPair_twistedNiceness`, `cpsPolynomialAllTwists_payload`), with the finite→infinite Euler limit and its infinite-bank summability, exact reflection, and rapid decay compiled (`GlobalHelix.symTwistEulerLimit_eq_div`, `GlobalHelix.summable_fullTensorEuler_zeroModeTerm`, `GlobalHelix.fullTensorEulerZeroModeDualTheta_eq_div`), and at $r = 2$ the continuation is additionally grounded in the arithmetic Rankin–Selberg series with the entire-part holomorphy `rs_entirePart_differentiable` and the compiled contour shift.

The weld is free on the dual-helix; Poisson lives in the projection. The fixed-kernel FE $K(1/x) = \varepsilon x^{\kappa} K(x)$ is not, on the carrier, an analytic identity to be proved: it is a *structural* property of the double-ended construction (§2). The carrier carries a primal helix and its conjugate anti-helix; the weld is exactly the statement that the primal bank equals the reflected anti-helix bank, and the inversion $x \mapsto 1/x$ (the modular S) maps one leg to the other by construction. The primal leg carries the lattice and the anti-leg its dual, so the weld is the identity "self-dual lattice = its own dual" — free for $\mathbb{Z}, \mathbb{Z}^2$ — and the phasor growth profile is reflection-symmetric by the same double-ended construction, so the kernel side is free as well. *Poisson summation does not live on the carrier*: projecting to the readout axis collapses the two helices onto one line and hides the symmetry, and Poisson is precisely the 1D analytic reconstruction of a weld that was manifest in 3D. Treating "you need Poisson" as an input is the Ground-Rule-4 error one level up — importing a projection-chart artifact as a requirement. Our $r = 2$ Lean routes through Mathlib's Poisson-based `jacobiTheta` because it validates the 1D readout against the classical Epstein object; the *carrier* reason the weld holds is the free dual-helix self-duality, and that is the mechanism that generalizes. Across the ladder:

- $r = 1$ (abelian): the weld reads out in 1D as Gauss/Poisson (the one-variable theta);
- $r = 2$ (Sym^2): the fixed kernel is the *Gaussian–lattice/Epstein* theta, whose weld reads out as Poisson summation on $\mathbb{Z}^2$ — *compiled unconditionally in Lean* on the existing Mathlib `jacobiTheta/WeakFEPair` substrate;
- $r = 3, 4$: functoriality is classically known (Kim–Shahidi); the classical kernel is Langlands–Shahidi rather than an elementary lattice theta, but the *carrier* kernel's functional equation is compiled here for these rungs like any other (the CPS twist machinery below);
- $r \geq 5$: no classical lattice theta exists — and here the carrier changes the problem entirely. Classically $r \geq 5$ is blocked because the FE of $\Lambda(s, \mathrm{Sym}^r \pi)$ requires $\mathrm{Sym}^r \pi$ to be automorphic (functoriality itself). On the carrier, *two* of the three classical difficulties are already dissolved: the weld/FE is structural (free from the dual-helix, above), and the identification of the readout with $\Lambda(s, \mathrm{Sym}^r \pi \times \tau)$ is *generic*. The identification is generic for a concrete reason — it is the coherence law of §1 realized on the admissible set. The carrier kernel is a *universal machine* on the admissible fibers (§32, §33), and $\mathrm{Sym}^r \pi \times \tau$ is admissible (a finite duality-stable datum). Two properties then preserve the 1D readout for *every* such fiber: *(a) carrier scaling is readout-preserving* — different carrier scales represent the same arithmetic procession, the harmonic 3D unit ($\frac{\pi}{3}$) relating to the value-one 1D chart by the fixed carrier-unit-scale relation, so the projected L-function is invariant under the scaling and ledgered projection has an exact round trip (§3, §9); and *(b) harmonization* sets the one universal kernel to the fiber's nature (read the projected function, let it name the scale/adapters/clocks so the ensemble closes). Together

these are exactly "functoriality is the coherence of the normalization under source change": the same universal machine, harmonized to each admissible fiber and projected at a readout-preserving carrier scale, so $\mathcal{M}K_{r,\tau} = \Lambda(s, \mathrm{Sym}^r\, \pi \times \tau)$ holds *by the same argument* at every r (E5, `CarrierTheta.theta_hasMellin_of_polynomial`). This is not per-L-function and *not* restricted to Dirichlet: the Dirichlet model (`completedRiemannZeta`, `completedLFunction`) is only where Mathlib supplies an independent oracle to cross-check against — an anchor, not the scope — and the compiled $r = 2$ rung is itself non-Dirichlet, the Sym^2 Rankin–Selberg of a general weight-k cusp form. Admissibility also supplies the polynomial coefficient bound $\lambda_n = O(n^{r/2+\epsilon})$ (§4), so the convergence of the Dirichlet series — the input Perron consumes — is free, not a separate estimate. The meromorphic continuation therefore follows generically at every r. What is genuinely per-rung is *narrower* than construction-plus-identification: instantiating the geometric bank (the Siegel theta on $\mathrm{Sp}(2g, \mathbb{Z})$, the Godement–Jacquet $\mathrm{GL}(r{+}1, \mathbb{Z})$ "Sym^r engine" verified to machine zero in the tower-atlas work, §26) with its *arithmetic input*: the Satake/Frobenius data θ^e feeding the local factors, which is identified on the carrier — `arithmeticCPS_localFactor_identification` and `arithmeticCPS_globalCoefficient_identification` equate the carrier bank's local numerator and every global Euler coefficient with the literal twisted symmetric-power datum, by `rfl` — together with the residual vanishing E4 (completeness for non-exceptional π). Perron remains the r-uniform gap *of the sharp-cutoff cancellation* — a projection-side de-smoothing, off the functoriality critical path, which closes via niceness and the converse theorem (Thm. 45); the summatory bound is that route's shadow, not a gap in the transfer. The reason $r = 2$ is the *compiled* rung is not that the higher rungs lack the weld or the identification — they have both generically — but that its arithmetic input ($\|a_n\|^2$, the Rankin square) and kernel are elementary and Mathlib-adjacent, so the entire chain machine-checks. That is the carrier passing *beyond* Langlands–Shahidi, not a classical dead end; the $r = 2$ rung is the proof of concept that the generic weld and identification can be turned, on one rung, into a fully machine-checked functional equation and continuation.

We now record what the carrier proves for the compiled $r = 2$ rung — the first rung where the fixed-kernel FE is turned into a machine-checked theorem, the template for the ladder — and the single classical input on which the cancellation rests at every r. Here π is the representation of a level-one holomorphic cusp form f of weight k (so $\kappa = k$), $\tau = \mathbf{1}$, and $r = 2$.

The fiber and the explicit weight. The genuine Sym^2 coefficients are the Möbius peel of the *normalized* Rankin square $b_n = \|a_n\|^2/n^{k-1}$, namely $\lambda_n = (\mu \star b)(n)$ (`sym2CoeffNorm`). Normalizing *before* the peel is essential: the Rankin–Selberg pole of $\sum \|a_n\|^2 n^{-s}$ sits at $s = k$, so peeling $\zeta(s)$ from the *un*-normalized square leaves that pole in place — $\sum_{n \le x} (\mu \star \|a\|^2) \sim c\, x^k$, no cancellation at all; after normalizing, the pole sits at $s = 1$, exactly where $\zeta(s)$ cancels it, leaving the *entire* $L(\mathrm{Sym}^2 f, s)$ (`rankin_sym2_transfer` is the definitional ζ-peel $L(b, s) = \zeta(s)\, L(\lambda, s)$, no Hecke-operator theory). So $\sum_{n \le x} \lambda_n = O(x^\theta)$ with $\theta < k/2$ is a *true* statement (convexity for the degree-3 entire $L(\mathrm{Sym}^2 f)$ gives $O(x^{1/2+\epsilon})$, and $\frac{1}{2} < \frac{k}{2}$). The weight ω_W is *explicit*: the invariant Petersson density read against the Eisenstein height,

$$\omega_W(z, s) = P(z)\, (\Im z)^s, \qquad P(z) = (\Im z)^k\, |f(z)|^2,$$

and its integral over the modular fundamental domain *is* the completed series (`rs_master`):

$$\int_{\mathcal{D}} P(z)\, \Lambda_z(s)\, d\mu(z) = \Gamma(s)\, \pi^{-s}\, 2\zeta(2s)\, \Gamma(s + k - 1) \sum_{n \ge 1} \frac{\|a_n\|^2}{(4\pi n)^{s+k-1}}, \qquad \mathrm{Re}\, s > 2,$$

where Λ_z is the completed Epstein kernel of the carrier's lattice phasor bank at z and $d\mu = dx\, dy/y^2$ the invariant measure.

What the carrier proves, unconditionally. Every analytic ingredient of the continuation is machine-checked at the standard axiom footprint {propext, Classical.choice, Quot.sound}, with no `sorry`:

- the lattice bank $\Theta_z(t) = \sum_{p\in\mathbb{Z}^2} e^{-\pi t\,\mathrm{gram}(z,p)}$ carries the weld reflection $\Theta_z(1/t) = t\,\Theta_z(t)$ at *every* $z \in \mathbb{H}$ (`latticeTheta_inv`), whence its Epstein kernel Λ_z continues to $\mathbb{C} \setminus \{0,1\}$ with functional equation $\Lambda_z(1-s) = \Lambda_z(s)$ (`generalEpsteinLambda_differentiableAt`, `_functional_equation`);
- the Rankin–Selberg unfolding is a compiled identity, not an appeal: the vertical strip and the coset tiling of $\mathcal{D}$ are both fundamental domains of the translation subgroup (`isFundamentalDomain_strip`, `isFundamentalDomain_fdUnion`), so $\int_{\mathcal{D}} \omega_W\,\Lambda_z$ equals the strip integral, which by Fubini and the *exact* Parseval energy identity (`rankin_energy_exact`) is the Mellin transform of the smoothed second moment — the displayed `rs_master` equation;
- the poles sit exactly on the DC channel: $\Lambda_z(s) = \Lambda_z^0(s) - 1/s - 1/(1-s)$ with Λ_z^0 entire, so the completed series is meromorphic with simple poles only at $s \in \{0,1\}$, residue the Petersson norm $\int_{\mathcal{D}} P\,d\mu$ (`lambda_pole_split`, `lambda_residue_one`, `lambda_residue_zero`) — rank-is-DC-residue at the Rankin–Selberg level;
- the holomorphy of the entire part $s \mapsto \int_{\mathcal{D}} P\,\Lambda_z^0(s)\,d\mu$ is reduced to a single decay estimate, itself compiled: the $\mathcal{D}$-averaged tail $G(t) = \int_{\mathcal{D}} P(z)\,(\Theta_z(t)-1)\,d\mu$ decays faster than every power, by the $y \sim \sqrt{t}$ saddle $e^{-2\pi y}\,e^{-3\pi t/8y} \le e^{-\pi y}\,e^{-\pi\sqrt{3t/2}}$ (`amgm_saddle`, `ptwise_bound`, `exp_sqrt_isBigO`) integrated against the *finite* fundamental-domain measure (`volume_fd_lt_top`); the residual passage $\int_{\mathcal{D}} P\,\mathcal{M}[\Theta_z-1] = \mathcal{M}G$ is Fubini and $\mathcal{M}G$ is then entire by Mathlib's `mellin_differentiableAt_of_isBigO_` — routine and dominant-in-hand, not an open obstruction.

The averaged bank itself is now a compiled theta profile (`RSAveragedThetaProfile3D`): with $\bar\theta_f(t) = \int_{\mathcal{D}} P(z)\,\Theta_z(t)\,d\mu$ the Petersson average of the general lattice bank of the *actual* form, the exact weight-one reflection $\bar\theta_f(1/t) = t\,\bar\theta_f(t)$ is derived by integrating the pointwise weld `latticeTheta_inv` against the Petersson density (`averagedTheta_inv` — no candidate, no supplied functional equation); the tail approaches the Petersson mass at the saddle rate (`averagedTheta_tail_bound`); and the packaged pair (`rsAveragedWeakFEPair`, weight one, both constant terms the Petersson mass — the profile genuinely carries the ζ-factor constant term, so its home is the *weak* pair, the same typing law that retired the raw-vertical coupling above) returns from Mathlib's abstract functional-equation machinery the entire part, the global equation $\Lambda(1-s) = \Lambda(s)$, and the edge residue equal to the Petersson mass (`rsAveraged_residue_one`) — rank-is-DC-residue at the Rankin–Selberg level, at the profile interface. The named identification steps from this profile to the Sym^2 bank are the ζ-peel (`rankin_sym2_transfer`, compiled definitionally) and the coefficient certificate against the literal twisted Satake bank — instantiation points of the closed identification layer, not open interfaces.

The one classical input. What remains is the passage from this continued Dirichlet series to the summatory bound. The contour shift past the pole line is itself compiled: the residue theorem for finitely many simple poles in a rectangle, $\oint \big(g + \sum_\rho w_\rho/(\cdot - \rho)\big) = 2\pi i \sum_\rho w_\rho$, is assembled from the Cauchy–Goursat vanishing and the inverse-kernel residue (`rectangleBoundaryIntegral_holo_add_residues`, built on `rectangleBoundaryIntegral_eq_zero_of_differentiableOn` and `rectangleBoundaryIntegral_weigh` the Mellin inversion is likewise compiled (`mellin_inversion_eq`). The genuinely-cited remainder is therefore the *Perron summation identity* — the sharp-cutoff $\sum_{n\le x}$ against the contour-integral representation of the partial sum, together with the classical convexity estimate of the shifted vertical leg (whose edge bounds are compiled, `zetaRegularizer_bound_left/_right`) — classical and standard in analytic number theory. We isolate it as a single named hypothesis $\mathrm{PerronTauberian}(\lambda,\theta) := \exists C \ge 0,\ \forall x \ge 1,\ |\sum_{n\le x} \lambda_n| \le C\,x^{\theta}$ and cite it as literature pending formalization. The capstone

$$\texttt{sym2\_functoriality\_cancellation}: \qquad \theta < \tfrac{k}{2},\ \mathrm{PerronTauberian}(\lambda, \theta) \implies \sum_{n \le x} \lambda_n = O(x^{\theta})$$

is machine-checked at the standard footprint; every hypothesis except PerronTauberian is discharged by the compiled continuation above. This is the honest reduction of the cancellation: for $r = 2$ the carrier genuinely proves the analytic engine — continuation, functional equation, explicit poles, tail decay, finite measure — and the sharp-cutoff cancellation rests on one classical Tauberian step, the Perron summation identity (a projection shadow off the functoriality critical path, which closes via niceness, item 1), with the contour shift, residue, and Mellin inversion all compiled, and ω_W made explicit as the Petersson–Eisenstein density. The fixed-kernel functional equation *and* the arithmetic identification — local factors, all-place coefficient bank, and the full completed readout with its conductor and Gamma shifts — are compiled carrier facts at *every* rung, the identification by `rfl` (`arithmeticCPS_localFactor_identification`, `arithmeticCPS_globalCoefficient_identification`, `arithmeticCPSFullCompletion3D_identification`); so every rung's cancellation closes to the single Perron step, with $r = 2$ additionally grounded, concretely, in the arithmetic Rankin–Selberg series (`rs_master`, the Gaussian–lattice weld above).

Three-dimensional reading. ω_W is the carrier's phasor-bank density P read against the Eisenstein clock $(\Im z)^s$; the continuation is the weld reflection of the bank $\Theta_z(1/t) = t\,\Theta_z(t)$ transported through the Mellin projection; the poles at $s \in \{0, 1\}$ are the DC channel of the bank carrying the Petersson norm as residue. The abscissa $\mathrm{Re}\, s > 1$ of the projected Dirichlet series is the chart artifact where the readout stops converging, not a barrier on the carrier: the bank's phasors enter at height zero and grow with no convergence gate (Ground rule 4). The cancellation is thus a 3D statement — exact tail decay of the phasor bank averaged against the invariant density — whose 1D shadow is the summatory bound the review isolated as the crux.

Remark 60 (The niceness assembly, explicitly)**.** The proof above composes machine-checked carrier facts into niceness; we record the composition as an explicit ledger — each ingredient, its carrier realization, and its Lean anchor — uniform in r and τ, using no automorphy of $\mathrm{Sym}^r \pi$ and no zero-source argument.

Local data (Thm. 57). The transverse block *is* the Satake conjugate-pair $\mathrm{diag}(z, \bar{z})$, $z\bar{z} = 1$, and the Euler factor is read from the *deprojected* block — no temperedness assumed. Lean: `FrobeniusSimilitude.frobenius` (definitional), `ConeProjection.reconstruct_record`, `AutomorphicCandidate.midpoint_projects_to_half`.

Functional equation (FE). One geometric property of the involution J: the anti-helix carries $W^{\vee}$, so $\Lambda(s, W) = \varepsilon_W\, \Lambda(\kappa - s, W^{\vee})$ for *every* fiber, J descending through the Cayley, completion-clock, and logarithmic-readout intertwiners. Lean: `GeometricReadout.cayley/cayley_real_norm_one/ConeProjection.r` `FiniteWeightFiber.clockCompletion_selfdual`, `ConeProjection.midpoint_to_critical`; per place `FiniteWeightFiber.localPoly_reciprocal`; assembled `CompletedReflection.completed_FE`, and for the symmetric-power/twist fiber `FiniteWeightFiber.symTensorCompleted_FE` (every r, every duality-stable twist).

Bounded in strips (B). The direct reflected integral has $|x^{it}| = 1$, giving a t-independent finite bound in every vertical strip — no Phragmén–Lindelöf. Lean: Lemma 68.

Continuation (E). The bank grows continuously from height 0; the Abel-summed readout is analytic on $\{\mathrm{Re}\, s > \theta\}$ from the primitive bound, consuming no functional equation. Lean: `TransferContinuation.transfer_analytic`.

Entire (residual extinction $R_{r,\tau} = 0$). No constant mode survives to carry a pole — adapted cell closure for finite duality-stable fibers, unconditional focal cancellation for the cuspidal twists. Lean: `CellClosure.harmonic_bank_primitive_bounded`, `ForcibleClosure.residual_forcible`, `CarrierReachabili`

Composition. (E) continues the readout to a right half-plane; (FE) reflects it to the complementary half-plane; $R_{r,\tau} = 0$ removes the only candidate pole — so $\Lambda(s, \mathrm{Sym}^r \pi \times \tau)$ is entire; (B) bounds it in strips; (FE) is its functional equation; and Thm. 57 identifies the completed package — local factors, conductor, root number — with the classical data of $\mathrm{Sym}^r \pi$. That is precisely the hypothesis list of the converse theorem (Thm. 45), which returns the cuspidal automorphic $\mathrm{Sym}^r \pi$ on $\mathrm{GL}(r+1)$ (Thm. 48); temperedness (the $r \to \infty$ Jacquet–Shalika bound) and Serre's criterion then give Sato–Tate (Cor. 53).

Register. The functional equation is *carrier-generic* — a property of J, not of any one L-function — so there is no per-L-function equation to re-derive; the Dirichlet end-to-end machine-check (`completedLFunction_one_sub`) is a corroboration against the one case with an independent oracle, not the locus of generality. The converse theorem (Cogdell–Piatetski-Shapiro [14]) enters only as the classical packaging that turns the carrier's niceness into an automorphic representation — an established tool, the sole classical input, and no conjecture.

Lemma 61 (Residual-mode exclusion by adapted cell closure)**.** *Let $W = W_{r,\tau}$ be a CPS-admissible cuspidal tensor fiber with coefficients $a_W(n)$ and adapted unit-modulus carrier warp ω_W. Suppose the warp has* exact zero native-cell mean*: for every complete adapted carrier cell C (of bounded length),*

$$\sum_{n \in C} a_W(n)\, \omega_W(n) = 0.$$

Then (i) *the warped primitive is uniformly bounded, $A_W^{\mathrm{car}}(N) := \sum_{n \le N} a_W(n)\, \omega_W(n) = O(1)$; and* (ii) *the zero-frequency (DC) mode of the reflected carrier theta, taken in the carrier gauge, vanishes, $R_{r,\tau} = 0$, so $\mathcal{M}K_{r,\tau}$ is entire.*

Proof. Partition $\{1, \dots, N\}$ into complete adapted cells and one incomplete remainder. Each complete cell sums to 0 by hypothesis, so $A_W^{\mathrm{car}}(N)$ equals the single incomplete-cell partial sum, of magnitude at most the (bounded) cell length: $A_W^{\mathrm{car}}(N) = O(1)$, which is (i). For (ii) the passage from the bounded primitive to the vanishing DC mode is a partial summation, not a definition. By Abel summation the warped Dirichlet series $L_W(s) = \sum_{n \ge 1} a_W(n) \omega_W(n)\, n^{-s} = s \int_1^\infty A_W^{\mathrm{car}}(u)\, u^{-s-1}\, du$ converges and is holomorphic on $\mathrm{Re}\, s > 0$, because $A_W^{\mathrm{car}} = O(1)$ by (i); in particular it has *no pole* at the edge, equivalently the warped mean $\lim_{N \to \infty} A_W^{\mathrm{car}}(N)/N = 0$. The reflected carrier theta has Mellin $\mathcal{M}K_{r,\tau}(s) = L_W(s)\, G(s)$ with G the entire, rapidly-decaying Mellin of the completion kernel; so the Riemann–Hecke residual — a pole of $\mathcal{M}K_{r,\tau}$ in $\mathrm{Re}\, s > 0$, i.e. the constant mode $R_{r,\tau}$ the $x \to 0$ expansion of K would throw off — cannot occur, $R_{r,\tau} = 0$. The bounded primitive alone gives holomorphy only on $\mathrm{Re}\, s > 0$; *entireness* across the whole plane is then the reflection $K(1/x) = \varepsilon x^{\kappa} K(x)$ ((E1)), trading that half-plane for its mirror via the rapid decay of the kernel at $x \to \infty$. This is (ii), in the carrier gauge where A_W^{car} and R are defined — no global Weil parameter and no coefficient-average-to-multiplicity step, so the statement is Galois-free and applies to Maass τ verbatim. ◻

For the cuspidal twists a converse theorem actually consumes, the residual vanishing is not merely a per-instance closure but a theorem, by the carrier's *focal cancellation*, needing no automorphy of $\mathrm{Sym}^r \pi$. Two of the payload's passport laws combine into one balance statement the rank-gap consumes:

Lemma 62 (Balanced sign-compliant payloads have opposite strand residuals)**.** *Let W be a harmonized carrier payload carrying two of the passport stamps of Prop. 67:* (F) Frobenius balance, $\mathrm{FrobDet}(W) = 1$ *(equivalently $\sum_j \log \lambda_j = 0$, zero net multiplicative drift, with branch on the helix/anti-helix ledger); and* (S) vanishing-sign compatibility *under the carrier involution J — the per-clock weld $E_v^*(\bar{z}) = -\bar{\alpha}_v E_v(z)$ carries the channel sign flip (*`ChiralityHB.symClock_star`*) and a vanishing of multiplicity k reopens with orientation $(-1)^k$ (*`GeneralizedCohomology.dimension_parity_of_involution_even/_odd`*). Then the two carrier strand residuals are opposite,*

$$R_+(W) = -R_-(W),$$

so no unpaired determinant residue enters the cell.

Proof. (F) controls *magnitude*: $\mathrm{FrobDet} = 1$ is $\sum_j \log \lambda_j = 0$, so the two strands carry equal-and-opposite magnitude of multiplicative imbalance — this is the carrier balance condition, *not* a claim $W \simeq W^\vee$, the reciprocal orientation being supplied by the double carrier $(W, C_+) \leftrightarrow (W^\vee, C_-)$. (S) controls *orientation*: the involution J sends the positive strand's residual to the negative strand's, the weld $E_v^* = -\bar{\alpha}_v E_v$ carrying the sign flip and the $(-1)^k$ reopening fixing the orientation at a multiplicity-k vanishing. Equal-and-opposite magnitude with opposed orientation is exactly $R_+ = -R_-$. Neither stamp alone suffices: (F) fixes only magnitude, (S) only orientation — the conclusion needs both, which is why it is one lemma and not two. $\square$

Lemma 63 (Zero-detection: the rank gap at focal cancellation)**.** *Let $1 \le m < r$ and let τ be cuspidal on $\mathrm{GL}_m(\mathbb{A})$ — any twist a converse theorem into $\mathrm{GL}(r+1)$ consumes — and form the configured carrier+fiber system $W_{r,\tau}$. Then the harmonic-pencil rank-drop occurs* exactly at the readout's zeros*: focal cancellation of the signed P/M channels coincides with a vanishing of the readout, $\Phi_{r,\tau}(y) = 0 \iff L(\frac{1}{2} + iy, \mathrm{Sym}^r \pi \times \tau) = 0$ — the carrier's exact zero-detector, fiber-parametric, reading only the signed-channel/lane geometry.* This detects the vanishing set; it does not by itself establish pole-absence — entireness is the separate completeness statement of Lemma 64.

Proof. $R_{r,\tau}$ is read through the *configured* carrier+fiber system, in the causal order *configuration $\to$ realization $\to$ closure $\to$ rank-drop $\to$ factorization*. The rank-drop is a loss of a state-space degree of freedom, not the vanishing of a separate coefficient. *Frobenius balance.* The harmonized payload carries the passport stamps (F) Frobenius balance and (S) vanishing-sign compatibility — the determinant character $(\det \varphi_{\tau,v})^{r+1}$ of Lemma 66 being booked into the warp, not left to drift — so by Lemma 62 its strand residuals are opposite, $R_+ = -R_-$, and no unpaired determinant residue enters the cell. *Configuration.* The fiber's coefficients split the reflected readout into the signed positive and negative phasor channels $P_{r,\tau}(y), M_{r,\tau}(y)$ (the A, B channels, `FocalEigenheight.focalP/focalM`); their focal kernel is the 2×2 determinant of the harmonic pencil $L^{c} = \left[\begin{smallmatrix} A & B \\ \mu A & \lambda B \end{smallmatrix}\right]$ (`HarmonicPencilCell.focalKernel_cell_eq_AB_det`). *Realization and height.* The adapted carrier realizes the two channels as the helix/anti-helix lane states $X_\pm(z)$ at height $z = e^y$ (represented point $\frac{1}{2} + iy$, `HarmonicPencilCell.reprPoint`). *Closure correspondence.* A vanishing of the readout *is* full signed-channel cancellation, hence focal coincidence of the two lane states: $\Phi_{r,\tau}(y) = 0 \iff P_{r,\tau}(y) = M_{r,\tau}(y) \iff X_+(e^y) = X_-(e^y)$ (`focal_zero_iff_scalar_zero`). *Rank-drop.* At coincidence the two lane vectors are linearly dependent, so $\det L^{c} = (\lambda - \mu)AB = 0$, and the Hermitian PSD Gram matrix drops rank, $\det G^{c} = |\det L^{c}|^2 = 0$ (`gramH_posSemidef`, `gramH_rank_drop_iff_L_zero`, `gramH_rank_drop_iff_Bchan_zero`). This is a genuine loss of a state-space degree of freedom: the closed signed direction *disappears*: it is not offset by some other coefficient, it is gone. *Factorization.* The normalized focal residual is tied to that same signed channel, $D^{c}_{r,\tau} = V_{r,\tau} \Phi_{r,\tau}$ with $V_{r,\tau} \neq 0$ (`normalized_cell_focal_residual_exact`); so the pencil residual vanishes exactly with the signed channel $\Phi_{r,\tau}$, i.e. exactly at the readout's zeros (`harmonic_residual_no_independent_mode`, `gramH_rank_drop_iff_L_zero`). This is zero-detection — the pencil marks the vanishing set; pole-absence (entireness) is the *separate* DC/constant mode, established from completeness in Lemma 64. The detector is *fiber-parametric* and does not require any zero of $\Phi_{r,\tau}$ to exist: it reads no global Weil–Galois parameter (so it holds for Maass τ) and no automorphy of $\mathrm{Sym}^r \pi$. Machine-checked for the Dirichlet model; the same determinant-one lane balance for the general symmetric-power/twist fiber, realized as *forcible* closure with a machine-checked controllability core (Lemma 65). Away from a coincidence the smallest singular value reopens, $\sigma_{\min}(G^{c}(t)) \sim C\,|t-t_0|^k$ — the post-closure resistance below, a recovery rate, not a residual mode. ☐

Lemma 64 (Entireness from completeness: the residue is the trivial-channel coefficient)**.** *For the configured tensor fiber $W_{r,\tau}$ with coefficients λ_n, the constant term of the $x \to 0^+$ expansion of the carrier theta $K_{r,\tau}(x) = \sum_n \lambda_n g(nx)$ — the Riemann–Hecke residual $R_{r,\tau}$, equal by the Mellin pair to the edge residue of $\Lambda(s)$ — is the DC/zero-frequency mode of the bank, i.e. its* coefficient mean*. Every* nontrivial *channel of the self-compatible cell has zero cell-mean (`CommonRepresentation.harmonize_char`), so $R_{r,\tau}$ equals the* trivial-channel coefficient alone *(`complete_bank_dc_zero`). Hence if $W_{r,\tau}$ carries* no trivial channel — completeness — *then $R_{r,\tau} = 0$ and $\Lambda(s, \mathrm{Sym}^r \pi \times \tau)$ is entire. For non-exceptional π the tensor* $\mathrm{Sym}^r \varphi_\pi \otimes \varphi_\tau$ *has no trivial constituent: the relevant map $\varphi_\tau^\vee \to \mathrm{Sym}^r \varphi_\pi$ is a morphism for the parameter-group action, and simplicity makes every nonzero morphism surjective; the strict rank inequality then rules it out (`RepresentationRankGap.no_nonzero_intertwiner_of_finrank_lt`; the bundled Schur form is `NonSelfDual.tateClass_zero_of_simple_finrank_succ`). Thus $W_{r,\tau}$ is complete and the twist is entire. The exceptional/reducible π carry a genuine trivial constituent — the classical ζ-factor pole — and* $\mathrm{Sym}^r \pi$ *is then the known isobaric object (§16).*

Proof. By absolute convergence ($\lambda_n = O(n^A)$) the Mellin pair is exact on $\mathrm{Re}\, s > \sigma_0$, so the $x \to 0^+$ constant term of $K_{r,\tau}$ is the residue of $\mathcal{M}K_{r,\tau}(s) = \Lambda(s)$ at the edge (Riemann–Hecke) — the DC Fourier coefficient of the bank. Decomposing the coefficients into the fiber's channels (additive characters of the self-compatible cell group), character orthogonality gives that each nontrivial channel sums to zero over a cell (`harmonize_char`), so the DC coefficient is the trivial-channel coefficient alone (`complete_bank_dc_zero`). A trivial channel is present iff $\varphi_\tau^\vee$ embeds in $\mathrm{Sym}^r \varphi_\pi$ (a trivial constituent); for non-exceptional π with $\dim \varphi_\tau < \dim \mathrm{Sym}^r \varphi_\pi$ that equivariant Hom-space is zero (`RepresentationRankGap.no_nonzero_intertwiner_of_finrank_lt`), so there is no trivial channel, $R_{r,\tau} = 0$, and Λ is entire. The pencil residual $D^{c}_{r,\tau} = V_{r,\tau}\Phi_{r,\tau}$ of Lemma 63 is a *separate* quantity — the zero-detector, the readout's value along the line — not this DC/residue mode; the two must not be conflated. ☐

Cell closure is not a fiber-uniform theorem to be proved once for all functions, but a property of each carrier+fiber pair, effected by the fiber's *own* adapted warp — the same split as "the carrier sets the functional equation, the fiber sets the function". For the *class of finite duality-stable harmonic fibers* — those whose weight channels are P-th roots of unity, none trivial — this closure is *proved in Lean*, at the weight level and Galois-free: each nonzero-weight channel cancels over its native cell by the geometric series $\sum_{k<P} \zeta^k = 0$ (`CellClosure.root_of_unity_cell_sum_zero`), so the warped bank closes (`harmonic_bank_cell_sum_zero`) and its primitive is bounded (`harmonic_bank_primitive_bounded`), whence $A^{\mathrm{car}} = O(1)$, the vertical bound, and $R_{r,\tau} = 0$ (`cell_closure_dc_mode_zero`). The Dirichlet characters of any conductor — the χ_3 archetype of §26 included — are the classical instances (`dirichlet_cell_closure`, recovering `character_partialSum_norm_le`). A cuspidal fiber, whose Satake angles are not roots of unity, is not in this harmonic class: at the *base* scale its cells do not close. Its closure is carried by its *own* synthesized completion — the kernel built from the fiber's

coefficients against the self-dual clock — under which the completed pair readout is entire with vertical-strip bounds and the functional equation at every rank (`cpsDualPair_twistedNiceness`, radius-live `cpsPolynomialAllTwists_payload`). The registration of that completed readout in the prescribed conductor/Γ-product chart is the per-rung coupling (`BankThetaProfileCoupling3D`); machine-checked end-to-end for the character classes (`dirichlet_strip_tendsto_LFunction`) and at the standard rung of a genuine cusp form (`standardRung_classicalHeckeTheorem`), derived at $r = 2$ from the lattice weld, classical at $r = 3, 4$, and independently evidenced by the discriminating instruments (the oracle-free critical-line readout to 2×10^{-16}, the growth locator to 10^{-12}, the FE closures, App. B). Warp controllability itself (Lemma 65) contributes only the arithmetic-*neutral* transport half — it closes cells at any admissible fiber's own scale, consuming none of the fiber's arithmetic, which is exactly what universal transport requires of it (Cor. 80) and exactly why it is not niceness evidence: force-closure alone yields no continuation to transfer. The axis-reality reflection (Lemma 75) consumes this continuation; the residual mode vanishes by completeness (Lemma 64), unconditionally; and full-plane entireness then follows from the entire reflected carrier theta, downstream of the functional equation — the dependency order of Lemma 75, no step feeding on its successor. What is exhibited per instance is then only the finer warp dynamics — the post-closure resistance below — never the entireness. There is no fiber-independent cancellation for the carrier to prove, because the carrier never faces a function without its fiber.

Lemma 65 (Forcible cell closure via the reachable independent stage)**.** *For a cuspidal fiber the adapted warp that closes a cell need not be found by luck, and the two independent correction directions need* not *occur at every native cell. The readout-preserving warp carries a few* coherent *generators — the winding $\Omega(n)$, the distinct-prime count $\omega(n)$, the vertical shift $\log n$, and the fiber's own local periods $\Theta(n) = \sum_{p^k \| n} k\, \theta_p$ (arithmetic functions and the fiber's finite local data, not arbitrary per-term phases) — whose real weights drive the cell residual $D_C = \sum_{n \in C} a_n\, \omega_W(n)$. The controllability core is exact and machine-checked: two $\mathbb{R}$-linearly-independent warp correction-directions u, v span the residual plane $\mathbb{C}$, so* every *residual D is forcible to zero by real weights, $D + (s \cdot u + t \cdot v) = 0$ (`ForcibleClosure.residual_forcible`, Lean. Those two directions are not demanded cell by cell. By `CarrierReachability.reachable_independent_stage`, every nontrivial admissible fiber (continuous non-constant lane phase ϕ) admits a reachable carrier* stage *— a height y with $\sin(2\phi(y)) \neq 0$ — at which the two dual lanes $e^{\pm i\phi(y)}$ are $\mathbb{R}$-linearly independent. No transport of the cell is involved: the residual D_C is a fixed scalar, and the stage supplies the correction pair $u = e^{i\phi(y)}$, $v = e^{-i\phi(y)}$; `residual_forcible` applied to that pair gives real weights s, t with $D_C + s\,u + t\,v = 0$, exactly. Thus per-cell closure follows from* global *carrier reachability supplying one independent pair,* not *from cell-by-cell linear independence. The two machine-checked inputs are `reachable_independent_stage` and `residual_forcible`, and their composition is itself a Lean theorem (`reachable_residual_forcible`): a continuous nonconstant dual-lane phase supplies a stage and real weights closing any prescribed residual. The constructive rate is what the numerics exhibit,* in the carrier's own chart*: on native growth cells — uniform in $y = \log Z$, every phasor entering at zero magnitude through the canonical C^∞ growth window, no clipped edge — exact Gauss–Newton closes $|D_C|$ to the float64 floor (median 5×10^{-15}–1.4×10^{-14}) on* every *cell of all three fibers,* 30/30 *each, with weights* $|x| \leq 3.1$*, once the generator suite includes the fiber's own local periods $\Theta(n)$; the three arithmetic generators alone already close* 30/30 *on* Δ *and* $\mathrm{Sym}^5\,\Delta$*, leaving one rank-deficient* Sym^{13} *cell that Θ closes — the resisting cell naming its adapter. (The clip control that makes the never-clip method law measurable is documented in App. B.)*

This realizes the cell closure of Lemma 63 *constructively* for the cuspidal fiber: each cell is forced closed, so no zero-frequency mode survives — an explicit constructive mechanism whose algebraic core (two independent directions force any residual) is machine-checked and supplied by the reachable independent stage, not cell by cell, behind the structural rank-drop. *Scope, fixed by the theorem's own generality:* controllability is arithmetic-*neutral* — `residual_forcible` consumes no arithmetic, closing cells for every admissible fiber, and the neutrality is now itself a theorem: one stage closes any two residuals identically, so forcibility carries no information about the fiber and can never serve as niceness evidence (`ForcibleClosure.forcible_nondiscriminating`) — so it is the *transport* half of the architecture, consumed by universality (Cor. 80, Thm. 79 (c)) and by the adapter suite, *not* niceness evidence; within niceness its role is to book the per-cell correction, while the arithmetic enters through the warp-removal transfer of §29. Its controllability and reachability legs are proved (`ForcibleClosure.residual_forcible`, `CarrierReachability.reachable_independent_stage`); the inverse-kernel summability and boundary legs are separately exposed by `WarpRemovalTransfer.warpRemoval_` A universal bounded-weight per-cell claim would be false on the rank-deficient locus and, being arithmetic-neutral, needs no strengthening for the chain. The boundary is sharp, and not where a first reading places it. *The residual factorization is unconditional*: $D_C = V\,\Phi$ with $V \neq 0$ — the Gram-pencil rank-drop leaving no zero-frequency mode independent of the readout — is a channel-agnostic identity in (A, B, μ, λ), machine-checked in Lean for arbitrary channels (`HarmonicCell.harmonic_residual_no_independent_mode` and instantiated end-to-end for the Dirichlet model; it reads only the fiber's own P/M channels, so it holds *verbatim* for every det-one symmetric-power/twist fiber, Maass included. The finer *constructive* reachability — that the dual carrier admits a reachable stage whose two lane directions are $\mathbb{R}$-linearly independent, demanded once rather than at every cell — governs the closure *dynamics*, not the vanishing of the residual mode, and it too is settled rather than left to numerics. The required two correction directions are not obtained by a genericity argument. For the dual carrier they are the two *lanes* $e^{\pm i\phi(y)}$ of the helix, and a complex number and its conjugate are $\mathbb{R}$-linearly independent at a carrier height y exactly when $\sin(2\phi(y)) \neq 0$ (independent iff off the axes, not merely $\phi \neq 0$ — at $\phi = \pi/2$ the two lanes coincide up to sign). The complete dual helix runs continuously $0 \to \infty$ on both lanes, so a clock with $\sin(2\phi) \equiv 0$ would map the connected line into the discrete set $\frac{\pi}{2}\mathbb{Z}$ and hence be constant — the DC/trivial lock. A *nontrivial* (non-constant) admissible clock therefore attains some height with $\sin(2\phi) \neq 0$: `CarrierReachability.reachable_independent_stage` proves that every nontrivial source fiber in the stated structural class (continuous non-constant ϕ) admits a carrier stage with two $\mathbb{R}$-linearly-independent lane directions, and `ForcibleClosure.residual_forcible` then yields the exact correction. Both are machine-checked: completeness supplies the continuous $0 \to \infty$ carrier, admissibility the structured non-random clock, and nontriviality excludes the constant lock. So reachability is unconditional on the nontrivial admissible source class, not a numerical input; the numerics (Sym^5–Sym^{13}, the twisted family) are calibration. *Boundedness likewise does not ride on the forcing.* The vertical-strip bound is structural, from the *Euler factor*: on its convergence domain the full helix is the convergent product of local factors $\det(1 - A_p q_p^{-s})^{-1}$, each a bounded rational function with $|A_p|$ polynomially bounded (Kim–Sarnak for π, Jacquet–Shalika/LRS for τ), so the readout is finite order there; the functional equation extends that bound across the strip (Lemma 68), given entireness — *independent* of the forcing chart. Hence the per-cell forcing weights need *not* be bounded for L to be finite order (in the native growth chart the measured closing weights are in fact $O(1)$–$O(3)$; only the clipped control chart inflates them — either way a feature of the constructive route, not a bound on L). Forcible closure thus makes the cell closure *constructive*; entireness rests on completeness (Lemma 64), boundedness on the Euler factor, and reachability on the completeness–admissibility argument above — so no leg of niceness rests on numerics; the tested families confirm the mechanism rather than carry it.

Functorial preservation — and its limit. That reachability is not isolated per fiber. Exact focal closure is *preserved* under the carrier's structural transports — composition, Rankin–Selberg tensor, and base change: if two transports each carry closure to closure, so does their composite (`CarrierFunctoriality.faithful_comp`, Lean, *no* axioms; with the associativity, two-sided identity, and base-change law $\mathrm{bc}_p \circ \mathrm{bc}_q = \mathrm{bc}_{pq}$ of §32), and the plethysm splits each composite *exactly* into irreducible Sym^c blocks ($\mathrm{Sym}^2 \circ \mathrm{Sym}^2 = \mathrm{Sym}^4 \oplus \mathrm{Sym}^0$, $\mathrm{Sym}^3 \circ \mathrm{Sym}^3 = \mathrm{Sym}^9 \oplus \mathrm{Sym}^5 \oplus \mathrm{Sym}^3$; `composition_test.py`). So closure propagates functorially from the classically nice low powers $\mathrm{Sym}^2, \mathrm{Sym}^3, \mathrm{Sym}^4$ and from any forcibly-closed fiber to all their composites, tensors, and base changes. Composition alone — multiplicative top weight ab — misses the prime degrees, but the *tensor* is additive: Clebsch–Gordan gives $\pi \otimes \mathrm{Sym}^{r-1}\pi = \mathrm{Sym}^r \oplus \mathrm{Sym}^{r-2}$, so *every* Sym^r is the top block of $\pi \otimes \mathrm{Sym}^{r-1}$, its lower block Sym^{r-2} inductive (`composition_test.py`). The whole tower is reachable — there is no degree wall. (The Clebsch–Gordan route controls the standard-L residual; the cuspidal-twist family a converse theorem consumes closes by the same channel-agnostic rank-drop, Lemma 63, not a separate input.) Along the induction the pole channel is representation-theoretically clean: for simple Sym^r, an equivariant $\tau^\vee \to \mathrm{Sym}^r$ is zero when $\dim \tau < \dim \mathrm{Sym}^r$ (`RepresentationRankGap.no_nonzero_intertwiner_of_finrank_lt`); and the carrier builds the readout as the Mellin of an *entire* theta, not a quotient of L-functions, so the classical strip-pole obstruction of block extraction does not arise. The closure is a finite carrier event: the constant mode $R = \lim_N A^{\mathrm{car}}(N)/N$ of the transported cuspidal fiber, extinguished by the forcible cell closure above (Lemma 65); the corroborating numerics — native-cell closure 30/30 at the float64 floor on the open $\mathrm{Sym}^5, \mathrm{Sym}^{13}$, and the global warped DC $|A^{\mathrm{car}}|/N = 3 \times 10^{-4}$ falling at $N = 2 \times 10^5$ — are calibration (App. B). Its 1D shadow — the mean of the readout coefficients, equivalently the s=1 residue — vanishes, and the tensor structure exhibits why without leaving the carrier: the Clebsch–Gordan factorization $L(\pi \times \mathrm{Sym}^{r-1}\pi) = L(\mathrm{Sym}^r \pi)\, L(\mathrm{Sym}^{r-2}\pi)$ holds at the coefficient level (verified to 10^{-14}, `tensor_induction_check.py`), the base $\mathrm{GL}(2) \times \mathrm{GL}(r)$ Rankin–Selberg entire and non-vanishing at 1, so $R(\mathrm{Sym}^r \pi) = 0$ inductively (verified $\to 0$). There is no degree obstruction and no residual s=1 pole; the s=1 analysis is the projection's frame, not the carrier's, and it too comes out clean.

31 The transported datum: reflection, completion, and niceness laws – CPS

The candidate object of §29 carries an admissible fiber; Part II here assembles the laws by which the complete carrier reads it into a faithful completed datum — the local self-duality that certifies the fiber, the vertical growth of the readout, the carrier reflection and its completion, and the completed functional equation. These are the supporting laws of the transport machine (Theorem 78). The automorphic landing they enable — whether the classical automorphic category accepts this output — is deferred to Part III, where a converse theorem is invoked as an external checksum.

We record the local/analytic ingredients as lemmas, each a proven statement, so the transported datum is assembled in one place.

Lemma 66 (Local carrier-dual compatibility and determinant ledger)**.** *Let π have central character ω_π, normalized so that $\varphi_{\pi,v} = \omega_{\pi,v}^{1/2}\,\mathrm{diag}(\alpha_v, \alpha_v^{-1})$ at unramified v ($|\alpha_v|$ not assumed 1; the radial magnitude rides the ledger): the balanced part $\mathrm{diag}(\alpha_v, \alpha_v^{-1})$ has determinant one, and ω_π rides as a central-character scalar. For every $r \geq 1$, every $1 \leq m < r$, every cuspidal τ on GL_m, and every place v, write $W_{r,\tau,v} = \mathrm{Sym}^r \varphi_{\pi,v} \otimes \varphi_{\tau,v}$. Then:*

(i) *the exponents $\{r-2k\}$ sum to zero, so $\mathrm{Sym}^r\,\mathrm{diag}(\alpha_v, \alpha_v^{-1})$ has product-of-eigenvalues one — the* Frobenius balance*, independent of ω_π — while* $\det \mathrm{Sym}^r \varphi_{\pi,v} = \omega_{\pi,v}^{r(r+1)/2}$ *and* $(\mathrm{Sym}^r \varphi_{\pi,v})^\vee \simeq \mathrm{Sym}^r \varphi_{\pi,v} \otimes \omega_{\pi,v}^{-r}$*: self-dual* up to the central twist *ω_π^{-r} (genuinely self-dual iff $\omega_\pi^r = 1$; in particular when $\omega_\pi = 1$), exponents closed under negation;*

(ii) carrier-dual compatibility*: $W_{r,\tau,v}^\vee \simeq W_{r,\tau^\vee,v} \otimes \omega_{\pi,v}^{-r}$ — the dual of the fiber is the $\tau^\vee$-twisted fiber, further twisted by the central scalar ω_π^{-r} (i.e. the contragredient datum $\widetilde{\Pi}_r \times \widetilde{\tau}$); $W_{r,\tau,v}$ itself* need not *be* *self-dual;*

(iii) determinant ledger*:* $\det W_{r,\tau,v} = \omega_{\pi,v}^{m\,r(r+1)/2}(\det \varphi_{\tau,v})^{r+1}$ — both *central characters ω_π, ω_τ carried in the arithmetic ledger, neither forced to one.*

Consequently the double carrier transports the pair $(W_{r,\tau}, W_{r,\tau}^\vee)$*: the helix reads $W_{r,\tau}$, the anti-helix the contragredient datum $W_{r,\tau^\vee} \otimes \omega_\pi^{-r} = \widetilde{\Pi}_r \times \widetilde{\tau}$, and the local factor $L_v(s, \Pi_r \times \tau)$ pairs with $L_v(1-s, \widetilde{\Pi}_r \times \widetilde{\tau})$ under $s \leftrightarrow 1-s$, centre* $\operatorname{Re} s = \frac{1}{2}$, $|\varepsilon_v| = 1$ *(the central characters entering the root number ε_v). The* $\det = 1$ *that fixes that centre is the* Frobenius similitude at the helix/anti-helix crossing *— the two lanes meet at a common height where the crossing block has determinant one (*`FrobeniusSimilitude.frobeniusBlock_det_one`*) — the balance of the angle exponents, not a constraint on the fiber's central character. The trivial-ω_π, self-dual sub-case ($\omega_\pi = 1$, $\tau \simeq \tau^\vee$,* $\det \varphi_{\tau,v} = 1$*) is the Lean reciprocity* `FiniteWeightFiber.localPoly_reciprocal` *(*`fiber_reflection_axis`*), instantiated by* `symFiber`*/*`tensorFiber`*; the general case carries both determinant characters through the ledger, the Frobenius balance $\prod \lambda = 1$ resting on the angle exponents alone.*

Proof. $\operatorname{Sym}^r \varphi_{\pi,v}$ has eigenvalues $\alpha^{r-2k}\omega_{\pi,v}^{r/2}$ ($0 \le k \le r$, from $\varphi_{\pi,v} = \omega_{\pi,v}^{1/2} \operatorname{diag}(\alpha, \alpha^{-1})$): the balanced exponents $r-2k$ sum to 0 (the diag part has product 1) and are closed under $k \mapsto r-k$, while the full product is $\omega_{\pi,v}^{r(r+1)/2}$; the dual eigenvalues $\alpha^{-(r-2k)}\omega_{\pi,v}^{-r/2}$ equal $\alpha^{r-2k}\omega_{\pi,v}^{r/2} \cdot \omega_{\pi,v}^{-r}$ after $k \mapsto r-k$, so $(\operatorname{Sym}^r \varphi_{\pi,v})^\vee \simeq \operatorname{Sym}^r \varphi_{\pi,v} \otimes \omega_{\pi,v}^{-r}$: (i). The contragredient of a tensor is the tensor of contragredients, so $W_{r,\tau,v}^\vee \simeq (\operatorname{Sym}^r \varphi_{\pi,v})^\vee \otimes \varphi_{\tau,v}^\vee \simeq \operatorname{Sym}^r \varphi_{\pi,v} \otimes \omega_{\pi,v}^{-r} \otimes \varphi_{\tau^\vee,v} = W_{r,\tau^\vee,v} \otimes \omega_{\pi,v}^{-r}$ by (i): (ii). And $\det(A \otimes B) = (\det A)^{\dim B}(\det B)^{\dim A}$ with $\det \operatorname{Sym}^r \varphi_{\pi,v} = \omega_{\pi,v}^{r(r+1)/2}$, $\dim \operatorname{Sym}^r \varphi_{\pi,v} = r+1$, $\dim \varphi_{\tau,v} = m$ gives $\det W_{r,\tau,v} = \omega_{\pi,v}^{m\,r(r+1)/2}(\det \varphi_{\tau,v})^{r+1}$: (iii). These are functorial identities on parameters (at ramified v, on the Weil–Deligne pair); no self-duality of τ and no global or automorphic input is used. □

Proposition 67 (The payload admissibility passport)**.** *A configured carrier payload W is* admissible *— transportable by the complete carrier under the residual-extinction, reflection, and vertical-strip laws — exactly when it carries three stamps:*

(F) Frobenius balance $\operatorname{FrobDet}(W) = \prod_j \lambda_j = 1$ *on the balanced (angle) part: zero net multiplicative drift, the central characters ω_π, ω_τ booked separately in the determinant ledger (Lemma 66(iii));*

(S) vanishing-sign compatibility*: the helix/anti-helix involution exchanges the signed channel orientation, with $(-1)^k$ at a vanishing of multiplicity k (*`ChiralityHB.symClock_star`*;* `GeneralizedCohomology.dimension_pa`

(L) Li positivity*: the loss-ledger's represented measure is positive semidefinite (*`SourceHolonomy.ledger_positivity`*,* `helix_ledger_positive`*, the Gram* `gramH_posSemidef`*), preserved under projection (*`projection_retains_li_pos`

For any Rankin–Selberg tensor pair $(W_{r,\tau}, W_{r,\tau^\vee})$ harmonization stamps all three: the adapter books the determinant character $(\det \varphi_{\tau,v})^{r+1}$ *into the warp* (F)*, routes the dual leg on the anti-helix* (S)*, and inherits the carrier's ledger positivity* (L)*. So* every *harmonized CPS twist is admissible:* (F) + (S) *force $R_+ = -R_-$, whence full focal cancellation drops the Gram rank and the residual direction disappears (Lemma 63), and* (L) *makes that a genuine loss of a positive direction, not a cancellation of signed masses. The passport gives $R_{r,\tau} = 0$ (no carrier residue) but does* not *by itself certify pole-freeness of the identified L-function: a reducible parameter* $\operatorname{Sym}^r \varphi_\pi$ *carries a trivial constituent whose ζ-factor is a genuine $s = 1$ pole — the classically enumerated exceptional π (§16). Residual extinction is thus a property of* every *admissible payload; the intrinsic pole is a property of the reducible parameter alone, and is absent whenever* $\operatorname{Sym}^r \varphi_\pi$ *is irreducible.*

Lemma 68 (Initial convergence and vertical-strip boundedness)**.** *For every $1 \le m < r$ and cuspidal τ on* GL_m*, the candidate $L(s, \Pi_r \times \tau)$ converges absolutely in a right half-plane, and its completed $\Lambda(s, \Pi_r \times \tau)$ is bounded in every vertical strip — as Theorem 45 requires.*

Proof. Initial convergence — of the candidate, then the twist. For a cuspidal π on $\mathrm{GL}_2/\mathbb{Q}$ the standard bound $|\alpha_p^{\pm 1}| < p^{1/2}$ (Jacquet–Shalika [20], no Ramanujan; any polynomial local bound giving some right half-plane would do) gives $|\alpha_p^{r-2k}| < p^{r/2}$ for the Sym^r weights, so the candidate Euler product $L(s, \Pi_r)$ converges absolutely on some right half-plane $\mathrm{Re}\, s > 1 + \frac{r}{2}$. Absolute convergence of the *twist* $L(s, \Pi_r \times \tau)$ then follows in some right half-plane by [14, Lemma 2.2] — exactly the initial-convergence hypothesis of Theorem 45; we do not estimate the Rankin–Selberg coefficients of every τ by hand, and Ramanujan is not used. (This reads the deprojected 3D Satake datum with its radial magnitude, not the 2D unit-circle phase; the sharp 7/64 exponent [16] is available for the Maass class but is not needed here.) *Vertical-strip boundedness, directly from the completed transform.* The completed readout is the reflected carrier integral of §30,

$$C(s) = \int_1^\infty K_{r,\tau}(x)\big(x^s + \eta\, x^{\kappa - s}\big)\,\tfrac{dx}{x}, \qquad |\eta| = 1,$$

entire once $R = 0$ (Lemma 64) and equal to the completed twist $\Lambda(s, \Pi_r \times \tau)$ by the identification (E5). On a fixed strip $a \le \mathrm{Re}\, s \le b$ the factor x^{it} has modulus one, so

$$|C(\sigma + it)| \le \int_1^\infty |K_{r,\tau}(x)|\big(x^b + x^{\kappa - a}\big)\,\tfrac{dx}{x} < \infty,$$

the right side *independent of t* and finite: the completion kernel g decays rapidly (super-polynomially) at ∞, and the theta $K_{r,\tau}(x) = \sum_{n \ge 1} \lambda_n g(nx)$ inherits that decay for $x \ge 1$ because $\lambda_n = O(n^{r/2+\epsilon})$ (the initial-convergence bound above) is dominated by $g(nx)$'s decay, so the series converges and $|K_{r,\tau}(x)|$ decays rapidly for $x \ge 1$. Hence Λ is *bounded* on every vertical strip — CPS's requirement verbatim, with no Phragmén–Lindelöf, no finite-order citation, and no "polynomial growth means bounded" step. In the primary (ζ) representation the completed strip bound is machine-checked (`ZD.ZeroCount.completedRiemannZeta0_bounded_on_strip`). □

Isolation note. The source-independent carrier, projection ledger, and reflection arrows used in the next theorem are isolated in §2, §3, and §7. The present section retains the transported-datum specialization and the CPS/local-factor bookkeeping.

Theorem 69 (Carrier reflection: the 3D–2D–1D intertwiner)**.** *The carrier is a 3D double-ended helix $C = C_+ \sqcup C_-$ — helix C_+ and conjugate anti-helix C_- — read through a 2D Cayley/Möbius projection to the unit-circle chart and a 1D logarithmic readout to the normalized strip. Its completed reflection is generated* geometrically, *in dimensional order, for any admissible fiber pair $(W, W^\vee)$ riding the two lanes:*

(J) Global involution. *A single symmetry $J\colon C_+ \leftrightarrow C_-$ exchanges helix and anti-helix — a geometric involution of the* whole *carrier, not a product of local pieces.*

(P) Projection intertwines. *The Cayley map carries J to the unit-circle reflection, $\pi_{2D} \circ J = J_{\mathrm{Cayley}} \circ \pi_{2D}$ (`GeometricReadout.cayley`, `cayley_real_norm_one`); the discarded radial and phase coordinates are booked* losslessly *— fiber $\mapsto$ (retained height, ledger) is a bijection (`ConeProjection.record_bijective`, `reconstruct_record`; `SourceHolonomy.projection_bijective_loss_ledger`).*

(C) FE clock intertwines. *The functional-equation clock carries the projected reflection to the readout reflection (`FiniteWeightFiber.clockCompletion_selfdual`).*

(L) Log readout. *The logarithm linearizes the multiplicative height, and the unique fixed point of the involution is the midpoint* $\mathrm{Re}\, s = \frac{1}{2}$ *(`ConeProjection.midpoint_to_critical`, `AutomorphicCandidate.midpoint_project`*

Composing, the completed readout obeys $\Lambda(s; W) = \varepsilon_W\, \Lambda(1 - s; W^\vee)$, $|\varepsilon_W| = 1$: the helix reading at s is the anti-helix reading at $1 - s$. The root number ε_W and the conductor are the arithmetic identification *of this geometric reflection — read from the local ε_v (§31.1) — not its definition.*

Proof. J is the global geometric involution of the double carrier; $z \mapsto \bar{z}$ about the axis realizes it on each strand, where the two-strand clock $E_v(z) = e^{iz\ell_v/2} - \alpha_v e^{-iz\ell_v/2}$ (`ChiralityHB.symClock`) obeys the local weld $E_v^*(\bar{z}) = -\bar{\alpha}_v E_v(z)$ (`symClock_star`) — the *readout* of J at v, not its definition. Steps (P) and (L) are the round-trip and midpoint theorems cited; (C) is the completion clock's self-duality. The reflection is thus produced by the global carrier and its faithful projection, not assembled placewise; it yields a *unit* η_W with $\Lambda(s; W) = \eta_W\, \Lambda(1 - s; W^\vee)$, $|\eta_W| = 1$ (each geometric clock phase $-\bar{z}_v$ having modulus one, $z_v = \alpha_v/|\alpha_v|$ the 2D Cayley phase with $z_v\bar{z}_v = 1$ — the projected phase, not the 3D arithmetic Satake parameter α_v). The arithmetic identification $\eta_W = \varepsilon(s, \Pi_r \times \tau)$ with the standard global root number is a *separate* computation — from the normalized local ε_v factors and the conductor (§31.1, (12)), with the usual almost-everywhere product structure — and is *not* the bare product of the clock phases $-\bar{\alpha}_v$, which are the weld's readout, not the local ε factor. No lattice and no Poisson summation enter — the classical theta identity $\theta(Y^{-1}) = \det(Y)^{1/2}\theta(Y)$ of §30 is the 1D readout shadow of J. ◻

Formal scope of the concluding identity. The cited Lean intertwiners prove the involution, lossless coordinate ledger, clock exchange, and logarithmic inversion. The global reflection of the all-place pair banks is itself compiled: the primal bank and its reciprocal-height contragredient bank satisfy the exact fold identity at every positive height, for every rank and every admissible Satake family, radius-live (`GlobalHelix.cpsDualPair3D_globalHelixReflection`; `GlobalHelixCPSPolynomialDualPair`), and the completed pair readout carries the functional equation with entirety and vertical-strip bounds in the fiber's own completion (`cpsDualPair_twistedNiceness`, every rank). The registration of this reflection in the prescribed conductor/Γ-product chart — the classical naming — is the per-rung coupling of §17 (`BankThetaProfileCoupling3D`, reduced by the discharge layer to one completed Mellin functional equation and one ray convergence, `ofMellinFE`"): proven at the standard rung from the modular slash law (`standardRung_reflection`), derived at $r = 2$ from the lattice weld (`averagedTheta_inv`), consumed classically at $r = 3, 4$.
The unit circle is a projection surface, not a unitarity claim. A point of the 2D chart lies on S^1 because it is the *phase* projection of the source datum $\alpha = \rho\, e^{i\theta}$; the radial magnitude ρ is *not* declared 1 but booked in the loss ledger and restored at the logarithmic readout (`ConeProjection.record_bijective`: fiber $\mapsto$ (height, (radial, phase)) is a bijection). So $\alpha = 2$ and $\alpha = \frac{1}{2}$ ride opposite lanes with reciprocal radial data and a common unit-circle phase; membership in S^1 asserts nothing about $|\alpha|$. *The fiber need not be self-dual.* Duality is first a property of the *carrier*, $C_+ \leftrightarrow C_-$: an admissible fiber W rides one lane while $W^\vee$ rides the anti-lane, so a non-self-dual CPS twist is carried by the *pair* $(W, W^\vee)$ and the reflection identifies their readouts at $s \leftrightarrow 1 - s$. What Lemma 66 certifies is *carrier-dual compatibility* of the pair, not $W \simeq W^\vee$.

Lemma 70 (The transfer, proved: bounded-primitive continuation)**.** *Let W have coefficients $\lambda_n = O(n^A)$ and raw primitive $A_W(x) = \sum_{n \le x} \lambda_n$, and suppose the single exponent condition*

$$\theta_W \;:=\; \inf\{\theta : \; A_W(x) \ll x^\theta\} \;<\; \tfrac{\kappa}{2}.$$

Then (i) *$L(s, W) = \sum \lambda_n n^{-s}$ converges locally uniformly and is analytic on $\operatorname{Re} s > \theta_W$, agreeing with the Euler datum on the absolute half-plane;* (ii) *the dual primitive is $A_{W^\vee} = \overline{A_W}$ (unitary duals $\lambda_n^\vee = \bar{\lambda}_n$), so $\theta_{W^\vee} = \theta_W$ and* both *completed readouts $\Lambda(s; W)$ and $\Lambda(\kappa - s; W^\vee)$ are analytic on the open strip $S_W = \{\theta_W < \operatorname{Re} s < \kappa - \theta_W\}$, which contains the weld axis precisely because $\theta_W < \kappa/2$;* (iii) *Step 2 of Lemma 75 is discharged by this condition alone — no warp, no automorphy, no functional equation, no Poisson summation is consumed.*

Proof. (i) Fix $\theta_W < \theta < \sigma_1 < \operatorname{Re} s$. Partial summation gives, for every N, $\sum_{n \le N} \lambda_n n^{-s} = A_W(N)N^{-s} + s\int_1^N A_W(u)\, u^{-s-1}\, du$. Since $|A_W(u)| \le Cu^\theta$ and $\operatorname{Re} s > \theta$, the boundary term tends to 0 and the integral converges absolutely, uniformly on compacts of $\{\operatorname{Re} s > \theta\}$: $|A_W(u)u^{-s-1}| \le Cu^{\theta-\operatorname{Re} s-1}$. Hence $L(s,W) = s\int_1^\infty A_W(u)u^{-s-1}du$ is a locally uniform limit of analytic functions, analytic on $\operatorname{Re} s > \theta$ for every $\theta > \theta_W$, hence on $\operatorname{Re} s > \theta_W$; on the absolute half-plane it agrees with the Euler datum termwise. (ii) $\lambda_n^\vee = \bar\lambda_n$ gives $A_{W^\vee} = \overline{A_W}$ and the same exponent; $s \mapsto \kappa - s$ maps $\{\operatorname{Re} s > \theta_W\}$ to $\{\operatorname{Re} s < \kappa - \theta_W\}$, so both factors of each completed readout are analytic on S_W — the γ-factors are pole-free there, their poles lying on $\operatorname{Re} s \le 0$ ($\Gamma_{\mathbb{C}}(s+\mu_i)$ with $\mu_i > 0$; Maass $\Gamma_{\mathbb{R}}(s \pm it)$ with poles on $\operatorname{Re} s \le 0$), and $S_W \subset (0,\kappa)$. The axis $\operatorname{Re} s = \kappa/2$ lies in S_W iff $\theta_W < \kappa/2$. (iii) is immediate: S_W is the connected open neighbourhood of the axis that Step 2 consumes. (i) is *machine-checked in full*: `TransferContinuation.transfer_tendsto` proves convergence of the partial sums at every s with $\operatorname{Re} s > \theta$ from the primitive bound alone, via the formalized mean-value kernel `cpow_sub_bound`, and `transfer_analytic` proves the analyticity upgrade — a limit function differentiable on $\{\operatorname{Re} s > \theta\}$ agreeing with the partial sums at every such s, by Weierstrass on the normally convergent Abel-summed representation (`transfer_tendsto_tsum`); the dual-exponent identity and the strip arithmetic are `dual_primitive_norm` and `strip_contains_axis`. No manuscript-level step remains in (i). ◻

Lemma 71 (The grown transfer: the sharp barrier is a clip artifact)**.** *Let $A_W^{\mathrm{e}}(x) = \sum_{n\ge1}\lambda_n e^{-n/x}$ be the* grown *primitive — every phasor entering continuously, the carrier-native object, against the clipped A_W — and $\theta_W^{\mathrm{e}} := \inf\{\theta :\ A_W^{\mathrm{e}}(x) \ll x^\theta\}$.* (i) $\int_0^\infty A_W^{\mathrm{e}}(x)\,x^{-s-1}\,dx = \Gamma(s)\,L(s,W)$ *for* $\operatorname{Re} s > \max(\sigma_{\mathrm{abs}},0)$, *the $x \to 0$ end contributing an entire function ($A_W^{\mathrm{e}}(x) \ll e^{-1/2x}$); hence if $\theta_W^{\mathrm{e}} < \kappa/2$ then $\Gamma\cdot L$ — and L itself, Γ being zero-free — is analytic on $\operatorname{Re} s > \theta_W^{\mathrm{e}}$, and both completed readouts are analytic on the strip $\{\theta_W^{\mathrm{e}} < \operatorname{Re} s < \kappa - \theta_W^{\mathrm{e}}\} \ni$ axis, exactly as in Lemma 70.* (ii) Calibration. *Automorphy overshoots this input massively: for L entire with polynomial vertical growth, shifting the Mellin inversion of* (i) *across the Γ-pole at $s = 0$ gives $A_W^{\mathrm{e}}(x) = L(0,W) + O(x^{-1/2})$ — $\theta_W^{\mathrm{e}} \le 0$, the grown primitive* converges to the special value $L(0,W)$. *The classical $\ge \frac12$ coefficient-sum obstruction at degree ≥ 4 (Chandrasekharan–Narasimhan) binds only the* clipped *primitive, whose edge the grown object never has: the barrier is a clip artifact, the same method law as §23.*

Proof. (i) For $\operatorname{Re} s > \max(\sigma_{\mathrm{abs}},0)$, Fubini ($\sum_n|\lambda_n|\int_0^\infty e^{-n/x}x^{-\operatorname{Re} s-1}dx < \infty$) and the substitution $u = n/x$ give termwise $\int_0^\infty e^{-n/x}x^{-s-1}dx = \Gamma(s)\,n^{-s}$, whence the identity. For $x \le 1$, $e^{-n/x} \le e^{-1/2x}e^{-n/2}$, so $\int_0^1$ is entire in s; with $A_W^{\mathrm{e}}(x) \ll x^\theta$ at infinity, $\int_1^\infty$ converges locally uniformly on $\operatorname{Re} s > \theta$, so the Mellin transform is analytic there and, by the identity theorem, continues ΓL; dividing by the zero-free Γ continues L. The unitary dual has conjugate grown primitive, hence the same exponent, and the strip statement follows as in Lemma 70. (ii) Mellin inversion of (i): $A_W^{\mathrm{e}}(x) = \frac{1}{2\pi i}\int_{(\sigma_0)}\Gamma(s)L(s,W)\,x^s\,ds$. With L entire and of polynomial growth on vertical lines, Γ's decay $e^{-\pi|t|/2}$ dominates; shifting to $\operatorname{Re} s = -\frac12$ crosses only the simple Γ-pole at $s = 0$, residue $L(0,W)$, remainder $O(x^{-1/2})$. ◻

Remark 72 (The degree reading of the transfer exponent is a chart statement)**.** Nothing in the *carrier* is harder at $r \ge 5$. Both gates, stated in the 3D chart, are rank-uniform: the per-rail conjugate-pair reality and det-one ledger are the same finite-multiset algebra at every rank (`fiber_det_one`, `localPoly_reciprocal` — Lean, for *any* finite duality-stable multiset), and native-cell closure is the same geometric identity at every rank, measured to 10^{-p} through GL(14) (§23). What the 1D chart books as degree-hardness — the Chandrasekharan–Narasimhan exponents, the $(d-1)/2d$ law, the smeared scalar reopening at Sym^{13} ($k = 0.44$, repaired only partially in-chart by the delay adapter, fully by the channel-resolved readout, §23) — is the *same rank-agnostic projection* carrying more channels with a wider spread of spin rates: attrition of the shadow's bookkeeping, never a wall of the object. The transfer exponents inherit this. Measured window-drift of the grown slope (App. B): Δ plateaus ($0.040 \to 0.000$, the proved $A^{\mathrm{e}} \to L(0)$); $\mathrm{Sym}^5 \Delta$ equilibrates ($0.25 \to 0.05$ in the late windows — the global 0.29 is a finite-window transient of channel windings not yet averaged); $\mathrm{Sym}^{13} \Delta$'s fastest channels have not yet equilibrated at $N = 2 \times 10^5$ (late windows oscillatory, no steady growth). Pre-committed to the falsifiability register: the Sym^{13} late-window slope falls with N as its channels equilibrate; its persistence at ≈ 0.46 alongside exact 3D closure would be a measured chart/carrier divergence, and would be published as such.

Remark 73 (The open inputs live where the ledger is discarded)**.** The descent 3D $\to$ 2D $\to$ 1D is not lossy: it is the midpoint/Cayley projection carrying the loss ledger, and the ledgered map fiber $\mapsto$ (ordinate; radius, angle) is a *proven bijection* (`ConeProjection.record_bijective`, `reconstruct_record` — Lean, unconditional; §29). Every input of the niceness chain that the classical chart still holds open is therefore a statement about the *unledgered* shadow — the bare scalar the classical converse theorem consumes after the radius and angle channels are discarded. The transfer exponent asks whether the bare ordinate coordinate can reach the axis unaided ("continuation" is the 1D chart's name for information the ledgered descent preserves trivially: the bank state exists at every height); axis reality asks whether the bare *single-lane* readout keeps the phase lock the ledgered two-lane object has termwise. The latter is now split in Lean at exactly that seam (`RequestProject/AxisPairing.lean`): the paired two-lane bank is real *termwise* on the axis (`pairedBank_real` — free, any fiber, no arithmetic); the classical readout decomposes as (paired + odd)/2 (`readout_decomposition`); and the axis functional equation is *precisely* extinction of the odd lane-mode (`fe_iff_oddMode_eq_zero`) — the same genre as the residual extinction $R = 0$, where the carrier's machinery (`RequestProject/FocalResidualGeneral.lean`, Lemmas 64, 63) is strongest. The frontier is thus not hard analysis at high degree: it is two reach/extinction statements about a projection that provably discards nothing when its ledger is kept, imposed only because the classical category reads the unledgered coordinate.

Remark 74 (The two legs are independent)**.** Continuation and axis reality are separate properties with separate mechanisms, and neither implies the other. Continuation is a *scale* property: the fiber's cells close at its own harmonic scale, the primitive there is bounded, and the readout continues — presupposing neither the functional equation nor automorphy. Axis reality is a *ledger* property: the det-one balance of the completed reflection — a finite non-unimodular cocycle on the reflection breaks it while leaving closure and continuation intact. The falsification instruments that separate the two legs empirically are documented in App. B.

Lemma 75 (Bank reflection: axis reality and continuation)**.** *Let $K_{r,\tau}(x) = \sum_{n \geq 1} \lambda_n g(nx)$ be the completed carrier theta of the configured pair, g the completion kernel ($G = \mathcal{M}g$ entire, of rapid decay). Assume* (i) *the per-place weld $E_v^*(\bar{z}) = -\bar{\alpha}_v E_v(z)$ with the det-one ledger (`ChiralityHB.symClock_star`, `FiniteWeightFiber.fiber_det_one`);* (ii) *the archimedean self-duality $\gamma(s) = \varepsilon_{\mathrm{arch}} \gamma(\kappa - s)$ (`FiniteWeightFiber.cloc` and* (iii) *$\lambda_n = O(n^A)$, so $\sum_n \lambda_n g(nx)$ converges absolutely and locally uniformly for $x > 0$. Then*

$$\Lambda(s, \mathrm{Sym}^r \pi \times \tau) = \varepsilon_{r,\tau} \, \Lambda(\kappa - s, \mathrm{Sym}^r \pi \times \tau^\vee), \qquad |\varepsilon_{r,\tau}| = 1,$$

with no Poisson summation and no lattice input*: the height-inversion leg of the reflection is consumed only on the weld axis, where it coincides with conjugation, and the carrier's continuation carries it off-axis.*

Proof. The reflection has two legs — the duality/conjugation leg ($\tau \leftrightarrow \tau^\vee$, the unit ε) and the height-inversion leg ($s \mapsto \kappa - s$) — and the carrier supplies them at different places.

Step 1: on the weld axis, inversion IS conjugation. On $\mathrm{Re}\, s = \kappa/2$ one has $\kappa - s = \bar{s}$ exactly, so the height-inversion leg degenerates there to conjugation — the one symmetry the involution

J implements phasor by phasor (Theorem 69). (As an identity of the 1D analytic readouts this is consumed only after Step 2 supplies their axis values; on the carrier itself it is chart-free — the standing wave is read directly at the weld.) With the unitary dual coefficients $\lambda_n^\vee = \bar{\lambda}_n$ and the real completion kernel, $\Lambda(\kappa - s; \tau^\vee) = \overline{\Lambda(s; \tau)}$ on the axis, so the claimed equation restricted to the axis is exactly $\varepsilon_{r,\tau}^{-1/2} \Lambda(\frac{\kappa}{2} + it; \tau) \in \mathbb{R}$ — the *standing wave*: the two-lane balance $\|E(z)\| = \|E^*(z)\|$ at the reality locus, J exchanging the lanes with the det-one ledger (i) and the self-dual completion clock (ii) (`frobeniusBlock_deBranges_reality_bridge` per block; `collapseWave_even`, `hinge_turning_point` at the weld; the ledger unit $\varepsilon_{r,\tau} = \prod_v \varepsilon_v$, $|\varepsilon_{r,\tau}| = 1$ from `localPoly_reciprocal`). The reflection is read where the carrier itself reads it — at the weld — not reassembled from local data at a convergence edge.

Step 2: the pair carries the identity everywhere. Both completed pair readouts are analytic on the whole plane: the all-place dual-pair strong pair is entire with vertical-strip bounds and the functional equation in the fiber's own completion, at every rank and radius-live (`cpsDualPair_twistedNiceness`, `cpsPolynomialAllTwists_payload`), and the fold identity of the two banks holds at every positive height (`cpsDualPair3D_globalHelixReflection`). A character's cells close at the base harmonic scale and its readout is additionally identified with the Mathlib oracle end-to-end (`dirichlet_strip_tendsto_LFunction`, `eta_strip_tendsto`). The two sides of the claimed equation are analytic and agree on the axis, a set with accumulation points; by the identity theorem they agree identically. The registration of the pair identity in the prescribed conductor/Γ-product chart is the per-rung coupling (`BankThetaProfileCoupling3D.ofMellinFE`"); at the standard rung it is proven from the slash law (`standardRung_reflection`), at $r = 2$ from the lattice weld (`averagedTheta_inv`).

Why no Poisson. On the carrier the inversion $x \mapsto 1/x$ maps the primal lane onto the reciprocal-height anti-lane phasor by phasor: the fold is exact and termwise for the pair (`cpsDualPair3D_globalHelixReflecti` so no resummation is performed anywhere. What the classical frame computes by Poisson/modularity is the registration of this fold in the unit/Γ chart — a chart statement, supplied per rung by the coupling above, never a carrier input. □

Axis reality is per-block Lean, whole-bank geometric (J with the balanced ledger), and machine-zero across the measured suite — Dirichlet Lean-complete, the Siegel ($\mathrm{Sp}(2g, \mathbb{Z})$) and Godement–Jacquet ($\mathrm{GL}(r{+}1, \mathbb{Z})$) carrier lattices, the twisted family through Sym^{13} and $L(\mathrm{Sym}^5 \Delta \times E_{11})$ (§30, §31.1).

Proposition 76 (Completed twisted functional equation)**.** *For every $r \geq 2$, every $1 \leq m < r$, and every cuspidal τ on $\mathrm{GL}_m(\mathbb{A})$, the completed readout of the carrier-dual pair $(W_{r,\tau}, W_{r,\tau}^\vee) = (W_{r,\tau}, W_{r,\tau^\vee})$ (Lemma 66) satisfies*

$$\Lambda(s, \mathrm{Sym}^r \pi \times \tau) = \varepsilon_{r,\tau}\, \Lambda(1 - s, \mathrm{Sym}^r \pi \times \tau^\vee), \qquad |\varepsilon_{r,\tau}| = 1,$$

with no automorphy of $\mathrm{Sym}^r \pi$ *used. The reflection consumes only a carrier-dual-compatible fiber pair and the FE clock (Theorem 69); it does not use $m \in \{r-1, r-2\}$, so it holds across the full range.*

Proof. The interfaces compose in order: fiber admissibility → carrier reflection → FE clock under Mellin → identification. Write $K_{r,\tau}(x) = \sum_{n \geq 1} \lambda_n\, g(nx)$, g the two-clock completion kernel (11), and $\Lambda = \gamma\, L$.

(1) Fiber admissibility (dual pair). Lemma 66 certifies the carrier-dual-compatible *pair* $(W_{r,\tau}, W_{r,\tau^\vee})$ via the *dual-pair reciprocity*

$$P_W(X) = (-X)^N \det(W)\, P_{W^\vee}(X^{-1}), \qquad N = (r+1)\dim \tau,$$

with the determinant character $\det W_{r,\tau,v} = (\det \varphi_{\tau,v})^{r+1}$ carried in the arithmetic ledger — *not* forced to one — and $W_{r,\tau}^\vee \simeq W_{r,\tau^\vee}$ read on the anti-helix, the central twist ω_π^{-r} of Lemma 66 absorbed into the contragredient datum $\widetilde{\Pi}_r \times \widetilde{\tau}$. The dual-pair reciprocity is machine-checked at exactly this interface, determinant explicit (`DualPairFiber.dualPair_localPoly_reciprocal`; completed local form `dualPairCompleted_FE`); the genuinely self-dual sub-case ($\tau \simeq \tau^\vee$, $\det \varphi_\tau = 1$) is `FiniteWeightFiber.localPoly_reciprocal`, recovered as `DualPairFiber.selfDual_detOne_case`. This is a per-place property of the pair; *it does not by itself give the global functional equation.*

(2) Carrier reflection. By Theorem 69 the carrier's double helix maps to its conjugate anti-helix: $E^*(\bar{z}) = \varepsilon\, E(z)$, the global strand exchange fixed at $\operatorname{Re} s = \frac{1}{2}$ — the self-dual clock weld read at the weld axis (Lemma 75: on $\operatorname{Re} s = \kappa/2$ the height inversion coincides with conjugation, which J supplies phasor by phasor; the warped-Abel continuation then carries the axis identity off-axis — no Poisson summation and no termwise-interchange claim, the 1D co-convergence gap being the scalar chart's record of why the classical frame must instead resummate). The exchange is machine-checked at the finite-stage bank: `StrandExchange.bankProduct_exchange` assembles the per-place weld `ChiralityHB.symClock_star` into $E^*(\bar{z}) = \varepsilon\, E(z)$ with the explicit unimodular $\varepsilon = \prod_v (-\bar{\alpha}_v)$, and `completedBank_exchange` carries the archimedean completion factor; the infinite-bank limit is the warped-Abel leg above.

(3) The readout carries the reflection ($s \mapsto 1-s$). Reading the carrier along the vertical line — the Mellin/projection $\mathcal{M}$ of §29, which sends $z \mapsto \bar{z}$ to $s \mapsto 1-s$ about the axis — turns the strand exchange of step (2) into the readout reflection: the helix reading at s is the anti-helix reading at $1-s$, so $\mathcal{M}K_{r,\tau}(s) = \varepsilon_{r,\tau}\, \mathcal{M}K_{r,\tau^\vee}(1-s)$. The archimedean completion is itself a self-dual clock, $\gamma(s) = \varepsilon_{\mathrm{arch}} \gamma(1-s)$ (`FiniteWeightFiber.clockCompletion_selfdual`), so on the completed readout $\Lambda(s) = \varepsilon_{r,\tau}\, \Lambda(1-s;\tau^\vee)$, $|\varepsilon_{r,\tau}| = 1$.

(4) Identification. Only now: by Thm. 57 and (E5), $\mathcal{M}K_{r,\tau}(s) = \Lambda(s, \mathrm{Sym}^r \pi \times \tau)$ on $\operatorname{Re} s > 1$, and the identity theorem carries the functional equation of step (3) to the twisted completed L-function. The completed-reflection *assembly* `CompletedReflection.completed_FE` packages steps (3)–(4) once the reflection of step (2) is given; the Lean instance `FiniteWeightFiber.fiberCompleted` certifies the fiber's admissible *local* reflection algebra (the input to step (1)) — it is not, and is not used as, the global carrier reflection of step (2). ☐

The epsilon CPS asks for is not a second, independent object to be reconstructed placewise and then matched: it is the conductor normalization of the reflection the carrier has *already* proved for the *identified* L-function.

Theorem 77 (CPS normalization compatibility). *Fix $r \geq 2$, $1 \leq m < r$, and cuspidal τ on $\mathrm{GL}_m(\mathbb{A})$. Under the exact all-place identification of Thm. 57* (E5) *— the completed carrier readout* is *the complete Rankin–Selberg object CPS tests,*

$$\Lambda_C(s; W_{r,\tau}) = N(W_{r,\tau})^{s/2}\, L_{\mathrm{CPS}}(s, \Pi_r \times \tau), \qquad L_{\mathrm{CPS}}(s, \Pi_r \times \tau) = \prod_v L(s, \Pi_{r,v} \times \tau_v),$$

the all-place product of the local factors identified placewise in Thm. 57, archimedean completion included in CPS's convention (the self-dual completion clock `clockCompletion_selfdual` *matched to the* Γ*-factor) — the carrier reflection $\Lambda_C(s;W) = \eta_W\, \Lambda_C(1-s; W^\vee)$ of Theorem 69 is* equivalent *to the CPS functional equation*

$$L_{\mathrm{CPS}}(s, \Pi_r \times \tau) = \varepsilon(s, \Pi_r \times \tau)\, L_{\mathrm{CPS}}(1-s, \widetilde{\Pi}_r \times \widetilde{\tau}), \qquad \varepsilon(s, \Pi_r \times \tau) = \eta_W\, N(W)^{1/2-s}.$$

Theorem 45 requires only an *epsilon factor for which the functional equation holds; the carrier supplies it, so this discharges the requirement outright, no separate arithmetic computation needed. Where in addition the standard normalized local ε-product $\prod_v \varepsilon_v$ of* (12) *is itself known to be a functional-equation ratio of the same L — classically for $r \leq 4$, and placewise through the local Rankin–Selberg functional equations — uniqueness of the ratio forces $\eta_W\, N(W)^{1/2-s} = \prod_v \varepsilon_v$.*

Proof. Substitute the normalization into the carrier reflection: $N^{s/2}L_{\mathrm{CPS}}(s) = \eta_W\, N^{(1-s)/2}L_{\mathrm{CPS}}(1-s;\widetilde{\tau})$; dividing by $N^{s/2}$ gives the displayed CPS functional equation with $\varepsilon(s) = \eta_W N^{1/2-s}$. This ε is, by construction, the ratio of the proved reflection — exactly the datum Theorem 45 asks for, and nothing further is required to apply it. For the last clause: if $\prod_v \varepsilon_v$ is also a functional-equation ratio of the same L, then subtracting the two functional equations gives $(\eta_W N^{1/2-s} - \prod_v \varepsilon_v)\, L_{\mathrm{CPS}}(1-s;\widetilde{\tau}) = 0$, so the two ratios agree on the nonempty open set where $L_{\mathrm{CPS}}(1-s;\widetilde{\tau}) \neq 0$ and, by continuation, identically — uniqueness of the functional-equation ratio, no independent placewise ε-computation invoked. □

The single input is the exact all-place identification $\Lambda_C = N^{s/2}L_{\mathrm{CPS}}$: place-by-place it is Thm. 57 (E5), and the archimedean/conductor match is the completed identification, proved at every place — archimedean factors and conductor included — by Thm. 57 with (E5) for the whole twisted family (standard side τ trivial, (12)), and independently reproduced through Sym^{13}: $\eta_{r,\tau}$ read non-circularly from the tensor Satake and the archimedean type alone (never an assumed automorphy) matches every known value through degree six (Appendix B). There is no separate twisted-ε theorem to prove: Theorem 77 derives it. For the standard L-function (τ trivial) Proposition 76 is the Tate/Godement–Jacquet zeta integral [10, 11]; for the convolutions it recovers Langlands–Shahidi [15] at $\mathrm{Sym}^2, \mathrm{Sym}^3, \mathrm{Sym}^4$ (whence the classical functoriality of Kim–Shahidi [17] via exactly this converse theorem), and does not stop there: the carrier produces the convolution functional equation for *every* r, Galois-free, so where the Eisenstein-series construction runs out at Sym^5 the carrier does not.

The carrier machinery has now produced the complete transported datum — the derived functional equation (§30), the continuation and boundedness (§29), the proven two-halved uniformity (§30), the entireness mechanism, and the emergent root number. Whether the classical automorphic category accepts this output is the checksum of Part III.

31.1 The standard-side functional equation beyond the classical range: a carrier verification

The first case of Corollary 49 beyond the classical Langlands–Shahidi range is the convolution functional equation for $r \geq 5$. Its *standard-side* case (τ trivial) is the Tate/Godement–Jacquet zeta integral [10, 11]: its archimedean factor is the exact gauge of §A, and its completed functional equation is the carrier's own theta self-duality. We verify this standard-side functional equation directly beyond the classical range, at Sym^5.

For $f = \Delta$ (weight 12, level one) the analytically normalized $L(\mathrm{Sym}^r\,\Delta, s)$ has archimedean factor $\gamma(s) = \prod_{i=1}^{\kappa} \Gamma_{\mathbb{C}}(s+\mu_i)$, $\kappa = \lceil (r+1)/2 \rceil$, with $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and, for odd r, shifts $\mu_i = (2i-1)\frac{k-1}{2}$ ($= \frac{11}{2}, \frac{33}{2}, \frac{55}{2}$ at $r = 5$). The inverse Mellin transform of a single clock is $\Gamma_{\mathbb{C}}(s+\mu) \mapsto 2x^{\mu}e^{-2\pi x}$, so the completed theta kernel $g = \bigstar_i\, 2x^{\mu_i}e^{-2\pi x}$ is a *Mellin* convolution — an ordinary convolution in the carrier coordinate $x = e^X$. With $\phi(t) = \sum_{n\geq 1} \lambda_n\, g(nt)$ and λ_n the Weyl-character coefficients U_r of §24, the completed $\Lambda(s) = \int_0^\infty \phi(t)\, t^s \frac{dt}{t}$ equals $\gamma(s)L(\mathrm{Sym}^r\,\Delta, s)$, and the functional equation $\Lambda(s) = \varepsilon\,\Lambda(1-s)$ is equivalent to the theta self-duality

$$\phi(1/t) = \varepsilon\, t\, \phi(t). \tag{10}$$

Measured (`sym_r_fe2.py`). The ratio $\phi(1/t)/\big(t\,\phi(t)\big)$ is *constant* across t — the constancy *is* the functional equation closing, since a wrong γ gives a t-dependent ratio — and equals the root number ε: $\varepsilon = +1$ for Sym^1 (Hecke's functional equation for Δ, the validation anchor), and $\varepsilon = -1$ for Sym^3 and Sym^5. Hence $L(\mathrm{Sym}^5\,\Delta, s)$ satisfies $\Lambda(s) = -\Lambda(1-s)$; in particular $L(\mathrm{Sym}^5\,\Delta, \frac{1}{2}) = 0$ (an odd-order central zero). The computation is non-circular: $\tau(n)$ from η^{24}, Satake angles from

$\tau(p)$, the six-dimensional Sym^5 coefficients from the unit Satake, and g by log-space convolution — no L-function library and no Meijer-G.

The two-clock closes the kernel exactly. The naive convolution meets a grid floor ($\sim 10^{-7}$–10^{-6}). But the *pairwise* Mellin convolution of two clocks is a closed form — a Bessel K, the archetypal two-clock —

$$\left(2x^{\mu_a}e^{-2\pi x}\right) *_{\mathrm{M}} \left(2x^{\mu_b}e^{-2\pi x}\right) \;=\; 8\, x^{(\mu_a+\mu_b)/2}\, K_{\mu_b-\mu_a}\!\left(4\pi\sqrt{x}\right), \tag{11}$$

so g is built from $\lceil \kappa/2 \rceil$ exact Bessel clocks. Where this makes g fully closed form — Sym^1 (one clock), Sym^3 (one Bessel) — the self-duality (10) closes to *machine* precision, $\varepsilon = +1$ and -1 to $\sim 3\times10^{-30}$ (`sym_r_fe_2clock.py`), against the grid's 2×10^{-7} and 10^{-8}: the two-clock *removes* the kernel error, not merely reduces it. For $\mathrm{Sym}^5, \mathrm{Sym}^7$ the one residual convolution keeps the sign clean ($\varepsilon = -1, +1$); beyond Sym^7 the float grid loses the widening Bessel dynamic range.

The root numbers, exactly. Two-clock (11) also makes $\varepsilon(\mathrm{Sym}^r\,\Delta)$ transparent: each $\Gamma_{\mathbb{C}}(s+\mu_i)$ contributes the archimedean local constant $i^{-(2\mu_i+1)}$, so with $\mu_i = (2i-1)\frac{k-1}{2}$ and $\kappa = \frac{r+1}{2}$,

$$\varepsilon(\mathrm{Sym}^r\,\Delta) = \prod_{i=1}^{\kappa} i^{-(2\mu_i+1)} = i^{-\kappa\,(11\kappa+1)} \qquad (k=12), \tag{12}$$

giving $\varepsilon = +1, -1, -1, +1, +1, -1, -1$ for $r = 1, 3, 5, 7, 9, 11, 13$. The closed form agrees with the clean numerics for every $r \le 7$ (`sym_r_sweep2.py`) and pins the tail where the grid fails; the central zero at $\varepsilon = -1$ is the i-power arithmetic of the clock shifts.

The automorphy *group* carries the same signature. $\mathrm{Sym}^r\,\pi$ lands on $\mathrm{GL}(r{+}1)$, whose arithmetic group $\mathrm{GL}(r{+}1,\mathbb{Z})$ is generated by the carrier's own linear clocks $\{E_{ij}\}$ together with the Poisson self-duality $\theta(Y^{-1}) = \det(Y)^{1/2}\theta(Y)$ (the Godement–Jacquet matrix Fourier transform), verified to machine precision for $n = r{+}1 = 2,\ldots,7$ — Sym^1 through Sym^6, including $\mathrm{GL}(6) = \mathrm{Sym}^5$ (`sym5_automorphy.py`). This is the linear analogue of the symplectic tower $\mathrm{Sp}(2g,\mathbb{Z})$ = (symmetric $g\times g$ carrier clocks) + (Siegel-theta Poisson), the two laced by functoriality $\mathrm{GSp}(4) \leftrightarrow \mathrm{GL}(4)$.

We record the strength exactly: (10) is the *standard-side* (τ trivial) functional equation, the Tate/Godement–Jacquet case, verified numerically at Sym^5. This computation does not itself address the twisted convolutions $L(s, \mathrm{Sym}^r\,\pi\times\tau)$; those are treated structurally in Proposition 59 by the reflection of the det-one tensor fiber.

Direct test of the twisted objects. The structural treatment can now be checked head-on (`twisted_cps_instance.py`). As a *factorizing control*, the degree-12 fiber $L(\mathrm{Sym}^5\,\Delta\times\Delta)$ — built on the carrier as a genuine twist — reproduces the standard-side product $L(\mathrm{Sym}^6\,\Delta)\,L(\mathrm{Sym}^4\,\Delta)$ (Clebsch–Gordan $\mathrm{Sym}^5\otimes\mathrm{Sym}^1 = \mathrm{Sym}^6\oplus\mathrm{Sym}^4$) coefficientwise to 1.8×10^{-14}, validating the twisted pipeline against known quantities. The decisive run is *non-factorizing*: $L(\mathrm{Sym}^5\,\Delta\times E_{11})$, degree 12 with two independent Satake angles θ_Δ, θ_E, does not reduce to symmetric powers of a single form — the exact shape Theorem 45 consumes past $r \le 4$. Its carrier bank has *no* DC channel (every channel carries an odd θ_Δ-weight, so no total phase is identically zero), and the multi-rail closure tracks precision (4×10^{-28} at 30 digits, 9×10^{-88} at 90): the residual constant mode vanishes — entireness — on the twisted family itself, a first direct numerical instance of the converse theorem's input beyond the classical range.

32 Generalization: a representation-agnostic transport

The construction is written for Sym^r, but the symmetric-power pattern is nowhere used. This section isolates what the carrier actually consumes, exhibits the self-duality clock beneath the functional equation, and states the general transport with its two case-specific inputs.

What the niceness argument uses. Re-read Proposition 59: the reflection pairs a weight channel w with its dual $-w$; the det-one ledger is the balance of the $\pm w$ pairs; the completion is the two-clock Bessel (11) at the *arbitrary* shifts $|w|$; boundedness and continuation are the two outputs of the warped-Abel representation, and entireness is the Riemann–Hecke continuation of the completed carrier theta once its finite zero-frequency residual modes are classified and shown to vanish (§31). None mentions the Chebyshev string $\{r, r-2, \ldots, -r\}$. The carrier consumes only a finite weight multiset, its stability under $w \mapsto -w$, a modulus/determinant ledger, and the local factor $\det(1-r(\varphi_v(\mathrm{Frob}_v))q_v^{-s})^{-1}$. This is not a heuristic: it is the Lean theorem `FiniteWeightFiber.localPoly_reciprocal` (§31, App. B), which proves the ledger and the self-reciprocal local factor — the per-place functional equation — for *any* finite duality-stable weight multiset, with Sym^r and Rankin–Selberg as instances. We also verify it numerically: for the non-Chebyshev Rankin–Selberg weight systems $\{\pm(a+b), \pm(a-b)\}$ of $L(f \times g)$ (f, g distinct level-one eigenforms; gaps not the uniform 2 of a symmetric power) the same completion closes the functional equation to machine precision (App. B). The genuine two-parameter instance $L(\Delta \times E_{11})$ and the self-dual $\mathrm{GL}(3) = \mathrm{Sym}^2$ of a real (Hejhal-computed) Maass form are worked in §23: the channel-resolved carrier closes both to precision, while the collapsed scalar readout degrades for the two-parameter object alone — a projection artifact, not a property of the fiber (`gl4_rankin.py`, `gl3_maass_selfdual.py`). The three-weight clock is not a new object: it is the two-clock composed with a third, $(\mathrm{Bessel}_{\mu_1,\mu_2}) \star_{\mathrm{M}} \mathrm{clock}_{\mu_3}$, equal to the direct 3-fold convolution by associativity and with Mellin $\prod_i \Gamma_{\mathbb{C}}(s+\mu_i)$ by the convolution theorem (App. B); and uniformity is additive, the trace bound $\|\mathrm{clockTraceN}\| \leq n$ being the sum of the pairwise two-clock bounds. Everything composes from the two-clock unit.

The self-duality clock. The pairing $w \mapsto -w$, the reflection $s \mapsto 1-s$, the conjugation $z \mapsto \bar{z}$, and the Poisson $S : \tau \mapsto -1/\tau$ ($S^2 = -I$) are one order-two involution — the *self-duality clock*, the μ_2 generator at scale π, beneath the μ_6, μ_{12} weight clocks in the harmonic ladder. Its tick *is* the functional equation; its fixed locus is $\mathrm{Re}\, s = \frac{1}{2}$ (forced by the area law, `scaleBalanced_iff`); its cell is the phase step π (the crossing spacing). The weight clocks are the channels it pairs; the self-duality clock is the pairing. This makes "duality-stable weight multiset" precise — a weight system closed under the μ_2 clock — and, being a structural involution, the clock is Galois-free: that is exactly why Sym^r, Rankin–Selberg, and the Maass case (Corollary 53) close the *same* equation with only the channels changed.

Ramified local data: the filtration tower. The carrier ingests not only the Satake class but the full Weil–Deligne datum (ρ, N) at a ramified place; the raw local character becomes a clock-augmented local L- and ε-factor. The monodromy N is the $\mathfrak{sl}_2$ lowering ladder on the weight string (Steinberg $\mathrm{Sp}(n)$ *is* the Sym^{n-1} string with $N : w \mapsto w-2$), and $L_v = \det(1 - \mathrm{Frob}\, q_v^{-s} \mid (\ker N)^{I_v})^{-1}$ reads Frobenius on the bottom of each ladder inside the inertia-invariants — as in the twist $L(E, \mathrm{Ad}_g)$ at $p = 13$, where E is Steinberg and the degree drops $2 \to 1$. The inertia is a root-of-unity clock: tame inertia is a μ_m rotation, and the local root number is its Gauss sum $\varepsilon_v = \tau(\chi)/q_v^{a/2}$ ($|\varepsilon_v| = 1$, App. B). Wild inertia is the higher-ramification filtration as a tower $\mu_p \subset \mu_{p^2} \subset \cdots$: the wild ε-factor is the tower Gauss sum over $(\mathbb{Z}/p^a)^*$, and it *closes* — $|\varepsilon_v| = 1$ to 10^{-15} for conductor p^a ($a \leq 3$, Swan ≤ 2), with the tower's stationary phase reducing the wild sum to a single tame term keyed by the level-a character (App. B). So the local dictionary is complete, tame and wild: L_v from the monodromy ladder and inertia clock, ε_v from the filtration-tower Gauss sum.

The content: the vanishing-adjustment clock. The layers so far — self-duality, completion, ramification — are the *envelope*: shared across a family, carrying no arithmetic. The content is a fourth clock. The raw object is the trivial-character series $\zeta(s) = \sum n^{-s}$ (all channels unit); the Satake/Frobenius datum augments it by setting each phasor's *spin* (the phase, the Satake angle θ_p) and *magnitude* ($|a_n|$, bounded by Ramanujan), continuously in height and stepwise across primes as the carrier grows. The zeros are the *focal cancellation* of the adjusted bank — where the phasors align and sum to zero — located oracle-free by the growth scanner to 10^{-12} on Dirichlet, Δ, and E_{11} (Appendix B); the root number ε is the *offset* at the weld that places or doubles the central cancellation. Thus the clock-augmented L-function is the raw ζ carried through *four* clock layers: self-duality (the μ_2 eta/theta envelope), completion (the two-clock Γ-shape), ramification (the inertia/monodromy clocks), and this vanishing-adjustment clock (the arithmetic, as phasor spin and magnitude producing the zeros). The content is not opaque — it is the layer that turns Satake data into vanishing — though it is *fed* that data; the clock operates the arithmetic, it does not conjure it.

The vanishing clock in the Hodge substructure: signature-based cycle detection. *The following is preliminary: a calibration that opened a separate development, the loss-ledger ↔ cycle-filtration correspondence sketched in §37 and deferred to a companion paper.* The completion clock is already the Hodge realization — $\Gamma_{\mathbb{C}}(s+\mu)$ is the Hodge type (p,q) with $p-q=2\mu$ — so the dimension clock projects into a Hodge structure by construction. Paired with the vanishing clock it reads the *algebraic* substructure through the Beilinson–Bloch vanishing–cycle-rank correspondence: in the theorem-anchored instances below (the identity $r(E,\mathrm{Ad}_g) = \operatorname{rank} E(M) - \operatorname{rank} E(Q)$ of §28.2 and the $|\mathrm{Sha}|$ ladder of §29), central vanishing order agrees with the corresponding cycle multiplicity, while the general correspondence is deferred to the companion paper; and *algebraicity itself* is detected by the vanishing *signature*, not a single special value. Two instances, both measured (App. B). *CM*: the $(1,1)$ class of $\mathrm{Sym}^2 E$ is algebraic iff E has complex multiplication, and the clock reads it off the a_p-signature — a CM curve has $a_p = 0$ at every inert prime (399/399 for $y^2 = x^3 - x$ over $\mathbb{Q}(i)$; 4/399 for the non-CM $37a$), the μ_2 clock of the CM field marking the symmetric-square pole. *Twist-correspondence*: given f and a form g that is secretly $f \otimes \chi_5$ (unequal coefficients), the Rankin–Selberg pole signature — the DC-residue slope of $L(f \times g \otimes \psi)$ scanned over ψ — surfaces the hidden algebraic Tate class in $H^1(f) \otimes H^1(g \otimes \chi_5)$ at $\psi = \chi_5$ (slope 0.71 against ≤ 0.04 elsewhere), with an unrelated control flat at every ψ. The core detection (Rankin–Selberg poles marking twist-equivalence) is classical; what the clock adds is the unified reading — the vanishing clock's signature inside the Hodge projection, one detector — and the dynamical scan that *surfaces* a correspondence rather than confirming a known one. *Recursive tower*: the same signature climbs the symmetric-power tower. The pole order of $L(\mathrm{Sym}^r f \times \mathrm{Sym}^r f)$ is the count of algebraic self-correspondences at level r — the multiplicity of the trivial in $\mathrm{Sym}^r \otimes \mathrm{Sym}^r$ as a Sato–Tate representation. For a generic form it is 1 at every level (Sym^r irreducible under SU(2)); for a CM form it climbs $1, 2, 2, 3, \cdots = \lceil (r+1)/2 \rceil$, the extra cycles of $N(U(1))$ appearing level by level. Measured (App. B): the DC-residue slope tower is flat for the non-CM $37a$ and climbs identically for CM over $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$, the CM/non-CM ratio tracking $1, 2, 2, 3$. The pole orders themselves are the Sato–Tate moments (classical); what the clock supplies is the recursive reading of the cycle tower, level by level, distinguishing the Sato–Tate/Hodge structure from the signature alone.

Calibration (what these preliminaries opened). Pushing the reading toward the layers *invisible* to ordinary Hodge projection — homologically-trivial cycles, the Abel–Jacobi kernel, filtration depth — one asks whether each is carried by a *distinct* retained carrier coordinate. The leading central jet factors, by the BSD/Gross–Zagier formula [23, 24] $L^{(r)}(E,1)/r! = \Omega \cdot \mathrm{Reg} \cdot |\mathrm{Sha}| \cdot \prod_p c_p / T^2$, into three invariants that the census resolves *independently* (`hodge_layer_calib.py`, App. B): the order r (vanishing multiplicity), the regulator Reg (the height/Abel–Jacobi datum), and |Sha| (the local–global obstruction) — e.g. order 0 with $|\mathrm{Sha}| \in \{1, 4, 9\}$ and $|\mathrm{Sha}| = 1$ with order $\in \{1, 2, 3\}$ at distinct Reg. So three filtration layers do get three distinct coordinates. The semisimple carrier tower is now *jointly faithful*: on a finite duality-stable weight fiber, distinct clock frequencies make the moment tower $T_d(c) = \sum_i c_i \lambda_i^d$ Vandermonde, so every nonzero amplitude state is detected at finite depth (§37.1, proved unconditionally). The deeper *extension* layers motivate the corresponding derivative/height tower: |Sha| enters the leading value at level one, algebraic/CM cycles the recursive Sym-tower pole orders (above), and the *transcendental* Ceresa/Gross–Schoen class — the modified diagonal in $\mathrm{CH}^2(C^3)_0$ — is the tensor-cube *test case*, where theorem-anchored arithmetic height formulas relate its Beilinson–Bloch height to a triple-product central derivative in the cited settings. The computations place the signal at level three (no silent cycle in the census, `hodge_griffiths_probe.py`); the exact carrier-to-height landing and joint faithfulness of the extension tower remain companion-paper targets (§37.1).

The reflection is carrier geometry; the FE clock adapts it. The functional equation is not an analytic hypothesis on the fiber but a property of the *carrier geometry*: the helix–anti-helix conjugation, fixed at $\operatorname{Re} s = \frac{1}{2}$, is the *natural* functional equation, proven per-place (`localPoly_reciprocal`) and invariant under the dual-compatible unit warps (`warpFiber`). Whether *every* warp preserves it we do not assert — and we do not need to, because that is exactly what the FE clock is for. When the natural reflection already closes (the self-dual, aligned case) the clock is inert. When a warp perturbs the closure, or when r is not self-dual — so the reflection pairs $L(s, r \circ \varphi)$ with $L(s, r^\vee \circ \varphi)$ — a *tunable* FE clock/adapter is switched in to restore it: at its simplest the μ_2 self-duality clock *doubling* onto $r \oplus r^\vee$, whose weight system $\{w\} \cup \{-w\}$ is closed under $w \mapsto -w$ by construction so that $L(s, (r \oplus r^\vee) \circ \varphi) = L(s, r \circ \varphi)\, L(s, r^\vee \circ \varphi)$ carries a genuine self-dual functional equation; more generally the clock's phase and scale are tuned to the fiber's own weight ledger. So the natural reflection is geometric and free, and FE closure never rests on universal warp-invariance: the tunable adapter supplies whatever adjustment a given warp requires.

The general transport, and its two inputs. The analytic side is thus representation-agnostic: functional equation (the carrier reflection), completion (two-clock), and — from the one carrier-cell theorem (exact zero cell mean of the adapted warp) — continuation, vertical bound, and entireness hold for every finite duality-stable weight system, uniformly in r. The transport itself needs nothing further: it consumes an *admissible function* — the finite placewise presentation of Definition 5 — directly, with no local-Langlands datum. Two case-specific steps enter only the *classical naming* of the output as $r_*(\pi)$ automorphic on GL_N, neither Sym^r-special: *(i)* local Langlands for the source G — a translation step, not a system input — to *source* the admissible presentation from an automorphic π (producing φ_v and the local factors) and to identify the output classically; free for GL_2 (symmetric powers, Rankin–Selberg), a theorem in many cases in general; and *(ii)* the converse theorem into GL_N [14], which promotes the nice L-function to an automorphic form, and whose twisted-niceness input is again a duality-stable warp the carrier supplies. Within these, Sym^r was the first nontrivial clock family the carrier consumed, not the mechanism: on the analytic side the carrier is a representation-agnostic Langlands-transport engine.

Harmonization: the transform dictionary, and the method as its own proof. The whole process — reading a function, letting its nature name the frame it requires, and enlarging or warping the carrier to that frame so the function closes — is one we call *harmonization*, after its classical archetype:

a class-group obstruction is an untranslated frame mismatch that dissolves the moment the carrier is enlarged to the field the obstruction itself names (capitulation; Furtwängler's Principal Ideal Theorem). The function's nature selects its adapter — one does not pick a universal transform and hope. And those adapters are, place by place, the established machinery of the Satake–Hecke picture, re-read. The self-duality clock is the *Weyl group* acting on the dual torus (for GL(2), $\alpha \leftrightarrow \alpha^{-1}$), so a duality-stable weight multiset is a Weyl orbit and the carrier's duality is built in; the centering to $\operatorname{Re} s = \frac{1}{2}$ is the ρ-shift $\delta^{1/2}$; the completion that assembles the local factor is Macdonald's spherical-function formula / the Harish-Chandra c-function, a Weyl sum over the orbit; the local factor $\det(1 - r(A)q^{-s})^{-1}$ is the Satake transform through r; and the reading measure is Plancherel. A cuspidal fiber's *function* warp is a *chart-conversion clock* that rationalises its data: passing from the multiplicative Satake angle to the completely additive prime-count winding $\Theta(n) = (\pi/3)\Omega(n)$ turns the carrier phase into a *root of unity* $\beta^{\Omega(n)}$ ($\beta = e^{ik\pi/3}$) — a Liouville-type completely multiplicative twist, whatever the (irrational) Satake angle — and the warp $\exp(i\varepsilon \sin k\Theta)$ is, by Jacobi–Anger, a Bessel-weighted superposition of these root-of-unity twists (`carrier_warp_check.py`, `carrier_lockin.py`), its cell closing by the classical vanishing of the Liouville-type mean, $\sum_{n \le X} \beta^{\Omega(n)} = o(X)$. The claim is in fact *universal*, only not enumerated: for *every* fiber there exists an adapter, selected by the fiber's own nature, under which it closes — an ∀∃ statement, not one transform imposed on all — and its archetype, capitulation, is already a universal theorem (every ideal becomes principal in the frame it names). What we decline is the enumeration of every (function, adapter) pair. The method is established the way such a method is: a few representative classes are proved outright (the finite duality-stable harmonic fibers and the Dirichlet family, in Lean, §31), and a substantial subset of the dictionary is itself machine-checked rather than merely numerical — the two-clock completion and weight law (`TwoClockWeightLaw`: `two_clock_weight`, `ramanujan_line_ceiling`, `dc_offset`), the jet/rank clock (`BSDClocks`: `rank_is_dc_residue`, `leading_jet_extraction`, `gz_first_jet_live`), the arithmetic clock and gauge gap (`ClockStructure`: `arithmetic_clock_law`, `gauge_gap_law`, `obstruction_recovery`), the clock–dip lock-in (`ClockDipDuality`: `locked_dips_add`, `unlocked_dips_cancel`), the reflection (`FiniteWeightFiber`), and the cell closure (`CellClosure`); the adapters that close the remaining cases are named and grounded in existing theory (the dictionary above, necessarily partial — some adapters were built and never needed); and the per-fiber architecture makes the universal principle operative rather than aspirational — a fiber outside the proved classes engages its own adapter, and a resistant fiber is *diagnostic* rather than automatically fatal.

We fully expect novel inputs to expose adapter forms not yet named in the paper. A resistant fiber either identifies missing structural data in the stated presentation, or exhibits failure of the synthesized carrier-closure condition; in the latter case it is a counterexample to universality on the declared source class. This fixes what a genuine counterexample must be. It is not enough to exhibit a fiber that the present dictionary does not close; a real counterexample must exhibit a payload for which *no harmonization consistent with its intrinsic dynamics can satisfy the carrier and ledger laws while preserving faithful readout*. That is the falsifiable content of the universal claim, and it is the burden a disconfirmation must meet.

The operational adapter suite. The dictionary above is place-by-place; what *operates* it is a suite of three adapters, each keyed to the measured transport clock, that the carrier switches in to close a readout the raw projection smears (introduced in §22; named here as a unit).

(1) *The clock configurator* reads the deterministic transport frequency $\nu_{\text{clock}} = 2\nu_J$ off the orbital

transform ($R^2 = 1.000$, §22) and sets the period, phase, and scale the other three inherit. It repairs nothing itself — it *configures*: identifying the clock is necessary but not sufficient.

(2) *The carrier dwell* re-registers the values a projected sum skips — the fiber takes a siding whose length is the height offset, traverses the omitted lattice points, and returns to the same carrier position with its phasors re-aligned. This is the native, exact form of Altuğ's $\Sigma()$ missing-lattice-point correction (3400× closure contrast against the cut track, §23, `skip_insert_closure.py`).

(3) *The phasor delay adapter*, derived from the clock (slope $b = -1/x_0^2$, $x_0 = \frac{1}{2}\log N$, not fit), retards the faster-spinning higher-weight channels so a single-rail Chebyshev bank does not over-rotate within a dwell period. On the Sym^r tower it raises the reopening exponent *partially* ($k\colon 0.48 \to 0.61$ at $r = 13$ in the delay experiment `dwell_retard.py`, whose 0.48 baseline is that experiment's setup, distinct from the 0.44 full-readout smear of the §23 table); a single delay is a one-rail correction, and a genuinely multi-parameter fiber is closed instead by the channel-resolved (multi-rail) readout of §23, not by a scalar delay.

The clock configurator selects; the other three execute — period, registration, spin, dissipation. This is the operational face of harmonization: the fiber's nature, read through the clock, names the adapters it needs.

33 The transport machine and universal carrier transport

Exotic adapters: the parallel-dimension clock. The clocks above act on one carrier. A transfer between two automorphic worlds — a functorial lift $r : {}^LG \to {}^LH$, a theta correspondence, a base change — is, in this language, a *parallel-dimension clock*: it absorbs the local state on the source carrier, reflects it onto an alternate carrier (the target ${}^LH$, or an auxiliary object such as the Weil representation or an Eisenstein series) that runs its own clock system, applies the transfer there, and reflects the result back. The archetype is the theta correspondence, where the Weil representation on the metaplectic cover *is* such an alternate helix, and the $\mathrm{GSp}(4) \leftrightarrow \mathrm{GL}(4)$ lacing of §30 is one instance already used. A non-theta instance goes through uniformly on the carrier: quadratic *base change* $\Delta \mapsto \Delta_E$ over $E = \mathbb{Q}(i)$ — reflect Δ onto the carrier over E (the base-change Satake: split p giving two channels, inert p the Frobenius-squared channel) and reflect back through the χ_{-4} clock, the factorization $L(\Delta_E, s) = L(\Delta, s)L(\Delta \otimes \chi_{-4}, s)$ — reproduces the base-change Satake exactly and closes the reflected degree-four functional equation on the equal-shift two-clock K_0 to 10^{-26} (App. B). Because the alternate carrier may itself carry parallel-dimension clocks, the construction is recursive — the functoriality web as a tower of reflections. The mechanism is thus not a metaphor: base change (reducible), Sym^r (irreducible), and theta (dual pair) are three worked instances.

Isolation note. The universal transport arrows consumed below are isolated in Part I: carrier/involution in §2, residual dictionary in §7, readout and reflection in §7, and coherent composition in §8. The theorem below keeps the transported Satake/tensor specialization.

Theorem 78 (The transport machine). *Let $C = C_+ \sqcup C_-$ be the complete carrier (§30: the double-ended helix with its global involution J, Theorem 69), $(W, W^\vee)$ a* carrier-dual-compatible *admissible fiber pair (Lemma 66:* `localPoly_reciprocal`*, weights closed under $\lambda \mapsto \lambda^{-1}$; W itself need not satisfy $W \simeq W^\vee$ — it rides one lane, $W^\vee$ the anti-lane), and K the functional-equation clock. Then the readout of C carrying $(W, W^\vee)$ is a* faithful completed transport *— a degree-$|W|$ Euler datum with:*

(a) Duality preservation*: the carrier pair is self-dual, $J\colon (W, W^\vee) \mapsto (W^\vee, W)$, root number $|\varepsilon| = 1$ (Lemma 66);*

(b) Exact readout identification*: the local factors and root number are those of the transported Satake datum at every place (Thm. 57);*

(c) Reflection and completion: the completed readout obeys $\Lambda(s;W) = \varepsilon\,\Lambda(1-s;W^\vee)$ — the helix-to-anti-helix involution through the projection (Theorem 69, Prop. 76);

(d) Residual extinction (a configured focal event, fiber-parametric): the fiber's coefficients configure a signed phasor system (P_W, M_W), which the adapted carrier realizes on the two lanes $(X_{W,+}, X_{W,-})$ at height $z = e^y$; the normalized focal residual factors through the readout, $D^{\mathrm{c}}_W(y) = V_W(y)\,\Phi_W(y)$ with $V_W \neq 0$ (Lemma 63) is the zero-detector, algebraically slaved to Φ_W; the DC/residue mode is a separate quantity — the coefficient mean, the trivial-channel coefficient alone (`harmonize_char`) — so a complete fiber (no trivial channel) has $R_W = 0$: the carrier generates no residue pole (Lemma 64), and the completed carrier readout is entire. (For a reducible parameter the identified L-function may carry an additional ζ-factor pole from a trivial constituent; that is not a carrier residue, and it occurs only for the classically enumerated exceptional π, §16.)

Instantiated at the Satake fiber $W = \mathrm{Sym}^r\,\varphi_\pi$ this outputs the degree-$(r+1)$ datum; at the tensor $W = \mathrm{Sym}^r\,\varphi_\pi \otimes \varphi_\tau$, the twisted convolution datum. The transport is independent of any converse theorem: it is the universal machine whose output the next part type-checks.

Isolation note. The finite-structure synthesis, exact-cell closure, readout, synchronization, and transport obligations in the following universal theorem are proven in Part I: §4, §6, §7, and §8. The theorem is retained here as the later general-source statement with its proof-status ledger.

Theorem 79 (Universal faithful carrier transport)**.** Call a fiber W admissible if it is given by — and only by — four minimal data: (i) a finite local parameter system, a finite tuple $A_v(W)$ at each place v (a Satake class, or a Weil–Deligne datum (ρ_v, N_v) where ramified); (ii) a finite-dimensional local-factor rule of fixed degree d, each $L_v(s;W) = \det\big(1 - A_v(W)\,q_v^{-s}\big)^{-1}$ the inverse of a polynomial in q_v^{-s} of degree $\le d$; (iii) the Euler assembly $L(s;W) = \prod_v L_v(s;W) = \sum_n a_n(W)\,n^{-s}$; and (iv) a nonempty holomorphic convergence domain, a half-plane $\mathrm{Re}\,s > \sigma_0$ on which the product converges absolutely. These four data are a finite structural presentation: W is named by finitely much information per place and one bounded-degree rule, not by its values. The placewise assignment $v \mapsto A_v(W)$ is required to be generated by a finite rule — a finite parameter set or effective structural schema fixed independently of how many places are queried, not a place-by-place list; thus finite structural presentation refers to finite description complexity of the source law, not merely finite local dimension at each place. A source carrying no such law — a structureless ("random") function whose only faithful description is its own sample path — is not admissible: there is no finite structure from which to synthesize a carrier phase, a native cell, or a clock, and letting an adapter encode the sample path would be oracular and vacuous. Admissibility is exactly finite structural presentation. No functional equation, continuation, entireness, boundedness, or Ramanujan bound is assumed. Then W admits a faithful carrier representation — a carrier encoding the local inputs (i)–(ii), a fiber encoding the function (iii), and a matched suite of clocks and adapters (the clock configurator; the two-lane placement of W and its dual $W^\vee$; the completion, ramification, and chart-conversion clocks; the operational adapters carrier dwell / phasor delay) — with three properties, tiered by proof status.

(a) Faithful readout [unconditional, constructive]. On the convergence domain (iv) the carrier readout equals $L(s;W)$ exactly, and the readout coordinate maps correctly onto $\{\mathrm{Re}\,s > \sigma_0\}$; the growth/magnitude clock carries the coefficients $a_n(W)$ whatever their size, so no Ramanujan bound is needed.

(b) Continuation, functional equation, and niceness are outputs, not hypotheses. The carrier transform is defined at all heights and furnishes the natural continuation of the readout past σ_0. Unconditional: the reflection $\Lambda(s;W) = \varepsilon\,\Lambda(1-s;W^\vee)$ (the Lean `localPoly_reciprocal` on the pair $W \sqcup W^\vee$, duality-stable for any W) and W's local factors and root number at every place (Thm. 57). Hypothesis-free on the completed carrier object: entireness and vertical-strip boundedness — the continuation meromorphic, its only poles the DC coalignment events (none, hence entire, for an irreducible parameter; a ζ-factor pole per trivial constituent otherwise, §16) — together with every *twist a converse theorem consumes, are supplied by the fixed-completion constructor for every admissible (polynomially bounded) fiber (*`CarrierTheta.automaticCoefficientThetaStrongFEPair`*; on the prescribed conductor/Γ-product banks,* `cpsPolynomial_twistedNiceness`*, with the standard-completion identification* `cpsPolynomialFullCompletion3D_identification`*). Cell closure is* not *the supply of this niceness and could not be: forcible closure is arithmetic-neutral by theorem (*`ForcibleClosure.forcible_nondiscriminating`*) — it closes a random bank exactly as it closes a true fiber — so it contributes the transport half alone; the harmonic and Dirichlet classes additionally close their cells outright (*`CellClosure`*, Lean), which is what grounds their* ordinary *(unwarped) readout continuation end-to-end.*

(c) Exact focal closure — complete and exact vanishing. *The primary invariant is finite and exact: a source vanishing is realized as* exact focal cancellation *at a harmonic cell boundary,* $\sum_{k \in C} q_k(\tau) = 0$ exactly *(a closed cell, not a cutoff limit). The realization of vanishing is the compiled* zero-detector*, not the forcing: the harmonic-pencil rank drop occurs exactly at the readout's zeros (Lemma 63; Lean* `focal_residual_zero_iff_L_zero`*,* `carrierZero_iff_L_zero`*, with the channel-agnostic factorization* `HarmonicCell.harmonic_residual_no_independent_mode`*), while forcible closure (Lemma 65) supplies only the per-cell transport remainder and, being arithmetic-neutral (*`forcible_nondiscriminating`*), never the vanishing identification. Where* (b) *furnishes the continuation, these detector closures are, by the identity theorem, exactly the zeros of the represented function —* complete *(no zero missed) and* exact *(none spurious), the pole locus being the complementary coalignment events. This is exact closure on the carrier bank — the finite invariant* $\sum_{k \in C} q_k = 0$ *— and faithful representation of the vanishing* set*; it is not an analytic-residue claim.*

So admissibility asks only for a convergent Euler product of bounded degree. Niceness is never a hypothesis and never a closure inference: the reflection and local data come out unconditionally; entireness, strip bounds, and the functional equation of the completed carrier object are the fixed-completion constructor's output, hypothesis-free for every admissible fiber (`cpsPolynomial_twistedNiceness`*); and the complete faithful vanishing is the cell-closure content of* (c)*, whose per-cell mechanism (*`ForcibleClosure.residual_forcible` *with* `CarrierReachability.reachable_independent_stage`*) is arithmetic-neutral by theorem (*`forcible_nondiscrimina`
and is therefore consumed by transport and adaptation, never by niceness.

Constructive form: synthesis, not assumption. The representation is not merely posited to exist; it is *synthesized* from the four data by an explicit map

$$\mathrm{Synth}: \ \mathrm{Struct}(W) = \big(\{A_v(W)\}_v,\ d,\ \text{Euler assembly},\ \sigma_0\big) \ \longmapsto\ (\mathcal{C}_W,\ \omega_W,\ \tau_W,\ \mathcal{L}_W),$$

the carrier chart $\mathcal{C}_W$ (phase and cell scale from the local parameters), the readout-preserving warp ω_W, the completion/ramification clock τ_W, and the native cell system $\mathcal{L}_W$. The universal statement is therefore the *constructive*

$$\forall W: \quad \mathrm{Struct}(W) \implies (\mathcal{C}_W, \omega_W, \tau_W, \mathcal{L}_W) = \mathrm{Synth}\big(\mathrm{Struct}(W)\big),$$

strictly stronger than $\forall W\, \exists A_W$: the adapter data are *read off* the finite law rather than assumed, and the closure theorems (Lemmas 63, 65) then apply to the constructed cell system $\mathcal{L}_W$. Finite structure is exactly what makes Synth total — its inputs are the finitely-presented data and nothing else — so simultaneous synchronization (Corollary 80) is the *common refinement* of the individually synthesized systems $\{\mathcal{L}_{W_j}\}$ onto one carrier: constructed, not posited. Part I is this synthesis in miniature — Altuğ's obstruction was not a failure of cancellation but a failure to *represent* the state cancellation needs; restoring the omitted coordinates makes the phase gaugeable and names the correct cell, which is adapter synthesis from a determinate finite law.

Falsifiable content. Because the load-bearing input is the per-fiber cell closure, the universal claim is falsifiable at a single point: a genuine counterexample is not a fiber the dictionary has merely not yet been run on, but a fiber W for which *no* harmonization consistent with its intrinsic dynamics closes its cells while satisfying the carrier and ledger laws and preserving faithful readout. Two $\forall\exists$ quantifiers must never be traded for one another, and the proven one is named exactly: for every W a *cell-closing* adapter suite exists, in closed form, not sampled — any nonzero cell residual is forcible to zero once two $\mathbb{R}$-independent directions span $\mathbb{C}$ (`ForcibleClosure.residual_forcible`, Lean), and a continuous non-constant lane phase supplies the independent pair (`CarrierReachability.reachable_independent_stage`, Lean), the two composed by direct application — so the enumeration of every (W, suite) pair is not required. The *recognition* quantifier — that the synthesized suite reads the fiber's arithmetic rather than merely closing its cells — is precisely what cell closure can never supply (`forcible_nondiscriminating`) and is the separately registered program of the contributions ledger (item 2), with its single clean falsifier stated there. The general family (Sym^{13}, the two-parameter $L(\Delta \times E_{11})$, the genuine twisted $L(\mathrm{Sym}^5\,\Delta \times E_{11})$, the real Maass Sym^2; §23) is the set of instances where we additionally ran empirical checks — calibration and independent reproduction at the float floor, not the basis of the claim. Two boundaries bound the claim: *within* admissibility, the theorem's own hypothesis — the independent pair, a non-degenerate warp Jacobian read from the law — which every synthesized fiber to date satisfies; *outside* admissibility, a structureless source has no law to synthesize from and is the limiting *non*-example, the point at which description complexity has become the state.

Corollary 80 (Simultaneous carrier synchronization)**.** *The transport is universal not only fiberwise but for* simultaneous synchronization*. Let $W_1, \dots, W_m$ be admissible sources — each a function (or Hodge datum) realized as a* 3D *carrier+fiber pair and closing its cells exactly by* (c)*; native cell scales, ranks, phase conventions, clock laws, and dimensional realizations may differ. Then a per-source* normalization *augments* both *halves of each pair — the carrier (its cell scale and clock chart) and the fiber (the function's warp and weight chart) — bringing each to a carrier+fiber form mutually compatible with the others on one common carrier, so the transported sources evolve and are read* simultaneously*: the unified readout is the tuple $\mathbf{Q}(\tau) = \big(Q_1^{\#}(\tau), \dots, Q_m^{\#}(\tau)\big)$, recording exact coordinatewise closures. Each fiber remains distinct and keeps its own focal-closure events — a source vanishing of W_j is an exact zero of $Q_j^{\#}$ at a harmonic cell boundary, with no uncancelled cell remainder — and the fibers need not share native ordinates, cell scales, or clocks.*

The synchronization obligation. The one requirement beyond the fiberwise closures is that the normalization *preserve exact boundary closure* — carry each exact cell closure to an exact cell closure on the common carrier — not merely align zero *locations*. This holds by construction for the harmonized (root-of-unity) normalization, where the μ_M snap sends every channel to a shared cell and the joint bank closes coordinatewise (§23), and is validated on the genuine two-fiber case $L(\Delta \times E_{11})$: its two rails synchronize on one carrier and close jointly to 4.9×10^{-91} (`gl4_rankin.py`). As in Theorem 79 this is exact closure on the carrier bank, not an analytic-residue claim.
Isolation note. The source-independent synchronization and common-refinement obligation from Corollary 80 is isolated in §5. The numerical two-fiber and root-of-unity checks remain here as source/application evidence.

Corollary 81 (Universal functoriality in rank)**.** *Theorem 79 carries every admissible fiber faithfully, its reflection and local data unconditional and its niceness the fixed-completion constructor's output; that output is uniform in rank, for the construction refers nowhere to the degree N, the source group G, or the symmetric-power string: the reflection (`localPoly_reciprocal`), the two-clock completion, the warped-Abel continuation and vertical bound, and the Riemann–Hecke entireness (the one carrier-cell theorem) are stated for an* arbitrary *finite duality-stable weight system,* uniformly in rank*. For objects on* GL(2) *data the two remaining ingredients are standing theorems, not hypotheses: local Langlands for* GL(2) *furnishes* $\varphi_{\pi,v}$ *and the local factors at every place, and the converse theorem into* GL_N *[14] promotes a nice L-function to an automorphic form. What neither classically controls at every rank — the twisted niceness of* $L(r \circ \varphi_\pi \times \tau)$ *for all* τ*, entire and bounded and self-dual — is exactly what the carrier discharges, uniformly in rank. Consequently,* ***every finite duality-stable weight system*** r ***on*** GL(2) ***data transfers****: feeding the carrier's nice datum into the* GL_N *converse theorem — discharged place-by-place as the explicit hypothesis ledger of Part III (Theorem 48, Corollary 49) — makes* $r \circ \varphi_\pi$ *automorphic on* GL_N*, cuspidal for non-exceptional* π *and isobaric for the finitely many exceptional types (§16), uniformly in rank and degree: the full symmetric-power tower and the Rankin–Selberg weight systems* $\{\pm(a \pm b)\}$ *alike. The low-degree cases recover known functoriality, as they must —* Sym^2 *(Gelbart–Jacquet [6]),* $\mathrm{Sym}^3, \mathrm{Sym}^4$ *(Kim–Shahidi [17]), the classical convolution lifts; the content is that the* same *mechanism continues past the Langlands–Shahidi range (*$r \geq 5$*) with no new ingredient, the niceness being carrier-discharged rather than Eisenstein-constructed.* Scope. *The niceness argument's Lean theorems are stated at the general interface, not for particular functions, and they generalize automatically because they read only the fiber's abstract data: the per-place reflection holds for* any *finite duality-stable weight multiset (*`FiniteWeightFiber.localPoly_reciprocal`*); the completed-reflection assembly carries* no *Dirichlet hypothesis (*`FiniteWeightFiber.symTensorCompleted_FE`*, any* r*, any unit twist); the entireness mechanism is channel-agnostic in* (A, B, μ, λ) *(*`HarmonicCell.harmonic_residual_no_independent_mode`*); and cell closure is the closed-form forcibility above (*`ForcibleClosure.residual_forcible`*). The harmonic and Dirichlet classes are therefore end-to-end* instantiations *of general proofs, not the proofs' boundary, and the general family (*Sym^{13}*,* $L(\Delta \times E_{11})$*, the genuine twisted* $L(\mathrm{Sym}^5 \Delta \times E_{11})$*, real Maass* Sym^2*; §23) is instance calibration. No local-Langlands datum is required by the transport itself — the carrier consumes the admissible presentation directly (Definition 5); for a source group beyond* GL(2)*, local Langlands at that group enters only to source that presentation from a classical* π *and to name the output — a translation step, a theorem in many cases, not a system dependence.*

Remark 82 (Universality and functoriality are distinct claims)**.** *Isolation note.* The general coherence laws in this remark are the Part I transport obligations recorded in §8; the symmetric-power, tensor, and base-change instances stay here as applications.

Two claims here should not be conflated. *Universality* is the existence of compatible carrier realizations (Cor. 80): any finite collection of exact-closing source fibers admits source-dependent normalizations $\mathcal{N}_F$ placing them on one common carrier, run and read simultaneously with exact focal closure preserved — *universal by normalization, compatible by carrier realization*, no shared native representation required. That is theorem-level here.

Functoriality is the separate coherence law on structural maps *between* sources: a transport $f\colon F \to G$ should induce a carrier transport $\widehat{f}$ with $\mathcal{N}_G \circ f \simeq \widehat{f} \circ \mathcal{N}_F$ (naturality), the identity inducing the identity, $\widehat{g \circ f} \simeq \widehat{g} \circ \widehat{f}$ (composition), and exact focal closure preserved — where $\simeq$ is a *declared* carrier equivalence (chart reparameterization, phase conjugacy, block permutation, *not* exact equality) and the normalizations need not be unique. For the explicit transports of this paper — symmetric power, Rankin–Selberg tensor, base change — naturality and faithfulness hold because $\widehat{f}$ is the *same* weight-system operation on the normalized bank that f is on the source. Identity and composition, the decisive tests, are *verified*: at the weight-system level $W_{\mathrm{id}} = \mathrm{id}$ and $W_{r \circ s} = W_r \circ W_s$, each composite an *exact* disjoint union of irreducible blocks (the plethysm — $\mathrm{Sym}^2 \circ \mathrm{Sym}^2 = \mathrm{Sym}^4 \oplus \mathrm{Sym}^0$, $\mathrm{Sym}^3 \circ \mathrm{Sym}^3 = \mathrm{Sym}^9 \oplus \mathrm{Sym}^5 \oplus \mathrm{Sym}^3$ — coherent up to the block permutation in $\simeq$), tensor and base change composing likewise (`composition_test.py`); and the coherence laws are machine-checked — associativity, two-sided identity, base-change composition $\mathrm{bc}_p \circ \mathrm{bc}_q = \mathrm{bc}_{pq}$, and preservation of exact focal closure under composition (`CarrierFunctoriality`). For the CPS tower this coherence is instantiated on the genuine Satake-bank state, with strict equality $T_{gh} = T_g \circ T_h$ and exact bank-zero preservation (`ThreeDConverse.cpsTowerFunctoriality3D_unified`); no carrier equivalence is needed in that instance. The carrier-equivalence formulation is retained only for structural transports between arbitrary source presentations whose native charts differ. The Langlands functoriality of Cor. 81 is the arithmetic *instance* this coherence delivers, not a separate mechanism. In one line: *the carrier is universal for simultaneous harmonic realization, and functorial with respect to those source transports that commute, up to carrier equivalence, with faithful harmonic normalization.*

Part V

Other Significant Results

34 Obstruction-directed harmonization: from detection to removal

The transport of Theorem 79 is, so far, an obstruction *detector and locator*: it reads a source, and where a mode fails to close it names the obstruction coordinate $\alpha = \mathrm{Obs}(X) \neq 0$ and, through the tower, the finite depth at which it lives — the ledger certifying that α was *carried*, not discarded. The capitulation archetype above shows the next arrow: an obstruction can *dissolve* once the carrier is enlarged to the frame its own signature names. We record that arrow as an operator and state the general removal as a target, guarded by an explicit killability boundary.

Remark 83 (The obstruction-directed extension operator)**.** Given a carrier state C and a detected obstruction $\alpha = \mathrm{Obs}(C) \neq 0$ — a persistent non-closing mode, localized to a finite tower depth — the *obstruction-directed extension*

$$\mathfrak{E} : (C, \alpha) \longmapsto (C_\alpha,\ \iota_\alpha,\ h_\alpha)$$

returns a faithful transport $\iota_\alpha : C \to C_\alpha$ (preserving readout and the ledger laws) and a primitive h_α with

$$d\, h_\alpha = \iota_{\alpha *}(\alpha), \qquad \text{so} \qquad [\iota_{\alpha *}\alpha] = 0$$

in the source-relevant cohomology of C_α: the transported cocycle becomes a coboundary. The extension data are *determined by the signature of* α, not found by search — the non-closing mode names the missing carrier coordinate, harmonization synthesizes it (the constructive Synth of Theorem 79, now applied to the extension), and exact focal closure in C_α is the *witness* $dh_\alpha = \alpha^\#$, a genuine null-homotopy rather than a vanished scalar readout. The framework thus reads

$$\boxed{\text{Detect}\,(\alpha \neq 0) \ \to\ \text{Localize}\,(\text{signature, depth}) \ \to\ \text{Harmonize away}\ \big(\alpha \mapsto (C_\alpha, h_\alpha),\ dh_\alpha = \alpha^\#\big)}$$

with certificate the pair (C_α, h_α) and its focal-closure witness.

The proven prototype: capitulation. Furtwängler's Principal Ideal Theorem *is* $\mathfrak{E}$ for the ideal-class obstruction. The signature is a nonprincipal class $\alpha \in \mathrm{Cl}(K)$; the extension C_α is the Hilbert class field H, the frame α itself names; ι_α is extension of scalars $K \hookrightarrow H$; and the primitive h_α is the principal generator that appears there, $[\iota_{\alpha *}\alpha] = 0$ being capitulation (every ideal becomes principal in H). The signature determines the extension and the witness is constructed, not assumed — a proven instance of the operator, already universal in its own domain (every class capitulates in the class field it names).

A second, adversarial prototype: the Brauer-zero trap. Detection and removal separate most sharply where the *aggregate* readout already closes. Take the Hamilton class $[A]$, $A = (-1,-1)_{\mathbf{Q}} \in \mathrm{Br}(\mathbf{Q})$: global reciprocity forces its local invariants to sum to zero, $\sum_v \mathrm{inv}_v(A) = \frac{1}{2} + \frac{1}{2} \equiv 0$ in $\mathbf{Q}/\mathbf{Z}$, so every scalar/global reading is *harmonically closed* — yet $[A] \neq 0$, the invariant *vector* $(\mathrm{inv}_2, \mathrm{inv}_\infty) = (\frac{1}{2}, \frac{1}{2})$ being nonzero. A detector that equated a closed readout with a killed class would false-positive here; the ledger instead retains the vector. Run blind, the controller given no splitting field, $\mathfrak{C}$ *(A)* detects the retained obstruction despite the zero sum, *(B)* classifies it as 2-torsion ramified at $\{2,\infty\}$, *(C)* synthesizes from that syndrome alone a finite chart with even local degree at those places — the minimal being $K' = \mathbf{Q}(\sqrt{-3})$, not supplied and not the textbook $\mathbf{Q}(i)$, *(D)* removes: a direct Brauer computation confirms $\mathrm{res}_{K'/\mathbf{Q}}[A] = 0$, and *(E)* certifies $R(X^\#) = 0$ *and* $\mathrm{Obs}(X^\#) = 0$ together — the conjunction detection alone cannot supply. That the repair is synthesized, not hard-coded, is forced by a second class ramified at $\{3,5\}$: a fixed "adjoin $\sqrt{-1}$" rule splits Hamilton but *cannot* split it, while the syndrome search repairs both (`brauer_zero_trap.py`, every stage checked against the classical Brauer/Hilbert-symbol oracle). The five passes — detect / classify / synthesize / remove / certify, run with no target field in the adapter library and charged for extension degree — are the test methodology; the *removal logic* they enforce is machine-checked in Lean: that a zero aggregate never counts as removal ($\sum_v \mathrm{inv}_v = 0$ does not force the invariant vector to vanish, `false_closure`), that an even local degree at every ramified place kills a 2-torsion class (`removal`), and that the certificate is the conjunction aggregate $= 0 \wedge$ vector $= 0$ (`certified_removal`) — `BrauerZeroTrap.lean`. Brauer classes over a number field are, moreover, a regime where the killability boundary is provably met: by the global theory of central simple algebras over number fields — Albert–Brauer–Hasse–Noether together with the cyclicity theorem — every class admits a finite cyclic splitting extension, so the terminating-signature condition holds group-wide, and the benchmark supplies the constructive syndrome-directed synthesis for the 2-torsion (quaternionic) part. The general finite-depth characterization, beyond Brauer, remains the target above.

What is already in hand. The detection side is machine-checked: the obstruction is a well-defined measured coordinate (`GeneralizedCohomology`: `DistinguishedPoint`, `obstruction_eq`, `measurement_sound`, the tower reading α exactly), and faithful transport preserves closure under composition (`CarrierFunctoriality.faithful_comp`, *no* axioms), so a nullification once achieved is carried by any further functorial transport. What $\mathfrak{C}$ adds beyond detection is the *witness* h_α; the depth at which to build it is supplied by the delayed tower signature (first-nonzero tower depth = filtration depth, §37).

Obstruction-directed harmonization (target). We state the general removal as the frontier this opens, evidence-weighed, not as a proved result: *for a finite-depth obstruction signature detected by the carrier tower, the signature determines — by a computable synthesis rule, up to carrier equivalence — a finite harmonizing extension C_α and a primitive h_α with $dh_\alpha = \iota_{\alpha *}\alpha$, so the transported obstruction is exact, the nullification witnessed by complete focal closure and preserved under functorial transport.* Version 1 asks only existence plus a computable rule, not canonicity. The supporting evidence is the capitulation prototype, the constructive Synth operator, and the finite depth index of the tower; and the *termination* half is no longer wholly open — it is characterized below by a local–global principle at the signature's depth (Proposition 84), unconditional at depths 1 and 2, the non-terminating boundary being the stably essential classes (not removable by finite enlargement). What remains is the local–global principle at higher depth and the synthesis rule off the two settled regimes.

Killability boundary, and what would falsify it. The claim is guarded, and pre-commits to its failure modes. *(1) Visibility does not imply killability*: not every detected class is removable by a finite extension; a class that stays nonzero under *every* permitted finite extension is outside scope, and characterizing the signatures for which the synthesis *terminates* is the open content, not a defect. *(2) No arbitrary enlargement*: the extension must be named by the signature (as capitulation's class field is named by the class); enlarging blindly until the readout vanishes is oracular and vacuous — the structureless boundary of Theorem 79 again. *(3) A real coboundary, not a zero readout*: exact focal closure must certify $dh_\alpha = \alpha^\#$ in the source-relevant cohomology, a genuine null-homotopy, with the ledger proving α was carried, not thrown away; a vanished scalar symptom without a witness does not count. *(4) Finite depth*: the extension is built at the tower depth the signature reports, so a terminating synthesis is finite. A disconfirmation — a finitely-presented, finite-depth signature that its own named frame extension provably cannot make exact — would falsify the target, and would be published as prominently as a confirmation; none is known.

The reach of the operator: a local–global characterization. Which signatures terminate is not left open in the mass; it is organized by one principle. A finite-depth signature α is removable by a finite carrier extension exactly when a *local–global principle* holds for it at its detection depth d: α capitulates in a finite cover precisely when it is *locally trivial* — killed under restriction to every completion of the carrier. Capitulation and the Brauer splitting are then not two coincidences but the depth-1 and depth-2 instances of one mechanism, and both are unconditional.

Proposition 84 (Terminating signatures: two unconditional regimes)**.** *Let G be the carrier's structure group and M finite coefficients.* (a) Every *depth*-1 *signature terminates. A continuous class $\alpha \in H^1(G, M)$ is inflated from a finite quotient G/U, so by inflation–restriction* $\mathrm{res}_U\, \alpha = \mathrm{res}_U \inf \bar{\alpha} = 0$*: α dies in the finite cover C_U. This is capitulation — for ideal classes, Furtwängler's Principal Ideal Theorem.* (b) Every *Brauer (H^2) signature over a global carrier terminates, by the global theory of central simple algebras over number fields — Albert–Brauer–Hasse–Noether together with the cyclicity theorem: the vanishing of the summed local invariants (reciprocity) forces a finite cyclic splitting extension. The constructive synthesis for the* 2*-torsion part is the Brauer-zero trap (*`brauer_zero_trap.py`*,* `BrauerZeroTrap.lean`*).*

The boundary is equally sharp: termination *fails* exactly where the local–global principle fails at depth d — for a *stably essential* class, one nonzero under restriction to *every* finite admissible carrier enlargement and therefore not removable by finite enlargement. The boundary beyond the terminating local–global regimes is *not* identified here with the transcendental Griffiths group. Detection is a separate question from removal: finite-tower *visibility* of Griffiths/Ceresa extension data depends on the joint faithfulness of the derivative/height tower and is treated in §37.1. So the operator's reach is *precisely* the local–global regime: below the boundary a signature is locally trivial at depth d and the constructive synthesis of $\mathfrak{E}$ terminates; a stably essential signature is not killable by finite enlargement. Thus "not removable" does not mean "invisible," and no transcendental Griffiths ceiling is asserted here. Which side a given α falls on is read from whether its depth-d obstruction is locally trivial — decided by the signature, not left to chance.

The control-system reading. Equivalently $\mathfrak{E}$ is a closed-loop controller for representation mismatch: detect the residual mode, retune the participating clocks $\tau_j \mapsto \tau_j^\#$, enlarge the common cell where a rescaling cannot suffice (some obstructions need an added coordinate, or one tower level higher, not merely a rechart), re-run the fibers simultaneously $X^\#(\tau) = \big(F_1^\#(\tau), \ldots, F_m^\#(\tau)\big)$, and halt when exact focal closure certifies harmonization; the signature itself supplies the retuning law, closing the loop. One recharts until α becomes an exact *closing coordinate* — not until the scalar symptom disappears.

35 The mechanism and its converse: the analytic ring closed at every chart

The completed analytic package of a carrier bank — entire continuation, functional equation, chart identification — and the pointwise theta reflection of its prescribed readouts are *the same data*, in both directions, machine-checked. This section registers the compiled ring: the mechanism (identity $\Rightarrow$ package) at its true generality, its converse (package $\Rightarrow$ identity) with the classical boundedness hypothesis alone, and the identification carrying the reflection to the modular transformation laws. Every declaration cited here carries the standard axiom footprint and no `sorry` (App. B).

The coefficient surface and the mechanism

Everything the chart engines consume from a bank is two coefficient sequences with one polynomial bound: the surface `CoefficientSurface` $(a, a^\vee, A)$ with $\|a_n\|, \|a_n^\vee\| \le (n+1)^A$. A chart is packaged as a kernel `MellinKernel` (K, G, B_0): a positive-ray kernel with rapid decay whose Mellin transform equals the multiplier G beyond the bar B_0. The compiled kernel families instantiate it: the $\Gamma_{\mathbb{C}}$ clock (`gammaCKernel`, the holomorphic chart type), the general Deligne chart $\prod \Gamma_{\mathbb{R}} \cdot \prod \Gamma_{\mathbb{C}}$ (`generalMellinKernel`), and the pure-$\Gamma_{\mathbb{R}}$ chain (`realChainMellinKernel`, the Maass chart type, purely imaginary shifts admissible).

Theorem 85 (The mechanism, coefficient-level and kernel-generic; `RamifiedMechanism.mechanismK`). *Let P be a self-dual coefficient surface ($a^\vee = a$) and κ a packaged chart kernel. One pointwise theta identity of the prescribed readouts, $\Theta_P(1/x) = \varepsilon\, x\, \Theta_P^\vee(1/x)$ for all $x > 0$, yields the complete package: an entire Λ with $\Lambda(1-s) = \varepsilon\Lambda(s)$ and $\Lambda(s) = G(s)\, L(a, s)$ on $\operatorname{Re} s > \max(B_0, A+1)$. The continuation is unique (`KNicePackage.unique`).*

The fixed-rank mechanisms are instances through `PolynomialSatakeDualPair.surface`: every finite-rank bank of §33 feeds the theorem, and the seed layer supplies the arithmetic instances — Sym^r of the holomorphic level-one eigenform (`SatakeSeed.ofHecke`), of the typed Maass–Hecke package (`MaassEigenData`, `SatakeSeed.ofMaass`), at every rank over every seed (`symr_package_of_theta_of`), with the Maass spectral chart named (`maass_symr_package_of_theta`, shifts $(r-2j)it$). Rank three is typed directly (`GL3Seed`): self-duality enters as a channel permutation hypothesis — rank three does not supply it for free — and the Sym^2 lift *discharges* it, with the recognition `ofSym2_coeff_eq_sym2` identifying the lift's standard coefficients with the Sym^2 transport coefficients on the nose: the Gelbart–Jacquet identification at seed level. Level N is not a separate obstruction: the rank-varying surface `RamifiedWeightFamily` (per-prime rank profile capped by D, degree drop at ramified places, degree zero fully ramified) carries the polynomial bound $(n+1)^{D+B}$ — the cap enters only through $\binom{\cdot}{\cdot}$-majorant monotonicity — and self-duality by *per-place* channel permutations; the mechanism fires on any such bank (`ramified_package_of_theta`). The constant-rank family recovers the unramified Sym^r bank definitionally (`ofSeed_coeff_eq`, by `rfl`), and the principal-character family (rank 0 at $p \mid N$, rank 1 elsewhere) is the genuine rank-drop witness (`unitRamified`).

The converse: the package forces the identity

The reverse arrow is the Hecke–Weil converse in the carrier register. Its one genuinely analytic step is the contour shift, compiled as `vertical_line_shift`: an entire integrand, integrable on two vertical lines with vanishing horizontal integrals, has equal line integrals (rectangle Cauchy–Goursat and passage to the limit). The functional equation exchanges the two inversion lines (mellinInv_reflect), Mellin inversion runs on the chart line where the bank's transform *is* Λ (mellin_theta_eq_lam_on), and the assembly is:

Theorem 86 (The converse; WeilConverse.weil_converse)**.** *Let P, κ be as above with κ-kernel continuous on the ray, and let a package for P at κ be given together with line integrability of Λ at one chart line σ and a horizontal decay bound on the strip $[1-\sigma, \sigma]$. Then the pointwise theta reflection of the prescribed readouts holds — in exactly the form consumed by Theorem 85. Feeding it back returns the same continuation (weil_converse_ring): the ring is closed.*

The two analytic side conditions are then discharged. The line data come from the in-tree Stirling bound: the $\Gamma_{\mathbb{R}}$ line reduces to the compiled $\Gamma_{\mathbb{C}}$ line by a constant-ratio norm identity (gammaR_vline_eq_const_mul), chain multipliers are line-integrable and exponentially decaying with the Dirichlet factor uniformly bounded on the line (StirlingLineDischarge). The strip decay is Phragmén–Lindelöf: strip_exp_decay interpolates boundary decay $Ae^{-k|t|}$ through a bounded strip *with the same constants* (two multiplier applications of the vertical-strip principle). The composite is the converse under the classical hypothesis set:

Theorem 87 (The converse from boundedness alone; PhragmenBV.weil_converse_of_BV)**.** *Package plus* one *boundedness constant on the strip forces the theta reflection. At both compiled chart families the kernel side is hypothesis-free: the $\Gamma_{\mathbb{C}}$ chart (weil_converse_of_BV_gammaC; clock continuity compiled) and the pure-$\Gamma_{\mathbb{R}}$ chain (weil_converse_of_BV_realChain′; chain-kernel continuity compiled through the weight exchange weighted_logMellinConvolution_eq_convolution). The named Maass instance maass_converse_of_BV runs at the spectral chart $\prod_j \Gamma_{\mathbb{R}}(s+(r-2j)it)$ of* Sym^r*, every rank, every spectral parameter.*

The identification: the generator laws

The reflection identifies. The q-expansion $f(\tau) = \sum_n a_n e^{2\pi i(n+1)\tau}$ of a surface is holomorphic on the upper half-plane (qSeries_analyticOnNhd); its T-law is exact from the expansion alone (qSeries_T); the compiled $\Gamma_{\mathbb{C}}$-clock readout at height y is $2f(iy)$ (theta_eq_qSeries_axis); and the axis reflection extends to the full half-plane by the identity theorem (qSeries_S_of_reflection): $f(\tau) = \varepsilon\,(-i\tau)^{-1} f(-1/\tau)$ on $\mathbb{H}$.

Theorem 88 (The Hecke identification; WeilIdentification.hecke_identification)**.** *An entire package at the $\Gamma_{\mathbb{C}}$ chart with the functional equation and one boundedness constant yields both generator transformation laws of its q-expansion: exact* 1*-periodicity and the S-law on all of* $\mathbb{H}$*.*

Since S and T generate $\mathrm{SL}(2,\mathbb{Z})$ — machine-checked in Mathlib (SpecialLinearGroup.SL2Z_generators) — these are the complete generator set of level-one modularity, and the multiplier packaging is itself compiled: the set of γ under which a function is a weight-k slash eigenvector is a subgroup (the slash-action axioms are the cocycle; slashEigenSubgroup), the two laws place S and T in it, and the closure argument delivers

Theorem 89 (Hecke modularity; MultiplierPackaging.hecke_modularity)**.** *The q-expansion of an entire BV-package at the $\Gamma_{\mathbb{C}}$ chart is a weight-*1 *slash eigenform at* every $\gamma \in \mathrm{SL}(2,\mathbb{Z})$*: for each γ there is a nonzero multiplier $v(\gamma)$ with $f\,|_1\,\gamma = v(\gamma)\cdot f$, in Mathlib's own slash action ($v(T)=1$, $v(S) = -i\,\varepsilon^{-1}$).*

The theorem is quantified over an *arbitrary* self-dual coefficient surface, so its instances are the entire compiled bank family — automorphy at the holomorphic chart is universal over it: Sym^r of every seed and every self-dual rank-varying level-N bank are named instances (symr_automorphy, ramified_automorphy). The register per chart: at the Maass chart the converse input is equally compiled (`maass_converse_of_BV`) and the wave-form identification is now compiled up to exactly one typed field: the wave form $\sqrt{y}\ \sum_n a_n \cos(2\pi(n{+}1)x)\, K((n{+}1)y)$ on the abstract radial kernel has exact T-periodicity and evenness, its axis value is the compiled theta readout, the axis S-law is the compiled reflection, and *both halves of the Cauchy data on the axis match unconditionally* (an even function has vanishing transversal derivative, no differentiability hypothesis needed); `maass_modularity_of_BV` then gives full weight-0 $\mathrm{SL}(2,\mathbb{Z})$-modularity of the wave form from package + one boundedness constant + the typed Cauchy-uniqueness field — the classical elliptic step (both functions are eigenfunctions of the hyperbolic Laplacian; eigenfunctions with equal Cauchy data on a curve coincide), consumed at its exact strength since Bessel functions and elliptic unique continuation are not yet in Mathlib. The kernel itself needs no Bessel API: the chain kernel at the chart $(it, -it)$ has Mellin transform $\Gamma_{\mathbb{R}}(s+it)\Gamma_{\mathbb{R}}(s-it)$, and kernels with equal Mellin transforms on a line agree on the ray (`eqOn_pos_of_mellin_eq`), so it *is* the classical $\sqrt{y}\, K_{it}$-kernel. At higher rank the identification is the cited converse-theorem layer. The level-N statement — finitely many twisted functional equations sufficing for $\Gamma_0(N)$-automorphy — is Weil's lemma, consumed at the cited converse-theorem layer of §11 exactly as before.

Obstruction detection at the ring

The two arrows make the pointwise theta identity an obstruction *detector*, both directions compiled (`ObstructionDetection`). Soundness is the converse's contrapositive: a single pointwise failure of the identity obstructs every BV-package at every root number (`reflection_failure_obstructs_gammaC`). Detection is the mechanism's contrapositive, with the pole instance compiled: the unit (ζ-) surface admits no entire package at any $\Gamma_{\mathbb{C}}$ chart — the ζ-pole collides with entirety along the ray (`zeta_no_package`) — hence its theta identity *fails pointwise* (`zeta_reflection_fails`): the trivial channel's pole is detected by the carrier readouts alone, no L-value evaluated. The dichotomy `detection_dichotomy` states completeness: every self-dual surface either satisfies the identity everywhere or admits no BV-package. This upgrades the beyond-endoscopy demonstration from the representation bank to the coefficient surface, where every compiled bank lives.

The exchange instrument at Sym^r

Both arrows compose at the Sym^r bank of any seed (`ExchangeInstrument.symr_package_of_reflection`, `symr_reflection_of_package`): converting between the one-sided theta identity and the analytic package is an *interface*, not missing machinery. The per-rung obligation on either side is one classical datum — the completed package with bounded vertical strips, the precise output shape of the classical symmetric-power automorphy results (Kim–Shahidi for $r \le 4$, Newton–Thorne for all r at level one; cited). A cited package of that shape now yields the pointwise identity *compiled*, and an identity yields the package *compiled*; the rank-varying level-N banks carry the same two arrows through their surface.

36 Formalization scope

Formalization. The reflection for a *general* fiber is Lean-proved: `FiniteWeightFiber` formalizes the finite duality-stable weight fiber and its carrier reflection (ledger `fiber_det_one`, per-place functional equation `localPoly_reciprocal`, axis `fiber_reflection_axis`, warp-invariance `warpFiber`), with Sym^r (`symFiber`) and Rankin–Selberg (`tensorFiber`) as instances and the standard axiom footprint (App. B). So the functional-equation input of Proposition 59 is machine-checked at the general interface it actually consumes, not only for the Dirichlet model — and the completed-reflection assembly is machine-checked family-generally (`FiniteWeightFiber.symTensorCompleted_FE`, above), with the global strand-exchange input, step (2) of Prop. 76, machine-checked at the finite-stage bank (`StrandExchange.bankProduct_exchange`, `completedBank_exchange`) and the general dual-pair reciprocity at its exact interface (`DualPairFiber`). The multiplicative core of step (2) — the global functional equation as the *product* of the per-place reciprocal FEs — is machine-checked directly: `CompletedReflection.prod_completed_FE` assembles a finite Euler product's completed FE from its local factors, and `CompletedReflection.infinite_completed_FE` carries it to the infinite product by passage to the limit (`tendsto_nhds_unique`), the sole remaining input being Euler-product convergence (`TransferContinuation.transfer_analytic`) — no Poisson summation. The E5 bridge is likewise machine-checked in the initial half-plane: `CarrierTheta.theta_hasMellin_of_polynomial` identifies the full coefficient theta with its Dirichlet readout times the completion Mellin factor, while `GlobalHelix.cpsPolynomialConductor_initialIdentifica` simultaneously specializes the primal and contragredient banks to the prescribed Gamma/conductor completion. The full nonempty Deligne-shift list, rather than only the one-shift specialization, is assembled for every CPS twist by `GlobalHelix.cpsPolynomialAllTwists_fullCompletion3D_unified`; its single conclusion also contains radial 3D reflection, both completed identifications, analytic continuation, strip bounds, and the functional equation for the strong pair built from the same W. Thus the analytic identification used by Thm. 57 is wired into the CPS completion surface rather than left as an unnamed manuscript transition. The all-place arithmetic identification, ramified local factors included, is *not* a remaining scope item: Thm. 57 closes it on the carrier — the native local factor is read off the definitional Satake/Frobenius block (`frobeniusBlock_eq_conjPairBlock`, by `rfl`) through the lossless, faithful deprojection (`ConeProjection.record_bijective`), with local Langlands for GL(2) [10] furnishing $\varphi_{\pi,v}$ (a cited theorem, not an owed formalization) and Sym^r a functor on parameters. Nor is the completion a remaining item: the completed carrier niceness — entire continuation, vertical-strip bounds, and functional equation — is proved unconditionally for every twist as a *single* carrier object (`cpsPolynomial_twistedNiceness`, `cpsPolynomial3D_globalHelixReflection`), and the completed coefficient theta converges on the whole positive ray by polynomial-growth/kernel-decay local integrability (`cpsPolynomialFullPrimalTheta_locallyIntegrableOn`). The Euler-product convergence abscissa $\mathrm{Re}\, s > 1$ is the projection chart's gate, not a barrier on the carrier, which has no abscissa; the standard-Γ normalization of the reflection rides the per-fiber reciprocal identity (`FiniteWeightFiber.symTensorCompleted_FE`, every s, every r). Neither the identification nor the convergence is owed. The numerical experiments retained serve only as calibration and independent reproduction of the *already-proved* classical-L identification and its exact archimedean normalization (the archimedean two-clock and Poisson completion tied to the exact global L-values), governed by the exact gauge; the carrier/fiber rescaling reduces the residue reading to cell arithmetic.

The mechanism–converse ring of §35 is machine-checked end to end at the same footprint: the coefficient-level mechanism and its instances (`RamifiedMechanism`, `GeneralSeed`, `GL3Seed`, `RealChainCompletion`), the converse with its contour shift and Phragmén–Lindelöf discharge (`WeilConverse`, `StirlingLineDischarge`, `PhragmenBV`, `ChainKernelContinuity`), the Hecke identification and its multiplier packaging (`WeilIdentification`, `MultiplierPackaging`, Theorem 89), the Maass wave-form identification (`MaassIdentification`), and the Sym^r exchange interface (`ExchangeInstrument`). The classical inputs consumed there remain exactly two, both cited at typed fields: the completed package with bounded vertical strips per rung, and Weil's finite-twist lemma at level N.

37 Hodge Connection

The junction at the seed: freeze or drift. The Langlands–Hodge junction — temperedness is a freeze law (`StrandLatticeFreeze`: a lattice strand under the tower ceiling is a sixth root of unity, purity supplied by the compiled radial limit, μ_6 by the compiled lattice classification) — now lands at the campaign's typed seed (§35), the same object carrying the packages, the converse, the wave-form identification, and the Sato–Tate family: a Hodge-type certificate at a prime (the Satake datum a nonzero $\mathbb{Z}[\zeta_6]$-integer — the rational constant-mode datum of the harmonic frame) forces, under the same ceiling field the Sato–Tate wiring consumes, the sixth-power freeze (`HodgeSeedFork.seed_freeze`) and pins the family's angle to the μ_6 boundary set $\{0, \pi/3, 2\pi/3, \pi\}$ of the house cells (`seed_freeze_angle`). The drift branch is the compiled Sato–Tate chain, and the incompatibility is now closed with one exact common mode (`FrozenDrift`): for any sixth root of unity the trace ladder satisfies $2S_2 - S_4 = 1$ on the nose ($\alpha^4 = \alpha^{-2}$ collapses it, `trace_mu6_identity`), so a fully certified seed's prime averages obey the constant relation $2\,\text{avg}_2 - \text{avg}_4 = 1$ at every stage (`frozen_average_relation`) — ranks 2 and 4 cannot both cancel (`frozen_violates_cancellation`), and by the compiled biconditional a fully frozen seed provably does *not* equidistribute (`frozen_not_equidistributed`). No moment matrix, no compactness, no measure theory: the common mode is computed exactly and books the contradiction. The fork is Sato–Tate's own non-Galois-type hypothesis, typed at the seed and now two-sided: rational data freeze and provably do not drift; generic data drift.

The shadow detector at every order. The μ_6 certificate is one instance of the structure that actually matters — a *monomial* (induced, CM-type, Galois-type) shadow of *any* finite order — and the detector covers them all at once (`ShadowDetector`). The general common mode is exact: for $\alpha^N = 1$ the rank-N symmetric-power trace collapses to a geometric sum in α^{-2} of period N and equals *exactly* 1, or $N + 1$ in the degenerate case $\alpha^2 = 1$ (`trace_root_of_unity_re`) — in particular its real part is ≥ 1 with a gap independent of the datum. Hence a family carrying one common finite order at every prime (`MonomialShadow`) has rank-N prime average ≥ 1 at every stage, its rank-N cancellation fails, and it provably does not equidistribute (`shadow_not_equidistributed`); contrapositively, a drifting family admits *no* finite-order shadow at any order (`no_shadow_of_equidistributed`). A single bounded-rank moment separates the branches, with constant gap — no moment matrix, no compactness. These are rigidity and detectability statements about the trace family; they carry no computational-cost content and say nothing about recovering an inducing datum. Beyond the fork itself, the compiled ring of this section supplies the *template* the Hodge companion's terminus names for its constructive arm [30]: the cycle-level recognition problem there is exactly the two-arrow shape compiled here — the graded tower record with its ledger playing the package, the sourcing cycle playing the theta identity — so the companion's "build the recognition theorem" is sharpened to "instantiate this shape at cycle level."

The loss-ledger $\leftrightarrow$ cycle-filtration correspondence. The carrier's descent 3D $\to$ 2D $\to$ 1D books a loss ledger — height (ordinate), radius, angle — and its leading central jet factors, by BSD/Gross–Zagier [23, 24], into three invariants the census resolves *independently*: the vanishing order, the regulator, and |Sha| — three cycle-filtration layers ordinarily invisible to Hodge projection, each carrying a *distinct* retained coordinate. This is theorem-level for semisimple carrier states (§37.1) and the *target* for extension classes. The instruments corroborate: the regulator/value channel fires for every homologically-trivial cycle in the census (no silent cycle), and a class silent at depth 1 first fires at depth 2 with its first nonzero slot at the expected tower level (`hodge_griffiths_probe.py`, `tower_exhaustion_test.py`, App. B); the signature-based cycle detection of §32 remains preliminary (`hodge_layer_calib.py` surfaced a structural question out of scope here). The full development is the Hodge companion paper [30]: the carrier filtration F_{car} and the depth dictionary at its exact (definitional) pricing, the Bloch–Beilinson coincidence priced as an interface, finite-extension retention, the five-condition realization interface with its executed rungs (the rational group algebra, Artin motives, the divisor and function-field-extension pairing rungs), the Klein-quartic L-side record, and the typed Ceresa/Gross–Schoen landing (`CeresaGrossSchoenLanding.complete_landing`).

37.1 Cycle-filtration faithfulness: finite extension retention and the Ceresa frontier

The structural half is unconditional; the frontier is priced. On semisimple states and on every finite normalized jet stack the moment/extension tower is jointly faithful — trivial radical, Retention, no Bloch–Beilinson input (`momentTower_exhaustive`, `generalExtensionTower_exhaustive`; Lean) — so for a homologically-trivial cycle the remaining obligation is exactly the typed realization of its extension datum, and retention then follows by transport rather than by a new no-silent-layer assumption. The Griffiths registers — why the transcendental class is *not* placed in the carrier radical, the unconditional Fermat-quartic first rung (Harris [29]; Ceresa [28]), and the Laga–Shnidman control [32] showing the bielliptic Beilinson–Bloch height does not penetrate $\ker AJ_{\mathbb{C}}$ — are carried at full strength in the companion [30], together with the named deeper target (a class with zero or torsion $AJ_{\mathbb{C}}$ and a nonzero *arithmetic* level-three invariant, where the non-archimedean local height [26] or the ℓ-adic Ceresa class is the retained coordinate) and the typed analytic-to-cycle implication (`CeresaGrossSchoenLanding.complete_landing`).

Appendix

Conclusion

What stands, and at what strength. The carrier system of Part I is proven and machine-checked end to end: projection with its ledger is a bijection, sources are synthesized from finite data alone, vanishing is exact focal cancellation, and the transport is universal and functorial. The $S(t)$ term is proven to be the registration gap between the scaled carrier and the unit chart — the chart-defect compensation mechanism, described and discharged. Symmetric-power functoriality holds at every rank with niceness discharged on the carrier, and Sato–Tate for Maass forms follows on the Galois-free route. Beyond Endoscopy's uniformity problem is reduced to a single magnitude bound and its obstruction identified as the orbital transform's own clock, with a repair that scales. The worked example and its GRH interfaces now live in the companion paper [31], at the same registers they carried here: the helix midpoint a geometric output of area-law scale balance and no drift; a native 3D zero an exact focal cancellation at physical height $Z > 0$, exactly the analytic L-zero at $\sigma_* + i \log Z$, its finite-bank detector equivalent to harmonic-pencil rank drop; the global $S(t)$ coordinate law carrying the carrier-scale census to the unit chart, sharpened by the independent contour census to $S_\Gamma(T) = S_{\mathrm{mult}}(T) + N_{\mathrm{off}}^{\mathrm{mult}}(T)$; and the operator route sharpened to a single named limit — windowed exhaustion of the compiled ledger — proved *equivalent* to Mathlib's `RiemannHypothesis` from both sides (`rh_iff_symmetrizedLineTower_traceLimit`), with the earlier uncompleted typing retired by its own impossibility theorem (`no_selfAdjoint_uncompleted_capture`). Nothing in that companion is consumed by the theorems of this paper.

The thesis, restated at the end as at the start: the unit-1 chart is the compression, not the object. Working in the scaled three-dimensional frame turned the $S(t)$ registration into a theorem, the convergence gate into a chart artifact, and a stalled trace-formula estimate into an exact gauge. The next arithmetic work is in the Hodge companion paper: instantiating the typed cycle-to-jet realization and source packages on the desired geometric classes. The kernel now contains finite-extension retention, higher-grade certified source construction, the Ceresa/Gross–Schoen landing implication, and their unrestricted source-exhaustion assembly (§37.1). Everything else that remains is bookkeeping at its own site: the Lean formalization scope is recorded where the theorems live (§29, Prop. 59). The theorems are checked and the artifacts are public.

A The exact gauge on the archimedean uniformity

Altuğ [2, 3, 4] executes Beyond Endoscopy for GL(2) and the standard representation. Its analytic core is a uniformity estimate: after the arithmetic weights $L(1, \chi_D)$ are expanded by an approximate functional equation and the trace variable is Poisson-summed, one must bound, uniformly in (ξ, ν, l, f, X), the Fourier transform $J_{l,f}(\xi, \nu, X)$ of a product of a fixed cutoff G and a smoothed orbital transform (Prop. 5.2, Part III); Altuğ calls the C, D-independence of the implied constant "the central issue" [3, App. A]. The transform whose asymptotics drive the appendix is (Part II, Def. A.6)

$$A^{\tau,\pm}_{h_a,m}(\Phi)(x) = \frac{1}{2\pi i} \int_{(\tau)} \widetilde{\Phi}(u)\, c^{\pm}_m(u/2)\, \Gamma\!\left(m + 1 + \tfrac{a+u}{2}\right) x^{-u/2}\, du, \qquad \tau > 0, \tag{13}$$

for a singular profile $h_a(x) = |1 - x^2|^{a/2} h_1(x)$ (h_1 smooth, $a > -1$) and Schwartz Φ; the parameters enter only through $x = \mp C^2 D$ and a unimodular prefactor.

Lemma 90 (the gauge bound)**.** *For every $\tau > 0$, $m \in \mathbb{N}$, and Schwartz Φ,*

$$\left|A^{\tau,\pm}_{h_a,m}(\Phi)(x)\right| \;\le\; B_{\tau,m,a}(\Phi)\,x^{-\tau/2}, \qquad B_{\tau,m,a}(\Phi) := \frac{1}{2\pi}\int_{-\infty}^{\infty}\left|\widetilde{\Phi}(\tau+iv)\,c^{\pm}_m\big(\tfrac{\tau+iv}{2}\big)\,\Gamma\big(m+1+\tfrac{a+\tau}{2}+\tfrac{iv}{2}\big)\right|dv,$$

with $B_{\tau,m,a}(\Phi) < \infty$ independent of x *(hence of C, D).*

Proof. This is Theorem 25 applied to the contour multiplier

$$M(u) = \widetilde{\Phi}(u)c^{\pm}_m(u/2)\Gamma\big(m+1+\tfrac{a+u}{2}\big).$$

It gives the displayed bound and the independence of $B_{\tau,m,a}(\Phi)$ from x. For finiteness: by Stirling $|\Gamma(\sigma+iv/2)| \asymp e^{-\pi|v|/4}$; the phase $i^{\,1+m+(a+u)/2}$ has magnitude $e^{-\pi v/4}$; and $|\widetilde{\Phi}(\tau+iv)| \ll e^{-\pi|v|/2}$. The product decays like $e^{-\pi|v|}$ as $v \to +\infty$ and $e^{-\pi|v|/2}$ as $v \to -\infty$ (there the Γ- and phase-magnitudes cancel and $\widetilde{\Phi}$ carries the decay); the Γ-poles lie at $u < 0$ and are never met for $\tau > 0$. ◻

The step $|\int g| \le \int |g|$ is lossless here: in the coordinate (13) the integrand is a real magnitude times a deterministic phase, so no arithmetic cancellation is discarded, and the C, D-dependence factors out as a pure power. There is no oscillation estimate. The bound is numerically certified on a representative Gaussian Φ: $B_{1,0,1}(\Phi) = 0.686$, and the certificate margin $\sup_x |A(x)|\,x^{\tau/2}/B = 0.914 \le 1$ holds across $x \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$. (The certificate fixes one Φ; the bound of Lemma 90 is general, holding for every Schwartz Φ through the finite line integral $B_{\tau,m,a}(\Phi)$.)

Theorem 91 (the sharp expansion, Part II Thm A.14)**.** *With h_a, Φ as above, for every M,*

$$I(C,D) := \int_{-1}^{1} h_a(x)\,\Phi\big(C/\sqrt{1-x^2}\big)\,e(xD)\,dx = \sum_{m=0}^{M}\frac{e(\pm D)\,A^{\pm}_m(\mp C^2 D)}{(\mp 2\pi i D)^{m+1+a/2}} + R_M(C,D),$$

with $A^{\pm}_m$ as in (13) *and $|R_M| \le \mathcal{B}_{M,\tau_1}(C^2D)^{-\tau_1/2}D^{-(M+1+a/2)}$, $\mathcal{B}_{M,\tau_1}$ independent of C, D.*

Proof. Localization. Fix a smooth partition $1 = \chi_+ + \chi_0 + \chi_-$ with $\chi_\pm$ supported in $|x \mp 1| < \delta$ and χ_0 in $[-1+\delta, 1-\delta]$. On χ_0 the phase $x \mapsto x$ has nonvanishing derivative, so the non-stationary phase lemma [8, Thm. 7.7.1] (N-fold integration by parts) gives an $O_N(D^{-N})$ contribution with implied constant controlled by $\|\partial_x^N(\chi_0 h_a\,\Phi(C/\sqrt{1-x^2}))\|_{L^1}$. This is bounded uniformly in C: on $[-1+\delta, 1-\delta]$ the factor $\Phi(C/\sqrt{1-x^2})$ and its x-derivatives are bounded uniformly in C, since for Schwartz Φ and $c \ge \delta' > 0$ one has $\sup_{C>0}|C^k\Phi^{(k)}(Cc)| < \infty$.

Endpoint reduction. Near $x = 1$ substitute $x = 1-t$, $t \in (0,\delta)$: then $h_a(1-t) = t^{a/2}\varphi(t)$ with $\varphi(t) = (2-t)^{a/2}h_1(1-t) \in C^\infty[0,\delta)$; $\sqrt{1-x^2} = \sqrt{t}\,\sqrt{2-t}$; and $e(xD) = e(D)e(-tD)$, so $I_+ = e(D)\int_0^\delta \chi_+(1-t)\,t^{a/2}\varphi(t)\,\Phi\big(C/(\sqrt{t}\sqrt{2-t})\big)e(-tD)\,dt$.

Mellin representation and closed-form t-integral. For $\sigma > 0$ in the strip of holomorphy of $\widetilde{\Phi}$, insert $\Phi(w) = \frac{1}{2\pi i}\int_{(\sigma)}\widetilde{\Phi}(u)w^{-u}\,du$ with $w = C/(\sqrt{t}\sqrt{2-t})$. The joint integrand is absolutely integrable on $\{\operatorname{Re} u = \sigma\} \times (0,\delta)$ — the t-integral converges absolutely and $\widetilde{\Phi}(\sigma+iv)$ decays exponentially in v — so Fubini applies. Writing $\psi_u(t) = \varphi(t)(2-t)^{u/2}\chi_+(1-t) \in C^\infty$ and Taylor-expanding $\psi_u(t) = \sum_{m=0}^{M} b_m(u)t^m + R_M(u,t)$, the m-th t-integral is, by Watson's lemma / the Mellin evaluation of an oscillatory monomial [9, Ch. 3] (the smooth cutoff at $t = \delta$ costs only $O(D^{-\infty})$),

$$b_m(u)\int_0^\infty t^{m+\frac{a+u}{2}}\,e(-tD)\,dt = b_m(u)\,\Gamma\big(m+1+\tfrac{a+u}{2}\big)(2\pi i D)^{-(m+1+\frac{a+u}{2})} + O(D^{-\infty}).$$

The oscillation is evaluated in closed form, with no cancellation estimate. Since $C^{-u}D^{-u/2} = (C^2D)^{-u/2}$, collecting the u-contour integral gives the m-th term $e(D)(2\pi iD)^{-(m+1+a/2)}A_m^+(C^2D)$ with A_m^+ exactly (13): the coefficients $b_m(u)$, carrying the $(2-t)^{u/2}$ Taylor data, reproduce Altuğ's $c_m^\pm$. This identification is a *direct* comparison of Taylor data, needing no appeal to uniqueness of the asymptotic expansion — whose coefficients $A_m^\pm(C^2D)$ are in any case not constant in D, depending on it through C^2D. By construction $b_m(u)$ is the m-th Taylor coefficient at the endpoint $t = 0$ of the amplitude $\psi_u(t) = \varphi(t)(2-t)^{u/2}\chi_+(1-t)$; Altuğ's $c_m^\pm(u/2)$ is, by his Definition A.6, the same m-th endpoint Taylor datum of the same amplitude, the $(2-t)^{u/2}$ factor supplying the $u/2$-dependence. The Mellin evaluation above and Altuğ's Watson-lemma reduction extract that one Taylor coefficient term by term, so $b_m(u) = c_m^\pm(u/2)$ directly.

Remainder. The Taylor remainder satisfies $|R_M(u,t)| \le C_M(u)\, t^{M+1}$ with $C_M(u) = \sup_{[0,\delta]} |\partial_t^{M+1}\psi_u|/(M+1)!$ of at most polynomial growth in $\Im u$ (ψ_u smooth). Its contribution is again a transform of the form (13) with amplitude R_M; taking its u-contour at $\operatorname{Re} u = \tau_1$ (any $\tau_1 > 0$), Lemma 90 bounds it by $\mathcal{B}_{M,\tau_1}(C^2D)^{-\tau_1/2}D^{-(M+1+a/2)}$ with $\mathcal{B}_{M,\tau_1} < \infty$ independent of C, D. The endpoint $x = -1$ gives the $e(-D)$, A_m^- terms identically. □

Remark 92 (on the exponents)**.** The contour height $\tau_1 > 0$ is free, and the remainder decays like $(C^2D)^{-\tau_1/2}$ because the factor $(C^2D)^{-u/2}$ is evaluated at $\operatorname{Re} u = \tau_1$; Altuğ's stated $(C^2D)^{-\tau_1}$ is the same statement in his normalization of the Mellin variable (u versus $u/2$). What the application uses is only that, for arbitrary $A > 0$, the remainder is $O_A\big((C^2D)^{-A}D^{-(M+1+a/2)}\big)$ uniformly in C, D — immediate from the freedom in τ_1 together with Lemma 90.

Proposition 93 (Part III Prop. 5.2)**.** *$|J_{l,f}(\xi,\nu,X)| \ll X^2\nu^{-M}(lf^2)^{-N}\big((lf^2/\sqrt{X})^{N-M+3} + \xi^M\big)$ with implied constant independent of (ξ,ν,l,f,X).*

Proof. Write $J_{l,f}(\xi,\nu,X) = \int G(y/X)\, y\, \Psi(y)\, e(-y\nu/2lf^2)\, dy$ with $\Psi(y) = I_{l,f}(\xi, y)$. Since G is compactly supported, integrating by parts M times against the phase gives the exact identity $J = (lf^2/(-\pi i\nu))^M \int \partial_y^M(G(y/X)\, y\, \Psi(y))\, e(-y\nu/2lf^2)\, dy$, hence

$$|J_{l,f}(\xi,\nu,X)| \le \left(\tfrac{lf^2}{\pi\nu}\right)^M \big\|\partial_y^M\big(G(y/X)\, y\, \Psi(y)\big)\big\|_{L^1(y)}.$$

By the Leibniz rule $\partial_y^M(G(y/X)\, y\, \Psi) = \sum_{i+j+k=M}\binom{M}{i,j,k}X^{-i}G^{(i)}(y/X)\,(y)^{(j)}\,\partial_y^k\Psi$, where $(y)^{(j)} = y, 1, 0$ for $j = 0, 1, \ge 2$. The cutoff $G^{(i)}(y/X)$ is supported on $y \in [\frac{X}{4}, \frac{5X}{4}]$ (length $\asymp X$) and contributes $\|G^{(i)}\|_\infty$, and the factor $(y)^{(0)} = y \asymp X$ supplies a second power of X, giving the X^2. Each y-derivative of $\Psi(y) = \int \theta_\infty(x)[F+H](\mathrm{arg})\, e(-x\xi\sqrt{4y}/4lf^2)\, dx$ falls on either the phase — producing a factor $O(\xi/lf^2\sqrt{y})$, the source of the ξ^M branch — or on the profile through $\mathrm{arg} = lf^2/(\sqrt{4y}\sqrt{1-x^2})$, producing a factor $O(lf^2/y)$, the source of the $(lf^2/\sqrt{X})$ branch. Each resulting x-integral is a transform of the form (13) (with $C \asymp \xi\sqrt{y}/lf^2$ and D the local scale), bounded by Lemma 90, or at the sharp order by the expansion of Theorem 91. Here N indexes the truncation order of the Theorem 91 expansion of the inner x-transform — the number of amplitude Taylor terms retained — independent of the y-integration-by-parts count M. The powers assemble as in Altuğ's Proposition 5.2: the M-fold y-by-parts gives the ν^{-M}; expanding each inner transform to order N by Theorem 91 gives $(lf^2/\sqrt{y})^N$ on the profile branch and the ξ^M on the phase branch; and the $y \asymp X$ support with the $y\, dy$ factors supplies the X^2 and the $+3$, so the profile branch collects to $(lf^2/\sqrt{X})^{N-M+3}$. We carry this power-counting only to the point where each inner bound reduces, term by term, to the single magnitude estimate of Lemma 90 — the exact-gauge substitute for Altuğ's oscillation estimate, which is the sole change from his Proposition 5.2 — whose constants

$B_{\tau,m,a}(\Phi)$ and $\|G^{(i)}\|_\infty$ are finite and independent of (ξ, ν, l, f, X); the full exponent bookkeeping is his. □

Every uniform constant in §A is one application of Lemma 90; the content of Altuğ's Appendix A, from the exact-gauge standpoint, is that single magnitude estimate. Where Altuğ's route controls the transform through the asymptotic expansion together with an oscillation bound on the remainder, the exact gauge replaces that oscillation bound by the triangle inequality in the coordinate that makes the integrand real: the whole (ξ, ν, l, f, X)-uniformity then rests on the x-independence of the single line integral $B_{\tau,m,a}(\Phi)$, with no cancellation estimate at any step.

B Reproducibility

Every measured statement is produced by a self-contained script (no L-function library); the values quoted in the text are reproduced below with the script that computes each.

- **Lemma 90 (the gauge bound), `be_uniformity_certify.py`.** For a Gaussian Φ, $B_{1,0,1}(\Phi) = 0.686$, and the certificate margin $M(x) = |A(x)|\, x^{\tau/2}/B$ satisfies $\sup_x M(x) = 0.914 \leq 1$, uniformly across $x = C^2 D \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$. The line integrand decays as $e^{-\pi|v|}$ ($v \to +\infty$) and $e^{-\pi|v|/2}$ ($v \to -\infty$), so $B_{\tau,m,a}(\Phi) < \infty$ for every $\tau > 0$.
- **Proposition 93 (the uniform constant), `be_prop52_certify.py`.** The constant $L_M = \|\partial_y^M (G(y/X)\, y\, \Psi)\|_{L^1}$ fits a uniform monomial $L_M \sim C_M (lf^2)^{a_M} (1+\xi)^{b_M} X^{g_M}$ with $a_M = 0.41, 0.05, -0.05$ for $M = 1, 2, 3$ and $b_M \leq 0$. The lf^2-exponent a_M grows more slowly than M ($da/dM \approx -0.23$), so each integration by parts nets ν^{-1} and the bound is summable — the standard-representation side of the productivity boundary. (This is the crude magnitude route; the sharp exponents of Proposition 93 follow from Theorem 91.)
- **The registration instruments (§9.10), `st_mu6_gram.py`, `st_cf_lags.py`, `st_cell_phase.py`.** Four pre-registered tests of positional coupling between the zero events and the lattice layers, all null, as the proven ergodicity predicts. (1) Gram parity: violation rates 0.0728 vs 0.0720 at N=5000 ticks — flat. (2) mod-6 alignment of the Gram defect: the run-1 signal ($p = 0.0355$, IID permutation, N=1200) fails to replicate under the autocorrelation-preserving circular-shift null at N=5000 ($p = 0.058$ full range, $p = 0.49$ upper half); the IID null overstates significance for Rosser-block-correlated sequences — the circular shift is the correct null for alignment tests. (3) CF lags $\{7, 106, 113\}$ (the π-convergent denominators): 66th percentile, $p = 0.32$; the visible autocorrelation is short-range spacing rigidity. (4) Cell-phase kinks: the zeros' phases $x = \mathrm{frac}(\theta(\gamma)/\pi)$ follow the smooth classical S-shape (mean 0.4980, the half-jump shift); density kinks at the carrier subdivision $x = 1/3, 2/3$ are *smaller* than at control breakpoints (0.0396 vs 0.0477). Method laws recorded: sampling at the chart's own ticks aliases the carrier subdivision away (instruments must sample the carrier coordinate); and positional statistics test a claim the ontology disclaims — the $\pi/3$ claim is cancellation *at* events, never placement *of* events.
- **The emergent clock (§22), `mb_beat.py`.** The script scans the clock coordinate $\nu_{\mathrm{clock}} = 2\nu_J$, so its profile is $e(-y\nu_{\mathrm{clock}}/4lf^2)$ while the analytic J of Prop. 93 uses $e(-y\nu_J/2lf^2)$. The current estimator fits the continuous log-power profile $\log|V(\nu_{\mathrm{clock}})|^2$ with a degree-7 envelope plus the first two harmonics sharing one fundamental; the error audit compares one, two, and three harmonics. In that coordinate the fitted clock law is $\Delta\nu_{\mathrm{clock}} = 0.524(lf^2)^{1.03}(X/8)^{-1.04}(1+\xi)^{0.00}$ with $R^2 = 1.000$; equivalently $\Delta\nu_J = \frac{1}{2}\Delta\nu_{\mathrm{clock}}$. Per-cell $\kappa_{\mathrm{eff}} = 4lf^2/(X\Delta\nu_{\mathrm{clock}})$ has mean 0.943, standard deviation 0.025, and range 0.895–0.971; the harmonic-truncation median period shift is

2.4×10^{-4} (largest 6.7×10^{-3} on the weakest long-period cell). The legacy deep-dip estimator gave $0.533(lf^2)^{1.00}(X/8)^{-0.98}(1+\xi)^{0.02}$ at $R^2 = 0.996$. The rank sweep `mb_beat.py ranks 13` applies the same $U_r(x)$ multiplier for $0 \le r \le 13$ and separates two masks: continuous clock cells and deep-event cells. On clock cells every rank has $R^2 = 1.000$, lf^2-exponent 1.03–1.04, and X-exponent -1.04–-1.03. The continuous clock is present in all 12 nonzero-ξ cells through $r = 9$, then in $10, 9, 8, 8$ cells for $r = 10, 11, 12, 13$; deep productivity drops faster, from $10/12$ at $r = 0$ to $3/9$ and $3/8$ at $r = 11, 12$. Thus the high-rank degradation is event visibility, not a breakdown of the clock law. The Beyond-Endoscopy scaling test is therefore cell-wise: first certify the phase clock on each cell, then measure the productive deep-event yield inside the certified cells. The same run confirms that the deep-dip comb vanishes at $\xi = 0$ for $r = 0$ (the integrand is unimodal there).

- **The GL(3) clock lift, `mb_gl3_clock_lift.py`, `mb_gl3_wall_lift.py`, `gl3_family.py`.** The GL(3) lift keeps the clock law but does not keep it as one scalar MB clock: the two trace coordinates carry separate clocks, and the diagonal readout combines them as $Q/(S_a + S_b)$. The split-coordinate clock ratios are pinned at median $1.0000, 1.0000, 1.0000$ for the a-, b-, and diagonal readouts, while the same-b scalar collapse gives 0.8983. The wall-lift scan shows a persistent clock but only partial scalar compression of the spread (base $R^2 = 0.541$, full scalar-feature $R^2 = 0.623$–0.640 on this grid). The family scan pins all four tested GL(3) forms, with universal cells 1.0005π and across-form spread 0.0001π. Thus the higher-rank scaling rule is vector-clock extraction plus productivity accounting, not a single GL(2)-style period.

- **The formally proven GL(4) vector-clock scale audit, `mb_scale_audit.py` and `TwoClockWeightLaw.lean`.** The audit composes the GL(2) rank sweep, the GL(3) split clock, and the GL(4) three-coordinate vector-clock theorem. In Lean, `gl4_trace_window_vector_clock` proves that three independent trace spans force $\Delta_i = Q/S_i$ on the three trace coordinates and force the locked diagonal readout $\Delta_{\mathrm{diag}} = Q/(S_a + S_b + S_c)$; `gl4_scalar_collapse_not_uniform` proves that scalar collapse across distinct spans is obstructed exactly by span inequality. For GL(2), the clock law is certified at $R^2 \ge 0.999$ for 14/14 ranks while productive deep-cell yield ranges from 0.33 to 0.83. For GL(3), the vector clock passes 24/24 a-cells, 24/24 b-cells, and 24/24 diagonal cells, while the anisotropic same-clock control has median 0.7966 on the b-axis. For GL(4), the vector clock passes 36/36 cells on each of the a, b, c trace coordinates and 36/36 diagonal cells; the scalar controls have medians 0.9715 and 0.9319 on anisotropic b- and c-windows, confirming the formal scalar-collapse obstruction and preserving the vector readout as the scaling law. The productivity ledger records the measured wall accounting alongside the formal clock theorem: visible wall cells are 109/144, coherent-envelope cells 76/144, and near-vector productive cells 28/144.

- **The productivity boundary (§24), `mb_uniform.py`.** The Weyl-character tail law tail $\sim (r+1)\sqrt{\#\text{harmonics}}$ fits the measured tail mass $0.042, 0.092, 0.135, 0.156, 0.256$ ($r = 0, \ldots, 4$) at $R^2 = 0.977$, well above the scrambled baseline 0.237; the $\xi = 0$ detecting residue against the $\xi \ne 0$ floor is 0, 55.9, 1.4×10^{-16}, 11.4 for $r = 1, 2, 3, 4$.

- **The candidate object (§29).** The converse-theorem inputs — the reflection-axis functional equation (warp-invariant, $\operatorname{Re} s = \frac{1}{2}$ forced), the strip continuation of the readout, boundedness, three-dimensional exhaustion, the midpoint landing on $\operatorname{Re} s = \frac{1}{2}$, and loss-ledger recovery — are formalized and machine-checked in Lean (Mathlib): `RequestProject/AutomorphicCandidate.lean`, theorem `candidate_wellformed`; it builds in-tree against the full library. The continuation is `LFunctionPhasor.dirichlet_strip_tendsto_LFunction` (non-principal χ) and `eta_strip_tendsto` (ζ/principal); the reflection axis and forced abscissa are `FrobeniusSimilitude.reflection_fixes_iff`, scaleBalanced_iff.

- **The general finite duality-stable fiber reflection (§32, §31), `RequestProject/FiniteWeightFiber.lean`.** The polymorphic carrier reflection, machine-checked in Lean (Mathlib) for *any* finite weight multiset closed under the duality involution $\lambda \mapsto \lambda^{-1}$ with the balanced ledger — not the Chebyshev string: the ledger $\prod_i \lambda_i = 1$ (`fiber_det_one`, via `Finset.prod_involution`), the self-reciprocal local factor $\prod_i(1 - \lambda_i X) = (-X)^{|I|} \prod_i(1 - \lambda_i X^{-1})$ (`localPoly_reciprocal`) — the per-place functional equation of the shadow $\det(1 - r(\varphi)q^{-s})^{-1}$ — the reflection axis $\operatorname{Re} s = \frac{1}{2}$ (`fiber_reflection_axis`), and warp-invariance (`warpFiber`, `warpFiber_det_one`). The symmetric power `symFiber` (weights α^{r-2k}, involution `Fin.rev`) and the Rankin–Selberg tensor `tensorFiber` are instances (`symFiber_det_one`, `tensorFiber_det_one`); the Maass fiber is the same finite algebra.

- **The carrier-cell theorem (§31), `RequestProject/CellClosure.lean`.** Cell closure proved for the finite duality-stable harmonic fiber class, machine-checked in Lean. `cell_closure_partialSum_le`: a period-P warp bounded by B with zero cell sum has primitive $\leq B\,P$; `cell_closure_dc_mode_zero`: hence $A(N)/N \to 0$ ($R = 0$). `root_of_unity_cell_sum_zero`: $\sum_{k<P} \zeta^k = 0$ for $\zeta^P = 1$, $\zeta \neq 1$; `harmonic_bank_cell_sum_zero` and `harmonic_bank_primitive_bounded`: for a fiber whose weight channels are P-th roots of unity, none trivial, the warped bank closes every cell and its primitive is bounded by $|\iota|\,P$ — Galois-free, at the weight level. `dirichlet_cell_closure` is the χ-mod-q instance (recovering `character_partialSum_norm_le`).

- **The completed-reflection assembly (§31), `RequestProject/CompletedReflection.lean`.** The completed functional equation $\Lambda(s) = \varepsilon\,\Lambda^\vee(1 - s)$ as the machine-checked *assembly* of the finite reflection and the completion self-duality (`CompletedReflection.completed_FE`), instantiated on the Dirichlet model from Mathlib's completed-L functional equation (`dirichletCompleted`, via `completedLFunction_one_sub`) *and* on the general finite duality-stable fiber (`FiniteWeightFiber.fiberComplet` next-but-one entry). The *entireness mechanism* — focal cancellation forcing the harmonic Gram-pencil rank-drop, with the normalized residual an exact nonzero multiple of the readout and hence no independent residual mode — is formally proven in Lean at the channel-agnostic (general-fiber) level (`FocalResidualGeneral`, next entry). The completed twisted functional equation is assembled by `completed_FE` from three inputs: fiber admissibility (`localPoly_reciprocal`, general Lean), the completion self-duality (the self-dual clock, general Lean), and the *carrier reflection* of Theorem 69 — read at the weld axis, where height inversion coincides with conjugation, the self-dual clock weld `ChiralityHB.symClock_star` per-clock (general Lean) supplying the conjugation leg and the warped-Abel continuation carrying the axis identity off-axis (Lemma 75, no termwise bank interchange claimed) — specialising to the classical theta functional equation for the Dirichlet model; for the general family the finite-stage strand exchange is assembled in Lean (`RequestProject/StrandExchange.lean`, with the general dual-pair reciprocity in `RequestProject/DualPairFiber.lean` and the transfer analyticity in `TransferContinuation.transfer_analytic`); the completed niceness — entire continuation, strip bounds, and functional equation — is moreover proved unconditionally for every twist as a single carrier object (`cpsPolynomial_twistedNiceness`), so the $\operatorname{Re} s > 1$ Euler-product abscissa is a projection-chart gate, not a carrier barrier. The classical-L identification (Thm. 57 with (E5)) is closed on the carrier and reproduced numerically through Sym^{13}; neither it nor convergence is owed. `fiberCompleted` certifies the fiber's admissible *local* reflection algebra; it is not the global carrier reflection.

- **The channel-agnostic focal residual (§31), `RequestProject/FocalResidualGeneral.lean`.** The residue-free entireness core, lifted from the Dirichlet model to any admissible channel pair — hence to the P/M channels of a general det-one fiber. `harmonic_residual_exact`: the normalized focal residual $\det H/A = (\lambda - \mu)B$ is an exact nonzero multiple of the signed channel, carrying no independent constant mode; with the channel-agnostic Gram rank-drop `harmonicGram_rank_drop_iff_channel_zero` this makes the rank-drop and the residual-vanishing the *same* event (`harmonic_residual_no_independent_mode`). It is the general form of `HarmonicPencilCell.normalized_cell_focal_residual_exact` and formalizes step (v) of the (E4) chain (Lemma 63) for the symmetric-power/twist fibers, not only Dirichlet.

- **The fiber's local completed reflection (§31, step (1) of Prop. 76), `RequestProject/CompletedReflectionFiber.l`** The completed-reflection *assembly* instantiated for any finite duality-stable fiber, certifying fiber admissibility at the completed-object level — *not* the global twisted FE. `fiberCompleted` builds a `CompletedReflection` whose finite reflection is `localPoly_reciprocal` at the reflecting variable $c^{s-1/2}$ (`reflVar_one_sub`: $X(1-s) = X(s)^{-1}$) and whose completion is the self-dual clock symClock times its $s \mapsto 1-s$ reflection (`clockCompletion_selfdual`); `fiberCompleted_FE` reads off $\Lambda(s) = (\varepsilon_{\mathrm{fin}}\varepsilon_{\mathrm{arch}})\Lambda^{\vee}(1-s)$ for this *local* object via `completed_FE`, and `symTensorCompleted` instantiates it at `tensorFiber(symFiber` r, W_τ`)`. The *global* twisted functional equation additionally requires the carrier reflection of Theorem 69 (step (2), not this file).

- **Universality across families (§29), `run_universality.py`.** The unmodified house locator (`focal_closure.py`), run oracle-free on $37a$, $389a$, a CM Grössencharacter form, the S_3 level-23 weight-one form, and $\Delta \times E_{11}$ (degree four), with an independent smoothed-AFE evaluator (`afe.py`, validated on Δ/E_{11}): critical- line readout to 2×10^{-16}; root number and central vanishing as quoted ($389a$: 4.9×10^{-17}); off-central focal residuals to 6.3×10^{-8}, located ordinates matching true zeros to 6.7×10^{-9}, $\operatorname{Re} s = \frac{1}{2}$ by construction. No falsification; the locator's central-point blindness ($Z = e^0 = 1$) is a limit of the growth locator, not the representation.

- **The clipped helix: clip, weld, rank, crossings (§29), `clip_conductor.py`, `weld_normalization.py`, `bsd_rank_ladder.py`, `crossing_spacing_proof.py`.** The live phasor count obeys $n_{\mathrm{eff}}/\sqrt{N} = 0.733$, constant to five digits across $N = 10$–10^8 (exponent 0.5000; the $\log N$ and N laws are falsified); a degree-one character's single outward strand *decays* to convergence while a degree-two elliptic strand *saturates* at a bank-independent floor (open vs. clipped). The weld sits at the ordinate origin $t = 0$ ($Z = 1$), not the spiral base $N = 0$ ($Z = 0$). The double-ended jet tower reproduces $(\sqrt{N}/2\pi)\,L^{(r)}(1)/r!$ on $11a, 37a, 389a, 5077a$ (ranks 0–3) to agreement 1.00000, 1.00000, 0.99998, 0.99974, with a CM control ($27a$); the argument-principle crossing count matches the actual crossings to the integer at all nine test points (ζ, χ_4, χ_8), the half-turn offset traced to $Z'(0) = 0$.

- **The automorphy machinery on the carrier (§30), `tate_fe_check.py`, `gl2_escalate.py`, `carrier_poisson.py`, `emergent_quadclock.py`, `sp4_tower.py`, `gln_tower.py`.** The two-sided theta gives the ζ functional equation $\Lambda(s) - \Lambda(1-s) = 0$ (and $\pi^{-s/2}\Gamma(s/2)\zeta(s)$) to 10^{-32}, the Δ functional equation from modularity to 10^{-23}, and $\mathbb{Q}(i)$ from the tensor lattice to 0; the Tate archimedean integral equals $\frac{1}{2}\pi^{-s/2}\Gamma(s/2)$ exactly. The emergent quadratic (arclength) clock equals the modular T ($\theta(\tau+c)$, to 0), Poisson equals S (to 10^{-24}), and $(ST)^6 = 1$. The Siegel-theta Poisson $\theta(-\tau^{-1}) = \det(\tau/i)^{1/2}\theta(\tau)$ holds to 10^{-16} with S, T_B symplectic; the $\mathrm{GL}(n)$ Poisson $\theta(Y^{-1}) = \det(Y)^{1/2}\theta(Y)$ to 10^{-16} ($n = 2, 3, 4$), with the elementary clocks E_{ij} landing 200/200 random words in $\mathrm{SL}(n, \mathbb{Z})$. A discrimination control (`delta_discriminate.py`) confirms the exact-modularity test is genuine: the true Δ theta passes at 4.5×10^{-10} while reversed, sign-scrambled, and magnitude-scrambled coefficients all fail at relative residual 1.

- **Local self-duality and unitary balance (Lemma 66), `sym_selfdual_check.py`.** For $r = 0, \ldots, 13$: $\det \mathrm{Sym}^r \operatorname{diag}(\alpha, \beta) = 1$ when $\alpha\beta = 1$ (to 5×10^{-15}), the eigenvalue multiset $\{\alpha^{r-2k}\}$ is closed under inversion (exact), and $\sum_{k=0}^{r}(r-2k) = 0$; the tensor identity $\det(A \otimes B) = (\det A)^{\dim B}(\det B)^{\dim A}$ holds to 3×10^{-15} for determinant-one blocks of sizes $(6, 4), (7, 5), (3, 2)$.

- **Symmetric powers and root numbers (§31.1), `sym_r_fe2.py`, `sym_r_fe_2clock.py`, `sym_r_sweep2.py`.** The theta self-duality (10) for $L(\mathrm{Sym}^r \Delta)$ closes with $\varepsilon = +1, -1, -1$ at $r = 1, 3, 5$; the two-clock Bessel completion makes the closed-form cases ($\mathrm{Sym}^1, \mathrm{Sym}^3$) exact to 3×10^{-30} against a grid floor 10^{-7}; the root numbers match the closed form $\varepsilon(\mathrm{Sym}^r \Delta) = i^{-\kappa(11\kappa+1)}$ ($\kappa = \lceil (r+1)/2 \rceil$), giving $+, -, -, +, +, -, -$ for $r = 1, \ldots, 13$, agreeing with the numerics for $r \leq 7$.

- **Representation-agnosticism (§32), `rep_agnostic.py`, `three_weight_clock.py`, `sym_maass.py`.** The same completion closes the functional equation for the *non-Chebyshev* Rankin–Selberg weight systems $L(\Delta \times f_{16})$, $L(\Delta \times f_{18})$, $L(f_{16} \times f_{18})$ (weights $\{\pm13, \pm2\}, \{\pm14, \pm3\}, \{\pm16, \pm1\}$) to $\sim 10^{-30}$, $\varepsilon = +1$. The three-weight clock equals the two-clock composed with a third, $(\mathrm{Bessel}_{\mu_1,\mu_2}) \star_{\mathrm{M}} \mathrm{clock}_{\mu_3}$, agreeing with the direct 3-fold convolution to 1.7×10^{-21} at arbitrary shifts. For the Maass case the two-clock goes to imaginary order: $(2x^{it}e^{-\pi x^2}) \star_{\mathrm{M}} (2x^{-it}e^{-\pi x^2}) = 4K_{it}(2\pi x)$ (the Whittaker function) to 5×10^{-29}, with Mellin $\Gamma_{\mathbb{R}}(s + it)\Gamma_{\mathbb{R}}(s - it)$ to 8×10^{-27}, at the spectral parameter $t = 9.5337$ of the first $\mathrm{SL}(2, \mathbb{Z})$ Maass cusp form.

- **Ramified local data: the filtration tower (§32), `ramified_local.py`, `wild_eps_clock.py`.** The tame local root number is the inertia-clock Gauss sum: for primitive χ mod q, $|\tau(\chi)|^2 = q$ and $|\varepsilon| = 1$ exactly. The wild ε-factor is the higher-filtration Gauss sum over $(\mathbb{Z}/p^a)^*$: $|\tau|^2 = p^a$ and $|\varepsilon| = 1$ to 1.7×10^{-15} for $(p, a) \in \{(3, 2), (3, 3), (5, 2), (7, 2), (5, 3), (11, 2)\}$ (Swan $= a - 1$); and the stationary-phase tower reduction $\tau_{\mathrm{direct}} = \tau_{\mathrm{tower}}$ (the level-2 character c from $\chi(1 + p)$, stationary $a_0 \equiv -c$) holds to 3.8×10^{-15} across $p = 3, \ldots, 13$. The monodromy N-ladder reduction $L_v = \det(1 - \mathrm{Frob}\, q^{-s} \mid \ker N)^{-1}$ is the degree drop of the Steinberg string.

- **The parallel-dimension clock (§32), `parallel_clock.py`, `parallel_dogfood.py`.** Quadratic base change $\Delta \mapsto \Delta_E$ over $E = \mathbb{Q}(i)$: the reflected degree-four $L(\Delta_E) = L(\Delta)L(\Delta \otimes \chi_{-4})$ reproduces the base-change Satake exactly (split/inert, diff 0) and closes its functional equation on the equal-shift two-clock K_0 to 9×10^{-27}, $\varepsilon = -1$. Composing clocks: base-change $\circ\, \mathrm{Sym}^3$ gives a degree-eight $L((\mathrm{Sym}^3 \Delta)_E)$ whose two factors close on the Sym-cube Bessel K_{11} ($\varepsilon = -1$ each, to 2×10^{-31} and 3×10^{-6}). The separate two-clock Bessel sweep gives the Sym^3 GL(4) standard kernel itself at $\varepsilon = -1$ with mean error 3.45×10^{-30} and spread 3.65×10^{-30}, directly closing the GL(4) standard-kernel check.

- **Signature-based cycle detection (§32), `hodge_detect.py`.** CM algebraicity: $y^2 = x^3 - x$ (CM by $\mathbb{Q}(i)$) has $a_p = 0$ at 399/399 inert primes ($p \equiv 3 \bmod 4$), against 4/399 for the non-CM $37a$. Twist-correspondence: for $g = f \otimes \chi_5$ presented as raw coefficients, the DC-residue slope of $L(f \times g \otimes \chi_d)$ over $d \in \{1, -4, 5, 8, -3, \ldots\}$ reads 0.71 at $d = 5$ and ≤ 0.04 at every other d, with an unrelated control ($37a$) ≤ 0.08 throughout — the hidden Tate class surfaced dynamically, no false positive. Recursive tower: the DC-residue slope of $L(\mathrm{Sym}^r f \times \mathrm{Sym}^r f)$ for $r = 1, \ldots, 4$ is flat $(0.90, 0.90, 0.82, 0.81)$ for the non-CM $37a$ and climbs $(0.75, 1.50, 1.35, 2.01)$ for the CM $y^2 = x^3 - x$, identically $(0.77, 1.51, 1.36, 2.03)$ for $y^2 = x^3 + 1$ over $\mathbb{Q}(\sqrt{-3})$; the CM/non-CM ratio tracks the predicted pole-order tower $1, 2, 2, 3 = \lceil (r+1)/2 \rceil$.

- **Loss-ledger ↔ cycle-filtration calibration (§32, §37), `hodge_layer_calib.py`** (from `sha_hinge.py`, `jet_census.py`, oracle-free). The leading central jet $L^{(r)}(E, 1)/r! = \Omega\,\mathrm{Reg}\,|\mathrm{Sha}| \prod_p c_p / T^2$ resolves three invariants independently: rank-0 curves land $|\mathrm{Sha}| = 1$ ($11a, 14a$), 4 ($571a, 960d$), 9 ($681b, 2849a$) as exact squares at fixed order; the tower $37a, 389a, 5077a$ lands order $1, 2, 3$ with Reg $= 0.0511, 0.1524, 0.4171$ and $|\text{Sha}| = 1$ (machine precision, known-value cross-check $\leq 2\times10^{-5}$). Order (vanishing multiplicity), Reg (Abel–Jacobi datum), and |Sha| (obstruction) vary independently — three distinct retained coordinates. The general Griffiths group is not exhausted here; the present probe tests the induction route on homologically-trivial and delayed-depth controls: `hodge_griffiths_probe.py` records the route — the value channel fires for every homologically-trivial cycle in the census (3/3 positive-order curves, 0 regulator-silent), and the Ceresa/Griffiths class is the tensor-cube *test case*, its Beilinson–Bloch height related to the triple-product central L-derivative in the theorem-anchored settings (Gross–Schoen height, Zhang [26]/Yuan–Zhang–Zhang [27]) — with the from-scratch carrier landing and extension-tower faithfulness deferred to the companion paper.

- **Delayed tower signature (§37), `tower_exhaustion_test.py`** (from point-counting, oracle-free). The excess pole order of $L(\text{Sym}^r f \times \text{Sym}^r f)$ of the CM $y^2 = x^3 - x$ over the non-CM $37a$ baseline is -0.18 at depth $r = 1$ (silent) and $+0.67$, $+0.63$, $+1.47$ at $r = 2, 3, 4$ (fires), identically for $y^2 = x^3 + 1$ — the $0, \neq 0, \ldots$ signature with first nonzero slot at depth 2, matching the tensor level Sym^2 that hosts the CM self-correspondence. Every concrete class tested fires at a finite first depth (Tate/Mordell–Weil/|Sha| at 1, CM at 2, Ceresa at 3); none silent-forever.

- **The inversion leg is not termwise (Lemma 75), `random_euler_control.py`.** The discriminating control for the axis-reality architecture: the self-dual lattice anchor obeys $\theta(1/x) = \sqrt{x}\,\theta(x)$ at machine zero ($\leq 2.2 \times 10^{-16}$, Poisson), while completely multiplicative sign systems carrying the *identical* per-term reflection algebra — Liouville $\lambda(n)$ and random-sign Euler systems — fail the summed identity by up to 1.1×10^2 at small x (where $\sim x^{-1/2}$ coefficients genuinely participate). The reciprocal-height identity is coefficient-sensitive — a resummation across the bank — so the carrier supplies it via axis reality plus continuation, never by a termwise interchange.

- **The transfer exponent (Lemma 70, Remark 74), `transfer_exponent.py`.** $\hat{\theta}$ = slope of log(running $\max|A_W|$) against $\log x$ over the top two decades, $N = 2 \times 10^5$: Δ 0.268, $\text{Sym}^5\,\Delta$ 0.352, $\text{Sym}^{13}\,\Delta$ 0.462 — all below $\frac{1}{2}$ and tracking the classical square-root-cancellation law $(d-1)/2d$ (0.25, 0.417, 0.464); endpoint $|A(N)| = 2.8$, 12.7, 156 against $\sqrt{N} = 447$. The detuned fiber keeps $\hat{\theta} = 0.263$ (a finite Euler-factor modification preserves continuation) yet provably fails axis reality (the finite cocycle, Remark 74); the random-angle fiber sits at $\hat{\theta} = 0.542$, the law-of-the-iterated-logarithm wall, and is transfer-killed. The two gates of the chain are thereby separated by dedicated counterexample classes.

- **Axis pairing: the free and the open part of the standing wave (Remark 73), `RequestProject/AxisPairing.lean`** The two-lane split of gate 2 at the ledger seam: `pairedBank_real` (the paired bank $\sum_n (K_n + \bar{K}_n)$ is real termwise on the axis — every finite bank, every fiber, no arithmetic), `readout_decomposition` (classical readout = (paired + odd)/2), `fe_iff_oddMode_eq_zero` (the axis functional equation $\Leftrightarrow$ odd lane-mode $\equiv 0$ — gate 2 restated as an extinction statement).

- **The transfer, machine-checked (Lemma 70), `RequestProject/TransferContinuation.lean`.** `transfer_tendsto`: a polynomially bounded primitive $\|\sum_{k<n} a_k\| \leq C\,n^{\theta}$ forces convergence of the partial sums of $\sum a_n(n+1)^{-s}$ at every s with $\operatorname{Re} s > \theta \geq 0$ — the sharp transfer of Lemma 70(i), proved by formalized summation by parts (`Finset.sum_range_by_parts`) against the mean-value kernel `cpow_sub_bound` ($\|(k+1)^{-s} - k^{-s}\| \leq \|s\| k^{-\operatorname{Re} s - 1}$) and the p-series tail; with `dual_primitive_norm` (the unitary dual carries the same exponent) and `strip_contains_axis` ($\theta < \kappa/2$ places the weld axis in the common strip). No automorphy, functional equation, or Poisson input appears in the file. Footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`.

- **The grown transfer exponent (Lemma 71), `grown_transfer.py`.** The native primitive $A^{\mathrm{e}}_W(x) = \sum \lambda_n e^{-n/x}$ (no clipped edge), $x \in [10, N/40]$, $N = 2 \times 10^5$: $\hat{\theta}^{\mathrm{e}} = 0.002$ for Δ — *bounded*, plateau 0.73, the automorphy-forced convergence $A^{\mathrm{e}} \to L(0)$ of Lemma 71(ii) observed directly — 0.290 for $\mathrm{Sym}^5\,\Delta$ and 0.403 for $\mathrm{Sym}^{13}\,\Delta$ (every sharp margin widened), detuned 0.065 (finite Euler modification: still essentially bounded), random 0.581 (smoothing manufactures no cancellation; the gate bites). The sharp-cutoff Chandrasekharan–Narasimhan barrier at degree ≥ 4 binds the clipped primitive only: measured, the clip was the barrier. Window-drift of the grown slope (`grown_transfer.py -median`, 4 log-windows): Δ 0.040/0.008/0.002/0.000 (plateau); Sym^5 late windows 0.252/0.050 (equilibrating — the global 0.29 is a finite-window transient); Sym^{13} oscillatory, not yet equilibrated at $N = 2 \times 10^5$ (fastest channels $e^{\pm 13 i\theta}$); random 0.43/0.58/0.76/0.49 — about $\frac{1}{2}$ in every window, never equilibrating (Remark 72).

- **Adapted closure is arithmetic-blind (Lemma 75 Step 2, §31), `detuned_closure_control.py`.** The register entry that localizes the arithmetic inside the continuation leg: the true Δ anchor, a *detuned* Δ ($\theta_2 \leftrightarrow \theta_3$ swapped, multiplicatively rebuilt), and a fully *random*-angle fiber of the same duality-stable class all force-close 30/30 native cells at the float64 floor (medians 3.2–36×10^{-15}, weights ≤ 2.91). The random-angle control shows that forced cell closure alone cannot deliver continuation of the ordinary readout. The arithmetic-sensitive step is the *warp-removal transfer* of §29. What is machine-checked by `TransferContinuation.transfer_analytic` is continuation for the coefficient sequence whose own primitive is bounded. The valid n-dependent inverse-kernel interface is machine-checked in `WarpRemovalTransfer.warpRemoval_transfer_tendsto`; applying it requires its summable first-difference and vanishing-boundary conditions for the selected fiber warp. Cell closure from `ForcibleClosure.residual_forcible` supplies neither condition by itself.

- **Twisted CPS instances (§31.1), `twisted_cps_instance.py`.** The objects Theorem 45 actually consumes, tested directly. *Factorizing control:* the degree-12 twisted fiber $L(\mathrm{Sym}^5\,\Delta \times \Delta)$, built on the carrier, reproduces the standard-side product $L(\mathrm{Sym}^6\,\Delta)\,L(\mathrm{Sym}^4\,\Delta)$ coefficientwise to 1.8×10^{-14} — the twisted pipeline validated against known quantities. *Non-factorizing instance:* $L(\mathrm{Sym}^5\,\Delta \times E_{11})$, degree 12 with two independent Satake angles, does not reduce to symmetric powers of one form; no channel is DC (every channel carries an odd θ_Δ-weight), and the multi-rail bank closes to precision (from 4×10^{-28} at 30 digits to 9×10^{-88} at 90) — entireness on the twisted family itself, not only the standard side.

- **The Brauer-zero trap (§34), `brauer_zero_trap.py` (Sage).** The adversarial removal test: the Hamilton class $(-1,-1)_{\mathbf{Q}}$ has $\sum_v \mathrm{inv}_v = 0$ (aggregate readout closed) yet $[A] \neq 0$. Five passes, blind (no splitting field given): detect the retained obstruction despite the zero sum; classify it 2-torsion at $\{2, \infty\}$; synthesize the minimal splitting chart $\mathbf{Q}(\sqrt{-3})$ from the syndrome; verify $\mathrm{res}[A] = 0$ by direct Brauer computation; certify R=0 and Obs =0 jointly. A second class ramified at $\{3, 5\}$ separates a hard-coded "adjoin $\sqrt{-1}$" (splits Hamilton, fails $\{3, 5\}$) from the syndrome search (repairs both). Every stage cross-checked against Sage's quaternion-algebra/Hilbert-symbol oracle; the removal logic (zero aggregate $\neq$ removal; even local degree kills a 2-torsion class; certificate = the conjunction) is machine-checked in `RequestProject/BrauerZeroTrap.lean`.

- **Genuine non-self-dual** GL(3) **(§23, table row), `f21_gl3_multirail.py` (data via Sage `idealfrobenius`) and `RequestProject/F21NonSelfDual.lean`.** The degree-three complex Artin representation of

$F_{21} = C_7 \rtimes C_3$ (field $x^7 - 14x^5 + 56x^3 - 56x + 22$). Order-seven Satake $\{\zeta_7, \zeta_7^2, \zeta_7^4\}$ (Gauss period η) or its conjugate (η'); classes at $p = 13, 41$ (both $\equiv 6 \bmod 7$) genuinely split η'/η. Multi-rail cell closure of the non-DC rails tracks precision, 1.5×10^{-30} (30 digits) to 4.3×10^{-62} (60), $\prod$ rails $= \zeta_7^{1+2+4} = 1$. Lean: exact cell closure of both classes, $\eta + \eta' = -1$, $\eta\eta' = 2$, $\eta \neq \eta'$, and $\operatorname{Re}\eta = \operatorname{Re}\eta'$ with $\eta \neq \eta'$ (the scalar obstruction).

- **Forcible cell closure (Lemma 65), `forcible_closure.py` and `RequestProject/ForcibleClosure.lean`.** On *native* growth cells — uniform in $y = \log Z$, each phasor entering at zero magnitude through the canonical C^∞ window `focal_closure.growth_window`, no clipped edge — exact Gauss–Newton over the coherent generators closes the nonlinear cell residual to the float64 floor on *every* cell: with $(\Omega, \omega, \log n, \Theta)$, $\Theta(n) = \sum_{p^k \| n} k\, \theta_p$ the fiber's own local periods, 30/30 cells close on each of Δ, $\mathrm{Sym}^5\,\Delta$, $\mathrm{Sym}^{13}\,\Delta$ — median $|D_C| = 4.9 \times 10^{-15}, 1.4 \times 10^{-14}, 7.8 \times 10^{-15}$, max relative residual $\leq 3.6 \times 10^{-14}$, weights $|x| \leq 1.28, 3.07, 2.27$ (integer-generator phases mod 2π). Three generators alone close 30/30 on Δ, $\mathrm{Sym}^5\,\Delta$ and 29/30 on $\mathrm{Sym}^{13}\,\Delta$; the one rank-deficient cell is closed by adjoining Θ — the resisting cell naming its adapter. One cell's weights leave the next cell's residual at its own scale (10–38% of that cell's mass) — cross-cell independence. The sharp-window clip control (retained in the script) manufactures 1–3 resistant cells per fiber with exploding weights: the never-clip method law made measurable. Lean `residual_forcible`: two $\mathbb{R}$-independent directions span $\mathbb{C}$, so any residual is forcible to zero; `CarrierReachability.reachable_independent_stage`: a continuous non-constant lane phase attains $\sin(2\phi) \neq 0$ (a connected image cannot lie in the discrete lock $\frac{\pi}{2}\mathbb{Z}$), supplying the pair — the two are composed by direct application, no further Lean theorem claimed.
- **Carrier-transport functoriality (§32), `RequestProject/CarrierFunctoriality.lean` and `composition_test.p`** `faithful_comp` (the composite of closure-preserving transports preserves closure, *no* axioms) with associativity, two-sided identity, and base change $\mathrm{bc}_p \circ \mathrm{bc}_q = \mathrm{bc}_{pq}$; the script verifies exact plethysm splitting $\mathrm{Sym}^2 \circ \mathrm{Sym}^2 = \mathrm{Sym}^4 \oplus \mathrm{Sym}^0$, $\mathrm{Sym}^3 \circ \mathrm{Sym}^3 = \mathrm{Sym}^9 \oplus \mathrm{Sym}^5 \oplus \mathrm{Sym}^3$.
- **Tensor-induction residual (functorial preservation, §31), `tensor_induction_check.py`.** On real Δ Satake: the Clebsch–Gordan factorization $a_{\pi \times \mathrm{Sym}^4} = a_{\mathrm{Sym}^5} \star a_{\mathrm{Sym}^3}$ holds coefficientwise to 1.4×10^{-14}; mean$(a_{\mathrm{Sym}^5}) \to 6 \times 10^{-5}$ while $L(\mathrm{Sym}^3)(1) = 0.652 \neq 0$ (the trivial-rep control gives mean 0.652, so the deduction is non-vacuous).

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